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Par **Bastien HUBERT**

Cartographie et Navigation Autonome sur Champs Physiques par Filtrage Particulaire à Processus Gaussiens

Autonomous Navigation and Mapping over Physical Fields
by Gaussian Process-based Particle Filtering

Sous la direction de : **Audrey GIREMUS**
la co-direction de : **Karim DAHIA**
et l'encadrement de : **Nicolas MERLINGE**

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Membres du jury :

Manon KOK	Professeure associée	TU Delft	Rapporteure
Carl MILNER	Professeur associé	ENAC	Rapporteur
Vincent FRÉMONT	Professeur	Centrale Nantes	Examinateur
Guillaume BOURMAUD	Professeur associé	Bordeaux INP	Examinateur
Pierre MINVIELLE	Ingénieur de recherche	CEA	Examinateur
Audrey GIREMUS	Professeure	Université de Bordeaux	Directrice
Karim DAHIA	Ingénieur de Recherche	ONERA	Co-directeur
Nicolas MERLINGE	Ingénieur de Recherche	ONERA	Encadrant

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ABSTRACTS

Français

La navigation autonome d'un véhicule repose généralement sur la fusion de mesures inertielles avec des données de recalage externes. Lorsque le positionnement par satellite est impossible ou peu fiable, les champs physiques environnants peuvent constituer une source d'information complémentaire pour estimer la position du véhicule. Leur exploitation est toutefois difficile en raison des non-linéarités et ambiguïtés de la fonction d'observation, ainsi que d'une connaissance imparfaite du champ considéré.

Cette thèse étudie la navigation autonome sur champs physiques dans ce contexte. L'approche proposée combine la robustesse du filtrage particulaire bayésien face aux distributions multimodales avec la capacité des processus gaussiens à modéliser le champ observé tout en quantifiant l'incertitude associée à sa reconstruction.

Deux cas sont distingués selon la connaissance a priori du champ. Lorsque celui-ci n'est accessible qu'au travers de données éparses et bruitées, le krigeage permet de reconstruire la fonction d'observation. Des stratégies de sous-échantillonnage sont développées afin d'adapter la reconstruction au champ local tout en réduisant son coût de calcul. D'autre part, en l'absence de donnée préalable, le champ est reconstruit conjointement à la localisation du véhicule.

Ces travaux présentent ainsi de nouvelles méthodes pour la navigation autonome sur champs physiques, prenant en compte à la fois les ambiguïtés et la méconnaissance du modèle de champ, depuis l'exploitation de données dégradées jusqu'à la reconstruction du champ au cours de la navigation.

English

Autonomous vehicle navigation generally relies on the fusion of inertial measurements with external positioning data. When satellite-based positioning is unavailable or unreliable, surrounding physical fields can provide a complementary source of information for estimating the vehicle's position. However, exploiting such fields is challenging due to the non-linearities and ambiguities of the observation function, as well as imperfect knowledge of the field itself.

This thesis focuses on autonomous navigation over physical fields in this context. The proposed approach combines the robustness of Bayesian particle filtering

to multimodal distributions with the ability of Gaussian processes to model the observed field while quantifying the uncertainty associated with its reconstruction.

Two cases are considered depending on the availability of prior knowledge of the field. When it is only accessible through sparse and noisy data, kriging can be used to reconstruct the observation function. Subsampling strategies are developed to adapt the reconstruction to the local field while reducing its computational cost. Conversely, in the absence of prior data, the field is reconstructed jointly with the vehicle's localisation.

These contributions introduce new methods for autonomous navigation over physical fields that account for both ambiguities and incomplete knowledge of the field model, ranging from the exploitation of degraded data to field reconstruction during navigation.

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NOMENCLATURE

Abbreviations

AK	Adaptive Kriging
AKKF	Adaptive Kernel Kalman Filter
APF	Auxiliary Particle Filter
ARMSE	Averaged Root Mean Squared Error
a.s.	almost surely
BLUE	Best Linear Unbiased Estimator
CAK	Clustered Adaptive Kriging
CDF	Cumulative Distribution Function
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
EKF	Extended Kalman Filter
flop	Floating-Point Operations
GP	Gaussian Process
HCA	Hierarchical Clustering Analysis
ICAK	Independently Clustered Adaptive Kriging
IEKF	Iterative Extended Kalman Filter
i.i.d.	independent and identically distributed
KKF	Kernel Kalman Filter
KL	Karhunen-Loève
KME	Kernel Mean Embedding
MC	Monte Carlo
MCAK	Merged Clustered Adaptive Kriging
MAP	Maximum *A Posteriori*
MISE	Mean Integrated Squared Error
ML	Maximum Likelihood
MMAE	Minimum Mean Absolute Error
MMSE	Minimum Mean Squared Error
NNAK	Nearest-Neighbour Adaptive Kriging
OK	Ordinary Kriging
PDF	Posterior Probability Density Function
PF	Particle Filter
RBPF	Rao-Blackwellised Particle Filter
RFF	Random Fourier Features

RKHS . . . . . Reproducing Kernel Hilbert Space
RLS . . . . . Recursive Least Squares
RMSE . . . . . Root Mean Squared Error
RPF . . . . . Regularised Particle Filter
RS . . . . . Random Subsampling
SE . . . . . Squared Exponential
SIR . . . . . Sampling Importance Resampling
SIS . . . . . Sequential Importance Sampling
SK . . . . . Simple Kriging
SLAM . . . . . Simultaneous Localisation and Mapping
SPD . . . . . Symmetric Positive Definite
SPSD . . . . . Symmetric/Self-adjoint Positive Semi-Definite
UK . . . . . Universal Kriging

Symbols

$\triangleq$ . . . . . Equality by definition
$\propto$ . . . . . Proportionality
$\sim$ . . . . . "is distributed according to"
$\overset{i.i.d.}{\sim}$ . . . . . Independently and identically distributed according to
$\xrightarrow{a.s.}$ . . . . . Almost sure convergence
$\xrightarrow{\mathcal{L}}$ . . . . . Convergence in law
$\cdot \succeq 0$ . . . . . Symmetric positive semi-definite
$\cdot \succ 0$ . . . . . Symmetric positive definite
$\langle \cdot | \cdot \rangle, \langle \cdot | \cdot \rangle_{\mathcal{E}}$ . . . . . Inner product, subscript: of the space $\mathcal{E}$
$\langle \cdot, \cdot \rangle$ . . . . . Measure/function duality pairing, $\langle \mu, \varphi \rangle = \int \varphi \, d\mu$
$\| \cdot \|, \| \cdot \|_2, \| \cdot \|_F, \| \cdot \|_{\mathrm{op}}$ . . . Generic, Euclidean, Frobenius and operator norms
$\cdot^\top$ . . . . . Transpose of a matrix
$\cdot^{-1}$ . . . . . Inverse of a matrix
$\cdot^\dagger$ . . . . . Moore-Penrose pseudo-inverse of a matrix
$*$ . . . . . Convolution operator
$\circ$ . . . . . Function composition operator
$\otimes$ . . . . . Kronecker product, tensor product of Hilbert spaces
$\mathbb{1}$ . . . . . Indicator function
$[\![\cdot, \cdot]\!]$ . . . . . Discrete interval
$\cdot_{m|n}$ . . . . . "at time m conditioned on the observations from 1 to n"
$\alpha, \hat{\alpha}_k$ . . . . . Map coefficients and their estimate
$\alpha_{\mathrm{reg}}, \alpha_{\mathrm{cov}}$ . . . . . Regularisation and window dilation coefficients
β . . . . . Forgetting factor
Γ . . . . . Gamma function
δ_x . . . . . Dirac measure at x
$\varepsilon, \varepsilon_i$ . . . . . Query and training noises
θ . . . . . Kernel hyperparameters
$\lambda_k^{(i)}$ . . . . . Auxiliary weights
$\lambda_{\min}, \lambda_{\max}, \lambda_i$ . . . . . Smallest, largest and i-th eigenvalues of a matrix
$\lambda_x, \lambda_z, \lambda_{\mathrm{RLS}}$ . . . . . Regularisation coefficients
μ . . . . . Target probability measure, of density p

$\mu_k^{(i)}$ . . . . . Auxiliary particle
$\mu_X, \mu_{XZ}, \mu_{Z|x}$ . . . . . KME, joint KME and conditional KME
$\mu_{Z|X}$ . . . . . Conditional embedding operator
$\mu_{k|k-1}, \mu_{k|k}$ . . . . . Embeddings of the predicted and filtering densities
ν . . . . . Proposal (importance) measure, of density q
π_k . . . . . Projection of a trajectory onto its last component
Π_{pos} . . . . . Projection matrix onto the position subspace
ρ . . . . . Distance on a space of estimators
$\Sigma_{k|k-1}, \Sigma_{k|k}$ . . . . . Covariances of the embedded weights
φ . . . . . Mapping of the state
$\hat{\psi}_{\text{MMSE}}, \hat{\psi}_{\text{MMAE}}, \hat{\psi}_{\text{MAP}}, \hat{\psi}_{\text{ML}}$ . . . . . State estimators
ψ_i, Ψ, Λ . . . . . Mercer eigenfunctions and eigenvalue matrix
Φ . . . . . Feature map of an RKHS
ω_j, b_j . . . . . Frequencies and phases of the basis functions
argmin, argmax . . . . . Argument of the minimum, of the maximum
$b_k^{(e)}, b_k^{(o)}$ . . . . . Evolution and observation bias vectors
C_μ, C_ν . . . . . Normalising constants of μ and ν
cdf . . . . . Cumulative distribution function
d_x, d_z, d_φ . . . . . Dimensions of the state, observation, and mapping
d_w, d_n . . . . . Dimensions of the evolution and observation noises
d_{pos} . . . . . Dimension of the position subspace
D, D_k . . . . . Residual weights of the embedded transition
$\mathfrak{D}_G$ . . . . . Gram inverse difference matrix
det . . . . . Determinant of a matrix
e_r, e_f, E_r . . . . . Subsampling errors and error covariance
$\mathbb{E}[\cdot]$, $\mathbb{E}_p[\cdot]$ . . . . . Expectation, subscript: under the law p
exp . . . . . Exponential function
$f_k, f_k^l, f_k^n, \overline{f}_k^l$ . . . . . Evolution functions
$F_k, F_k^l, F_k^n, \overline{F}_k^l$ . . . . . Evolution matrices
g, g_l . . . . . Trend functions
$\mathcal{GP}(m, k)$ . . . . . Gaussian process
G, G_θ . . . . . Gram matrix
$\mathcal{G}, \mathcal{G}_h$ . . . . . Bordered and hollowed-out Gram matrices
h_k . . . . . Observation function
H_k . . . . . Observation matrix
I, I_n . . . . . Identity matrix, subscript: of size n
$\mathbf{1}_n$ . . . . . Vector of ones of size n
J_f . . . . . Jacobian matrix of a function f
$k(\cdot,\cdot), k_{\text{SE}}, k_\nu, k_{\text{sep}}, k_{\text{MS}}, k_B, k_{r_{\text{KL}}}$ . . . . . Covariance kernels
$K, K_h, K_{\text{Epa}}, K_{\text{Gauss}}$ . . . . . Regularisation kernels
K_ν . . . . . Modified Bessel function of the second kind
$k_{d|r}, \tilde{\mathbf{z}}_{d|r}, S_{d|r}$ . . . . Residual covariance function, observation vector and Schur complement
K_k . . . . . Kalman gain
$\mathcal{K}, \mathcal{K}_{r_{\text{KL}}}$ . . . . . Mercer operator and its reduced-rank truncation
$\mathcal{L}^0(E,F), \mathcal{L}^p(E,F)$ Sets of measurable and p-integrable functions from E to F
m . . . . . Mean function of a Gaussian process
$\hat{m}_{\text{OK}}, \hat{m}_{\text{UK}}$ . . . . . Estimated ordinary and universal-kriging means

M . . . . . Covariance of the estimated trend coefficients
$\mathcal{M}_{m,n}(\mathbb{R})$ . . . . . Set of real $m \times n$ matrices
n_k . . . . . Observation noise
$N, \tilde{N}$ . . . . . Number of particles, of training points
$\tilde{N}^r, \tilde{N}_{\min}$ . . . . . Number of retained training points and its floor
$\hat{N}_{\text{eff}}, N_{\text{eff}}$ . . . . . Estimated and real effective sample size
N_{th} . . . . . Resampling threshold
$\mathbb{N}, \mathbb{N}^*$ . . . . . Sets of natural and strictly positive natural integers
$\mathcal{N}(m, P)$ . . . . . Gaussian distribution
$o(\cdot)$ . . . . . Landau symbol for asymptotic negligibility
O . . . . . Observation matrix of the embedded weights
$\mathcal{O}(\cdot)$ . . . . . Landau symbol for asymptotic boundedness
p . . . . . Target probability density
p_0 . . . . . Probability density of the initial state
$p_k^{(e)}, p_k^{(o)}$ . . . . . Probability densities of the evolution and observation noises
$p_{A|B}(a|b)$ . . . . . Conditional density of A given B
$P_0, P_{k|k-1}, P_{k|k}$ . . . . . State covariance matrices
P_k^α . . . . . Covariance of the map coefficients
$\mathbb{P}$ . . . . . Probability of an event
q . . . . . Proposal (importance) density
$Q_k, Q_k^n, Q_k^l, Q_k^{nl}, \overline{Q}_k^l$ . . . . . Evolution noise covariances
$r_{\text{UK}}, r_{\text{KL}}, r_{\text{RFF}}$ . . Number of trend functions, KL rank and number of features
$r_{\text{cov},k}, r_{\min,k}$ . . . . . Covariance and minimum-count window radii
rank . . . . . Rank of a kernel or operator
$R, R_k, \tilde{R}$ . . . . . Observation and training noise covariances
$R_{\text{SK}}, R_{\text{OK}}, R_{\text{UK}}, R_{\text{K}}, R_{\hat{m}}$ . . . . . Kriging error covariances
$\mathbb{R}, \mathbb{R}_+, \mathbb{R}_+^*$ . . . . Sets of real, positive real, and strictly positive real numbers
$S^N(\mu), S^N(\mu, \nu), \tilde{S}^N, S_{K_h}^N$ . . . . . Monte Carlo measures
S_k . . . . . Innovation covariance
S_ν, p_ν . . . . . Spectral density of k_ν and its normalisation
$\mathcal{S}_d^+, \mathcal{S}_d^{++}$ . . . . . Sets of SPSD and SPD matrices
supp . . . . . Support of a measure or function
tr . . . . . Trace of a matrix
T, T_k . . . . . Transition matrices of the embedded weights
U . . . . . Weights of the estimated trend coefficients
$\mathcal{U}$ . . . . . Input set of a stochastic process
$\mathcal{U}_X$,$\mathcal{U}_Z$ . . . . . Sets of values of the random variables X and Z
$\mathcal{U}([a, b])$ . . . . . Uniform distribution
V . . . . . Residual covariance of the embedded transition
$\mathbb{V}[\cdot]$, $\mathbb{V}_\mu[\cdot]$ . . . . . Variance, subscript: under the measure μ
$w_k, w_k^l, w_k^n, \overline{w}_k^l$ . . . . . Evolution noises
$w_k^{(i)}, \tilde{w}_k^{(i)}$ . . . . . Normalised and unnormalised particle weights
$\mathbf{w}_{k|k-1}, \mathbf{w}_{k|k}, \mathbf{w}_k^+$ . . . . Predicted, corrected and resampled embedded weights
$x_k, x_k^n, x_k^l, \mathbf{x}_{0:k}$ . . . . . System states and trajectory
$\mathbf{x}, \mathbf{x}', \tilde{\mathbf{x}}, \tilde{\mathbf{x}}_k^r$ . . . . . Input tuples, training inputs and their retained subset
$\tilde{\mathbf{x}}^-$ . . . . . Training states preceding $\tilde{\mathbf{x}}$
$x_k^{(i)}$ . . . . . Particle
$\mathbf{x}_k, \mathbf{x}_k^+, \mathbf{z}_k$ . . Propagated and resampled particle sets, simulated observations

$x_{k|k-1}, x_{k|k}, \hat{x}_k$ . . . . . Predicted, corrected and estimated state
y_k . . . . . Innovation vector
$z_k, \mathbf{z}_{1:k}$ . . . . . Observation vector and sequence
$z, \tilde{\mathbf{z}}$ . . . . . Query and training observations
$\hat{z}, \hat{z}_{\mathrm{SK}}, \hat{z}_{\mathrm{OK}}, \hat{z}_{\mathrm{UK}}, \hat{z}_{\mathrm{K}}$ . . . . . Kriging estimators

CHAPTER 1
GENERAL INTRODUCTION

1.1 Motivation and Context

In order for an unmanned vehicle to move autonomously from one point to another, the integration of several disciplines is required, ranging from motor control to trajectory planning. This work focuses on autonomous navigation, which corresponds to the estimation of the vehicle's state, including its position, angular orientation and velocity, using only the information available to the vehicle at any given time.

One of the simplest ways to do this is through the use of inertial measurement units or wheel encoders, when applicable, to estimate the odometry of the vehicle over short periods of time. While this solution is acceptable for short missions, the inertial measurements are imperfect due to bias or scale factors. Since inertial navigation integrates these measurements once or twice, the resulting solution drifts over time, and cannot be used for longer missions. In order to keep the inertial solution bounded, data fusion algorithms can use an additional source of information to periodically correct the estimated state of the vehicle.

In open environments, this is usually achieved by using the Global Navigation Satellite System (GNSS). In many situations, however, such as underwater, underground or indoors, GNSS data may be unavailable. Moreover, signal interference, jamming or spoofing may prevent reliable GNSS-based navigation algorithms.

In GNSS-denied scenarios, navigation can instead be aided by the physical fields through which the vehicle travels, such as sea floor depth, terrain or anomalies in the ambient gravitational or magnetic field. Given a map of such a field, the values measured by an embedded sensor along the trajectory can be compared with the map to estimate the likely position of the vehicle.

Physical field-aided navigation falls under the scope of Bayesian estimation methods, the most famous of which is the Kalman filter. In practice, however, two main issues prevent the use of this algorithm. First, the observed physical field may be strongly non-linear and non-injective, so that a single observation may remain compatible with several, possibly distant, hypotheses about the vehicle state. This can result in highly multimodal state distributions that cannot be adequately represented by a Kalman filter. Second, the initial state of the vehicle may be poorly known at the beginning of the mission, which may prevent localised approximations

from converging towards the true state.

An alternative to Kalman filtering are particle filters, which use Monte Carlo approximations to represent the probability density of the vehicle state throughout the mission. While these methods are robust to multimodal densities and high initial uncertainty, they suffer from two problems, known as weight degeneracy and particle impoverishment, potentially limiting their ability to operate reliably over long missions. More advanced particle filters, used throughout this work, are thus required.

Although particle filters provide a reliable solution to the navigation problem, they require an embedded map of the environment. When no such map is available, the observation model is itself unknown and must be inferred during the mission. In this work, we examine two scenarios depending on the amount of prior information available about the environment.

The first one arises when the environment is poorly known, and the vehicle only has access to a degraded map consisting of sparse and noisy measurements of the field. In this case, Gaussian processes provide a probabilistic framework for representing the field and quantifying the associated uncertainty. The available data can then be used to reconstruct its spatial distribution through kriging. However, potential non-stationarity in the observed field and the computational cost of standard kriging methods pose significant challenges to their use in real-time outdoor navigation.

The second scenario arises when no prior data are available at all, so that the field must be mapped during the mission itself, jointly with the localisation of the vehicle on it. Additional assumptions, such as local stationarity, can be made on the field to allow parametric representations of the observation function. The associated coefficents can then be learnt online, alongside the estimation of the vehicle's state.

The contributions of this thesis are thus threefold:

- The exploration of Gaussian process-based representations of physical fields and the quantification of the resulting approximation errors.
- The use of this representation to develop a computationally tractable particle filter-based navigation solution in poorly known environments.
- The extention of these methods to navigation without prior knowledge of the field, through its online reconstruction concurrently with the estimation of the vehicle's state.

1.2 Thesis structure

This document is organised into two parts and six chapters. Aside from the general introduction, the first two chapters present mathematical concepts relative to Bayesian estimation and Gaussian processes respectively.

In particular, chapter 2 first introduces point estimators of probability densities, then the optimal recursive Bayesian filter. The Kalman filter, corresponding to the linear-Gaussian case, is detailed next, along with the approximations that extend it to non-linear systems. The chapter then turns to Monte Carlo methods to handle more general distributions, by approximating the posterior by a weighted sample.

Finally, it presents advanced particle filters, including the regularised and Rao-Blackwellised variants used throughout this work.

Chapter 3 starts by describing fundamental properties of Gaussian processes and introducing commonly used covariance kernels. It also presents Gaussian process regression under increasingly relaxed assumptions on the mean function, and examines reduced-rank approximations of the kernel. Finally, reproducing kernel Hilbert spaces are introduced as a complementary perspective on these methods.

The second part applies these tools to the navigation problem and is structured according to the level of prior knowledge available about the observed field.

Chapter 4 considers the case where a limited set of data is available to the vehicle to characterise the observed field, and uses this data to embed a kriged observation function within a particle filter. It then develops adaptive subsampling strategies that rebuild the field from the relevant data alone, lowering the computational cost while improving the local reconstruction. Finally, it establishes theoretical bounds on the error introduced by this subsampling.

In turn, chapter 5 addresses the more challenging case in which no prior training data are available, requiring the field to be estimated jointly with the vehicle's state. The proposed approach combines a reproducing kernel Hilbert space formulation of the Kalman filter with a random Fourier features approximation of the observation function. This method is then compared with an existing Rao-Blackwellised framework.

Finally, chapter 6 concludes this document by summarising the contributions of the thesis and discussing perspectives.

1.3 Publications

The work presented in this thesis has led to the following publications:

[1] B. Hubert, K. Dahia, N. Merlinge, and A. Giremus. "Adaptive Kriging Particle Filter and its Application to Terrain-Aided Navigation". In: *2024 27th International Conference on Information Fusion (FUSION)*. 2024, pp. 1–7.

[2] B. Hubert, N. Merlinge, K. Dahia, and A. Giremus. "A Clustered Approach to Adaptive Kriging Particle Filter for Magnetic Field-Aided Navigation". In: *2025 33rd European Signal Processing Conference (EUSIPCO)*. 2025, pp. 1–5.

[3] B. Hubert, N. Merlinge, K. Dahia, and A. Giremus. "Learning Observation Model for SLAM via Random Fourier Features in a Kernel Kalman Filter". In: *2026 29th International Conference on Information Fusion (FUSION)*. 2026, pp. 1–8.

Part I

Mathematical Foundations

CHAPTER 2

BAYESIAN ESTIMATION

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2.1 Introduction

The goal of this chapter is to formally introduce the tools necessary to solve the general filtering problem. More precisely, given a dynamical system whose state evolves according to a known stochastic model and is partially observed through noisy measurements, we seek to sequentially estimate the system state at each time step in an optimal sense. We first define what optimality means in a rigorous setting, then derive exact and approximate solutions to this problem. We also discuss the limitations of such methods.

Formally, for $d_x, d_z, d_\varphi \in \mathbb{N}^*$, we consider a hidden Markov chain $(x_k)_{k\geq 0}$ taking values in $\mathbb{R}^{d_x}$ and a series of observations $(z_k)_{k\geq 1}$ taking values in $\mathbb{R}^{d_z}$, each associated with the unknown state x_k. The quantity of interest is a mapping of the state under a known function $\varphi : \mathbb{R}^{d_x} \to \mathbb{R}^{d_\varphi}$, which allows us to target any derived quantity of the state (*e.g.* position, velocity, or a subset of coordinates). When solving a navigation problem, the state and observation processes are linked by the state-space model [1]:

$$\begin{cases} x_k = f_k(x_{k-1}, w_k) \\ z_k = h_k(x_k, n_k) \end{cases} \tag{2.1}$$

where the processes $w_k \in \mathbb{R}^{d_w}$ and $n_k \in \mathbb{R}^{d_n}$ denote the evolution and observation noises respectively, and are assumed to be white noises (*i.e.* temporally independent, centered noises of finite variance), mutually independent and independent of x_0.

The known data of this problem are:

$$\begin{cases} p_0 & \text{the probability density of the initial state } x_0 \\ f_k & \text{the evolution function} \\ p_k^{(e)} & \text{the probability density of the evolution noise } w_k \\ z_k & \text{the current noisy observation} \\ h_k & \text{the observation function} \\ p_k^{(o)} & \text{the probability density of the observation noise } n_k \end{cases} \tag{2.2}$$

Remark 2.1 (Control input)**.** *In many navigation applications, a control input u_k is also available, and the evolution function takes the form $f_k(x_{k-1}, u_k, w_k)$. For the sake of clarity, we omit u_k from the general formulation as it can always be absorbed into f_k or treated as a known deterministic input.*

Depending on how observations are used relative to the time index of the state to estimate, three problems arise [2]:

- **Filtering:** estimate $\varphi(x_k)$ given $\mathbf{z}_{1:k} \triangleq (z_1, \dots, z_k)$. It is the most common case in real-time navigation.
- **Smoothing:** estimate $\varphi(x_l)$ for $l < k$ given $\mathbf{z}_{1:k}$. It corresponds to refining past estimates using future observations.
- **Prediction:** estimate $\varphi(x_l)$ for $l > k$ given $\mathbf{z}_{1:k}$. This amounts to anticipating future states from current observations.

This thesis focuses on the filtering problem. The rest of this chapter is organised as follows:

- Section 2.2 rigorously defines what it means to optimally estimate $\varphi(x_k)$ from the observations $\mathbf{z}_{1:k}$, and shows that all classical optimality criteria reduce to computing the posterior probability density function (PDF) $p(x_k|\mathbf{z}_{1:k})$.
- Section 2.3 derives analytical filtering algorithms to propagate this density over time, and discusses their limitations.
- Section 2.4 then presents asymptotically optimal numerical methods based on Monte Carlo (MC) sampling that alleviate such flaws.
- Finally, section 2.5 presents advanced particle filtering methods that improve upon standard MC techniques.

2.2 State Estimation

The first issue to tackle is to define precisely the meaning of "best estimator" of $\varphi(x_k)$ given $\mathbf{z}_{1:k}$ [3]. Formally, given a functional metric space $(\mathcal{F}, \rho)$, it amounts to finding a measurable function $\psi \in \mathcal{F}$ that minimises the expected loss:

$$\underset{\psi\in\mathcal{F}}{\operatorname{argmin}}(\rho(\psi(\mathbf{z}_{1:k}), \varphi(x_k))) \tag{2.3}$$

Depending on the considered space and distance, different optimal estimators arise. We present three classical ones below. For the sake of simplicity, we write Z for a generic observation vector and X for the underlying state, as opposed to the time-indexed processes x_k and z_k. At a fixed time step k, (2.3) thus becomes:

$$\underset{\psi\in\mathcal{F}}{\operatorname{argmin}}(\rho(\psi(Z), \varphi(X))) \tag{2.4}$$

For the remainder of this document, we let, for $p \in \mathbb{N}^*$, E a measure space and F a normed vector space:

$$\begin{aligned} \mathcal{L}^0(E,F) &\triangleq \{f : E \to F \mid f \text{ measurable}\} \\ \mathcal{L}^p(E,F) &\triangleq \left\{ f \in \mathcal{L}^0(E,F) \;\middle|\; \int_E \|f\|^p < \infty \right\} \end{aligned} \tag{2.5}$$

where almost everywhere equal functions with respect to the Lebesgue measure on E are identified.

2.2.1 Posterior Expectation Estimator

Let:

$$\mathcal{F} \triangleq \left\{ \psi \in \mathcal{L}^0\left(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi}\right) \;\middle|\; \mathbb{E}\left[\|\psi(Z)\|^2\right] < \infty \right\} \tag{2.6}$$

To simplify notations, we identify each estimator $\psi \in \mathcal{F}$ with the random variable $\psi(Z)$ it produces. Under this identification, $\mathcal{F}$ is equal to $\mathcal{L}^2\left(\mathbb{R}^{d_z}, \mathbb{R}^{d_\varphi}\right)$, and can be equipped with the inner product:

$$(A, B) \in \mathcal{F}^2 \mapsto \langle A|B\rangle \triangleq \mathbb{E}\left[A(Z)^\top B(Z)\right] \tag{2.7}$$

where $\mathbb{E}[X]$ denotes the expectation of the random variable X.

Let $\|\cdot\|$ be the norm associated with $\langle\cdot|\cdot\rangle$. When $\varphi(X)$ is square-integrable, that is $\mathbb{E}\left[\|\varphi(X)\|^2\right] < \infty$, the minimisation problem (2.4) is equivalent to:

$$\operatorname*{argmin}_{\psi\in\mathcal{F}} \left(\mathbb{E}\left[\|\psi(Z)-\varphi(X)\|^2\right]\right) \tag{2.8}$$

An estimator solving (2.8) is called a minimum mean squared error (MMSE) estimator.

Proposition 2.1 (MMSE estimator)**.** *The unique solution to* (2.8) *is the posterior expectation:*

$$\hat{\psi}_{\text{MMSE}}(z) \triangleq \mathbb{E}[\varphi(X)|Z=z] \tag{2.9}$$

Proof. Let $\hat{\psi}(z) \triangleq \mathbb{E}[\varphi(X)|Z=z]$ and $\psi \in \mathcal{F}$. It comes that:

$$\begin{aligned}
\rho(\psi(Z),\varphi(X))^2 &= \langle\psi(Z)-\varphi(X)|\psi(Z)-\varphi(X)\rangle \\
&= \langle\psi(Z)-\hat{\psi}(Z)+\hat{\psi}(Z)-\varphi(X)|\psi(Z)-\hat{\psi}(Z)+\hat{\psi}(Z)-\varphi(X)\rangle \\
&= \langle\psi(Z)-\hat{\psi}(Z)|\psi(Z)-\hat{\psi}(Z)\rangle + \langle\hat{\psi}(Z)-\varphi(X)|\hat{\psi}(Z)-\varphi(X)\rangle \\
&\quad + 2\langle\psi(Z)-\hat{\psi}(Z)|\hat{\psi}(Z)-\varphi(X)\rangle
\end{aligned} \tag{2.10}$$

The cross term vanishes by the tower property of conditional expectation:

$$\begin{aligned}
\langle\psi(Z)-\hat{\psi}(Z)|\hat{\psi}(Z)-\varphi(X)\rangle &= \mathbb{E}\left[(\psi(Z)-\hat{\psi}(Z))^\top(\hat{\psi}(Z)-\varphi(X))\right] \\
&= \iint (\psi(z)-\hat{\psi}(z))^\top(\hat{\psi}(z)-\varphi(x))\,p(dx,dz) \\
&= \iint (\psi(z)-\hat{\psi}(z))^\top(\hat{\psi}(z)-\varphi(x))\,p(dx|z)\,p(dz) \\
&= \int (\psi(z)-\hat{\psi}(z))^\top \left(\int (\hat{\psi}(z)-\varphi(x))\,p(dx|z)\right) p(dz) \\
&= \int (\psi(z)-\hat{\psi}(z))^\top \left(\hat{\psi}(z)-\mathbb{E}[\varphi(X)|Z=z]\right) p(dz) \\
&= \int (\psi(z)-\hat{\psi}(z))^\top \times 0\, p(dz) \\
&= 0
\end{aligned} \tag{2.11}$$

by definition of $\hat{\psi}(z)$.

Therefore:

$$\begin{aligned}
\rho(\psi(Z),\varphi(X))^2 &= \rho(\psi(Z),\hat{\psi}(Z))^2 + \rho(\hat{\psi}(Z),\varphi(X))^2 \\
&\geq \rho(\hat{\psi}(Z),\varphi(X))^2
\end{aligned} \tag{2.12}$$

with equality if and only if $\|\psi(Z)-\hat{\psi}(Z)\|^2 = 0$, *i.e.* $\psi = \hat{\psi}$ almost surely (a.s.). □

Remark 2.2 (Geometric interpretation)**.** *The MMSE estimator is the orthogonal projection of $\varphi(X)$ onto $\mathcal{F}$, viewed as the set of square-integrable $\mathbb{R}^{d_\varphi}$-valued random variables that are measurable with respect to Z. The disappearance of the cross term in the proof above is precisely the orthogonality condition characterising this projection.*

Remark 2.3 (MMSE assumptions)**.** *Square-integrability of $\varphi(X)$ is only required for the minimisation (2.8) to be well-posed, not the estimator itself. The posterior expectation $\mathbb{E}[\varphi(X)|Z=z]$ remains well-defined as soon as $\varphi(X)$ is integrable, that is $\mathbb{E}[\|\varphi(X)\|] < \infty$, but it only minimises the mean squared error when $\varphi(X)$ is square-integrable.*

2.2.2 Posterior Median Estimator

Because the MMSE estimator minimises a quadratic loss, it is highly sensitive to extreme values and may perform poorly for heavy-tailed distributions. To mitigate this sensitivity to outliers, one can use an absolute loss instead. In dimension $d_\varphi = 1$, let:

$$\mathcal{F} \triangleq \left\{\psi \in \mathcal{L}^0\left(\mathbb{R}^{d_z}, \mathbb{R}\right) \,\middle|\, \mathbb{E}[|\psi(Z)|] < \infty\right\} \tag{2.13}$$

As previously, $\mathcal{F}$ is identified as a subspace of $\mathcal{L}^1\left(\mathbb{R}^{d_z}, \mathbb{R}\right)$, and is equipped with the following non-Euclidean distance:

$$(A, B) \in \mathcal{F}^2 \mapsto \rho(A, B) \triangleq \mathbb{E}[|A(Z) - B(Z)|] \tag{2.14}$$

The minimisation problem (2.4) thus becomes:

$$\underset{\psi \in \mathcal{F}}{\operatorname{argmin}}(\mathbb{E}[|\psi(Z) - \varphi(X)|]) \tag{2.15}$$

While the terminology is less standard than MMSE, an estimator solving (2.15) is often called a minimum mean absolute error (MMAE) estimator.

Proposition 2.2 (MMAE estimator, simplified assumptions)**.** *Assume that, for every z, the conditional distribution of $\varphi(X)$ given $Z = z$ is absolutely continuous with a strictly positive density. Then, the unique solution to (2.15) is the posterior median of $\varphi(X)$ given $Z = z$, i.e. the value $\hat{\psi}_{\mathrm{MMAE}}(z) \in \mathbb{R}$ that satisfies:*

$$\mathbb{P}\left(\varphi(X) \geq \hat{\psi}_{\mathrm{MMAE}}(z) | Z = z\right) = \mathbb{P}\left(\varphi(X) < \hat{\psi}_{\mathrm{MMAE}}(z) | Z = z\right) = \frac{1}{2} \tag{2.16}$$

Proof. By the law of total expectation, for any $\psi \in \mathcal{F}$, we have:

$$\mathbb{E}[|\psi(Z) - \varphi(X)|] = \mathbb{E}\left[\int |\psi(z) - \varphi(x)| \, p(dx|Z=z)\right] \tag{2.17}$$

Since the outer expectation is over non-negative quantities, this is minimised by minimising the inner integral pointwise for each z, *i.e.* by solving:

$$\underset{u \in \mathbb{R}}{\operatorname{argmin}} \int |\varphi(x) - u| \, p(dx|Z=z) \tag{2.18}$$

As $u \mapsto |\varphi(x) - u|$ is convex, so is the integral, hence any stationary point is a global minimum. Since the conditional law is absolutely continuous, the integral is differentiable in u. By differentiating this expression with respect to $u \in \mathbb{R}$, it

follows that:

$$\begin{aligned}
& \frac{d}{du}\left(\int |\varphi(x)-u|\, p(dx|Z=z)\right)=0 \\
\iff \quad & \frac{d}{du}\left(\int \left((\varphi(x)-u)\,\mathbb{1}_{\varphi(x)\geq u}(x)-(\varphi(x)-u)\,\mathbb{1}_{\varphi(x)<u}(x)\right)\, p(dx|Z=z)\right)=0 \\
\iff \quad & \int \left(-\mathbb{1}_{\varphi(x)\geq u}(x)+\mathbb{1}_{\varphi(x)<u}(x)\right)\, p(dx|Z=z)=0 \\
\iff \quad & \mathbb{P}(\varphi(X)\geq u|Z=z)=\mathbb{P}(\varphi(X)<u|Z=z)
\end{aligned} \tag{2.19}$$

Since the total probability is one, this is equivalent to $u=\hat{\psi}_{\mathrm{MMAE}}(z)$ as defined in (2.16), with the strict positivity of the density ensuring uniqueness. □

Remark 2.4 (Median assumptions)**.** *The absolute continuity and positivity assumptions are only required for the clean statement above, not for the existence of a solution. Since the inner integral is convex, continuous and coercive in u, a minimum always exists for any integrable conditional distribution. In full generality, the set of solutions is the closed interval of medians:*

$$\left\{u\in\mathbb{R}\;\middle|\;\mathbb{P}(\varphi(X)\leq u|Z=z)\geq\frac{1}{2}\ \textit{and}\ \mathbb{P}(\varphi(X)\geq u|Z=z)\geq\frac{1}{2}\right\} \tag{2.20}$$

The absence of atoms then guarantees that this interval is characterised by the equality (2.16), while the strict positivity of the density ensures that it reduces to a single point.

Remark 2.5 (Generalised median)**.** *The extension to higher dimensions $d_\varphi>1$ is non-trivial. The geometric median, which minimises the mean absolute error, is the most common generalisation. In the remainder of this thesis, we focus on the MMSE and MAP estimators (presented below), which extend naturally to any dimension.*

2.2.3 Maximum *a Posteriori* Estimator

As they produce a single point estimate that may lie in a low-probability region, both the MMSE and the MMAE estimators may not be suited for multimodal posteriors. In such cases, a natural alternative is to select a maximum of the posterior density.

Let $\mathcal{F}\triangleq\mathcal{L}^0\left(\mathbb{R}^{d_z},\mathbb{R}^{d_\varphi}\right)$, with no integrability constraint. The maximum *a posteriori* (MAP) estimator is defined as the global mode of the posterior density of $\varphi(X)$:

$$\hat{\psi}_{\mathrm{MAP}}\triangleq\operatorname*{argmax}_{\psi\in\mathcal{F}}\left(\mathbb{E}\left[p_{\varphi(X)|Z}(\psi(Z)|Z)\right]\right) \tag{2.21}$$

As the objective is separable over z, the maximisation can be carried out pointwise, yielding for each z:

$$\hat{\psi}_{\mathrm{MAP}}(z)=\operatorname*{argmax}_{u\in\mathbb{R}^{d_\varphi}}\left(p_{\varphi(X)|Z}(u|z)\right) \tag{2.22}$$

Using Bayes' theorem, this is equivalent to:

$$\hat{\psi}_{\mathrm{MAP}}(z)=\operatorname*{argmax}_{u\in\mathbb{R}^{d_\varphi}}\left(p_{Z|\varphi(X)}(z|u)\,p_{\varphi(X)}(u)\right) \tag{2.23}$$

since the normalising constant $p_Z(z)$ is independent of u.

Although the MAP estimator avoids low-density regions, it still reduces the posterior to a single point. In particular, in multimodal situations, only one mode is retained, while the remaining ones are discarded.

Remark 2.6 (MAP optimisation problem)**.** *The MAP estimator can be seen as the solution of an optimisation problem of the form* (2.4)*, associated with the threshold loss:*

$$\forall \varepsilon > 0,\ (A, B) \in \mathcal{F}^2 \mapsto \rho_\varepsilon(A, B) \triangleq \mathbb{1}_{\|A(Z)-B(Z)\|>\varepsilon} \tag{2.24}$$

in the limit $\varepsilon \to 0^+$*:*

$$\hat{\psi}_{\text{MAP}} = \lim_{\varepsilon \to 0^+} \operatorname*{argmin}_{\psi \in \mathcal{F}} (\rho_\varepsilon(\psi(Z), \varphi(X))) \tag{2.25}$$

Maximum Likelihood Estimator

In the special case where no prior distribution on $\varphi(X)$ is assumed, (2.23) reduces to the maximum likelihood (ML) estimator:

$$\hat{\psi}_{\text{ML}}(z) \triangleq \operatorname*{argmax}_{u \in \mathbb{R}^{d_\varphi}} \left(p_{Z|\varphi(X)}(z|u)\right) \tag{2.26}$$

As it does not take any prior information on $\varphi(X)$ into account, this estimator is not a Bayesian estimator and may be poorly suited when the available data is scarce. While it coincides with the MAP estimator under a uniform prior, it will not be considered in the rest of this document.

2.3 Analytical Density Estimation

As established in section 2.2, all three estimators ultimately rely on the PDF $p(x_k|\mathbf{z}_{1:k})$, from which an estimate of $\varphi(x_k)$ can be derived. Point estimators provide a concise and directly interpretable representation of the state, which is often desirable for our applications. However, reducing the posterior density to a single point estimate may discard valuable information, particularly when it is heavy-tailed or multimodal.

The goal of the following section is thus to present algorithms that propagate the PDF itself at each discrete time step k in order to solve the inference problem presented by (2.1).

2.3.1 Optimal Bayesian Filter

In the Bayesian filtering framework, the PDF can be propagated recursively through two steps. However, each step relies on a specific conditional independence hypotheses as presented by Assumptions 2.1 and 2.2.

Assumption 2.1 (Markov property)**.** *The state sequence* $(x_k)_{k\geq 0}$ *is a Markov chain. As such, it verifies:*

$$p(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}) = p(x_k|x_{k-1}) \tag{2.27}$$

with $\mathbf{x}_{0:k} \triangleq (x_0, \ldots, x_k)$.

Assumption 2.2 (Conditional observation independence)**.** *The current observation z_k only depends on the current state x_k:*

$$p(z_k|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}) = p(z_k|x_k) \tag{2.28}$$

Both assumptions follow from the state-space model (2.1) and, in particular, from the whiteness of w_k and n_k and their independence from x_0. Building on them, the optimal estimation algorithm computes the PDF in two steps: prediction and correction.

Prediction. The prior density at time k can be obtained by marginalising the previous PDF:

$$\begin{aligned} p(x_k|\mathbf{z}_{1:k-1}) &= \int p(x_k, x_{k-1}|\mathbf{z}_{1:k-1})\, dx_{k-1} \\ &= \int p(x_k|x_{k-1}, \mathbf{z}_{1:k-1})\, p(x_{k-1}|\mathbf{z}_{1:k-1})\, dx_{k-1} \end{aligned} \tag{2.29}$$

Assumption 2.1 simplifies (2.29) into the Chapman-Kolmogorov equation [4]:

$$p(x_k|\mathbf{z}_{1:k-1}) = \int p(x_k|x_{k-1})\, p(x_{k-1}|\mathbf{z}_{1:k-1})\, dx_{k-1} \tag{2.30}$$

where $p(x_k|x_{k-1})$ is the evolution, or transition, density induced by the state equation $x_k = f_k(x_{k-1}, w_k)$ and the density $p_k^{(e)}$ of the evolution noise w_k.

Correction. Upon receiving the new observation z_k, Bayes' theorem gives:

$$p(x_k|\mathbf{z}_{1:k}) = \frac{p(z_k|x_k, \mathbf{z}_{1:k-1})\, p(x_k|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \tag{2.31}$$

Applying Assumption 2.2 to (2.31) yields:

$$p(x_k|\mathbf{z}_{1:k}) = \frac{p(z_k|x_k)\, p(x_k|\mathbf{z}_{1:k-1})}{\displaystyle\int p(z_k|x_k)\, p(x_k|\mathbf{z}_{1:k-1})\, dx_k} \tag{2.32}$$

where the likelihood $p(z_k|x_k)$ depends on the observation model $z_k = h_k(x_k, n_k)$ and the density $p_k^{(o)}$ of the observation noise n_k.

These two equations allow for a recursive algorithm to compute the PDF of the system at each time step. This algorithm, usually referred to as the optimal Bayesian filter [5], is given by Algorithm 2.1.

Algorithm 2.1 Optimal Bayesian Filter

Compute $p(x_k\|\mathbf{z}_{1:k-1})$ using (2.30)	▷ Prediction
Compute $p(x_k\|\mathbf{z}_{1:k})$ using (2.32)	▷ Correction

While theoretically optimal, this algorithm has no closed-form solution in general. The integrands in both (2.30) and (2.32) are analytically intractable for arbitrary f_k, h_k.

2.3.2 Kalman Filter

While the optimal estimation algorithm cannot be used in the form given by Algorithm 2.1, a closed-form solution can be obtained by introducing additional hypotheses on (2.1), known as the Kalman assumptions [6]. In this section, we assume that:

$$\begin{cases} x_0 & \sim \mathcal{N}(\overline{x}_0, P_0) \\ f_k & : x_{k-1} \mapsto F_k\, x_{k-1} + b_k^{(e)} \\ w_k & \sim \mathcal{N}(0, Q_k), \text{ independent of } x_0 \\ z_k & \text{is known} \\ h_k & : x_k \mapsto H_k\, x_k + b_k^{(o)} \\ n_k & \sim \mathcal{N}(0, R_k), \text{ independent of } x_0 \text{ and } w_k \end{cases} \tag{2.33}$$

where $F_k \in \mathcal{M}_{d_x,d_x}(\mathbb{R})$ and $H_k \in \mathcal{M}_{d_z,d_x}(\mathbb{R})$ are known matrices, and $b_k^{(e)} \in \mathbb{R}^{d_x}$ and $b_k^{(o)} \in \mathbb{R}^{d_z}$ are known bias vectors. $P_0 \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$ is the initial state covariance, and $Q_k \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$ and $R_k \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ are the evolution and observation noise covariance matrices. Here, $\mathcal{S}^{++}$ denotes the set of symmetric positive definite (SPD) matrices [7], as introduced by Definition 2.1.

Definition 2.1 (Symmetric positive definite matrix)**.** *Let $n \in \mathbb{N}^*$ and $M \in \mathcal{M}_{n,n}(\mathbb{R})$. M is symmetric positive semi-definite (SPSD), denoted $M \succeq 0$, if it is:*

$$\begin{cases} M^\top = M & \textit{symmetric} \\ \forall a \in \mathbb{R}^n,\ a^\top M a \geq 0 & \textit{positive semi-definite} \end{cases} \tag{2.34}$$

If the inequality is strict for every non-zero vector, then M is SPD. We write $\mathcal{S}_n^+(\mathbb{R})$ and $\mathcal{S}_n^{++}(\mathbb{R})$ respectively for the sets of SPSD and SPD matrices of size n.

In particular, the state-space model (2.1) becomes:

$$\begin{cases} x_k = F_k\, x_{k-1} + b_k^{(e)} + w_k \\ z_k = H_k\, x_k + b_k^{(o)} + n_k \end{cases} \tag{2.35}$$

Finally, x_0, $(w_k)_{k\geq 0}$ and $(n_k)_{k\geq 0}$ are assumed to be mutually independent. Under these assumptions, the true state x_k follows a Gaussian distribution at each time step as an affine transformation of Gaussian random variables.

Proposition 2.3 (Gaussian closure under affine assumptions)**.** *For $m, n \in \mathbb{N}$, let the index $m|n$ denote "at time m conditioned on the observations from 1 to n". Under the hypotheses of (2.33), if $p(x_{k-1}|\mathbf{z}_{1:k-1}) = \mathcal{N}\left(\hat{x}_{k-1|k-1}, P_{k-1|k-1}\right)$, then:*

$$\begin{cases} p(x_k|\mathbf{z}_{1:k-1}) = \mathcal{N}\left(\hat{x}_{k|k-1}, P_{k|k-1}\right) \\ p(x_k|\mathbf{z}_{1:k}) = \mathcal{N}\left(\hat{x}_{k|k}, P_{k|k}\right) \end{cases} \tag{2.36}$$

where the predicted and corrected states $\hat{x}_{k|k-1}$ and $\hat{x}_{k|k}$, and covariances $P_{k|k-1} \succ 0$ and $P_{k|k} \succ 0$ are given by Algorithm 2.2.

Proof. See section B.1. □

The resulting algorithm, known as the Kalman filter [8], is depicted by Figure 2.1 and presented in Algorithm 2.2.

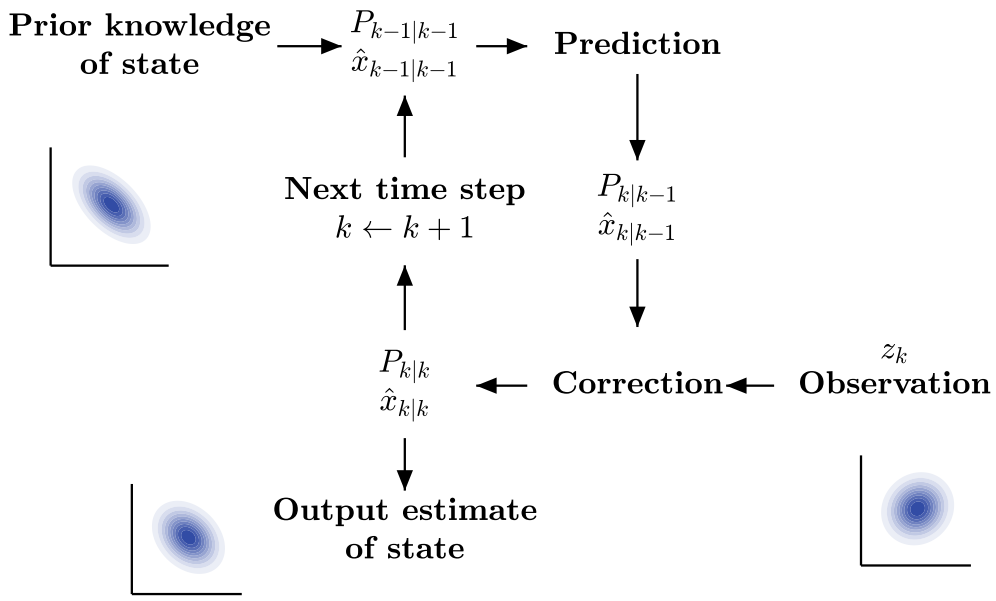


Figure 2.1: Schematic representation of the Kalman filter prediction/correction cycle. Each ellipsoid represents a confidence region associated with its corresponding Gaussian density.

Algorithm 2.2 Kalman Filter

$\hat{x}_{k\|k-1} = F_k\, \hat{x}_{k-1\|k-1} + b_k^{(e)}$	▷ State prediction
$P_{k\|k-1} = F_k\, P_{k-1\|k-1}\, F_k^\top + Q_k$	▷ Covariance prediction
$y_k = z_k - \left(H_k\, \hat{x}_{k\|k-1} + b_k^{(o)}\right)$	▷ Innovation
$S_k = H_k\, P_{k\|k-1}\, H_k^\top + R_k$	▷ Innovation covariance
$K_k = P_{k\|k-1}\, H_k^\top\, S_k^{-1}$	▷ Kalman gain
$\hat{x}_{k\|k} = \hat{x}_{k\|k-1} + K_k\, y_k$	▷ State correction
$P_{k\|k} = (I - K_k\, H_k)\, P_{k\|k-1}$	▷ Covariance correction

Remark 2.7 (Kalman state estimation)**.** *Under the Kalman assumptions, the PDF is Gaussian, and so every Bayesian estimator presented in section 2.2 coincides with* $\hat{x}_{k|k}$*:*

$$\hat{\psi}_{\mathrm{MMSE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\mathrm{MMAE}}(\mathbf{z}_{1:k}) = \hat{\psi}_{\mathrm{MAP}}(\mathbf{z}_{1:k}) = \hat{x}_{k|k} \tag{2.37}$$

However, since the above equations use a predictive prior, the ML estimator does not generally correspond to the corrected state.

Remark 2.8 (Precomputation of gains)**.** *Under the assumptions of* (2.33)*, the matrices* $P_{k|k-1}$*,* K_k *and* $P_{k|k}$ *depend only on the system matrices* F_k*,* H_k*,* Q_k*,* R_k *and the initial covariance* P_0*, but not on the observations* z_k*. They can therefore be computed before any measurement is received.*

Joseph's Formula

In finite-precision arithmetic, the standard update $P_{k|k} = (I - K_k\, H_k)\, P_{k|k-1}$ may lose symmetry or positive definiteness due to rounding errors. Joseph's formula [9, 10] gives an alternative way to compute $P_{k|k}$ while ensuring that this property remains numerically satisfied:

$$\begin{aligned} P_{k|k} &= (I - K_k\, H_k)\, P_{k|k-1} \\ &= (I - K_k\, H_k)\, P_{k|k-1} - K_k\, S_k\, K_k^\top + K_k\, S_k\, K_k^\top \\ &= (I - K_k\, H_k)\, P_{k|k-1} - (P_{k|k-1}\, H_k^\top\, S_k^{-1})\, S_k\, K_k^\top + K_k\, (H_k\, P_{k|k-1}\, H_k^\top + R_k)\, K_k^\top \\ &= (I - K_k\, H_k)\, P_{k|k-1} - P_{k|k-1}\, H_k^\top\, K_k^\top + K_k\, H_k\, P_{k|k-1}\, H_k^\top\, K_k^\top + K_k\, R_k\, K_k^\top \\ &= (I - K_k\, H_k)\, P_{k|k-1} - (I - K_k\, H_k)\, P_{k|k-1}\, H_k^\top\, K_k^\top + K_k\, R_k\, K_k^\top \\ &= (I - K_k\, H_k)\, P_{k|k-1} (I - K_k\, H_k)^\top + K_k\, R_k\, K_k^\top \end{aligned} \tag{2.38}$$

Under this more robust form, $P_{k|k}$ is SPSD as the sum of two SPSD matrices.

2.3.3 Extended Kalman Filter

The hypothesis that the evolution and observation functions are affine in x_k is often too restrictive in practice. The extended Kalman filter (EKF) [11] relaxes this assumption by locally linearising f_k and h_k around the current state estimate via a first-order Taylor expansion.

Let $m, n \in \mathbb{N}^*$, let $f : \mathbb{R}^n \to \mathbb{R}^m$ be differentiable at some $a \in \mathbb{R}^n$. f admits the following first-order expansion:

$$\forall x \in \mathbb{R}^n,\ f(x) = f(a) + J_f(a)\,(x - a) + o(\|x - a\|_2) \tag{2.39}$$

where $J_f(a) \in \mathcal{M}_{m,n}(\mathbb{R})$ is the Jacobian matrix of f evaluated at a, and defined as:

$$J_f \triangleq \frac{\partial f}{\partial x} = \begin{bmatrix} \dfrac{\partial f^{(1)}}{\partial x^{(1)}} & \cdots & \cdots & \dfrac{\partial f^{(1)}}{\partial x^{(n)}} \\ \vdots & \ddots & \ddots & \vdots \\ \dfrac{\partial f^{(m)}}{\partial x^{(1)}} & \cdots & \cdots & \dfrac{\partial f^{(m)}}{\partial x^{(n)}} \end{bmatrix} \tag{2.40}$$

and $f^{(i)}$ and $x^{(j)}$ respectively denote the i-th and j-th component of f and x.

Assuming that f_k and h_k are differentiable along the state trajectory, we can apply the first-order expansion (2.39) to the evolution and observation model. Successive linearisations at each time step can then be used to apply the Kalman filter equations locally. The resulting algorithm is given by Algorithm 2.3.

While less restrictive than the standard Kalman filter, the EKF is suboptimal for three reasons:

- First, f_k and h_k must be differentiable along the state trajectory, which excludes discontinuous or piecewise-defined models.
- Second, the linearisation in (2.39) neglects higher-order terms of the Taylor expansion, which introduces an approximation error that grows with the degree of nonlinearity of f_k and h_k. This error increases over time as it propagates through the equations of the Kalman filter due to the sequential formulation.

Algorithm 2.3 Extended Kalman Filter

$\tilde{F}_k = J_{f_k}\left(\hat{x}_{k-1|k-1}\right)$ ▷ Evolution Jacobian

$\hat{x}_{k|k-1} = f_k\left(\hat{x}_{k-1|k-1}\right) + b_k^{(e)}$ ▷ State prediction

$P_{k|k-1} = \tilde{F}_k\, P_{k-1|k-1}\, \tilde{F}_k^\top + Q_k$ ▷ Covariance prediction

$\tilde{H}_k = J_{h_k}\left(\hat{x}_{k|k-1}\right)$ ▷ Observation Jacobian

$y_k = z_k - \left(h_k\left(\hat{x}_{k|k-1}\right) + b_k^{(o)}\right)$ ▷ Innovation

$S_k = \tilde{H}_k\, P_{k|k-1}\, \tilde{H}_k^\top + R_k$ ▷ Innovation covariance

$K_k = P_{k|k-1}\, \tilde{H}_k^\top\, S_k^{-1}$ ▷ Kalman gain

$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\, y_k$ ▷ State correction

$P_{k|k} = (I - K_k\, \tilde{H}_k)\, P_{k|k-1}$ ▷ Covariance correction

- Finally, it relies on Gaussian assumptions and thus fails to handle multimodal or heavy-tailed distributions, as does its affine counterpart.

2.3.4 Other Suboptimal Kalman Filters

Iterative and Higher-Order EKF

The iterative EKF (IEKF) [12] improves upon the EKF correction step by iteratively re-linearising h_k around successive state estimates within the same measurement update, while keeping the prediction step unchanged. Starting from the initialisation $\hat{x}_{k|k}^{(0)} = \hat{x}_{k|k-1}$, each iteration produces a new $\hat{x}_{k|k}^{(i)}$ by repeating the correction step from the predicted state, with the Jacobian evaluated at the latest estimate. This is done until:

$$\left\|\hat{x}_{k|k}^{(i)} - \hat{x}_{k|k}^{(i-1)}\right\|_2 \leq \varepsilon \tag{2.41}$$

for some $\varepsilon > 0$, or a maximum number of iterations is reached. The corrected state and covariance matrix are then defined as the result of the last loop.

This procedure is equivalent to performing a Gauss-Newton descent [13] on the posterior density maximisation problem, and converges to the MAP estimator under Gaussian assumptions. It reduces the linearisation error of the correction step when h_k is strongly non-linear, at the cost of several Jacobian evaluations per time step.

The second-order EKF (SO-EKF) [14] can also be used to extend the linearisation (2.39) to second-order terms via the Hessians of f_k and h_k, which reduces the approximation error for highly nonlinear models. However, it requires computing and storing the Hessian matrices of f_k and h_k at each time step, making it computationally prohibitive in high dimensions.

Unscented Kalman Filter

Rather than linearising the model, the unscented Kalman filter (UKF) [15] propagates a small deterministic set of points through the non-linear functions themselves. Let $d \in \mathbb{N}^*$, and let X be a random vector of $\mathbb{R}^d$ with mean μ and covariance $P \succ 0$.

The $2d+1$ sigma points associated with X, and their weights, are:

$$\begin{cases} \mathcal{X}^{(0)} \triangleq \mu, & W^{(0)} \triangleq \dfrac{\lambda}{d+\lambda} \\ \mathcal{X}^{(\pm i)} \triangleq \mu \pm \sqrt{d+\lambda}\,(A)_i, & W^{(\pm i)} \triangleq \dfrac{1}{2(d+\lambda)},\ i \in [\![1, d]\!] \end{cases} \tag{2.42}$$

where A is a Cholesky factor [16] of P, $(A)_i$ is its i-th column, and $\lambda > -d$ is a scaling parameter. In particular, the weights are independent of X and add up to 1.

For a function φ, the first and second moments of $\varphi(X)$ can then be approximated by the weighted moments of the images of the sigma points:

$$\begin{cases} \hat{\varphi} \triangleq \displaystyle\sum_{i=-d}^{d} W^{(i)}\, \varphi\left(\mathcal{X}^{(i)}\right) \\ \hat{P} \triangleq \displaystyle\sum_{i=-d}^{d} W^{(i)} \left(\varphi\left(\mathcal{X}^{(i)}\right) - \hat{\varphi}\right) \left(\varphi\left(\mathcal{X}^{(i)}\right) - \hat{\varphi}\right)^{\top} \end{cases} \tag{2.43}$$

This procedure is known as the unscented transform [17], and is applied twice per time step by the UKF. It is first used with $\mu = \hat{x}_{k-1|k-1}$, $P = P_{k-1|k-1}$ and $\varphi = f_k$ to compute the predicted state, then with $\mu = \hat{x}_{k|k-1}$, $P = P_{k|k-1}$ and $\varphi = h_k$ to obtain the predicted observation and thus the innovation. In both cases, Q_k and R_k are added to the transformed covariance $\hat{P}$, as in Algorithm 2.3, to account for evolution and observation uncertainty.

This removes the linearisation shortcomings of the EKF, at the cost of $2d_x + 1$ evaluations of f_k and h_k per time step. A square-root formulation [18] propagates a factor of the covariance rather than the covariance itself, which preserves its symmetry and positive definiteness throughout the recursion. However, the UKF still relies on the Gaussian assumption, the posterior remaining summarised by a mean and a covariance matrix.

Ensemble Kalman Filter

The ensemble Kalman filter (EnKF) [19] avoids both the local linearisation of the EKF and the deterministic sigma-point approximation of the UKF. Instead, the state distribution is represented by a set of $N \in \mathbb{N}^*$ samples $\left(x_{k|k}^{(1)}, \dots, x_{k|k}^{(N)}\right)$, which are propagated through the true evolution and observation functions.

For two sets $\mathbf{X} \triangleq (X^{(1)}, \dots, X^{(N)})$ and $\mathbf{Z} \triangleq (Z^{(1)}, \dots, Z^{(N)})$, we let:

$$\begin{cases} \hat{X} \triangleq \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} X^{(i)}, & \hat{Z} \triangleq \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} Z^{(i)} \\ \hat{P}_{\mathbf{XZ}} \triangleq \dfrac{1}{N-1}\displaystyle\sum_{i=1}^{N} \left(X^{(i)} - \hat{X}\right)\left(Z^{(i)} - \hat{Z}\right)^{\top} \end{cases} \tag{2.44}$$

Each predicted sample $x_{k|k-1}^{(i)}$ is obtained by propagating the corresponding sample $x_{k-1|k-1}^{(i)}$ through f_k. The predicted observation $z_k^{(i)}$ is then computed using h_k.

The approximate Kalman gain is defined as:

$$\hat{K}_k \triangleq \hat{P}_{\mathbf{X}_{k|k-1}\mathbf{Z}_k} \left(\hat{P}_{\mathbf{Z}_k\mathbf{Z}_k} + R_k\right)^{-1} \tag{2.45}$$

where $\mathbf{X}_{k|k-1}$ is the predicted sample set, and $\mathbf{Z}_k$ is the predicted observation set. Finally, each corrected sample $x_{k|k}^{(i)}$ is computed from its associated predicted sample through a standard Kalman update, using $\hat{K}_k$ in place of the exact gain.

The computational cost of the EnKF is primarily determined by the value of N instead of the dimensions d_x and d_z. It is therefore particularly attractive for high-dimensional systems, such as those encountered in meteorological and oceanographic data assimilation [20].

It should be noted that, while the EnKf is more flexible than its counterparts, the correction step remains a Kalman update based on the empirical mean and covariance matrix of the sample set. As a result, it may provide a poor approximation when the PDF is strongly non-Gaussian.

2.4 Monte Carlo Methods

While Kalman-based extensions can alleviate some of the limitations of the original Kalman filter, they all enforce the PDF to be Gaussian at each time step. This assumption is often violated in practice, as navigation problems frequently involve highly non-linear and non-injective functions. In turn, these functions induce multimodal or heavy-tailed distributions, which cannot be adequately characterised by Gaussian hypotheses.

Monte Carlo (MC) methods [21, 22] offer an alternative approach that remains consistent under very mild assumptions. Their accuracy improves asymptotically with the number of samples, at the expense of an increased computational cost. This section introduces the main concepts for MC-based algorithms to solve the navigation problem (2.1).

2.4.1 Monte Carlo Integration

Even though MC methods originally stem from the computation of expectations, they approximate the entire distribution [23], from which every estimator of section 2.2 can be derived. Being itself an expectation, the MMSE estimator is simply the most immediate, and is where we begin.

Let X be a random variable taking values in $\mathbb{R}^{d_x}$ with probability density p, and let $\varphi \in \mathcal{L}^0(\mathbb{R}^{d_x}, \mathbb{R}^{d_\varphi})$ such that $\varphi(X)$ is integrable. We wish to compute the expectation of $\varphi(X)$:

$$\mathbb{E}_p[\varphi(X)] = \int \varphi(x)\, p(x)\, dx \tag{2.46}$$

Let $\mu(dx) \triangleq p(x)\, dx$ be the probability measure with density p with respect to the Lebesgue measure. (2.46) can be rewritten as:

$$\langle \mu, \varphi \rangle \triangleq \int \varphi(x)\, \mu(dx) = \mathbb{E}_p[\varphi(X)] \tag{2.47}$$

Since μ is a continuous measure, evaluating (2.47) exactly is intractable in most practical situations. The MC method aims to approximate μ by an empirical measure induced by N independent samples, as presented by Definition 2.2 and Proposition 2.4.

Definition 2.2 (Empirical measure)**.** *Let $N \in \mathbb{N}^*$ and $X^{(1)}, \dots, X^{(N)}$ be independent and identically distributed (i.i.d.) random variables drawn from μ. The empirical measure associated with $\left(X^{(i)}\right)_{i \in [\![1,N]\!]}$ is defined as:*

$$S^N(\mu) \triangleq \frac{1}{N} \sum_{i=1}^{N} \delta_{X^{(i)}} \tag{2.48}$$

where δ_x denotes the Dirac measure at x.

Proposition 2.4 (Monte Carlo estimator)**.** *The MC estimator:*

$$\left\langle S^N(\mu), \varphi \right\rangle = \frac{1}{N} \sum_{i=1}^{N} \langle \delta_{X^{(i)}}, \varphi \rangle = \frac{1}{N} \sum_{i=1}^{N} \varphi\left(X^{(i)}\right) \tag{2.49}$$

is unbiased and consistent:

$$\mathbb{E}_p\left[\left\langle S^N(\mu), \varphi \right\rangle\right] = \mathbb{E}_p[\varphi(X)] = \langle \mu, \varphi \rangle \tag{2.50}$$

$$\left\langle S^N(\mu), \varphi \right\rangle \xrightarrow{a.s.} \langle \mu, \varphi \rangle \tag{2.51}$$

If moreover $\varphi(X)$ is square-integrable, its mean squared error is:

$$\mathbb{E}_p\left[\left|\left\langle S^N(\mu) - \mu, \varphi \right\rangle\right|^2\right] = \frac{1}{N} \operatorname{tr} \mathbb{V}_\mu[\varphi] \tag{2.52}$$

where $\mathbb{V}_\mu[\varphi] \triangleq \left\langle \mu, (\varphi - \langle \mu, \varphi \rangle)(\varphi - \langle \mu, \varphi \rangle)^\top \right\rangle$ is the covariance matrix of φ under μ, and its rescaled error converges in distribution to a centred Gaussian:

$$\sqrt{N} \left\langle S^N(\mu) - \mu, \varphi \right\rangle \xrightarrow{\mathcal{L}} \mathcal{N}(0, \mathbb{V}_\mu[\varphi]) \tag{2.53}$$

Proof. By linearity of the expectation and the transfer theorem, it comes that:

$$\begin{aligned} \mathbb{E}_p\left[\left\langle S^N(\mu), \varphi \right\rangle\right] &= \frac{1}{N} \sum_{i=1}^{N} \mathbb{E}_p\left[\varphi\left(X^{(i)}\right)\right] \\ &= \frac{1}{N} \sum_{i=1}^{N} \langle \mu, \varphi \rangle \\ &= \langle \mu, \varphi \rangle \end{aligned} \tag{2.54}$$

Furthermore, by independence of the $X^{(i)}$:

$$\begin{aligned} \mathbb{E}_p\left[\left|\left\langle S^N(\mu) - \mu, \varphi \right\rangle\right|^2\right] &= \mathbb{E}_p\left[\left|\frac{1}{N} \sum_{i=1}^{N} \left(\varphi\left(X^{(i)}\right) - \langle \mu, \varphi \rangle\right)\right|^2\right] \\ &= \frac{1}{N^2} \sum_{i=1}^{N} \mathbb{E}_p\left[\left|\varphi\left(X^{(i)}\right) - \langle \mu, \varphi \rangle\right|^2\right] \\ &= \frac{1}{N^2} \sum_{i=1}^{N} \mathbb{E}\left[|\varphi(X) - \langle \mu, \varphi \rangle|^2\right] \\ &= \frac{1}{N} \operatorname{tr} \mathbb{V}_\mu[\varphi] \end{aligned} \tag{2.55}$$

Finally, the a.s. convergence and the convergence in distribution follow respectively from the strong law of large numbers and the central limit theorem, applied to the i.i.d. variables $\varphi\left(X^{(i)}\right)$. □

Informally, this means that:

$$S^N(\mu) = \frac{1}{N}\sum_{i=1}^{N} \delta_{X^{(i)}} \approx \mu \tag{2.56}$$

in the sense that, for any measurable function φ such that $\varphi(X)$ is integrable:

$$\left\langle S^N(\mu), \varphi \right\rangle = \frac{1}{N}\sum_{i=1}^{N} \varphi\left(X^{(i)}\right) \approx \langle \mu, \varphi \rangle \tag{2.57}$$

2.4.2 Importance Sampling

The standard MC method requires sampling directly from μ, which can be complex in practice. Importance sampling [24] circumvents this issue by drawing from a chosen importance, or proposal, distribution ν with density q.

Assuming that $\text{supp}(q) \subseteq \text{supp}(p)$, and for $x \in \text{supp}(q)$, it follows from the identity $p(x) = \frac{p(x)}{q(x)}\, q(x)$ and the transfer theorem that:

$$\mathbb{E}_p[\varphi(X)] = \int \left(\varphi(x)\,\frac{p(x)}{q(x)}\right) q(x)\, dx = \mathbb{E}_q\left[\varphi(X)\,\frac{p(X)}{q(X)}\right] \tag{2.58}$$

Using measure theory notations, this is equivalent to writing:

$$\langle \mu, \varphi \rangle = \left\langle \nu, \frac{d\mu}{d\nu}\,\varphi \right\rangle \tag{2.59}$$

The density $\frac{d\mu}{d\nu}$ is known as the Radon-Nikodym derivative. This motivates the introduction of the weighted empirical measure, given by Definition 2.3.

Definition 2.3 (Weighted empirical measure)**.** *Let $\nu(dx) = q(x)\,dx$ with $\text{supp}(\mu) \subseteq \text{supp}(\nu)$, and let $X^{(1)}, \ldots, X^{(N)} \overset{i.i.d.}{\sim} \nu$. The associated weighted empirical measure is:*

$$S^N(\mu, \nu) \triangleq \frac{d\mu}{d\nu}\, S^N(\nu) = \frac{1}{N}\sum_{i=1}^{N} \frac{p(X^{(i)})}{q(X^{(i)})}\, \delta_{X^{(i)}} \tag{2.60}$$

Pairing this measure with a test function φ yields the weighted MC estimator of Proposition 2.5.

Proposition 2.5 (Weighted MC estimator)**.** *The weighted MC estimator:*

$$\left\langle S^N(\mu, \nu), \varphi \right\rangle = \frac{1}{N}\sum_{i=1}^{N} \frac{p\left(X^{(i)}\right)}{q\left(X^{(i)}\right)}\, \varphi\left(X^{(i)}\right) \tag{2.61}$$

is unbiased and consistent:

$$\mathbb{E}_p\left[\left\langle S^N(\mu, \nu), \varphi \right\rangle\right] = \mathbb{E}_p[\varphi(X)] \tag{2.62}$$

$$\left\langle S^N(\mu, \nu), \varphi \right\rangle \xrightarrow{a.s.} \mathbb{E}_p[\varphi(X)] \tag{2.63}$$

Furthermore, when $\mathbb{V}_\nu\left[\frac{d\mu}{d\nu}\,\varphi\right] < \infty$*, its mean squared error and rescaled error satisfy:*

$$\mathbb{E}_q\left[\left|\left\langle S^N(\mu,\nu) - \mu, \varphi\right\rangle\right|^2\right] = \frac{1}{N}\operatorname{tr}\mathbb{V}_\nu\left[\frac{d\mu}{d\nu}\,\varphi\right] \tag{2.64}$$

$$\sqrt{N}\left\langle S^N(\mu,\nu) - \mu, \varphi\right\rangle \xrightarrow{\mathcal{L}} \mathcal{N}\left(0, \mathbb{V}_\nu\left[\frac{d\mu}{d\nu}\,\varphi\right]\right) \tag{2.65}$$

Proof. This is direct using the results of Proposition 2.4. □

For $\varphi \equiv 1$, the total mass:

$$\left\langle S^N(\mu,\nu), 1\right\rangle = \frac{1}{N}\sum_{i=1}^{N}\frac{p\left(X^{(i)}\right)}{q\left(X^{(i)}\right)} \tag{2.66}$$

converges a.s. to 1, but each finite sample is not necessarily normalised. Furthermore, p and q are often known only up to normalising constants $\tilde{p} = C_\mu p$ and $\tilde{q} = C_\nu q$. To alleviate these issues, Definition 2.4 introduces the self-normalised measure [25], whose estimator remains consistent despite these unknown constants, as proven by Proposition 2.6.

Definition 2.4 (Self-normalised measure)**.** *Let* $X^{(1)}, \ldots, X^{(N)} \overset{i.i.d.}{\sim} \nu$*, with* $\tilde{\mu} = C_\mu\mu$ *and* $\tilde{\nu} = C_\nu\nu$*. The self-normalised measure is:*

$$\tilde{S}^N(\mu,\nu) \triangleq \sum_{i=1}^{N} w^{(i)}\,\delta_{X^{(i)}} \tag{2.67}$$

$$w^{(i)} \triangleq \tilde{w}^{(i)} \Bigg/ \sum_{j=1}^{N}\tilde{w}^{(j)} \tag{2.68}$$

$$\tilde{w}^{(i)} \triangleq \frac{\tilde{p}\left(X^{(i)}\right)}{\tilde{q}\left(X^{(i)}\right)} = \frac{d\tilde{\mu}}{d\tilde{\nu}}\left(X^{(i)}\right) \tag{2.69}$$

Proposition 2.6 (Self-normalised estimator)**.** *The self-normalised estimator:*

$$\left\langle \tilde{S}^N(\mu,\nu), \varphi\right\rangle = \sum_{i=1}^{N} w^{(i)}\,\varphi\left(X^{(i)}\right) = \frac{\left\langle S^N(\tilde{\mu},\tilde{\nu}), \varphi\right\rangle}{\left\langle S^N(\tilde{\mu},\tilde{\nu}), 1\right\rangle} \tag{2.70}$$

is biased at finite N*, but consistent as soon as* $\varphi(X)$ *is integrable and* $\operatorname{supp}(\mu) \subseteq \operatorname{supp}(\nu)$*:*

$$\left\langle \tilde{S}^N(\mu,\nu), \varphi\right\rangle \xrightarrow{a.s.} \langle \mu, \varphi\rangle \tag{2.71}$$

Proof. Although the bias of $\tilde{S}^N(\mu,\nu)$ admits no simple closed-form expression in general, it can be computed for $N = 1$. In this situation, the self-normalisation forces the single weight to 1, such that:

$$\left\langle \tilde{S}^1(\mu,\nu), \varphi\right\rangle = \varphi\left(X^{(1)}\right) \tag{2.72}$$

and thus:

$$\mathbb{E}\left[\left\langle \tilde{S}^1(\mu,\nu), \varphi\right\rangle\right] = \langle \nu, \varphi\rangle \neq \langle \mu, \varphi\rangle \tag{2.73}$$

in general.

Furthermore, applying the strong law of large numbers separately to both the numerator and denominator of the self-normalised estimator yields:

$$\left\langle S^N(\tilde{\mu},\tilde{\nu}),\varphi\right\rangle = \frac{1}{N}\sum_{i=1}^{N}\tilde{w}^{(i)}\varphi\left(X^{(i)}\right) \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}}\,\varphi\right\rangle = \frac{C_\mu}{C_\nu}\left\langle \mu,\varphi\right\rangle \tag{2.74}$$

and:

$$\left\langle S^N(\tilde{\mu},\tilde{\nu}),1\right\rangle = \frac{1}{N}\sum_{j=1}^{N}\tilde{w}^{(j)} \xrightarrow{a.s.} \left\langle \nu, \frac{d\tilde{\mu}}{d\tilde{\nu}}\right\rangle = \frac{C_\mu}{C_\nu}\left\langle \mu,1\right\rangle = \frac{C_\mu}{C_\nu} \neq 0 \tag{2.75}$$

Combining both expressions finally gives the desired result. □

2.4.3 Sequential Importance Sampling

In the filtering context, the target distribution is the joint posterior $p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$ over the entire state trajectory. As new observations arrive, this distribution grows in dimension, making a static importance sampling approach impractical. Sequential importance sampling (SIS) [26, 27] addresses this by updating the weights recursively rather than re-approximating the full trajectory at every time step.

Assumption 2.3 (SIS hypotheses)**.** *In addition to Assumptions 2.1 and 2.2, we assume that the proposal density of the previous state does not depend on the current observation:*

$$q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k}) = q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \tag{2.76}$$

This assumption enables the following factorisation:

$$\begin{aligned} q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) &= q(x_k,\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k}) \\ &= q(x_k|\mathbf{x}_{0:k-1},\mathbf{z}_{1:k})\, q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k}) \\ &= q(x_k|\mathbf{x}_{0:k-1},\mathbf{z}_{1:k})\, q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \end{aligned} \tag{2.77}$$

While Assumption 2.3 is sufficient to derive the SIS algorithm, an additional assumption is often considered:

$$q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) = q(x_k|x_{k-1},z_k)\, q(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \tag{2.78}$$

so that a new particle $x_k^{(i)}$ can be propagated from $x_{k-1}^{(i)}$ and z_k alone.

Using Assumptions 2.1 and 2.2 at time k, the PDF factorises as:

$$\begin{aligned} p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) &= p(\mathbf{x}_{0:k}|z_k,\mathbf{z}_{1:k-1}) \\ &= \frac{p(z_k|\mathbf{x}_{0:k},\mathbf{z}_{1:k-1})\, p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k|\mathbf{x}_{0:k},\mathbf{z}_{1:k-1})\, p(x_k,\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k|\mathbf{x}_{0:k},\mathbf{z}_{1:k-1})\, p(x_k|\mathbf{x}_{0:k-1},\mathbf{z}_{1:k-1})\, p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &= \frac{p(z_k|x_k)\, p(x_k|x_{k-1})\, p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1})}{p(z_k|\mathbf{z}_{1:k-1})} \\ &\propto p(z_k|x_k)\, p(x_k|x_{k-1})\, p(\mathbf{x}_{0:k-1}|\mathbf{z}_{1:k-1}) \end{aligned} \tag{2.79}$$

Let $\mathbf{x}_{0:k}^{(i)}$ denote the sequence $\left(x_0^{(i)}, \dots, x_k^{(i)}\right)$. The recursive weight update is obtained by combining (2.79) with Assumption 2.3:

$$\begin{aligned} w_k^{(i)} &\propto \frac{\tilde{p}\left(\mathbf{x}_{0:k}^{(i)} | \mathbf{z}_{1:k}\right)}{\tilde{q}\left(\mathbf{x}_{0:k}^{(i)} | \mathbf{z}_{1:k}\right)} \\ &\propto \frac{\tilde{p}\left(z_k | x_k^{(i)}\right)\ \tilde{p}\left(x_k^{(i)} | x_{k-1}^{(i)}\right)\ \tilde{p}\left(\mathbf{x}_{0:k-1}^{(i)} | \mathbf{z}_{1:k-1}\right)}{\tilde{q}\left(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}\right)\ \tilde{q}\left(\mathbf{x}_{0:k-1}^{(i)} | \mathbf{z}_{1:k-1}\right)} \\ &\propto w_{k-1}^{(i)} \frac{\tilde{p}\left(z_k | x_k^{(i)}\right)\ \tilde{p}\left(x_k^{(i)} | x_{k-1}^{(i)}\right)}{\tilde{q}\left(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}\right)} \end{aligned} \tag{2.80}$$

with the further simplification under the hypothesis of (2.78):

$$w_k^{(i)} \propto w_{k-1}^{(i)} \frac{\tilde{p}\left(z_k | x_k^{(i)}\right)\ \tilde{p}\left(x_k^{(i)} | x_{k-1}^{(i)}\right)}{\tilde{q}\left(x_k^{(i)} | x_{k-1}^{(i)}, z_k\right)} \tag{2.81}$$

While the current formulation allows us to approximate the density of the full trajectory $p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$ through the self-normalised measure of Definition 2.4, we often only want to estimate the current state x_k. This can be done naturally by letting π_k denote the projection of a $(k+1)$-tuple onto its last element:

$$\pi_k : \begin{cases} \left(\mathbb{R}^{d_x}\right)^{k+1} \to \mathbb{R}^{d_x} \\ (x_0, \dots, x_k) \mapsto x_k \end{cases} \tag{2.82}$$

and computing the self-normalised estimator for $\varphi = \pi_k$, which approximates the MMSE estimator of section 2.2:

$$\hat{x}_k \triangleq \left\langle \tilde{S}^N(\mu_k, \nu_k), \pi_k \right\rangle = \sum_{i=1}^{N} w_k^{(i)}\ x_k^{(i)} \approx \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}[x_k] \tag{2.83}$$

where $\mu_k(dx) \triangleq \tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})\ dx$ and $\nu_k(dx) \triangleq \tilde{q}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})\ dx$. The corresponding empirical covariance matrix is:

$$\hat{P}_k \triangleq \sum_{i=1}^{N} w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right) \left(x_k^{(i)} - \hat{x}_k\right)^\top \tag{2.84}$$

The other estimators of section 2.2 remain accessible, though they are not as immediate:

- For a scalar state, the MMAE estimator only requires the cumulative distribution function, which $\tilde{S}^N(\mu_k, \nu_k)$ does define, and is obtained by accumulating the weights of the sorted particles until the total exceeds $\frac{1}{2}$. Its extension to a vector state is however not straightforward.

- The MAP estimator, on the other hand, calls for a density, which the particle approximation does not possess. In particular, $\hat{\psi}_{\text{MAP}}(\mathbf{z}_{1:k})$ cannot be computed using the weights, since they are importance ratios rather than density values. The heaviest particle maximises the Radon-Nikodym derivative rather than the PDF itself, and maximising $\tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})$ over the particles only locates a mode among the sampled locations.

As for the optimal and Kalman filters, the SIS algorithm follows the prediction/correction loop. The prediction step corresponds to the sampling of the new particle set through the proposal density, and is often called particle propagation. Conversely, the correction step is the weight update of (2.80). The SIS algorithm is given by Algorithm 2.4.

Algorithm 2.4 Sequential Importance Sampling

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim q\left(\cdot|x_{k-1}^{(i)}, z_k\right)$ ▷ Particle propagation

$\forall i \in [\![1, N]\!],\ \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \dfrac{\tilde{p}\left(z_k|x_k^{(i)}\right)\ \tilde{p}\left(x_k^{(i)}|x_{k-1}^{(i)}\right)}{\tilde{q}\left(x_k^{(i)}|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}\right)}$ ▷ Weight update

$\forall i \in [\![1, N]\!],\ w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

Remark 2.9 (Feynman-Kac distribution)**.** *The recurrence* (2.79) *can be expanded:*

$$p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k}) \propto p(x_0) \prod_{j=1}^{k} p(x_j|x_{j-1}) \times \prod_{j=1}^{k} p(z_j|x_j) \tag{2.85}$$

to show that the PDF is a Feynman-Kac distribution [28, 29], with the transitions $p(x_j|x_{j-1})$ *acting as the mutation and the likelihoods* $p(z_j|x_j)$ *as the selection function. Rather than being an artifice used to cancel the unknown constants, the self-normalisation introduced in Definition 2.4 is thus, more fundamentally, the normalisation of this distribution.*

Numerical correction of the covariance matrix

For numerical reasons, the estimated covariance matrix $\hat{P}_k$ may not satisfy SPD properties in practice. While an analytic reformulation existed for the Kalman filter, the numerical nature of MC methods prevents a similar solution. Formally, we want to find the closest matrix $M_+^* \succ 0$ to $\hat{P}_k$ in the sense of the Frobenius norm:

$$A \in \mathcal{M}_{d_x,d_x}(\mathbb{R}) \mapsto \|A\|_F \triangleq \sqrt{\sum_{i=1}^{d_x} \sum_{j=1}^{d_x} {A_{i,j}}^2} \tag{2.86}$$

If $d_x = 1$, the positive real $m_+ \in \mathbb{R}_+$ closest to some real $m \in \mathbb{R}$ is:

$$m_+ = \frac{m + |m|}{2} \tag{2.87}$$

While this only applies to scalars, a similar operation can be performed in higher dimensions by using the SPSD term of the polar decomposition. The singular values decomposition is used to simplify the computation of the polar decomposition.

Formally, the correction algorithm [30] is as follows:

- Symmetrisation of $\hat{P}_k$: $M \triangleq \dfrac{\hat{P}_k + \hat{P}_k^\top}{2}$
- Singular values decomposition: $M = UDV^\top$
- Polar decomposition: $M = \Omega\,\Sigma$ with $\Omega \triangleq U\,V^\top$ and $\Sigma \triangleq V\,D\,V^\top$
- Computation of M_+: $M_+ \triangleq \dfrac{M + \Sigma}{2}$

With this algorithm, M_+ is ensured to be SPSD. However, we wish to achieve strict positive definiteness. To solve this issue, we recall that a symmetric matrix is SPD if and only if all its eigenvalues are strictly positive, and we introduce a small correction $\varepsilon > 0$. Let $\lambda_{\min}$ denote the smallest eigenvalue of M_+. Let:

$$\delta \triangleq \max(0, -\lambda_{\min}) + \varepsilon \tag{2.88}$$

where the first term is introduced to compensate any numerical errors introduced during the symmetrisation and SVD steps.

Since M_+ and δI commute, they are simultaneously diagonalisable, and the eigenvalues of:

$$M_+^* \triangleq M_+ + \delta I \tag{2.89}$$

are thus:

$$\lambda_i(M_+^*) = \lambda_i(M_+) + \delta \geq \lambda_{\min} + \delta > 0 \tag{2.90}$$

For small enough values of ε, M_+^* is the desired matrix.

2.4.4 Weight Degeneracy

A fundamental limitation of SIS is that the variance of the importance weights accumulates along the trajectory [31] as shown by Proposition 2.7.

Proposition 2.7 (Weight degeneracy)**.** *Let $\mathbf{x}_{0:k} \sim \nu_k$, and let $\tilde{w}_k$ denote the non-normalised weight associated with the entire trajectory:*

$$\tilde{w}_k \triangleq \frac{\tilde{p}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}{\tilde{q}(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})} \tag{2.91}$$

Its variance decomposes as:

$$\mathbb{V}_{\nu_k}[\tilde{w}_k] = \mathbb{E}_{\nu_k}\left[\mathbb{V}_{\nu_k}[\tilde{w}_k|\mathbf{x}_{0:k-1}]\right] + \mathbb{V}_{\nu_k}\left[\mathbb{E}_{\nu_k}[\tilde{w}_k|\mathbf{x}_{0:k-1}]\right] \tag{2.92}$$

where the first term is non-negative, and vanishes if and only if $\tilde{w}_k$ is a.s. determined by $\mathbf{x}_{0:k-1}$.

Proof. (2.92) is the law of total variance. Being the expectation of a non-negative quantity, its first term vanishes if and only if $\mathbb{V}_{\nu_k}[\tilde{w}_k|\mathbf{x}_{0:k-1}]$ is a.s. zero, that is if and only if $\tilde{w}_k$ is a.s. determined by $\mathbf{x}_{0:k-1}$. □

Given $\mathbf{x}_{0:k-1}$, the only randomness remaining in (2.92) is the draw of x_k. The inner variance measures its effect, while the inner expectation integrates it out and leaves a function of $\mathbf{x}_{0:k-1}$. The first term therefore corresponds to the variance injected by x_k and the second is the variance inherited from the past.

As each step adds to the previous ones, the weight distribution typically concentrates on a small number of particles, and so (2.67) becomes a poor approximation of the PDF. While degeneracy occurs in most cases, Proposition 2.8 exhibits a proposal density that eliminates the additional variance introduced at the current step.

Proposition 2.8 (Optimal proposal density)**.** *When the proposal density is the posterior transition:*

$$q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k|x_{k-1}, z_k) \tag{2.93}$$

the trajectory weight is determined by $\mathbf{x}_{0:k-1}$*, so that the variance introduced at step* k *vanishes:*

$$\mathbb{V}_{\nu_k}\left[\tilde{w}_k|\mathbf{x}_{0:k-1}\right] = 0 \tag{2.94}$$

Proof. Plugging (2.93) into (2.80) yields:

$$\begin{aligned}
\tilde{w}_k &\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k|x_k)\,\tilde{p}(x_k|x_{k-1})}{\tilde{q}(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k})} \\
&\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k|x_k)\,\tilde{p}(x_k|x_{k-1})}{\tilde{p}(x_k|x_{k-1}, z_k)} \\
&\propto \tilde{w}_{k-1} \frac{\tilde{p}(z_k|x_k)\,\tilde{p}(x_k|x_{k-1})}{\frac{\tilde{p}(z_k|x_k, x_{k-1})\,\tilde{p}(x_k|x_{k-1})}{\tilde{p}(z_k|x_{k-1})}} \\
&\propto \tilde{w}_{k-1}\,\tilde{p}(z_k|x_{k-1})\,\frac{\tilde{p}(z_k|x_k)}{\tilde{p}(z_k|x_k, x_{k-1})} \\
&\propto \tilde{w}_{k-1}\,\tilde{p}(z_k|x_{k-1})
\end{aligned} \tag{2.95}$$

Both factors being functions of $\mathbf{x}_{0:k-1}$ alone, $\tilde{w}_k$ is determined by the conditioning, and its conditional variance is therefore zero. □

Using Proposition 2.8, the draw of x_k thus leaves $\tilde{w}_k$ untouched, but reaches $\tilde{w}_{k+1}$ through $\tilde{p}(z_{k+1}|x_k)$. Its contribution is deferred by one step rather than suppressed, and so this density delays the degeneracy rather than completely removing it. Furthermore, sampling from the posterior transition requires computing the marginal likelihood:

$$p\left(z_k|x_{k-1}^{(i)}\right) = \int p\left(z_k|x_k^{(i)}\right) p\left(x_k^{(i)}|x_{k-1}^{(i)}\right)\, dx_k^{(i)} \tag{2.96}$$

which is generally intractable.

A common alternative is to use the prior, or transition, density:

$$q(x_k|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k}) = p(x_k|x_{k-1}) \tag{2.97}$$

which is easier to compute, but ignores the current observation during propagation.

The prior density also simplifies the weight update (2.80) to:

$$w_k^{(i)} \propto w_{k-1}^{(i)}\,\tilde{p}\left(z_k|x_k^{(i)}\right) \tag{2.98}$$

However, because the particles are propagated without regard to z_k, their incremental weight is simply the likelihood $\tilde{p}\left(z_k|x_k^{(i)}\right)$. When this likelihood is peaked, most particles land where it is negligible. Only a few particles then carry most of the weight, which causes the degeneracy to accelerate. The prior is thus a less effective proposal in highly informative observation regimes.

2.4.5 Sampling Importance Resampling

Weight degeneracy is intrinsic to SIS and worsens with time unless the proposal distribution is chosen extremely close to the optimal proposal density. The sampling importance resampling (SIR) filter [32], also known as the bootstrap filter [33], mitigates this by introducing a resampling step after each weight update. Resampling replaces the weighted particle set $\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^N$ by an unweighted set $\left\{x_k^{*(j)}, 1/N\right\}_{j=1}^N$ drawn according to a chosen resampling algorithm:

$$\left\{x_k^{*(j)}, 1/N\right\}_{j=1}^N = \text{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^N\right) \tag{2.99}$$

It also uses the transition density as proposal (2.97), which further reduces (2.98) to:

$$w_k^{(i)} \propto \tilde{p}\left(z_k|x_k^{(i)}\right) \tag{2.100}$$

since all weights are reset to $1/N$ after resampling. This makes the SIR applicable under very weak assumptions, only requiring the ability to sample from $p(\cdot|x_{k-1})$ and compute the likelihood $p(z_k|x_k)$. Its pseudo-code is given by Algorithm 2.5.

Algorithm 2.5 Sampling Importance Resampling

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim p\left(\cdot|x_{k-1}^{(i)}\right)$ ▷ Particle propagation

$\forall i \in [\![1, N]\!],\ \tilde{w}_k^{(i)} = \tilde{p}\left(z_k|x_k^{(i)}\right)$ ▷ Weight update

$\forall i \in [\![1, N]\!],\ w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^N \tilde{w}_k^{(j)}$ ▷ Normalisation

$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^N \leftarrow \text{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^N\right)$ ▷ Resampling

Resampling Algorithms

Several resampling algorithms [34, 35] can be used in (2.99). We present three algorithms, which all select a new particle set from the particle approximation of the PDF:

$$p(x_k|\mathbf{z}_{1:k})\, dx_k \approx \sum_{i=1}^N w_k^{(i)}\, \delta_{x_k^{(i)}} \tag{2.101}$$

Drawing a particle from this discrete measure amounts to drawing its index $i \in [\![1, N]\!]$ from the categorical distribution of the weights $\left(w_k^{(i)}\right)_{i \in [\![1,N]\!]}$ through its

cumulative distribution function (CDF):

$$\begin{cases} \mathrm{cdf}_i \triangleq \sum_{j=1}^{i} w_k^{(j)}, & i \in [\![1, N]\!] \\ \mathrm{cdf}_0 = 0 \end{cases} \tag{2.102}$$

As illustrated by Figure 2.2, a draw of $u_j \in [0, 1]$ uniquely determines the new particle $x_k^{*(j)}$ as the parent particle $x_k^{(i)}$ of index i satisfying:

$$\mathrm{cdf}_{i-1} \leq u_j < \mathrm{cdf}_i \tag{2.103}$$

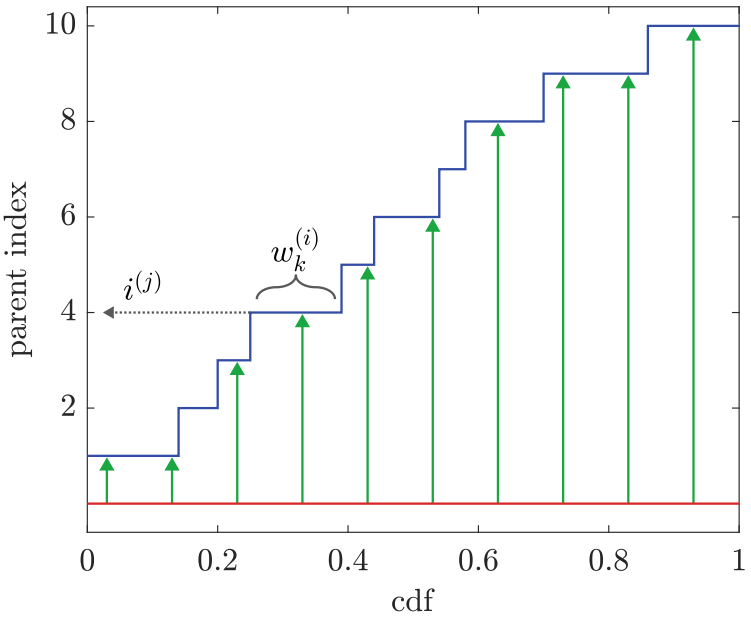


Figure 2.2: An example CDF of the particle weights. The width of each step is equal to the corresponding weight, so heavier particles are more likely to be selected.

The main difference between the three algorithms lies in how the N samples $(u_j)_{j \in [\![1,N]\!]}$ are generated:

- **Multinomial resampling.** All the draws are independent:

$$u_j \overset{i.i.d.}{\sim} \mathcal{U}([0, 1]) \tag{2.104}$$

  This is the most natural method, but has the highest resampling variance.

- **Systematic or Kitagawa resampling [36].** A single draw generates all samples via:

$$\begin{cases} u_1 \sim \mathcal{U}([0, 1/N]) \\ u_j = u_1 + (j-1)/N, \ j \in [\![2, N]\!] \end{cases} \tag{2.105}$$

  This significantly reduces the variance of the resampling step and is generally preferred in practice.

- **Deterministic resampling.** All samples are deterministic such that, for some $m \in [0, 1]$:

$$u_j = (j - m)/N,\ j \in [\![1, N]\!] \tag{2.106}$$

This eliminates resampling variance entirely at the cost of a deterministic bias.

Some advanced filters, such as the auxiliary particle filter, presented in section 2.5, require keeping track of the parent index $i^{(j)}$ of each resampled particle. This value is thus recorded at the creation of each new particle. Algorithm 2.6 presents a linear implementation of the Kitagawa resampling algorithm, using the fact that both the CDF values and the Kitagawa samples are increasing to compute the parent indices sequentially.

Algorithm 2.6 Kitagawa Resampling

$\text{cdf}_0 = 0$ ▷ Computation of the CDF
$\forall i \in [\![1, N]\!],\ \text{cdf}_i = \text{cdf}_{i-1} + w_k^{(i)}$
$u_1 \sim \mathcal{U}([0, 1/N])$ ▷ Kitagawa samples
$\forall j \in [\![2, N]\!],\ u_j = u_1 + (j - 1)/N$
$i = 1$
for $j = 1 : N$ **do**
  while $u_j \geq \text{cdf}_i$ **do** ▷ Computation of the parent index
    $i \leftarrow i + 1$
  end while
  $x_k^{*(j)} = x_k^{(i)}$ ▷ Generation of the new particle
  $w_k^{*(j)} = 1/N$
  $i^{(j)} = i$
end for

Remark 2.10 (Loss of independence)**.** *Because resampling duplicates particles, the resulting set is no longer independent, and the i.i.d. results of Propositions 2.4 and 2.5 no longer apply. However, the resampling is unbiased and the propagation step draws the particles independently conditionally on the past. The Feynman-Kac formalism proves [37, 38] that consistency and asymptotic Gaussianity still hold in this scenario, at the price of a variance larger than in the independent case and which accumulates over time.*

2.5 Advanced Particle Filters

Even though the SIR applies under very mild assumptions, it has several caveats. Indeed, while resampling at every step eliminates the weight degeneracy of the SIS, it wastes computational load, and erodes particle diversity. Furthermore, the number of particles required grows quickly with the state dimension, a phenomenon known as the curse of dimensionality [39].

The SIR admits several refinements [40], each targeting a specific issue. We first present the generic particle filter, which reduces the SIR's computational cost by

only resampling when necessary. We then introduce the regularised and auxiliary particle filters to restore particle diversity and improve on the prior proposal. Finally, we derive the equations of the Rao-Blackwellised particle filter, which reduces the dimension of the explored space under linear-Gaussian substructure assumptions.

2.5.1 Generic Particle Filter

The SIR resamples at every time step, which eliminates weight degeneracy but comes with two drawbacks:

- It introduces unnecessary computational cost when weights are already well-distributed, and increases the variance of the estimated state, as mentioned by Remark 2.10.
- It also replicates the heaviest particles and discards the lightest, so the set progressively concentrates on a few distinct locations. This loss of diversity, known as particle impoverishment [41], is especially severe in low evolution noise or high-dimensional settings, where little diversity is re-introduced at the propagation step. It is illustrated by Figure 2.3.

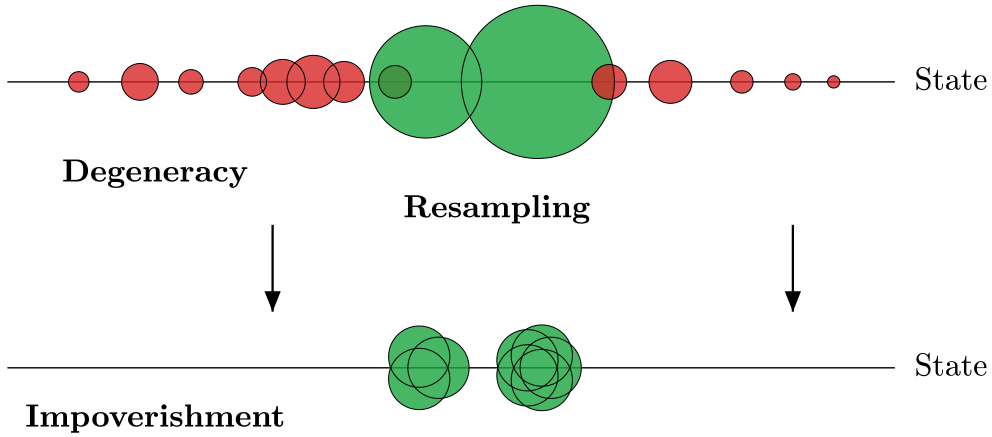


Figure 2.3: Particle impoverishment after resampling. Discarding non-resampled particles progressively shrinks the support of the particle approximation, degrading the filter's ability to explore the state-space.

To limit both issues, the generic particle filter (PF) [42] resamples only when the weight distribution has degraded significantly. This degradation is measured by the effective sample size [31]:

$$N_{\text{eff}} \triangleq \frac{N}{1 + \mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})}\left[\dfrac{p(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}{q(\mathbf{x}_{0:k}|\mathbf{z}_{1:k})}\right]} \tag{2.107}$$

Its denominator increases with the variance of the importance weights, whose growth over time corresponds to the weight degeneracy characterised by Proposition 2.7. When all the weights are uniform, the variance is zero, and thus $N_{\text{eff}} = N$.

Conversely, the more unevenly they are distributed, the smaller N_{eff}, with $N_{\text{eff}} \to 1$ when the mass concentrates on a single particle.

Since the true importance weights are unknown, N_{eff} is estimated by replacing $\mathbb{V}_{q(\cdot|\mathbf{z}_{1:k})}[\cdot]$ with the empirical spread of the normalised weights [43], which gives:

$$\hat{N}_{\text{eff}} \triangleq \frac{1}{\sum_{i=1}^{N} \left(w_k^{(i)}\right)^2} \leq N \tag{2.108}$$

Like N_{eff}, this quantity ranges from N for uniform weights, down to 1 in the fully degenerate case. The PF then performs the resampling step only when this estimate falls below a threshold:

$$\hat{N}_{\text{eff}} < N_{\text{th}} \tag{2.109}$$

where N_{th} is a predetermined value, typically chosen proportional to N.

Unlike the SIR, which traditionally uses the transition density as its proposal distribution, the generic PF is not typically restricted to any particular choice of proposal distribution. It is presented by Algorithm 2.7.

Algorithm 2.7 Generic Particle Filter

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim q\left(\cdot|x_{k-1}^{(i)}, z_k\right)$ ▷ Particle propagation

$\forall i \in [\![1, N]\!],\ \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \dfrac{\tilde{p}\left(z_k|x_k^{(i)}\right)\ \tilde{p}\left(x_k^{(i)}|x_{k-1}^{(i)}\right)}{\tilde{q}\left(x_k^{(i)}|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}\right)}$ ▷ Weight update

$\forall i \in [\![1, N]\!],\ w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

$\hat{N}_{\text{eff}} = 1 \Big/ \sum_{i=1}^{N} \left(w_k^{(i)}\right)^2$ ▷ Resampling criterion

if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**

$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^{N} \leftarrow \text{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^{N}\right)$ ▷ Resampling

end if

2.5.2 Regularised Particle Filter

Since each resampling step still collapses diversity, the PF only mitigates impoverishment, but does not prevent it. Indeed, the self-normalised measure of Definition 2.4 approximates the PDF by a mixture of Diracs:

$$p(x_k|\mathbf{z}_{1:k})\, dx_k \approx \sum_{i=1}^{N} w_k^{(i)}\, \delta_{x_k^{(i)}} \tag{2.110}$$

Since this measure is singular, resampling from it can only reproduce the existing particles. The regularised particle filter (RPF) [44] addresses this issue through

kernel density estimation [45, 46, 47], by replacing each Dirac with a smooth kernel to obtain the absolutely continuous measure:

$$p(x_k|\mathbf{z}_{1:k})\,dx_k \approx \sum_{i=1}^{N} w_k^{(i)}\, K_h\left(x_k - x_k^{(i)}\right) dx_k \tag{2.111}$$

New particles are then drawn from this continuous density rather than from the discrete set, which restores diversity. This is illustrated by Figure 2.4 and formalised by Definition 2.5 and proposition 2.9. Being absolutely continuous, this approximation also supplies the density that (2.110) lacks, and thus makes the MAP estimator of section 2.2 well-defined at the cost of a numerical maximisation.

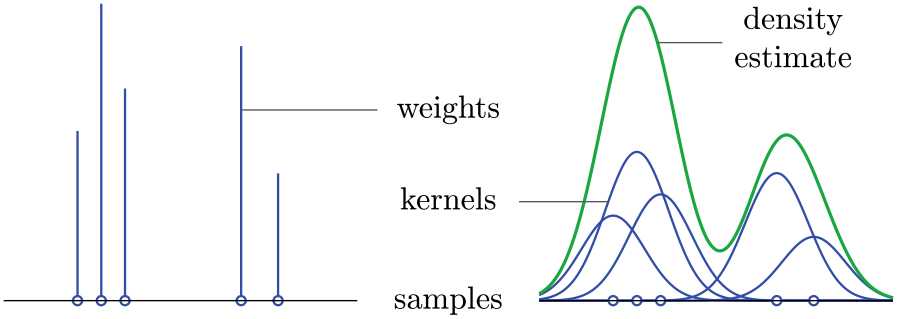


Figure 2.4: An example Monte Carlo approximation of the PDF (2.110) on the left and its kernel regularisation (2.111) on the right, for a given particle set. The latter approximation bridges the gaps between particles and spans a larger region of the state space.

Definition 2.5 (Regularisation kernel)**.** *Let $K : \mathbb{R}^{d_x} \to \mathbb{R}$. K is a regularisation kernel if:*

$$\begin{cases} \forall x \in \mathbb{R}^{d_x},\ K(x) \geq 0 & \textit{positive} \\ \forall x \in \mathbb{R}^{d_x},\ K(-x) = K(x) & \textit{symmetric} \\ \displaystyle\int K(x)\,dx = 1 & \textit{normalised} \\ \displaystyle\int \|x\|^2 K(x)\,dx < \infty & \textit{of finite variance} \end{cases} \tag{2.112}$$

Furthermore, for K a regularisation kernel and $h > 0$, we define the rescaled kernel K_h of bandwidth h as:

$$K_h : \begin{cases} \mathbb{R}^{d_x} \to \mathbb{R} \\ \quad x \mapsto \dfrac{1}{h^{d_x}} K\left(\dfrac{x}{h}\right) \end{cases} \tag{2.113}$$

Definition 2.6 (Regularised measure)**.** *Let $*$ denote the convolution operator. The regularised measure $S^N_{K_h}$ of some empirical measure S^N with respect to a regularisation kernel K_h is:*

$$S^N_{K_h}(\mu) \triangleq K_h * S^N(\mu) \tag{2.114}$$

In particular, the regularised self-normalised measure verifies:

$$\tilde{S}^N_{K_h}(\mu,\nu) = \sum_{i=1}^{N} w^{(i)}\ (K_h * \delta_{X^{(i)}}) \tag{2.115}$$

Proposition 2.9 (Regularised estimator)**.** *The regularised self-normalised estimator is:*

$$\left\langle \tilde{S}^N_{K_h}(\mu,\nu), \varphi \right\rangle = \sum_{i=1}^{N} w^{(i)}\ (K_h * \varphi)\left(X^{(i)}\right) = \left\langle \tilde{S}^N(\mu,\nu), K_h * \varphi \right\rangle \tag{2.116}$$

It is a biased and inconsistent estimator in general:

$$\left\langle \tilde{S}^N_{K_h}(\mu,\nu), \varphi \right\rangle \xrightarrow{a.s.} \langle \mu, K_h * \varphi \rangle \tag{2.117}$$

However, it becomes consistent when φ is linear. In particular:

$$\left\langle \tilde{S}^N_{K_h}(\mu_k,\nu_k), \pi_k \right\rangle \xrightarrow{a.s.} \langle \mu_k, \pi_k \rangle = \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}[x_k] \tag{2.118}$$

Proof. Let $i \in [\![1, N]\!]$. By definition of $*$, we have:

$$(K_h * \delta_{X^{(i)}})(dx) = K_h(x - X^{(i)})\, dx \tag{2.119}$$

and thus, by symmetry of K_h:

$$\begin{aligned} \langle K_h * \delta_{X^{(i)}}, \varphi \rangle &= \int \varphi(x) K_h(x - X^{(i)})\, dx \\ &= \int \varphi(x) K_h(X^{(i)} - x)\, dx \\ &= (K_h * \varphi)(X^{(i)}) \end{aligned} \tag{2.120}$$

In turn, this result gives:

$$\left\langle \tilde{S}^N_{K_h}(\mu,\nu), \varphi \right\rangle = \sum_{i=1}^{N} w^{(i)}\ (K_h * \varphi)\left(X^{(i)}\right) = \left\langle \tilde{S}^N(\mu,\nu), K_h * \varphi \right\rangle \tag{2.121}$$

and Proposition 2.6, applied to $K_h * \varphi$, yields the a.s. limit $\langle \mu, K_h * \varphi \rangle$.

Assume that φ is linear, that is $\varphi : x \mapsto A\,x$ for some $A \in \mathcal{M}_{d_\varphi, d_x}(\mathbb{R})$. Then, by normalisation and symmetry of K_h, for all $x \in \mathbb{R}^{d_x}$:

$$\begin{aligned} (K_h * \varphi)(x) &= \int K_h(y) \varphi(x - y)\, dy \\ &= \varphi(x) \int K_h(y)\, dy - A \int y K_h(y)\, dy \\ &= \varphi(x) \end{aligned} \tag{2.122}$$

Hence $K_h * \varphi = \varphi$, and the limit reduces to $\langle \mu, \varphi \rangle$. In particular, for the linear projection π_k, it becomes $\langle \mu_k, \pi_k \rangle = \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k})}[x_k]$. □

The choice of bandwidth h is critical to the performance of the RPF. If h is too small, K_h concentrates around the Dirac measure, and the regularisation has no practical effect on particle diversity. Conversely, the kernel density estimate over-smooths the posterior when h is too large, artificially spreading particles across regions of low probability, and degrading filter accuracy. In this section, we present two classical choices for K_h:

- **Epanechnikov kernel.** This kernel minimises the mean integrated squared error (MISE) among all regularisation kernels [48], and the associated bandwidth is optimal when the density is Gaussian with identity covariance. It is defined by:

$$\begin{cases} K_{\text{Epa}} : x \mapsto \dfrac{d_x + 2}{2c_{d_x}} \left(1 - \|x\|^2\right) \mathbb{1}_{\|x\| \leq 1}(x) \\ h_{\text{Epa}} = \left(\dfrac{8(d_x + 4)\,(2\sqrt{\pi})^{d_x}}{c_{d_x}} \right)^{\frac{1}{d_x+4}} N^{-\frac{1}{d_x+4}} \end{cases} \tag{2.123}$$

  where:

$$c_{d_x} \triangleq \frac{\pi^{d_x/2}}{\Gamma(d_x/2 + 1)} \tag{2.124}$$

  is the volume of the $\mathbb{R}^{d_x}$ unit ball, and Γ is the Gamma function.

- **Gaussian kernel.** While suboptimal with respect to the Epanechnikov kernel, it is simpler to compute [49]. Its expression is given by:

$$\begin{cases} K_{\text{Gauss}} : x \mapsto (2\pi)^{-d_x/2} \exp\left(-\dfrac{\|x\|^2}{2}\right) \\ h_{\text{Gauss}} = \left(\dfrac{4}{d_x + 2}\right)^{\frac{1}{d_x+4}} N^{-\frac{1}{d_x+4}} \end{cases} \tag{2.125}$$

It must be noted that the assumptions for the Epanechnikov, and consequently the Gaussian, regularisation are not usually met in practice, since MC methods are precisely used when the PDF is not Gaussian.

However, the two conditions do not play the same role. The Gaussian hypothesis is only used to compute the optimal bandwidth, whose expression would otherwise depend on an unknown PDF. It is thus a calibration convention rather than a modelling assumption. In particular, it tends to overestimate h when the PDF is multimodal.

On the other hand, the unit covariance is a scaling requirement, and can be enforced exactly by a whitening step before the regularisation. This amounts to replacing every particle by $A_k^{-1}\, x_k^{(i)}$, where A_k is a Cholesky factor of $\hat{P}_k$:

$$\hat{P}_k = A_k\, A_k^\top \tag{2.126}$$

which ensures that the covariance matrix of the new particle set is the identity. While seemingly computationally heavy, this reduces to using the following regularisation kernel on the standard particle set:

$$\tilde{K}_h : x \mapsto \frac{\det(A_k)^{-1}}{h^{d_x}} K\left(\frac{1}{h} A_k^{-1} x\right) \tag{2.127}$$

Moreover, the bandwidth h is multiplied by a modulation factor $\alpha_{\text{reg}} \in [0, 1]$ to avoid over-smoothing in cases where the PDF is multimodal. The pseudo-code for the RPF is given by Algorithm 2.8.

Algorithm 2.8 Regularised Particle Filter

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim q\left(\cdot | x_{k-1}^{(i)}, z_k\right)$ ▷ Particle propagation

$\forall i \in [\![1, N]\!],\ \tilde{w}_k^{(i)} = w_{k-1}^{(i)} \dfrac{\tilde{p}\left(z_k | x_k^{(i)}\right)\ \tilde{p}\left(x_k^{(i)} | x_{k-1}^{(i)}\right)}{\tilde{q}\left(x_k^{(i)} | \mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k}\right)}$ ▷ Weight update

$\forall i \in [\![1, N]\!],\ w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

$\hat{N}_{\text{eff}} = 1 \Big/ \sum_{i=1}^{N} \left(w_k^{(i)}\right)^2$ ▷ Resampling criterion

if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**

$\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)}$ ▷ Whitening

$\hat{P}_k = \sum_{i=1}^{N} w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right) \left(x_k^{(i)} - \hat{x}_k\right)^\top$

A_k s.t. $\hat{P}_k = A_k\, A_k^\top$

$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^{N} \leftarrow \text{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^{N}\right)$ ▷ Resampling

$\forall j \in [\![1, N]\!],\ \varepsilon^{(j)} \sim K$ ▷ Regularisation

$\forall j \in [\![1, N]\!],\ x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k\, \varepsilon^{(j)}$

end if

2.5.3 Auxiliary Particle Filter

The auxiliary particle filter (APF) [50, 51] addresses the degeneracy problem at its root by exploiting a different weakness of the SIR. Since particles are propagated before the new observation z_k is taken into account, particles that will ultimately receive low likelihood waste computational effort. The APF thus introduces a look-ahead step by using a prediction of their future likelihood to resample promising parent particles before propagation.

Formally, the APF targets the joint distribution $p(x_k, i | \mathbf{z}_{1:k})$, where $i \in [\![1, N]\!]$ indexes the parent particle at time $k-1$. Using Bayes' theorem and the chain rule gives:

$$p(x_k, i | \mathbf{z}_{1:k}) \propto p(z_k | x_k, i, \mathbf{z}_{1:k-1})\ p(x_k | i, \mathbf{z}_{1:k-1})\ p(i | \mathbf{z}_{1:k-1}) \tag{2.128}$$

Each term of this decomposition can be simplified as follows:

- Assumption 2.2 states that the current observation depends only on the current state:

$$p(z_k | x_k, i, \mathbf{z}_{1:k-1}) = p(z_k | x_k) \tag{2.129}$$

- Since the parent index uniquely determines the ancestor trajectory, conditioning on i is equivalent to conditioning on $\mathbf{x}_{0:k-1}^{(i)}$. Moreover, Assumption 2.1 establishes that $(x_k)_{k\geq 0}$ is a Markov chain, and thus:

$$p(x_k|i, \mathbf{z}_{1:k-1}) = p\left(x_k|\mathbf{x}_{0:k-1}^{(i)}, \mathbf{z}_{1:k-1}\right) = p\left(x_k|x_{k-1}^{(i)}\right) \tag{2.130}$$

- Finally, assuming that the particle approximation is exact gives:

$$p(i|\mathbf{z}_{1:k-1}) \propto w_{k-1}^{(i)} \tag{2.131}$$

Combining these equations reduces (2.128) to:

$$p(x_k, i|\mathbf{z}_{1:k}) \propto p\left(z_k|x_k\right)\ p\left(x_k|x_{k-1}^{(i)}\right)\ w_{k-1}^{(i)} \tag{2.132}$$

However, $p(z_k|x_k)$ cannot be evaluated before x_k is drawn. It is thus approximated by $p\left(z_k|\mu_k^{(i)}\right)$, where:

$$\mu_k^{(i)} \sim p\left(\cdot|x_{k-1}^{(i)}\right) \tag{2.133}$$

is an auxiliary particle drawn from the transition density, and used to assess the consistency of the future particle $x_k^{(i)}$ with the new observation z_k. Alternatively, $\mu_k^{(i)}$ can be chosen as the mean of this density, a deterministic choice which avoids an additional draw.

In any case, the importance density is therefore:

$$q(x_k, i|\mathbf{z}_{1:k}) \propto p\left(z_k|\mu_k^{(i)}\right)\ p\left(x_k|x_{k-1}^{(i)}\right)\ w_{k-1}^{(i)} \tag{2.134}$$

Since they are drawn from the prior distribution, (2.98) ensures that the auxiliary weights $\lambda_k^{(i)}$ follow:

$$\lambda_k^{(i)} \propto w_{k-1}^{(i)}\, \tilde{p}\left(z_k|\mu_k^{(i)}\right) \tag{2.135}$$

In the hope that the evolution noise is sufficiently weak for the auxiliary particles to correctly represent the future particle set, the parent indices $\left\{i^{(j)}\right\}_{j=1}^{N}$ are resampled according to the auxiliary weights.

The new particle set is then propagated through a standard SIS algorithm using the transition density:

$$x_k^{(j)} \sim p\left(\cdot|x_{k-1}^{\left(i^{(j)}\right)}\right) \tag{2.136}$$

with their weights updating as:

$$\begin{aligned} w_k^{(j)} &\propto \frac{\tilde{p}\left(x_k, i^{(j)}|\mathbf{z}_{1:k}\right)}{\tilde{q}\left(x_k, i^{(j)}|\mathbf{z}_{1:k}\right)} \\ &\propto \frac{\tilde{p}\left(z_k|x_k^{(j)}\right)\ \tilde{p}\left(x_k^{(j)}|x_{k-1}^{\left(i^{(j)}\right)}\right)\ w_{k-1}^{\left(i^{(j)}\right)}}{\tilde{p}\left(z_k|\mu_k^{\left(i^{(j)}\right)}\right)\ \tilde{p}\left(x_k^{(j)}|x_{k-1}^{\left(i^{(j)}\right)}\right)\ w_{k-1}^{\left(i^{(j)}\right)}} \\ &\propto \frac{\tilde{p}\left(z_k|x_k^{(j)}\right)}{\tilde{p}\left(z_k|\mu_k^{\left(i^{(j)}\right)}\right)} \end{aligned} \tag{2.137}$$

The full APF algorithm is given in algorithm 2.9.

Algorithm 2.9 Auxiliary Particle Filter

$$\forall i \in [\![1, N]\!],\ \mu_k^{(i)} \sim p\left(\cdot | x_{k-1}^{(i)}\right) \qquad \triangleright \text{ Auxiliary propagation}$$

$$\forall i \in [\![1, N]\!],\ \tilde{\lambda}_k^{(i)} \propto w_{k-1}^{(i)}\, \tilde{p}\left(z_k | \mu_k^{(i)}\right) \qquad \triangleright \text{ Auxiliary update}$$

$$\forall i \in [\![1, N]\!],\ \lambda_k^{(i)} = \tilde{\lambda}_k^{(i)} \Bigg/ \sum_{j=1}^{N} \tilde{\lambda}_k^{(j)} \qquad \triangleright \text{ Normalisation}$$

$$\left\{-, -, i^{(j)}\right\}_{j=1}^{N} = \text{Resampling}\left(\left\{\mu_k^{(i)}, \lambda_k^{(i)}\right\}_{i=1}^{N}\right) \qquad \triangleright \text{ Resampling}$$

$$\forall j \in [\![1, N]\!],\ x_k^{(j)} \sim p\left(\cdot | x_{k-1}^{\left(i^{(j)}\right)}\right) \qquad \triangleright \text{ Particle propagation}$$

$$\forall j \in [\![1, N]\!],\ \tilde{w}_k^{(j)} \propto \frac{\tilde{p}\left(z_k | x_k^{(j)}\right)}{\tilde{p}\left(z_k | \mu_k^{\left(i^{(j)}\right)}\right)} \qquad \triangleright \text{ Weight update}$$

$$\forall j \in [\![1, N]\!],\ w_k^{(j)} = \tilde{w}_k^{(j)} \Bigg/ \sum_{i=1}^{N} \tilde{w}_k^{(i)} \qquad \triangleright \text{ Normalisation}$$

Remark 2.11 (Comparison between the RPF and the APF)**.** *Both the RPF and the APF aim to alleviate the degeneracy and impoverishment problems of the standard PF, but they target different sources:*

- *The RPF restores diversity through kernel density estimation, addressing the severe impoverishment caused by near-deterministic dynamics. It is therefore suited to low evolution noise scenarios.*
- *The APF instead pre-selects promising parents from the incoming observation. It is preferred when the observation noise is small, since a peaked likelihood triggers an important single-step weight degeneracy.*

These preferences are rules of thumb rather than theoretical results, and the APF's pre-selection also degrades under strong evolution noise. Acting at different stages, both filters are ultimately complementary rather than exclusive.

2.5.4 Rao-Blackwellised Particle Filter

The marginalised particle filter [52], also known as the Rao-Blackwellised particle filter (RBPF) [53, 54], reduces the dimension of the state-space explored by the particle filter by exploiting a partial linear-Gaussian structure in the state-space model. Suppose that the state can be decomposed as:

$$x_k = \left[x_k^n, x_k^l\right] \tag{2.138}$$

such that (2.1) is linear and Gaussian in x_k^l conditionally on x_k^n. The navigation system can be rewritten as:

$$(S): \begin{cases} x_k^n = F_k^n\left(x_{k-1}^n\right)\ x_{k-1}^l + f_k^n\left(x_{k-1}^n\right) + w_k^n \\ x_k^l = F_k^l\left(x_{k-1}^n\right)\ x_{k-1}^l + f_k^l\left(x_{k-1}^n\right) + w_k^l \\ z_k = H_k\left(x_k^n\right)\ x_k^l + h_k\left(x_k^n\right) + n_k \end{cases} \tag{2.139}$$

with:

$$w_k = \begin{bmatrix} w_k^n \\ w_k^l \end{bmatrix} \sim \mathcal{N}\left(0, \begin{bmatrix} Q_k^n & (Q_k^{nl})^\top \\ Q_k^{nl} & Q_k^l \end{bmatrix}\right), \qquad n_k \sim \mathcal{N}(0, R_k) \tag{2.140}$$

x_0^n of known distribution, and x_0^l Gaussian.

The idea behind the RBPF is to factorise the PDF as:

$$p\left(\mathbf{x}_{0:k}^l, \mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}\right) = p\left(\mathbf{x}_{0:k}^l | \mathbf{x}_{0:k}^n, \mathbf{z}_{1:k}\right)\ p\left(\mathbf{x}_{0:k}^n | \mathbf{z}_{1:k}\right) \tag{2.141}$$

where $p(\mathbf{x}_{0:k}^n|\mathbf{z}_{1:k})$ is estimated via particle filtering techniques, and $p(\mathbf{x}_{0:k}^l|\mathbf{x}_{0:k}^n, \mathbf{z}_{1:k})$ is analytically solved by a Kalman filter. Here, $\mathbf{x}_{0:k}^l$ and $\mathbf{x}_{0:k}^n$ are defined similarly to $\mathbf{x}_{0:k}$.

When $Q_k^{nl} \neq 0$, the noises w_k^l and w_k^n are correlated, and the Kalman filter cannot be applied directly. Proposition 2.10 decouples them by removing the component of w_k^l correlated with w_k^n.

Proposition 2.10 (Noise decorrelation)**.** *The decorrelated noise:*

$$\overline{w}_k^l \triangleq w_k^l - Q_k^{nl}\ (Q_k^n)^{-1}\ w_k^n \tag{2.142}$$

is centred, uncorrelated with w_k^n, and has covariance:

$$\overline{Q}_k^l \triangleq \mathbb{V}\left[\overline{w}_k^l\right] = Q_k^l - Q_k^{nl}\ (Q_k^n)^{-1}\ (Q_k^{nl})^\top \tag{2.143}$$

Proof. See section B.2. □

Let:

$$\overline{F}_k^l(x_{k-1}^n) \triangleq F_k^l(x_{k-1}^n) - Q_k^{nl}\ (Q_k^n)^{-1} F_k^n(x_{k-1}^n) \tag{2.144}$$

$$\overline{f}_k^l(x_{k-1}^n) \triangleq f_k^l(x_{k-1}^n) - Q_k^{nl}\ (Q_k^n)^{-1} f_k^n(x_{k-1}^n) \tag{2.145}$$

The system (2.139) can be rewritten as:

$$\left(\overline{S}\right): \begin{cases} x_k^n = F_k^n\left(x_{k-1}^n\right)\ x_{k-1}^l + f_k^n\left(x_{k-1}^n\right) + w_k^n \\ x_k^l = \overline{F}_k^l\left(x_{k-1}^n\right)\ x_{k-1}^l + \overline{f}_k^l\left(x_{k-1}^n\right) + Q_k^{nl}\ (Q_k^n)^{-1}\ x_k^n + \overline{w}_k^l \\ z_k = H_k\left(x_k^n\right)\ x_k^l + h_k\left(x_k^n\right) + n_k \end{cases} \tag{2.146}$$

which can in turn be decomposed as:

$$\begin{aligned} (S_1): & \begin{cases} x_{k-1}^l = x_{k-1}^l \\ x_k^n = F_k^n\left(x_{k-1}^n\right)\ x_{k-1}^l + f_k^n\left(x_{k-1}^n\right) + w_k^n \end{cases} \\ (S_2): & \begin{cases} x_k^l = \overline{F}_k^l\left(x_{k-1}^n\right)\ x_{k-1}^l + \overline{f}_k^l\left(x_{k-1}^n\right) + Q_k^{nl}\ (Q_k^n)^{-1}\ x_k^n + \overline{w}_k^l \\ z_k = H_k\left(x_k^n\right)\ x_k^l + h_k\left(x_k^n\right) + n_k \end{cases} \end{aligned} \tag{2.147}$$

Note the similarities between each of the systems (S_1) and (S_2), and the navigation equations under Kalman hypotheses (2.35). Indeed, (S_1) can be treated as if the propagation of the non-linear state x_k^n were an observation of x_{k-1}^l. Its absence of dynamics leads to only performing the (pre-)correction step of the Kalman filter. On the other hand, (S_2) can be handled by the usual Kalman prediction/correction.

In full generality, a standard RBPF is computed in five steps:

- **Particle propagation.** The non-linear state is propagated according to the proposal density:

$$x_k^{n(i)} \sim q\left(\cdot|{x_{k-1}^n}^{(i)}, z_k\right) \tag{2.148}$$

- **Kalman pre-correction.** The previous linear state is corrected according to the Kalman equations applied to (S_1):

$$\tilde{F}_k^{n(i)} = F_k^n\left(x_{k-1}^{n(i)}\right) \tag{2.149}$$

$$y_k^{n(i)} = x_k^{n(i)} - \left(\tilde{F}_k^{n(i)}\,\hat{x}_{k-1|k-1}^{l(i)} + f_k^n\left(x_{k-1}^{n(i)}\right)\right) \tag{2.150}$$

$$S_k^{n(i)} = \tilde{F}_k^{n(i)}\,P_{k-1|k-1}^{l(i)}\left(\tilde{F}_k^{n(i)}\right)^\top + Q_k^n \tag{2.151}$$

$$K_k^{n(i)} = P_{k-1|k-1}^{l(i)}\left(\tilde{F}_k^{n(i)}\right)^\top\left(S_k^{n(i)}\right)^{-1} \tag{2.152}$$

$$\hat{x}_{k-1|k-1}^{l(i)} \leftarrow \hat{x}_{k-1|k-1}^{l(i)} + K_k^{n(i)}\,y_k^{n(i)} \tag{2.153}$$

$$P_{k-1|k-1}^{l(i)} \leftarrow \left(I - K_k^{n(i)}\,\tilde{F}_k^{n(i)}\right)P_{k-1|k-1}^{l(i)} \tag{2.154}$$

- **Kalman prediction.** The current linear state is then predicted via the dynamics of (S_2):

$$\tilde{F}_k^{l(i)} = \overline{F}_k^l\left(x_{k-1}^{n(i)}\right) \tag{2.155}$$

$$\hat{x}_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)}\,\hat{x}_{k-1|k-1}^{l(i)} + \overline{f}_k^l\left(x_{k-1}^{n(i)}\right) + Q_k^{nl}\,(Q_k^n)^{-1}\,x_k^{n(i)} \tag{2.156}$$

$$P_{k|k-1}^{l(i)} = \tilde{F}_k^{l(i)}\,P_{k-1|k-1}^{l(i)}\,\left(\tilde{F}_k^{l(i)}\right)^\top + \overline{Q}_k^l \tag{2.157}$$

$$\tilde{H}_k^{(i)} = H_k\left(x_k^{n(i)}\right) \tag{2.158}$$

$$y_k^{l(i)} = z_k - \left(\tilde{H}_k^{(i)}\,\hat{x}_{k|k-1}^{l(i)} + h_k\left(x_k^{n(i)}\right)\right) \tag{2.159}$$

$$S_k^{l(i)} = \tilde{H}_k^{(i)}\,P_{k|k-1}^{l(i)}\left(\tilde{H}_k^{(i)}\right)^\top + R_k \tag{2.160}$$

- **Weight update.** The weights corresponding to each particle are updated with respect to the new observation:

$$\tilde{w}_k^{n(i)} = w_{k-1}^{n(i)}\,\frac{\tilde{p}\left(z_k|x_k^{n(i)}\right)\,\tilde{p}\left(x_k^{n(i)}|x_{k-1}^{n(i)}\right)}{\tilde{q}\left(x_k^{n(i)}|x_{k-1}^{n(i)}, z_k\right)} \tag{2.161}$$

$$w_k^{n(i)} = \tilde{w}_k^{n(i)}\Bigg/\sum_{j=1}^{N}\tilde{w}_k^{n(j)} \tag{2.162}$$

- **Kalman correction.** The linear state is corrected through the standard Kalman correction for (S_2):

$$K_k^{l(i)} = P_{k|k-1}^{l(i)}\left(\tilde{H}_k^{(i)}\right)^\top\left(S_k^{l(i)}\right)^{-1} \tag{2.163}$$

$$\hat{x}_{k|k}^{l(i)} = \hat{x}_{k|k-1}^{l(i)} + K_k^{l(i)}\,y_k^{l(i)} \tag{2.164}$$

$$P_{k|k}^{l(i)} = \left(I - K_k^{l(i)}\,\tilde{H}_k^{(i)}\right)P_{k|k-1}^{l(i)} \tag{2.165}$$

In the RBPF, only the non-linear state is sampled, while the conditional distribution of the linear substate is propagated analytically. The particle approximation is thus the mixture [55]:

$$p(x_k|\mathbf{z}_{1:k})\,dx_k \approx \sum_{i=1}^{N} w_k^{n(i)}\,\delta_{x_k^{n(i)}}\left(dx_k^n\right)\,\mathcal{N}\left(\hat{x}_{k|k}^{l(i)}, P_{k|k}^{l(i)}\right)\left(dx_k^l\right) \tag{2.166}$$

The MMSE estimator of section 2.2 can then be approximated by:

$$\hat{x}_k = \sum_{i=1}^{N} w_k^{n(i)}\,x_k^{(i)} \tag{2.167}$$

where

$$x_k^{(i)} \triangleq \begin{bmatrix} x_k^{n(i)} \\ \hat{x}_{k|k}^{l(i)} \end{bmatrix} = \mathbb{E}\left[x_k|\mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k}\right] \tag{2.168}$$

Furthermore, since the non-linear state is deterministic when conditioned on the sampled trajectory, we have:

$$\mathbb{V}\left[x_k|\mathbf{x}_{0:k}^{n(i)}, \mathbf{z}_{1:k}\right] = \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \tag{2.169}$$

Applying the law of total covariance finally gives the associated empirical covariance matrix:

$$\hat{P}_k = \sum_{i=1}^{N} w_k^{n(i)}\left(x_k^{(i)} - \hat{x}_k\right)\left(x_k^{(i)} - \hat{x}_k\right)^\top + \sum_{i=1}^{N} w_k^{n(i)} \begin{bmatrix} 0 & 0 \\ 0 & P_{k|k}^{l(i)} \end{bmatrix} \tag{2.170}$$

where the first term is the covariance of the conditional means, and the second is the average conditional covariance.

The full RBPF algorithm is summarised in Algorithm 2.10.

Algorithm 2.10 Rao-Blackwellised Particle Filter

$\forall i \in [\![1, N]\!]$, propagate $x_k^{n(i)}$	using (2.148)
$\forall i \in [\![1, N]\!]$, update $\hat{x}_{k-1\|k-1}^{l(i)}$ and $P_{k-1\|k-1}^{l(i)}$	using (2.149) through (2.154)
$\forall i \in [\![1, N]\!]$, predict $\hat{x}_{k\|k-1}^{l(i)}$ and $P_{k\|k-1}^{l(i)}$	using (2.155) through (2.160)
$\forall i \in [\![1, N]\!]$, update $w_k^{n(i)}$	using (2.161) through (2.162)
$\forall i \in [\![1, N]\!]$, compute $\hat{x}_{k\|k}^{l(i)}$ and $P_{k\|k}^{l(i)}$	using (2.163) through (2.165)

It must be noted that the version presented by Algorithm 2.10 uses a simple SIS to handle the particles, but any more advanced algorithm presented in this section can be used instead. Furthermore, the likelihood $\tilde{p}\left(z_k|x_k^{n(i)}\right)$ and the transition $\tilde{p}\left(x_k^{n(i)}|x_{k-1}^{n(i)}\right)$ in the weight update are white Gaussian densities, with covariances $S_k^{l(i)}$ and $S_k^{n(i)}$ respectively. This is why $S_k^{l(i)}$ is computed within the Kalman prediction step, before the weight update. However, the correction step itself must be performed after the weight update in situations where a resampling step is involved.

Remark 2.12 (Single Kalman filter)**.** *In general, one Kalman filter is required per particle, since* $\tilde{F}_k^{n(i)}$*,* $\tilde{F}_k^{l(i)}$ *and* $\tilde{H}_k^{(i)}$ *all depend on* $x_{k-1}^{n(i)}$ *and* $x_k^{n(i)}$*. However, if these matrices do not depend on the non-linear state, then the gains* $K_k^{n(i)}$ *and* $K_k^{l(i)}$*, and the covariances* $P_{k|k}^{l(i)}$ *are identical for all particles. In this scenario, a single Kalman filter can be used for the whole particle set, which substantially reduces the computational cost of the RBPF.*

Remark 2.13 (Triangular model)**.** *If* (2.139) *is such that* $F_k^n = 0$*, the system* (S) *is called triangular. In this scenario,* x_k^n *does not depend on* x_{k-1}^l*, which decouples the two filters entirely. The Kalman pre-correction step can thus be omitted, and the RBPF becomes a simple interleaving between a PF and a collection of Kalman filters.*

CHAPTER 3

GAUSSIAN PROCESSES

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3.1 Introduction

The analytical and Monte Carlo methods of chapter 2 all assume the observation function h_k to be fully known. However, many navigation scenarios depend on a physical field that is either unavailable or known only through scarce and noisy measurements.

A first way to fill this gap is to reconstruct the field directly from the measurements: **TODO: j'avais oublié, mais à reprendre + présenter des cas d'application au krigeage**

- Deterministic interpolation, through nearest-neighbour selection, multilinear interpolation over a regular grid, or spline fitting;
- Linear regression on a fixed basis, such as polynomials or sinusoids;
- More expressive models, such as neural networks.

All of them return a single value at the queried location, without quantifying how much it can be trusted. Their use in the filters of chapter 2, however, requires the observation likelihood $p(z_k|x_k)$, a full distribution rather than a point prediction. The reconstruction error must then be absorbed into the observation noise, usually as a covariance held constant over the whole domain, although this error is almost zero next to a measurement and grows in sparsely sampled regions.

This chapter instead models such fields as a Gaussian process (GP), an adaptable framework that encodes both a prior expectation of the field and its spatial correlation structure. Together with its uncertainty, the resulting estimator can then serve as a surrogate observation model [56, 57] for the filters of chapter 2.

The rest of this chapter is organised as follows:

- Section 3.2 defines GPs through their mean functions and covariance kernels. It discusses the choice of kernel.
- Section 3.3 then introduces kriging as a map reconstruction tool in situations where we are given preexisting data samples.
- Section 3.4 addresses the cost of inverting large Gram matrices in kriging by approximating k with a reduced-rank kernel, therefore lowering the effective complexity of kriging methods.
- Finally, section 3.5 derives two representations of the mean function over a dictionary of functions, used to reconstruct the field when no prior knowledge is available.

3.2 Definition and Properties

Formally, a stochastic process generalises the notion of random variable to a random function, *i.e.* a family $(f(x))_{x\in\mathcal{U}}$ of random variables indexed by a set $\mathcal{U}$, which may represent time $\mathbb{R}_+$, physical space $\mathbb{R}^2$ or $\mathbb{R}^3$, or any other domain. As shown by Definition 3.1, a GP is a special case of stochastic process [58].

Definition 3.1 (Gaussian process)**.** *For $d_z \in \mathbb{N}^*$, a stochastic process $(f(x))_{x\in\mathcal{U}}$ taking values in $\mathbb{R}^{d_z}$ is a GP if every finite linear combination of its values follows a Gaussian distribution. Formally, for all $n \in \mathbb{N}^*$, all $x_1, \dots, x_n \in \mathcal{U}$ and all $a_1, \dots, a_n \in \mathbb{R}^{d_z}$, there exist $\mu \in \mathbb{R}^{d_z}$ and $\Sigma \in \mathcal{S}_{d_z}^+(\mathbb{R})$ such that:*

$$\sum_{i=1}^{n} a_i^\top f(x_i) \sim \mathcal{N}(\mu, \Sigma) \tag{3.1}$$

The GP is said to be non-degenerate if $\Sigma \succ 0$ whenever the $x_1, \dots, x_n$ are pairwise distinct and the coefficients $a_1, \dots, a_n$ are not all zero.

A GP is associated with a mean function:

$$m : \begin{cases} \mathcal{U} \to \mathbb{R}^{d_z} \\ x \mapsto \mathbb{E}[f(x)] \end{cases} \tag{3.2}$$

and a covariance kernel:

$$k : \begin{cases} \mathcal{U} \times \mathcal{U} \to \mathcal{M}_{d_z,d_z}(\mathbb{R}) \\ x, x' \mapsto \mathbb{E}\left[(f(x) - m(x))(f(x') - m(x'))^\top\right] \end{cases} \tag{3.3}$$

3.2.1 Basic Properties

As shown by Proposition 3.1, m and k entirely characterise every finite-dimensional marginal of f. By Kolmogorov's extension theorem, they also uniquely determine the law of f. We thus write $f \sim \mathcal{GP}(m, k)$ for a GP with mean function m and covariance kernel k.

Proposition 3.1 (Finite-dimensional marginals)**.** *For $n \in \mathbb{N}^*$, let $x_1, \dots, x_n$ be pairwise distinct elements of $\mathcal{U}$. The joint distribution of $F \triangleq \left[f(x_1)^\top, \dots, f(x_n)^\top\right]^\top$ is the multivariate Gaussian distribution:*

$$\begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix}, \begin{bmatrix} k(x_1,x_1) & \cdots & k(x_1,x_n) \\ \vdots & \ddots & \vdots \\ k(x_n,x_1) & \cdots & k(x_n,x_n) \end{bmatrix}\right) \tag{3.4}$$

where each block $k(x_i, x_j) \in \mathcal{M}_{d_z,d_z}(\mathbb{R})$.

Proof. According to Definition 3.1, every linear combination:

$$a^\top F = \sum_{i=1}^{n} a_i^\top f(x_i), \qquad a \triangleq \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \in \mathbb{R}^{nd_z} \tag{3.5}$$

is Gaussian. By the Cramér-Wold theorem [59] F is thus also a Gaussian vector of $\mathbb{R}^{nd_z}$.

Such a vector is entirely determined by its mean and covariance, which we compute. In particular, we have:

$$\mathbb{E}[F] = \begin{bmatrix} \mathbb{E}[f(x_1)] \\ \vdots \\ \mathbb{E}[f(x_n)] \end{bmatrix} = \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \tag{3.6}$$

By definition of k, it comes that:

$$\begin{aligned}\mathbb{V}[F] &= \Big[\mathbb{E}\left[(f(x_i) - m(x_i))(f(x_j) - m(x_j))^\top\right]\Big]_{i,j=1}^{n} \\ &= \Big[k(x_i, x_j)\Big]_{i,j=1}^{n}\end{aligned} \tag{3.7}$$

which is exactly (3.4). □

For two tuples of inputs $\mathbf{x} \triangleq (x_1, \dots, x_n) \in \mathcal{U}^n$ and $\mathbf{x}' \triangleq (x'_1, \dots, x'_p) \in \mathcal{U}^p$ of sizes $n, p \in \mathbb{N}^*$, we introduce the following vectorised notation:

$$f(\mathbf{x}) \triangleq \begin{bmatrix} f(x_1) \\ \vdots \\ f(x_n) \end{bmatrix} \tag{3.8}$$

$$m(\mathbf{x}) \triangleq \begin{bmatrix} m(x_1) \\ \vdots \\ m(x_n) \end{bmatrix} \tag{3.9}$$

$$k(\mathbf{x}, \mathbf{x}') \triangleq \begin{bmatrix} k(x_1, x'_1) & \cdots & k(x_1, x'_p) \\ \vdots & \ddots & \vdots \\ k(x_n, x'_1) & \cdots & k(x_n, x'_p) \end{bmatrix} \tag{3.10}$$

Using these notations, Proposition 3.1 reads compactly:

$$\forall n \in \mathbb{N}^*,\ \forall \mathbf{x} \in \mathcal{U}^n \text{ pairwise distinct},\ f(\mathbf{x}) \sim \mathcal{N}\left(m(\mathbf{x}), k(\mathbf{x}, \mathbf{x})\right) \tag{3.11}$$

A GP is thus a consistent generalisation of the multivariate Gaussian distribution to an infinite and potentially continuous index set. Both of its parameters admit a direct interpretation. The mean function m encodes prior knowledge about the average behaviour of f, while the kernel k captures the correlation structure. The diagonal blocks $k(x, x)$ represent the covariance of $f(x)$, while the off-diagonal blocks $k(x, x')$, for $x \neq x'$, describe the cross-covariance between $f(x)$ and $f(x')$.

In order for the GP to be non-degenerate, the matrix $k(\mathbf{x}, \mathbf{x})$ must be SPD in the sense of Definition 2.1 for every finite tuple $\mathbf{x}$ of pairwise distinct inputs. Definition 3.2 recasts this as a condition on k alone [60], and Proposition 3.2 proves the equivalence between the two.

Definition 3.2 (Symmetric positive definite kernel)**.** *A kernel $k : \mathcal{U} \times \mathcal{U} \to \mathcal{M}_{d_z, d_z}(\mathbb{R})$ is symmetric positive semi-definite (SPSD) if it is:*

- ***Symmetric:***

$$\forall x, x' \in \mathcal{U},\ k(x', x) = k(x, x')^\top \tag{3.12}$$

- ***Positive semi-definite:***

$$\begin{aligned}&\forall n \in \mathbb{N}^*,\ \forall x_1, \dots, x_n \in \mathcal{U},\ \forall a_1, \dots, a_n \in \mathbb{R}^{d_z}, \\ &\sum_{i=1}^{n} \sum_{j=1}^{n} a_i^\top k(x_i, x_j)\, a_j \geq 0\end{aligned} \tag{3.13}$$

If the inequality is strict for every choice of pairwise distinct points $x_1, \ldots, x_n$ and coefficients $a_1, \ldots, a_n$ not all zero, then k is symmetric positive definite (SPD).

Proposition 3.2 (Definite kernel/matrix equivalence)**.** *The following equivalences hold:*

- *$k : \mathcal{U} \times \mathcal{U} \to \mathcal{M}_{d_z,d_z}(\mathbb{R})$ is a SPSD kernel if and only if, for every $n \in \mathbb{N}^*$ and every $\mathbf{x} \in \mathcal{U}^n$, $k(\mathbf{x}, \mathbf{x}) \in \mathcal{M}_{nd_z,nd_z}(\mathbb{R})$ is a SPSD matrix.*
- *k is a SPD kernel if and only if, for every such tuple $\mathbf{x}$ of pairwise distinct points, $k(\mathbf{x}, \mathbf{x})$ is a SPD matrix.*

Proof. See section B.3. □

Remark 3.1 (Necessity of distinct points)**.** *While pairwise distinctness does not intervene in the computation of the strict version above, it is necessary for the SPD condition to be satisfiable. Indeed, at coinciding inputs $x_1 = x_2$, the choice $a_1 = -a_2 = 1$, with the other coefficients set to zero, cancels the quadratic form for non-zero coefficients, so no kernel could be SPD.*

Remark 3.2 (Non-diagonal blocks)**.** *While Proposition 3.2 equates the SPDness of k with that of $k(\mathbf{x}, \mathbf{x})$ at distinct points, it does not make the off-diagonal blocks $k(x, x')$ SPD themselves. For $x \neq x'$, these may be non-positive or even non-symmetric.*

Throughout the rest of this document, we assume that the covariance kernel k is SPD and that the points of every collection $\mathbf{x} = (x_1, \ldots, x_n) \in \mathcal{U}^n$ are pairwise distinct. This ensures that we only work with non-degenerate GPs.

3.2.2 Common Kernels

Assume in this section that $\mathcal{U}$ is a finite-dimensional normed vector space, *i.e.* $\mathcal{U} = \mathbb{R}^{d_x}$ for some $d_x \in \mathbb{N}^*$, up to isomorphism. A natural way to build kernels on $\mathcal{U}$ is to let the correlation between the field values at two points x, x' depend only on the distance $\|x - x'\|$ between them. Kernels of this form are called radial kernels [61], and are formally defined by Definition 3.3.

Since f typically represents a physical field, observations tend to become less correlated as the distance between them increases. Definition 3.4 introduces a common family of radial kernels modelling this behaviour, called the Matérn kernels [62]. Proposition 3.3 then establishes their continuity and SPDness.

Definition 3.3 (Radial kernel)**.** *A kernel $k : \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \to \mathcal{M}_{d_z,d_z}(\mathbb{R})$ is radial if there exists a function $\kappa : \mathbb{R}_+ \to \mathcal{M}_{d_z,d_z}(\mathbb{R})$ and a norm $\|\cdot\|$ on $\mathbb{R}^{d_x}$ such that:*

$$\forall x, x' \in \mathbb{R}^{d_x},\ k(x, x') = \kappa(\|x - x'\|) \tag{3.14}$$

To simplify notations, we identify κ with k in the remainder of this document, such that we write:

$$\forall x, x' \in \mathbb{R}^{d_x},\ k(x, x') = k(\|x - x'\|) \tag{3.15}$$

Definition 3.4 (Matérn kernel). *Assume that $d_z = 1$, and let $\nu, l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel of smoothness ν, correlation length l, and characteristic variance σ_f^2 is the radial kernel defined as:*

$$k_\nu : \begin{cases} \mathbb{R}_+ \to \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \dfrac{2^{1-\nu}}{\Gamma(\nu)} \left(\dfrac{\sqrt{2\nu}\, r}{l} \right)^\nu K_\nu \left(\dfrac{\sqrt{2\nu}\, r}{l} \right) \end{cases} \tag{3.16}$$

where Γ is the Gamma function, and K_ν is the modified Bessel function of the second kind [63].

This family extends to $\nu = +\infty$ by taking the pointwise limit, yielding the squared exponential (SE) kernel:

$$k_{\text{SE}} \triangleq k_{+\infty} : \begin{cases} \mathbb{R}_+ \to \mathbb{R}_+ \\ r \mapsto \sigma_f^2 \exp\left(-\dfrac{r^2}{2\, l^2} \right) \end{cases} \tag{3.17}$$

Proposition 3.3 (Properties of Matérn kernels). *Let $\nu \in \mathbb{R}_+^* \cup \{+\infty\}$ and $l, \sigma_f^2 \in \mathbb{R}_+^*$. The Matérn kernel associated with ν, l and σ_f^2 is a continuous SPD radial kernel.*

Proof. See section B.4. □

The parameter ν controls the regularity. Indeed, for $p \in \mathbb{N}$, it can be shown [64] that a GP equipped with k_ν is p times mean-square differentiable if and only if $\nu > p$. For half-integer $\nu = p + \frac{1}{2}$, Matérn kernels can be written as the product of an exponential and a polynomial function of degree p:

$$k_{p+1/2}(r) = \sigma_f^2 \exp\left(-\frac{\sqrt{2p+1}r}{l} \right) \frac{\Gamma(p+1)}{\Gamma(2p+1)} \sum_{i=0}^{p} \frac{(p+i)!}{i!(p-i)!} \left(\frac{2\sqrt{2p+1}r}{l} \right)^{p-i} \tag{3.18}$$

In particular, the following kernels correspond to usual choices of ν, the SE kernel being recalled here for comparison:

- **Exponential kernel.**

$$k_{1/2}(r) = \sigma_f^2 \exp\left(-\frac{r}{l} \right) \tag{3.19}$$

- **Once-differentiable Matérn kernel.**

$$k_{3/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{3}\, r}{l} \right) \exp\left(-\frac{\sqrt{3}\, r}{l} \right) \tag{3.20}$$

- **Twice-differentiable Matérn kernel.**

$$k_{5/2}(r) = \sigma_f^2 \left(1 + \frac{\sqrt{5}\, r}{l} + \frac{5\, r^2}{3\, l^2} \right) \exp\left(-\frac{\sqrt{5}\, r}{l} \right) \tag{3.21}$$

- **Squared exponential kernel.**

$$k_{+\infty}(r) = \sigma_f^2 \exp\left(-\frac{r^2}{2\, l^2} \right) \tag{3.22}$$

Figure 3.1 illustrates the radial decay of these kernels, evaluated with unit variance $\sigma_f^2 = 1$ and correlation length $l = 1$ to facilitate comparison.

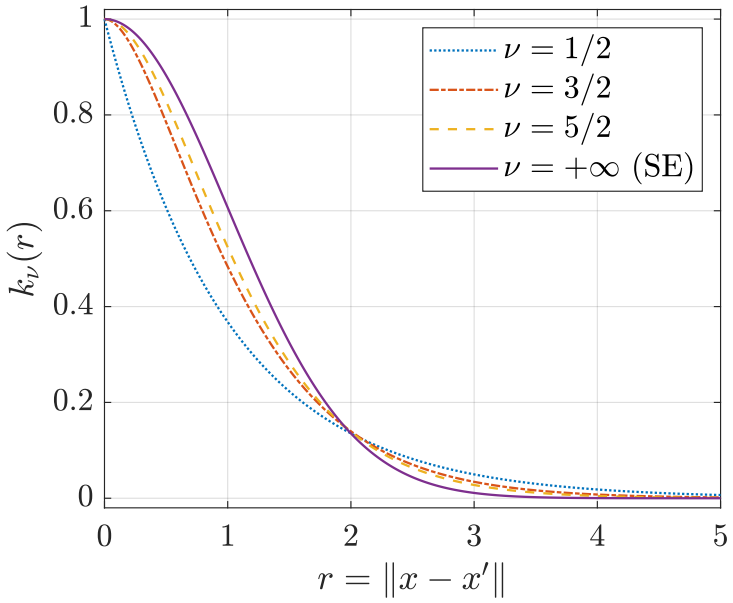


Figure 3.1: Radial profile of Matérn kernels with increasing smoothness, together with their squared exponential limit. Larger values of ν produce smoother kernels, with flatter behaviours near the origin and faster-decaying tails.

Remark 3.3 (Anisotropic kernel)**.** *The norm used in Definition 3.3 does not necessarily need to be the canonical norm of $\mathbb{R}^{d_x}$. In particular, for $M \in \mathcal{S}_{d_x}^{++}(\mathbb{R})$, the Mahalanobis norm:*

$$x \in \mathbb{R}^{d_x} \mapsto \|x\|_M \triangleq \sqrt{x^\top M^{-1} x} \tag{3.23}$$

may produce anisotropic radial kernels in situations where M is not proportional to the identity matrix.

The main limitation of Matérn kernels is that they are scalar, while our observation model takes values in $\mathbb{R}^{d_z}$. To alleviate this issue, Proposition 3.4 presents a method to create a vector-valued kernel from a scalar one.

Proposition 3.4 (Separable kernel)**.** *Let $k : \mathcal{U} \times \mathcal{U} \to \mathbb{R}$ be a scalar SPD kernel, and let $\Sigma \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$. Then:*

$$k_{\text{sep}} : \begin{cases} \mathcal{U} \times \mathcal{U} \to \mathcal{M}_{d_z,d_z}(\mathbb{R}) \\ x, x' \mapsto \Sigma\, k(x, x') \end{cases} \tag{3.24}$$

is a valid SPD kernel for an $\mathbb{R}^{d_z}$-valued GP.

Furthermore, when k is continuous, k_{sep} is also continuous.

Proof. The symmetry and potential continuity of k_{sep} are direct from that of k and Σ. The positive definiteness can be established by letting A be a Cholesky factor of the SPD matrix Σ, and considering, for every $n \in \mathbb{N}^*$ and every not all zero coefficients $a_1, \ldots, a_n \in \mathbb{R}^{d_z}$:

$$\forall i \in [\![1, n]\!],\ b_i \triangleq A^\top a_i \tag{3.25}$$

Since A is invertible, these coefficients are not all zero, and thus:

$$\begin{aligned}\sum_{i=1}^{n}\sum_{j=1}^{n} a_i^\top k_{\text{sep}}(x_i,x_j)\,a_j &= \sum_{i=1}^{n}\sum_{j=1}^{n} a_i^\top A\,A^\top a_j\,k(x_i,x_j)\\ &= \sum_{i=1}^{n}\sum_{j=1}^{n} b_i^\top b_j\,k(x_i,x_j)\\ &= \sum_{l=1}^{d_z}\sum_{i=1}^{n}\sum_{j=1}^{n} b_i^{(l)}\,b_j^{(l)}\,k(x_i,x_j)\end{aligned} \tag{3.26}$$

where $b_i^{(l)}$ denotes the l-th component of b_i. Each inner double sum is a quadratic form of k, hence non-negative, and at least one of them is strictly positive since at least one of the vectors $\left(b_1^{(l)},\dots,b_n^{(l)}\right)$ is non-zero. □

Remark 3.4 (Multi-scale kernels)**.** *For $n \in \mathbb{N}^*$, $\Sigma_1,\dots,\Sigma_n \in \mathcal{S}_{d_z}^{++}(\mathbb{R})$ and $k^{(1)},\dots,k^{(n)}$ Matérn kernels of correlation lengths $l_1 < \dots < l_n \in \mathbb{R}_+^*$, the multi-scale kernel is:*

$$k_{\text{MS}} : \begin{cases}\mathbb{R}_+ \to \mathcal{M}_{d_z,d_z}(\mathbb{R})\\ \quad r \mapsto \sum_{i=1}^{n}\Sigma_i\,k^{(i)}(r)\end{cases} \tag{3.27}$$

Each of its terms is separable in the sense of Proposition 3.4, hence SPD, and a sum of SPD kernels with positive coefficients is itself SPD. When the correlation lengths have different orders of magnitude, such kernels can be used to capture local fluctuations and global trends simultaneously.

Every kernel considered in this section is specified by a set of hyperparameters, such as ν, l and σ_f^2 for a Matérn kernel, Σ for its separable extension, or one such set per scale for a multi-scale model. Their estimation is a standard step in GP modelling, and is independent of bayesian estimation considerations. However, the specific characteristics of our applications can be used to improve the estimation of such hyperparameters. These aspects are thus discussed in section 4.2.2, and the hyperparameters are assumed to be fully specified for the rest of this chapter.

3.3 Gaussian Process Regression

Gaussian process regression, also known as kriging [65, 66, 67], estimates the value of a field at an unobserved location using noisy measurements of the same field at known locations. Formally, let $f \sim \mathcal{GP}(m,k) : \mathcal{U} \to \mathbb{R}^{d_z}$, and $x \in \mathcal{U}$. We wish to estimate the law of:

$$z \triangleq f(x) + \varepsilon \in \mathbb{R}^{d_z}, \qquad \varepsilon \sim \mathcal{N}(0,R), \qquad R \in \mathcal{S}_{d_z}^{+}(\mathbb{R}) \tag{3.28}$$

where ε is independent of f.

While kriging usually aims to reconstruct $f(x)$ itself, the query noise ε models the observation uncertainty, so that kriging estimates the noisy observation from (2.1). The standard formulation can nonetheless be recovered by taking $R = 0$.

Without further information regarding f, we would only be able to state that z follows a Gaussian distribution of expectation $m(x)$ and covariance $k(x,x)+R$. However, we are given $\tilde{N} \in \mathbb{N}^*$ distinct training points $\tilde{\mathbf{x}} \triangleq (\tilde{x}_1, \dots, \tilde{x}_{\tilde{N}}) \in \mathcal{U}^{\tilde{N}}$ and their corresponding noisy observations:

$$\tilde{\mathbf{z}} \triangleq \begin{bmatrix} f(\tilde{x}_1) + \varepsilon_1 \\ \vdots \\ f(\tilde{x}_{\tilde{N}}) + \varepsilon_{\tilde{N}} \end{bmatrix} \in \mathbb{R}^{\tilde{N} d_z}, \qquad \varepsilon_i \sim \mathcal{N}(0, \tilde{R}), \qquad \tilde{R} \in \mathcal{S}_{d_z}^+(\mathbb{R}) \tag{3.29}$$

where $\varepsilon_1, \dots, \varepsilon_{\tilde{N}}$ are mutually independent and independent of f.

The goal of this section is to derive the best linear unbiased estimator (BLUE) of z with respect to $\tilde{\mathbf{z}}$, *i.e.* the estimator:

$$\hat{z} = W^\top \tilde{\mathbf{z}} \tag{3.30}$$

parametrised by a weight matrix:

$$W \triangleq \begin{bmatrix} w_1 \\ \vdots \\ w_{\tilde{N}} \end{bmatrix} \in \mathcal{M}_{\tilde{N} d_z, d_z}(\mathbb{R}) \tag{3.31}$$

such that the error vector has an expectation of 0 and a minimum covariance. This setting is illustrated by Figure 3.2 for a $\mathbb{R}^2 \to \mathbb{R}$ field.

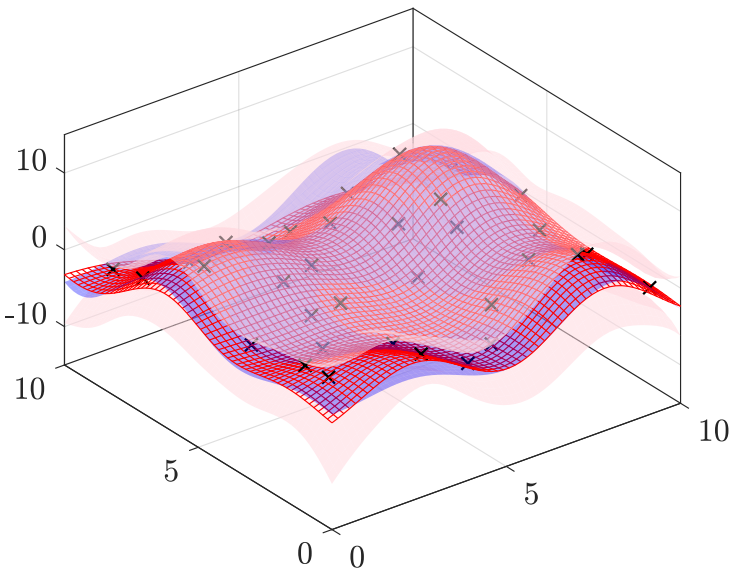


Figure 3.2: Ordinary kriging (red mesh) of an unknown field (blue surface) from noisy samples (black crosses). The confidence band (red envelope) is proportional to the square root of the interpolation covariance, and widens away from the samples.

Since $\hat{z} - z$ is a vector, its covariance is a matrix, and cannot be minimised as such. To alleviate this issue, we minimise its trace, which is exactly the mean squared error:

$$\operatorname{tr}(\mathbb{V}[\hat{z} - z]) = \mathbb{E}\left[\|\hat{z} - z\|^2\right] \tag{3.32}$$

In the following, we present three kriging estimates, based on increasingly relaxed assumptions made on our knowledge of the underlying GP. Each method relies on the inversion of the Gram matrix:

$$\begin{aligned} G(\tilde{\mathbf{x}}) &\triangleq k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R} \\ &= \begin{bmatrix} k(\tilde{x}_1, \tilde{x}_1) + \tilde{R} & \cdots & k(\tilde{x}_1, \tilde{x}_{\tilde{N}}) \\ \vdots & \ddots & \vdots \\ k(\tilde{x}_{\tilde{N}}, \tilde{x}_1) & \cdots & k(\tilde{x}_{\tilde{N}}, \tilde{x}_{\tilde{N}}) + \tilde{R} \end{bmatrix} \end{aligned} \tag{3.33}$$

of $\tilde{\mathbf{x}}$ with respect to f, where $\otimes$ is the Kronecker matrix product. Since k is SPD and the training points are distinct, it comes that:

$$G(\tilde{\mathbf{x}}) \in \mathcal{S}^{++}_{\tilde{N}d_z}(\mathbb{R}) \tag{3.34}$$

and so $G(\tilde{\mathbf{x}})$ is invertible, even in the noiseless case $\tilde{R} = 0$.

We also assume that the query point x differs from the training points $\tilde{\mathbf{x}}$, so that the joint law of $(z, \tilde{\mathbf{z}})$ is non-degenerate. This hypothesis is not restrictive and is only needed for the proofs. Indeed, the estimators are continuous in x and thus remain valid at the training points as soon as m and k are continuous, which we assume.

Moreover, $\mathbb{V}[\hat{z} - z]$ is a covariance matrix, and therefore SPSD whatever the weight matrix W. It is furthermore positive definite when x is not one of the training points, since $\hat{z} - z$ is a non-degenerate linear combination of z and $\tilde{\mathbf{z}}$.

3.3.1 Simple Kriging

Simple kriging (SK) is the simplest form of kriging, and assumes that the mean function m is known. Proposition 3.5 gives a closed-form expression of the SK estimator and its associated covariance matrix, while Proposition 3.7 shows that it coincides with the conditional distribution of z given $\tilde{\mathbf{z}}$.

Proposition 3.5 (Simple kriging estimator)**.** *Using the notations introduced in section 3.3, the SK estimator of z and its error covariance are:*

$$\begin{cases} \hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = m(x) + k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} \left(\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})\right) \\ \quad R_{\mathrm{SK}}(x, \tilde{\mathbf{x}}) = k(x, x) + R - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k\left(\tilde{\mathbf{x}}, x\right) \end{cases} \tag{3.35}$$

Proof. Since m is known, we may assume it to be identically equal to 0 by working with $f - m$ instead of f, without any loss of generality. In the centred case, (3.32) can be developed as follows:

$$\mathbb{V}[W^\top \tilde{\mathbf{z}} - z] = W^\top G(\tilde{\mathbf{x}})\, W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}})\, W + k(x, x) + R \tag{3.36}$$

By symmetry of $G(\tilde{\mathbf{x}})$, differentiating (3.32) with respect to W yields:

$$\begin{aligned} & \frac{\partial}{\partial W} \left(\mathrm{tr}\left(W^\top G(\tilde{\mathbf{x}})\, W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}})\, W + k(x, x) + R\right)\right) = 0 \\ \iff \quad & \frac{\partial}{\partial W} \left(\mathrm{tr}\left(W^\top G(\tilde{\mathbf{x}})\, W\right)\right) - 2 \frac{\partial}{\partial W} \left(\mathrm{tr}\left(W^\top k(\tilde{\mathbf{x}}, x)\right)\right) = 0 \\ \iff \quad & 2 G(\tilde{\mathbf{x}})\, W - 2 k(\tilde{\mathbf{x}}, x) = 0 \\ \iff \quad & W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{aligned} \tag{3.37}$$

The SK estimator is thus:

$$\hat{z}_{\mathrm{SK}} = W^\top \tilde{\mathbf{z}} = k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \tag{3.38}$$

and its corresponding covariance:

$$\begin{aligned} R_{\mathrm{SK}} &= W^\top G(\tilde{\mathbf{x}})\, W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}})\, W + k(x,x) + R \\ &= k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} G(\tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &\quad - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) + k(x,x) + R \\ &= k(x,x) + R - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \end{aligned} \tag{3.39}$$

The general case is obtained by adding m back into f. □

Remark 3.5 (Additive query noise)**.** *In Proposition 3.5, the term R appears because the noisy query z is predicted rather than the field value $f(x)$. The two share the same estimator, and differ in covariance only by an additive R term. Kriging the field can thus be done by subtracting R, or equivalently by taking $R = 0$ as mentioned in the introduction to kriging.*

Remark 3.6 (Error covariance minimisation)**.** *While the SK weight matrix $W = G^{-1} k(\tilde{\mathbf{x}}, x)$ was obtained by minimising the trace of $\mathbb{V}[\hat{z} - z]$, it minimises the whole covariance in the SPSD sense. Indeed:*

$$\begin{aligned} \mathbb{V}[\hat{z} - z] - R_{\mathrm{SK}} &= W^\top G(\tilde{\mathbf{x}})\, W - W^\top k(\tilde{\mathbf{x}}, x) - k(x, \tilde{\mathbf{x}})\, W + k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\ &= \big(W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)\big)^\top G(\tilde{\mathbf{x}}) \big(W - G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)\big) \\ &\succeq 0 \end{aligned} \tag{3.40}$$

with equality if and only if $W = G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x)$ since $G(\tilde{\mathbf{x}}) \succ 0$.

Proposition 3.6 (Exact interpolation in the noiseless case)**.** *Assume that $\tilde{R} = 0$, and let $i \in [\![1, \tilde{N}]\!]$. The SK estimator interpolates the data at $\tilde{x}_i$ exactly:*

$$\begin{cases} \hat{z}_{\mathrm{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) = \tilde{z}_i \\ R_{\mathrm{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}) = R \end{cases} \tag{3.41}$$

In particular, the error covariance reduces to the query noise R alone, and vanishes when $R = 0$. That degenerate case is the one set aside at the start of this section, and is reached by continuity in x. On the contrary, a positive training noise $\tilde{R} \succ 0$ turns kriging into a smoother rather than an interpolator [68].

Proof. Since $\tilde{R} = 0$, the Gram matrix reduces to $G(\tilde{\mathbf{x}}) = k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})$. Let $e_i \triangleq [0, \dots, 0, 1, 0, \dots, 0]^\top$ be the i-th canonical basis vector of $\mathbb{R}^{\tilde{N}}$. The matrix $e_i^\top \otimes I_{d_z}$ extracts the i-th block of a stacked matrix, such that:

$$\big(e_i^\top \otimes I_{d_z}\big)\, G(\tilde{\mathbf{x}}) = \big(e_i^\top \otimes I_{d_z}\big)\, k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) = k(\tilde{x}_i, \tilde{\mathbf{x}}) \tag{3.42}$$

The kriging weights thus reduce to this selector:

$$k(\tilde{x}_i, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} = \big(e_i^\top \otimes I_{d_z}\big)\, G(\tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} = e_i^\top \otimes I_{d_z} \tag{3.43}$$

Evaluating (3.35) at $x = \tilde{x}_i$ by the continuity yields:

$$\begin{aligned}\hat{z}_{\mathrm{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m) &= m(\tilde{x}_i) + k(\tilde{x}_i, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} \left(\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})\right) \\ &= m(\tilde{x}_i) + \left(e_i^\top \otimes I_{d_z}\right) \left(\tilde{\mathbf{z}} - m(\tilde{\mathbf{x}})\right) \\ &= m(\tilde{x}_i) + \tilde{z}_i - m(\tilde{x}_i) \\ &= \tilde{z}_i\end{aligned} \tag{3.44}$$

Similarly:

$$\begin{aligned}R_{\mathrm{SK}}(\tilde{x}_i, \tilde{\mathbf{x}}) &= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, \tilde{x}_i) \\ &= k(\tilde{x}_i, \tilde{x}_i) + R - \left(e_i^\top \otimes I_{d_z}\right) k(\tilde{\mathbf{x}}, \tilde{x}_i) \\ &= k(\tilde{x}_i, \tilde{x}_i) + R - k(\tilde{x}_i, \tilde{x}_i) \\ &= R\end{aligned} \tag{3.45}$$

□

Proposition 3.7 (Gaussian conditioning interpretation)**.** *The conditional distribution of z knowing $\tilde{\mathbf{z}}$ is Gaussian, parametrised by the SK estimator and its error covariance matrix:*

$$z \mid \tilde{\mathbf{z}} \sim \mathcal{N}\left(\hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, m), R_{\mathrm{SK}}(x, \tilde{\mathbf{x}})\right) \tag{3.46}$$

Proof. Since they are evaluations of the same GP under an additive Gaussian white noise, z and $\tilde{\mathbf{z}}$ are jointly Gaussian. Their joint distribution is:

$$\begin{bmatrix} z \\ \tilde{\mathbf{z}} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} m(x) \\ m(\tilde{\mathbf{x}}) \end{bmatrix}, \begin{bmatrix} k(x,x) + R & k(x, \tilde{\mathbf{x}}) \\ k(\tilde{\mathbf{x}}, x) & G(\tilde{\mathbf{x}}) \end{bmatrix}\right) \tag{3.47}$$

As noted at the start of this section, this joint covariance is SPD. The Gaussian conditioning formula (see Proposition A.1) can thus be used to give the desired result. □

This result is a special case of the more general result established in Proposition 2.1. Since the unbiased hypothesis of the BLUE is automatically verified by the SK estimator, the minimisation criterion (3.32) amounts to finding the MMSE estimator of z given $\tilde{\mathbf{z}}$, which is always the conditional expectation.

Remark 3.7 (Simultaneous estimation)**.** *The simultaneous estimation of f at several query points $\mathbf{x} = (x_1, \ldots, x_n) \in \mathcal{U}^n$ can be done by directly replacing x with $\mathbf{x}$ in (3.35). In particular, the $\mathcal{O}\left(\tilde{N}^3\right)$ cost of inverting $G(\tilde{\mathbf{x}})$ can be shared across the n estimations whether they are performed successively or simultaneously.*

However, computing only the n diagonal blocks of $R_{\mathrm{SK}}(\mathbf{x}, \tilde{\mathbf{x}})$ costs $\mathcal{O}(\tilde{N}^3 + n\tilde{N}^2)$, whereas forming the full matrix before discarding its off-diagonal blocks costs $\mathcal{O}(\tilde{N}^3 + n\tilde{N}^2 + n^2\tilde{N})$ due to the extra off-diagonal matrix products. The off-diagonal blocks describe the correlation between the estimation errors made at distinct query points, and should only be computed when that correlation is needed.

3.3.2 Ordinary Kriging

Assuming the mean function to be known is a strong assumption, and can be unrealistic in practical scenarios. While an approximated model could be used instead of the true m, this would cause the kriging estimator to be overly confident in its estimation. Ordinary kriging (OK) extends its simple counterpart by assuming that the mean is an unknown constant $m \in \mathbb{R}^{d_z}$, and estimating it alongside the weights.

The unbiased assumption of the OK estimator yields:

$$\begin{aligned} & \mathbb{E}[\hat{z} - z] = 0 \\ \iff \quad & W^\top \mathbb{E}[\tilde{\mathbf{z}}] - \mathbb{E}[z] = 0 \\ \iff \quad & \left(\sum_{i=1}^{\tilde{N}} w_i^\top - I_{d_z} \right) m = 0 \\ \iff \quad & C^\top W - I_{d_z} = 0 \end{aligned} \tag{3.48}$$

with:

$$C \triangleq \mathbf{1}_{\tilde{N}} \otimes I_{d_z} \in \mathcal{M}_{\tilde{N}d_z, d_z}(\mathbb{R}), \qquad \mathbf{1}_{\tilde{N}} \triangleq \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^{\tilde{N}} \tag{3.49}$$

The OK weights thus minimise (3.32) under the constraint (3.48). Using the Lagrange multiplier method [69], Proposition 3.8 derives the corresponding estimator and covariance matrix, and Remark 3.8 establishes the link between ordinary and simple kriging.

Proposition 3.8 (Ordinary kriging estimator)**.** *The OK estimator of z given $\tilde{\mathbf{z}}$ and its error covariance are:*

$$\begin{cases} \hat{z}_{\mathrm{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) + k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} \left(\tilde{\mathbf{z}} - C\, \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \right) \\ R_{\mathrm{OK}}(x, \tilde{\mathbf{x}}) = R_{\hat{m}_{\mathrm{OK}}}(x, \tilde{\mathbf{x}}) + k(x, x) + R - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k\left(\tilde{\mathbf{x}}, x\right) \end{cases} \tag{3.50}$$

with:

$$\begin{cases} \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \left(C^\top G(\tilde{\mathbf{x}})^{-1} C \right)^{-1} C^\top G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ R_{\hat{m}_{\mathrm{OK}}}(x, \tilde{\mathbf{x}}) = \left(I_{d_z} - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} C \right) \left(C^\top G(\tilde{\mathbf{x}})^{-1} C \right)^{-1} \\ \qquad\qquad \left(I_{d_z} - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} C \right)^\top \end{cases} \tag{3.51}$$

and $R_{\hat{m}_{\mathrm{OK}}}(x, \tilde{\mathbf{x}}) \succeq 0$.

Proof. See section B.5. □

Remark 3.8 (Comparison between simple and ordinary kriging)**.** *The formulas obtained in (3.50) closely resemble those of simple kriging. In fact, they can be rewritten as follows:*

$$\begin{cases} \hat{z}_{\mathrm{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) = \hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})) \\ R_{\mathrm{OK}}(x, \tilde{\mathbf{x}}) = R_{\hat{m}_{\mathrm{OK}}}(x, \tilde{\mathbf{x}}) + R_{\mathrm{SK}}(x, \tilde{\mathbf{x}}) \end{cases} \tag{3.52}$$

where, by abuse of notation, $\hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ denotes either the estimated mean vector or its replication for each of the training points $\tilde{\mathbf{x}}$, namely:

$$\underbrace{\begin{bmatrix} \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ \vdots \\ \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \end{bmatrix}}_{\tilde{N} \text{ times}} = C\, \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \tag{3.53}$$

Ordinary kriging can therefore be interpreted as simple kriging in which the mean m is concurrently estimated from the observations. The uncertainty associated with this second estimation is captured by the covariance term $R_{\hat{m}_{\mathrm{OK}}}(x, \tilde{\mathbf{x}}) \succeq 0$, and is added to the SK covariance.

3.3.3 Universal Kriging

Universal kriging (UK) [70] generalises the constant mean of ordinary kriging to a linear combination of $r_{\mathrm{UK}} \in \mathbb{N}^*$ known trend functions $g_1, \ldots, g_{r_{\mathrm{UK}}} : \mathcal{U} \to \mathbb{R}$:

$$m(x) = g(x)\, \beta = \sum_{i=1}^{r_{\mathrm{UK}}} g_i(x)\, \beta_i \tag{3.54}$$

where $g(x)$ is the trend matrix:

$$g(x) \triangleq \begin{bmatrix} g_1(x) & \cdots & g_{r_{\mathrm{UK}}}(x) \end{bmatrix} \otimes I_{d_z} \in \mathcal{M}_{d_z, r_{\mathrm{UK}} d_z}(\mathbb{R}) \tag{3.55}$$

and β gathers the unknown coefficients $\beta_i \in \mathbb{R}^{d_z}$:

$$\beta \triangleq \begin{bmatrix} \beta_1 \\ \vdots \\ \beta_{r_{\mathrm{UK}}} \end{bmatrix} \in \mathbb{R}^{r_{\mathrm{UK}} d_z} \tag{3.56}$$

The trend matrix at the training points stacks these evaluations:

$$g(\tilde{\mathbf{x}}) = \begin{bmatrix} g(\tilde{x}_1) \\ \vdots \\ g(\tilde{x}_{\tilde{N}}) \end{bmatrix} \in \mathcal{M}_{\tilde{N} d_z, r_{\mathrm{UK}} d_z}(\mathbb{R}) \tag{3.57}$$

so that $m(\tilde{\mathbf{x}}) = g(\tilde{\mathbf{x}})\, \beta$ lies in $\mathbb{R}^{\tilde{N} d_z}$, as does $\tilde{\mathbf{z}}$.

This matrix is assumed to have full column rank, which requires at least $\tilde{N} \geq r_{\mathrm{UK}}$ training points, and the trend functions to be linearly independent once restricted to the training points. Proposition 3.9 generalises Proposition 3.8 accordingly.

Proposition 3.9 (Universal kriging estimator)**.** *Assume that $g(\tilde{\mathbf{x}})$ has full column rank. The UK estimator of z given $\tilde{\mathbf{z}}$ and its error covariance are:*

$$\begin{cases} \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = g(x)\, \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) + k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} \left(\tilde{\mathbf{z}} - g(\tilde{\mathbf{x}})\, \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)\right) \\ R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, g) = R_{\hat{m}_{\mathrm{UK}}}(x, \tilde{\mathbf{x}}, g) + k(x, x) + R - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k\left(\tilde{\mathbf{x}}, x\right) \end{cases} \tag{3.58}$$

with:

$$\begin{cases} \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = \left(g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})\right)^{-1} g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \\ R_{\hat{m}_{\mathrm{UK}}}(x, \tilde{\mathbf{x}}, g) = \left(g(x) - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})\right) \left(g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})\right)^{-1} \\ \qquad\qquad \left(g(x) - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})\right)^\top \end{cases} \tag{3.59}$$

and $R_{\hat{m}_{\mathrm{UK}}}(x, \tilde{\mathbf{x}}, g) \succeq 0$.

Proof. See section B.6. □

Remark 3.9 (Link with ordinary and simple kriging)**.** *As mentioned in the beginning of this section, ordinary kriging is a special case of universal kriging, which arises when* $r_{\mathrm{UK}} = 1$ *and* $g_1 \equiv 1$*. In this scenario, the trend matrices reduce to* $g(x) = I_{d_z}$ *and* $g(\tilde{\mathbf{x}}) = C$*. This, in turn, yields:*

$$\begin{cases} \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 1) = \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ R_{\hat{m}_{\mathrm{UK}}}(x, \tilde{\mathbf{x}}, 1) = R_{\hat{m}_{\mathrm{OK}}}(x, \tilde{\mathbf{x}}) \end{cases} \tag{3.60}$$

and thus:

$$\begin{cases} \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 1) = \hat{z}_{\mathrm{OK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}) \\ R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, 1) = R_{\mathrm{OK}}(x, \tilde{\mathbf{x}}) \end{cases} \tag{3.61}$$

Similarly to Remark 3.8, universal kriging can be seen as the simultaneous simple kriging from the training points and mean estimation from the trend functions. Indeed, comparing (3.58) with Proposition 3.5 shows that the following relation:

$$\begin{cases} \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = \hat{z}_{\mathrm{SK}}\left(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g\, \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)\right) \\ R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, g) = R_{\hat{m}_{\mathrm{UK}}}(x, \tilde{\mathbf{x}}, g) + R_{\mathrm{SK}}(x, \tilde{\mathbf{x}}) \end{cases} \tag{3.62}$$

Remark 3.8 is the special case $r_{\mathrm{UK}} = 1$, $g_1 \equiv 1$ *of this reading, where the estimated mean function is constant.*

Remark 3.10 (Kriging with a coarse model)**.** *Universal kriging is particularly relevant when a coarse model of the field is available, such as a polynomial trend [71] or a simplified reference model [72]. In the simplest case, setting* $r_{\mathrm{UK}} = 1$*, and choosing* g_1 *as this model amounts to kriging the discrepancy between the field and its approximation. Universal kriging can then be viewed as correcting the approximation, rather than reconstructing the field entirely from the observations.*

Remark 3.11 (Cost of estimating the mean)**.** *In addition to the* $\mathcal{O}(\tilde{N}^3)$ *cost of inverting* $G(\tilde{\mathbf{x}})$*, and the* $\mathcal{O}(\tilde{N}^2)$ *required to query the simple kriging model, universal kriging incurs an additional cost to estimate the mean. This can be decomposed into an* $\mathcal{O}(r_{\mathrm{UK}}\tilde{N}^2 + r_{\mathrm{UK}}^2\tilde{N} + r_{\mathrm{UK}}^3)$ *computation of* $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})$*, and an additional* $\mathcal{O}(r_{\mathrm{UK}}\tilde{N} + r_{\mathrm{UK}}^2)$ *per query to compute the estimated mean and covariance matrix.*

In particular, when $\tilde{N} \gg r_{\mathrm{UK}}$*, universal kriging retains the same dominant complexity as simple kriging.*

Proposition 3.7 shows that simple kriging can be seen as a Gaussian conditioning, which ordinary and universal kriging do not admit as such, their estimators being BLUEs of an unknown mean rather than posterior expectations. This interpretation can nonetheless be recovered by using a Gaussian prior on the trend coefficients, as proven by Proposition 3.10 [73, 74].

Proposition 3.10 (Conditioning under a vague trend prior)**.** *Let $B \in \mathcal{S}^{++}_{r_{\mathrm{UK}} d_z}$, and assume that the UK coefficients follow:*

$$\beta \sim \mathcal{N}(0, B) \tag{3.63}$$

Then:

$$f \sim \mathcal{GP}(0, k_B) \tag{3.64}$$

where k_B is the augmented kernel:

$$k_B : x, x' \mapsto k(x, x') + g(x)\, B\, g(x')^\top \tag{3.65}$$

We write $\hat{z}_{\mathrm{SK},B}$ and $R_{\mathrm{SK},B}$ for the SK estimator and error covariance of Proposition 3.5 taken under k_B rather than k. The following limit holds:

$$z \mid \tilde{\mathbf{z}} \sim \mathcal{N}\left(\hat{z}_{\mathrm{SK},B}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0), R_{\mathrm{SK},B}(x, \tilde{\mathbf{x}})\right) \underset{B^{-1} \to 0}{\longrightarrow} \mathcal{N}\left(\hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g), R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, g)\right) \tag{3.66}$$

which gives an asymptotic equivalent of Proposition 3.7 for universal, and thus ordinary, kriging.

Proof. See section B.7. □

3.4 Reduced-Rank Kernel Approximation

Exact GP regression requires inverting the $\tilde{N} d_z \times \tilde{N} d_z$ Gram matrix $G(\tilde{\mathbf{x}})$, which scales as $\mathcal{O}\left(\tilde{N}^3\right)$. When the number of training points is large, this operation becomes prohibitive. Reduced-rank methods [75, 76] circumvent this issue by approximating the kernel as a linear combination of $r_{\mathrm{KL}} \in \mathbb{N}^*$ basis functions, with $r_{\mathrm{KL}} < \tilde{N}$. The inversion cost is reduced to $\mathcal{O}\left(r_{\mathrm{KL}}^3\right)$, bringing down the overall kriging cost to $\mathcal{O}\left(r_{\mathrm{KL}}^2 \tilde{N}\right)$.

Mercer's theorem [77], presented in Proposition 3.12, establishes the existence of such a decomposition as soon as k is a continuous SPD kernel, which Matérn kernels satisfy, as proven by Proposition 3.3, and $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$. The Woodbury inversion formula is then used to rewrite the kriging estimator in a computationally efficient expression. Finally, the Karhunen-Loève decomposition is introduced as a way to represent the structure of a GP without relying on training data.

3.4.1 Mercer's Theorem

We start by generalising Definition 3.2 to functional operators on $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$.

Definition 3.5 (Self-adjoint positive semi-definite operator)**.** *A functional operator T on $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ is self-adjoint positive semi-definite (SPSD) if:*

$$\begin{cases} \forall \phi, \phi' \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}),\ \langle T\phi | \phi' \rangle = \langle \phi | T\phi' \rangle & \textit{self-adjoint} \\ \forall \phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}),\ \langle \phi | T\phi \rangle \geq 0 & \textit{positive semi-definite} \end{cases} \tag{3.67}$$

where $\langle \cdot | \cdot \rangle$ is the following scalar product:

$$(\phi, \phi') \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})^2 \mapsto \langle \phi | \phi' \rangle \triangleq \int_{\mathcal{U}} \phi(x)^\top \phi'(x)\, dx \tag{3.68}$$

Proposition 3.11 (Mercer operator). *Assume that $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$ and k is a continuous SPD kernel. The Mercer operator:*

$$\mathcal{K} : \begin{cases} \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \to \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \\ \qquad \phi \mapsto \left(x \mapsto \int_{\mathcal{U}} k(x, y)\, \phi(y)\, dy \right) \end{cases} \tag{3.69}$$

is a compact SPSD functional operator.

Proof. See section B.8. □

Proposition 3.12 (Mercer's theorem). *Under the assumptions of Proposition 3.11, there exists a non-increasing sequence $(\lambda_i)_{i \in \mathbb{N}^*}$ of strictly positive eigenvalues of $\mathcal{K}$ converging to 0, and an associated orthonormal family $(\psi_i)_{i \in \mathbb{N}^*}$ of continuous $\mathbb{R}^{d_z}$-valued eigenfunctions, such that:*

$$\forall x, x' \in \mathcal{U},\ k(x, x') = \sum_{i=1}^{+\infty} \lambda_i\, \psi_i(x)\, \psi_i(x')^\top \tag{3.70}$$

Furthermore, this series converges absolutely and uniformly on $\mathcal{U} \times \mathcal{U}$.

Proof. See section B.9. □

Remark 3.12 (Definiteness without invertibility). *Since all the eigenvalues λ_i are strictly positive, the quadratic form:*

$$\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \mapsto \langle \phi | \mathcal{K}\phi \rangle = \sum_{i=1}^{+\infty} \lambda_i\, \langle \phi | \psi_i \rangle^2 \tag{3.71}$$

is strictly positive for every $\phi \neq 0$, hence positive definite. For a symmetric matrix, definiteness amounts to invertibility, but this equivalence fails in infinite dimension. In particular, since $\mathcal{K}$ is compact with $\lambda_i \to 0$, it admits no positive lower bound on the unit sphere, so 0 belongs to its spectrum. $\mathcal{K}^{-1}$ is thus unbounded, and so the Mercer operator is positive definite yet not invertible.

In finite dimension, strict positivity and invertibility coincide and jointly characterise SPD matrices, whereas this is not the case in infinite dimension. This is not restrictive, however, as only the finite Gram matrices induced by k are inverted throughout this document.

3.4.2 Efficient Kriging Estimation

For the sake of brevity, the space of the square-integrable matrix-valued kernels over $\mathcal{U} \times \mathcal{U}$ is written as:

$$\mathcal{F} \triangleq \mathcal{L}^2\left(\mathcal{U} \times \mathcal{U}, \mathcal{M}_{d_z, d_z}(\mathbb{R})\right) \tag{3.72}$$

Proposition 3.13 (Reduced-rank approximation). *For a continuous SPD kernel k, we define the rank of k as the dimension of the image of $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ by its associated Mercer operator:*

$$\operatorname{rank}(k) \triangleq \dim\left(\mathcal{K}\left(\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})\right)\right) \tag{3.73}$$

Under the assumptions of Proposition 3.11, the truncated kernel:

$$k_{r_{\mathrm{KL}}} : \begin{cases} \mathcal{U} \times \mathcal{U} \to \mathcal{M}_{d_z,d_z}(\mathbb{R}) \\ x, x' \mapsto \displaystyle\sum_{i=1}^{r_{\mathrm{KL}}} \lambda_i \, \psi_i(x) \, \psi_i(x')^\top \end{cases} \tag{3.74}$$

is a continuous SPSD kernel of rank at most $r_{\mathrm{KL}} \in \mathbb{N}^*$*. It converges uniformly towards* k *as* $r_{\mathrm{KL}} \to +\infty$*, and minimises* $\|k - k_{r_{\mathrm{KL}}}\|_{\mathcal{F}}$ *among all kernels of rank at most* r_{KL}*.*

Proof. Let $r_{\mathrm{KL}} \in \mathbb{N}^*$ and $\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. We have:

$$\mathcal{K}_{r_{\mathrm{KL}}} \phi = \sum_{i=1}^{r_{\mathrm{KL}}} \lambda_i \langle \phi | \psi_i \rangle \, \psi_i \tag{3.75}$$

where $\mathcal{K}_{r_{\mathrm{KL}}}$ is the Mercer operator associated with $k_{r_{\mathrm{KL}}}$. Every $\mathcal{K}_{r_{\mathrm{KL}}} \phi$ thus belongs to the subspace of continuous functions spanned by $\psi_1, \ldots, \psi_{r_{\mathrm{KL}}}$. The orthonormality of the basis ensures that this subspace is of dimension r_{KL}, and thus the rank of $k_{r_{\mathrm{KL}}}$ is at most r_{KL}. The uniform convergence of $(k_{r_{\mathrm{KL}}})_{r_{\mathrm{KL}} \in \mathbb{N}^*}$ has already been proven by Proposition 3.12.

Finally, applying the Eckart-Young-Mirsky theorem [78, 79] to $\mathcal{K}$ and using the orthonormality of $(\psi_i \, \psi_j^\top)_{i,j \in \mathbb{N}^*}$ yields:

$$\min_{\tilde{k} \in \mathcal{F} | \mathrm{rank}(\tilde{k}) \leq r_{\mathrm{KL}}} \|k - \tilde{k}\|_{\mathcal{F}}^2 = \sum_{j > r_{\mathrm{KL}}} \lambda_j^2 = \|k - k_{r_{\mathrm{KL}}}\|_{\mathcal{F}}^2 \tag{3.76}$$

□

In compact form, the eigenfunctions are collected into:

$$\Psi(x) \triangleq \begin{bmatrix} \psi_1(x) & \cdots & \psi_{r_{\mathrm{KL}}}(x) \end{bmatrix} \in \mathcal{M}_{d_z, r_{\mathrm{KL}}}(\mathbb{R}) \tag{3.77}$$

and the corresponding eigenvalues are gathered as:

$$\Lambda \triangleq \begin{bmatrix} \lambda_1 & & (0) \\ & \ddots & \\ (0) & & \lambda_{r_{\mathrm{KL}}} \end{bmatrix} \in \mathcal{M}_{r_{\mathrm{KL}}, r_{\mathrm{KL}}}(\mathbb{R}) \tag{3.78}$$

By Proposition 3.12, the eigenvalues λ_i are strictly positive, so that $\Lambda \succ 0$ and Λ^{-1} is well-defined. Proposition 3.13 can thus be rewritten as:

$$k(x, x') \approx \Psi(x) \, \Lambda \, \Psi(x')^\top \tag{3.79}$$

and then:

$$G(\tilde{\mathbf{x}}) \approx \Psi(\tilde{\mathbf{x}}) \, \Lambda \, \Psi(\tilde{\mathbf{x}})^\top + I_{\tilde{N}} \otimes \tilde{R}, \qquad \Psi(\tilde{\mathbf{x}}) = \begin{bmatrix} \Psi(\tilde{x}_1) \\ \vdots \\ \Psi(\tilde{x}_{\tilde{N}}) \end{bmatrix} \tag{3.80}$$

Proposition 3.14 (Reduced-rank simple kriging)**.** *Let:*

$$\tilde{G}(\tilde{\mathbf{x}}) \triangleq \Lambda^{-1} + \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{N}} \otimes \tilde{R}\right)^{-1} \Psi(\tilde{\mathbf{x}}) \in \mathcal{M}_{r_{\mathrm{KL}}, r_{\mathrm{KL}}}(\mathbb{R}) \tag{3.81}$$

be the reduced-rank Gram matrix. Under the reduced-rank approximation of (3.80)*, the centred SK estimator of Proposition 3.5 and its error covariance matrix reduce to:*

$$\begin{cases} \hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x)\, \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(\tilde{\mathbf{x}})^\top \left(I_{\tilde{N}} \otimes \tilde{R}\right)^{-1} \tilde{\mathbf{z}} \\ R_{\mathrm{SK}}(x, \tilde{\mathbf{x}}) \approx R + \Psi(x)\, \tilde{G}(\tilde{\mathbf{x}})^{-1} \Psi(x)^\top \end{cases} \tag{3.82}$$

which require only the inversion of $\tilde{G}(\tilde{\mathbf{x}})$*.*

When there exists $\sigma > 0$ *such that* $\tilde{R} = \sigma^2 I$*,* (3.82) *further simplify to:*

$$\begin{cases} \hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1}\right)^{-1} \Psi(\tilde{\mathbf{x}})^\top \tilde{\mathbf{z}} \\ R_{\mathrm{SK}}(x, \tilde{\mathbf{x}}) \approx R + \sigma^2 \Psi(x) \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1}\right)^{-1} \Psi(x)^\top \end{cases} \tag{3.83}$$

These expressions remain valid at $\sigma = 0$ *provided* $\Psi(\tilde{\mathbf{x}})$ *has full column rank.*

Proof. See section B.10. □

Remark 3.13 (Construction of the eigenpairs)**.** *While Mercer's theorem establishes the existence of the decomposition presented by* (3.70) *for any continuous SPD kernel defined on a compact subset of* $\mathbb{R}^{d_x}$*, it does not provide a closed-form expression for the eigenpairs* $(\lambda_i, \psi_i)_{i \in \mathbb{N}^*}$*. As they depend on* k *and* $\mathcal{U}$*, additional assumptions are usually made in order to derive an explicit approximation of the form of Proposition 3.13.*

3.4.3 Karhunen-Loève Decomposition

While section 3.4.1 uses the eigenpairs of the Mercer operator associated with k to decompose the kernel, they can also be used to represent the process itself. This representation is known as the Karhunen-Loève (KL) decomposition [80] and is presented by Proposition 3.15.

Proposition 3.15 (Karhunen-Loève decomposition)**.** *Let* $f \sim \mathcal{GP}(m, k)$*, where* $k : \mathcal{U} \times \mathcal{U} \to \mathcal{M}_{d_z, d_z}(\mathbb{R})$ *is a continuous SPD kernel, and* $\mathcal{U}$ *is a compact subset of* $\mathbb{R}^{d_x}$*. Let* $(\lambda_i, \psi_i)_{i \in \mathbb{N}^*}$ *be the Mercer eigenpairs of Proposition 3.12. Then, for all* $x \in \mathcal{U}$*, we have:*

$$f(x) = m(x) + \sum_{i=1}^{+\infty} \sqrt{\lambda_i}\, \xi_i\, \psi_i(x), \qquad \xi_i \triangleq \frac{1}{\sqrt{\lambda_i}} \langle f - m | \psi_i \rangle \overset{i.i.d.}{\sim} \mathcal{N}(0, 1) \tag{3.84}$$

where the series converges in mean square uniformly over $\mathcal{U}$*.*

Proof. See section B.11. □

Remark 3.14 (Almost sure convergence)**.** *When* f *has almost surely continuous sample paths, the convergence of* (3.84) *strengthens from mean square to almost sure and uniform over* $\mathcal{U}$*: by the Itô–Nisio theorem [80], the partial sums converge to* f *uniformly over* $\mathcal{U}$ *almost surely. A single event of probability one then carries* (3.84) *simultaneously for all* $x \in \mathcal{U}$*, rather than for each* x *separately.*

Conversely, the KL decomposition can be used to generate, from a family $(\xi_i)_{i\in\mathbb{N}^*}$ of i.i.d standard Gaussian variables, a GP of covariance kernel k. This is the object of Proposition 3.16.

Proposition 3.16 (Karhunen-Loève converse)**.** *Assume that there exist a continuous SPD kernel $k : \mathcal{U}\times\mathcal{U} \to \mathcal{M}_{d_z,d_z}(\mathbb{R})$ and a function $m : \mathcal{U} \to \mathbb{R}^{d_z}$, where $\mathcal{U}$ is a compact subset of $\mathbb{R}^{d_x}$. Let $(\lambda_i, \psi_i)_{i\in\mathbb{N}^*}$ be the Mercer eigenpairs associated with k and $\mathcal{U}$, and let:*

$$(\xi_i)_{i\in\mathbb{N}^*} \overset{i.i.d.}{\sim} \mathcal{N}(0,1) \tag{3.85}$$

The stochastic process:

$$f : \begin{cases} \mathcal{U} \to \mathbb{R}^{d_z} \\ x \mapsto m(x) + \sum_{i=1}^{+\infty} \sqrt{\lambda_i}\, \xi_i\, \psi_i(x) \end{cases} \tag{3.86}$$

is a GP with mean function m and covariance kernel k.

Proof. See section B.12. □

Remark 3.15 (Finite-dimensional parametrisation)**.** *As for section 3.4.2, Mercer's theorem can be used to represent the field by a finite approximation. Indeed, for $r_{\mathrm{KL}} \in \mathbb{N}^*$, Ψ defined as in (3.80) and:*

$$\alpha \triangleq \begin{bmatrix} \sqrt{\lambda_1}\xi_1 & \cdots & \sqrt{\lambda_{r_{\mathrm{KL}}}}\xi_{r_{\mathrm{KL}}} \end{bmatrix}^\top \tag{3.87}$$

we have:

$$\forall x \in \mathcal{U},\ f(x) \approx m(x) + \Psi(x)\,\alpha, \qquad \alpha \sim \mathcal{N}(0,\Lambda) \tag{3.88}$$

Up to this approximation, inferring f thus amounts to inferring the r_{KL} coefficients of α.

3.5 Reproducing Kernel Hilbert Spaces

Beyond regression, the covariance kernel also induces a functional Hilbert space, known as reproducing kernel Hilbert space (RKHS) [81, 82]. The RKHS representation is complementary to the probabilistic viewpoint described throughout this chapter. Indeed, while the GP describes a probabilistic distribution over functions, the RKHS associated with its covariance kernel provides a deterministic functional space in which such functions can be represented, manipulated, or approximated. For the sake of simplicity, we assume that $d_z = 1$, so that the reproducing property can be written in its most natural form.

3.5.1 Reproducing Property and Definition

Let $\mathcal{U}$ be an arbitrary set, and let $(\mathcal{H}, \langle\cdot\,|\,\cdot\rangle_{\mathcal{H}})$ denote a Hilbert space of functions from $\mathcal{U}$ to $\mathbb{R}$. Definition 3.6 and proposition 3.17 describe the conditions for this space to be an RKHS.

Definition 3.6 (Reproducing kernel Hilbert space)**.** *A Hilbert space $(\mathcal{H}, \langle \cdot \mid \cdot \rangle_{\mathcal{H}})$ of functions from $\mathcal{U}$ to $\mathbb{R}$ is an RKHS if there exists a function $k : \mathcal{U} \times \mathcal{U} \to \mathbb{R}$ such that:*

$$\begin{cases} \forall x \in \mathcal{U}, k(x, \cdot) \in \mathcal{H} \\ \forall x \in \mathcal{U}, \forall f \in \mathcal{H}, \ \langle f \mid k(x, \cdot) \rangle_{\mathcal{H}} = f(x) \end{cases} \tag{3.89}$$

This second equation is called the reproducing property of $\mathcal{H}$.

Proposition 3.17 (Reproducing kernel)**.** *When such a k exists, it is symmetric and unique. It is called the reproducing kernel of $\mathcal{H}$.*

Proof. Let k_1 satisfy Definition 3.6, and let $x, x' \in \mathcal{U}$. Then:

$$\begin{aligned} k_1(x, x') &= (k_1(x, \cdot))(x') \\ &= \langle k_1(x, \cdot) \mid k_1(x', \cdot) \rangle_{\mathcal{H}} \\ &= \langle k_1(x', \cdot) \mid k_1(x, \cdot) \rangle_{\mathcal{H}} \\ &= k_1(x', x) \end{aligned} \tag{3.90}$$

so k_1 is symmetric.

Assume that there exists another function k_2 that verifies Definition 3.6. It comes that:

$$\begin{aligned} k_2(x, x') &= (k_2(x, \cdot))(x') \\ &= \langle k_2(x, \cdot) \mid k_1(x', \cdot) \rangle_{\mathcal{H}} \\ &= \langle k_1(x', \cdot) \mid k_2(x, \cdot) \rangle_{\mathcal{H}} \\ &= k_1(x', x) \\ &= k_1(x, x') \end{aligned} \tag{3.91}$$

and so $k_2 = k_1$. □

It can be shown [83] that $\mathcal{H}$ is an RKHS if, and only if, the evaluation functional at every point of the domain:

$$e_x : f \in \mathcal{H} \mapsto f(x) \tag{3.92}$$

is continuous on $\mathcal{H}$ for every $x \in \mathcal{U}$. Its reproducing kernel is necessarily SPSD. Conversely, every SPSD kernel k admits a unique RKHS $\mathcal{H}_k$ for which k is the reproducing kernel. Such an RKHS admits an explicit construction, as detailed in Proposition 3.18.

Proposition 3.18 (Native RKHS)**.** *Let k be an SPSD kernel, let $\mathcal{H}_k$ be the completion of the set spanned by $\{k(x, \cdot), \ x \in \mathcal{U}\}$, and let $\langle \cdot \mid \cdot \rangle_{\mathcal{H}_k}$ be defined on the generators by:*

$$\forall x, x' \in \mathcal{U}, \ \langle k(x, \cdot) \mid k(x', \cdot) \rangle_{\mathcal{H}_k} \triangleq k(x, x') \tag{3.93}$$

Then $(\mathcal{H}_k, \langle \cdot \mid \cdot \rangle_{\mathcal{H}_k})$ is a Hilbert space and an RKHS. Its reproducing kernel is k, and it is the only RKHS for which k is the reproducing kernel. It is called the native RKHS of k.

Proof. See section B.13. □

The native RKHS is spanned by the functions $k(x, \cdot)$, so every $x \in \mathcal{U}$ is naturally represented in $\mathcal{H}_k$ by the $k(x, \cdot)$ it generates. This correspondence defines the feature map, introduced by Definition 3.7, and provides the representation on which the kernel methods of the following sections operate.

Definition 3.7 (Feature map)**.** *Let k be an SPSD kernel on $\mathcal{U}$, and $(\mathcal{H}_k, \langle \cdot \mid \cdot \rangle_{\mathcal{H}_k})$ be its native RKHS. The feature map of $\mathcal{H}_k$ is:*

$$\Phi : \begin{cases} \mathcal{U} \to \mathcal{H}_k \\ x \mapsto k(x, \cdot) \end{cases} \tag{3.94}$$

The reproducing property can thus be rewritten as:

$$\forall x \in \mathcal{U},\ \forall f \in \mathcal{H}_k,\ f(x) = \langle f \mid \Phi(x) \rangle_{\mathcal{H}_k} \tag{3.95}$$

In particular, the kernel trick:

$$\forall x, x' \in \mathcal{U},\ \langle \Phi(x) \mid \Phi(x') \rangle_{\mathcal{H}_k} = k(x, x') \tag{3.96}$$

can be used to compute inner products in $\mathcal{H}_k$ without explicitly constructing the feature map Φ or evaluating $\langle \cdot \mid \cdot \rangle_{\mathcal{H}_k}$.

3.5.2 Link with Gaussian Processes

In this section, we let k be an SPD kernel on $\mathcal{U}$, and $(\mathcal{H}_k, \langle \cdot \mid \cdot \rangle_{\mathcal{H}_k})$ be its native RKHS. Proposition 3.20 illustrates the relationship between the GP and RKHS viewpoints by showing that kriging using k is equivalent to solving a variational problem in $\mathcal{H}_k$. It is a particular case of the representer theorem [84, 85], presented by Proposition 3.19.

Proposition 3.19 (Representer theorem)**.** *Let $\tilde{N} \in \mathbb{N}^*$, $\tilde{\mathbf{x}} \triangleq (\tilde{x}_1, \ldots, \tilde{x}_{\tilde{N}}) \in \mathcal{U}^{\tilde{N}}$, $E : \mathbb{R}^{\tilde{N}} \to \mathbb{R}$ and $g : \mathbb{R}_+ \to \mathbb{R}$ strictly increasing.*

If the minimisation problem:

$$\operatorname*{argmin}_{\psi \in \mathcal{H}_k} (E(\psi(\tilde{x}_1), \ldots, \psi(\tilde{x}_{\tilde{N}})) + g(\|\psi\|_{\mathcal{H}_k})) \tag{3.97}$$

admits a solution, then it admits a solution in the set spanned by the features of $\tilde{\mathbf{x}}$:

$$\mathcal{V}_{\tilde{\mathbf{x}}} \triangleq \operatorname{span}\{\Phi(\tilde{x}_1), \ldots, \Phi(\tilde{x}_{\tilde{N}})\} \subseteq \mathcal{H}_k \tag{3.98}$$

In particular, this minimiser can be written as:

$$\psi_{\min} = \sum_{i=1}^{\tilde{N}} a_i\, \Phi(\tilde{x}_i) = a^\top \Phi(\tilde{\mathbf{x}}) \tag{3.99}$$

for some $a \triangleq [a_1, \ldots, a_{\tilde{N}}]^\top \in \mathbb{R}^{\tilde{N}}$.

Proof. For $\psi \in \mathcal{H}_k$, we decompose it orthogonally as:

$$\exists\, \psi_{\mathcal{V}} \in \mathcal{V}_{\tilde{\mathbf{x}}},\ \psi_\perp \in \mathcal{V}_{\tilde{\mathbf{x}}}^\perp \mid \psi = \psi_{\mathcal{V}} + \psi_\perp \tag{3.100}$$

where $\mathcal{V}_{\tilde{\mathbf{x}}}^\perp$ denotes the subspace of $\mathcal{H}_k$ orthogonal to $\mathcal{V}_{\tilde{\mathbf{x}}}$.

Let $i \in [\![1, \tilde{N}]\!]$. Since $\Phi(\tilde{x}_i) \in \mathcal{V}_{\tilde{\mathbf{x}}}$, the reproducing property gives

$$\psi_\perp(\tilde{x}_i) = \langle \psi_\perp \mid \Phi(\tilde{x}_i) \rangle_{\mathcal{H}_k} = 0 \tag{3.101}$$

and thus:

$$\psi(\tilde{x}_i) = \psi_{\mathcal{V}}(\tilde{x}_i) \tag{3.102}$$

Moreover, by orthogonality:

$$\|\psi\|_{\mathcal{H}_k} = \sqrt{\|\psi_{\mathcal{V}}\|_{\mathcal{H}_k}^2 + \|\psi_{\perp}\|_{\mathcal{H}_k}^2} \geq \|\psi_{\mathcal{V}}\|_{\mathcal{H}_k} \tag{3.103}$$

with equality if and only if $\psi \in \mathcal{V}_{\tilde{\mathbf{x}}}$.

Since E depends only on the values of ψ at $\tilde{x}_1, \dots, \tilde{x}_{\tilde{N}}$ and g is strictly increasing, it comes that:

$$E(\psi(\tilde{x}_1), \dots, \psi(\tilde{x}_{\tilde{N}})) + g(\|\psi\|_{\mathcal{H}_k}) \geq E(\psi_{\mathcal{V}}(\tilde{x}_1), \dots, \psi_{\mathcal{V}}(\tilde{x}_{\tilde{N}})) + g(\|\psi_{\mathcal{V}}\|_{\mathcal{H}_k}) \tag{3.104}$$

Therefore, any minimiser of (3.97) must belong to $\mathcal{V}_{\tilde{\mathbf{x}}}$, provided that such a minimiser exists. □

Proposition 3.20 (Regularised least-squares in an RKHS)**.** *Let $\tilde{N} \in \mathbb{N}^*$, $\tilde{\mathbf{x}} \triangleq (\tilde{x}_1, \dots, \tilde{x}_{\tilde{N}}) \in \mathcal{U}^{\tilde{N}}$ pairwise distinct, $\tilde{\mathbf{z}} \triangleq (\tilde{z}_1, \dots, \tilde{z}_{\tilde{N}}) \in \mathbb{R}^{\tilde{N}}$ and $\lambda \geq 0$. The regularised least-squares problem:*

$$\operatorname*{argmin}_{\psi \in \mathcal{H}_k} \left(\frac{1}{\tilde{N}} \sum_{i=1}^{\tilde{N}} (\psi(\tilde{x}_i) - \tilde{z}_i)^2 + \lambda \|\psi\|_{\mathcal{H}_k}^2 \right) \tag{3.105}$$

admits a solution, unique for $\lambda > 0$, and given by:

$$\psi_{\mathrm{RKHS}}(x) = k(x, \tilde{\mathbf{x}}) \left(k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda \tilde{N} I_{\tilde{N}} \right)^{-1} \tilde{\mathbf{z}} \tag{3.106}$$

Proof. Let $\psi \in \mathcal{H}_k$, and let $\psi_{\mathcal{V}}$ denote its orthogonal projection onto $\mathcal{V}_{\tilde{\mathbf{x}}}$. As established in the proof of Proposition 3.19, replacing ψ with $\psi_{\mathcal{V}}$ does not change the values $\psi(\tilde{x}_1), \dots, \psi(\tilde{x}_{\tilde{N}})$, and does not increase the norm. It is thus sufficient to minimise (3.105) over $\mathcal{V}_{\tilde{\mathbf{x}}}$.

Let $a \in \mathbb{R}^{\tilde{N}}$ such that $\psi = a^\top \Phi(\tilde{\mathbf{x}})$. By the reproducing property and the kernel trick, we have:

$$\begin{cases} \psi(\tilde{\mathbf{x}}) = k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})\, a \\ \|\psi\|_{\mathcal{H}_k}^2 = a^\top k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})\, a \end{cases} \tag{3.107}$$

and the objective function of (3.105) becomes:

$$J : a \in \mathbb{R}^{\tilde{N}} \mapsto \frac{1}{\tilde{N}} \|k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})\, a - \tilde{\mathbf{z}}\|_2^2 + \lambda\, a^\top k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})\, a \tag{3.108}$$

Since k is SPD and the training points are pairwise distinct, $k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \succ 0$. The function J is thus strictly convex, and differentiating it with respect to a yields:

$$\begin{aligned} & \frac{\partial J}{\partial a} = 0 \\ \iff \quad & \frac{2}{\tilde{N}} k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \left(\left(k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda \tilde{N} I_{\tilde{N}} \right) a - \tilde{\mathbf{z}} \right) = 0 \\ \iff \quad & a = \left(k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda \tilde{N} I_{\tilde{N}} \right)^{-1} \tilde{\mathbf{z}} \end{aligned} \tag{3.109}$$

which gives the expression of ψ_{RKHS}.

Finally, when $\lambda > 0$, the norm of ψ is strictly greater than that of $\psi_{\mathcal{V}}$ for every $\psi \notin \mathcal{V}_{\tilde{\mathbf{x}}}$, and so ψ_{RKHS} is the unique solution of (3.105). When $\lambda = 0$, however, adding any element of $\mathcal{V}_{\tilde{\mathbf{x}}}^{\perp}$ to ψ_{RKHS} leaves the objective function unchanged, and the solution is not unique in general. □

Remark 3.16 (Simple kriging in the RKHS)**.** *For:*

$$\lambda = \frac{1}{\tilde{N}}\,\tilde{R} \tag{3.110}$$

the least-squares estimator ψ_{RKHS} coincides with the centred SK estimator of Proposition 3.5. Simple kriging can therefore be interpreted as a regularised least-squares regression in $\mathcal{H}_k$ [86]. In the noiseless case $\tilde{R} = 0$, the SK estimator is also the solution of the minimum-norm interpolation problem:

$$\operatorname*{argmin}_{\psi \in \mathcal{H}_k} \left\{ \|\psi\|_{\mathcal{H}_k} \;\middle|\; \forall i \in [\![1, \tilde{N}]\!],\ \psi(\tilde{x}_i) = \tilde{z}_i \right\} \tag{3.111}$$

Remark 3.17 (GP and RKHS correspondence)**.** *The GP and RKHS viewpoints thus give two interpretations of the same estimator. The first one is probabilistic, as the mean of the conditional distribution of Proposition 3.7, while the second one is deterministic, as the solution of the variational problem (3.105). Moreover, under the assumptions of Proposition 3.12, it can be shown [87] that $\mathcal{H}_k$ is the set of the functions:*

$$\varphi = \sum_{i=1}^{+\infty} a_i\,\psi_i \tag{3.112}$$

such that:

$$\|\varphi\|_{\mathcal{H}_k}^2 = \sum_{i=1}^{+\infty} \frac{a_i^2}{\lambda_i} < +\infty \tag{3.113}$$

where $(\lambda_i, \psi_i)_{i \in \mathbb{N}^*}$ are the Mercer eigenpairs of k. Since the eigenvalues converge to 0, the coefficients a_i must decay fast enough for this norm to be finite, which imposes a regularity condition on the elements of $\mathcal{H}_k$.

Choosing a kernel thus amounts to choosing a regularity class for the unknown field. In particular, for $\mathcal{U} = \mathbb{R}^{d_x}$, the native RKHS of a Matérn kernel of smoothness ν is the Sobolev space of order $\nu + d_x/2$, up to an equivalent norm [88], whereas that of the SE kernel only contains infinitely differentiable functions.

Proposition 3.21 extends this correspondence to the estimation error, by showing that the error committed on a function of unit norm of $\mathcal{H}_k$ is at most the standard deviation of the same error for a GP of covariance kernel k.

Proposition 3.21 (Worst-case error interpretation)**.** *Let $\tilde{N} \in \mathbb{N}^*$, $\tilde{\mathbf{x}} \in \mathcal{U}^{\tilde{N}}$ pairwise distinct, $x \in \mathcal{U}$ and $W \in \mathbb{R}^{\tilde{N}}$. Let $\psi \in \mathcal{H}_k$ and $f \sim \mathcal{GP}(0, k)$. Then:*

$$\left| W^{\top}\psi(\tilde{\mathbf{x}}) - \psi(x) \right| \leq \|\psi\|_{\mathcal{H}_k} \sqrt{\mathbb{V}\left[W^{\top} f(\tilde{\mathbf{x}}) - f(x)\right]} \tag{3.114}$$

with equality when:

$$\psi = W^{\top}\Phi(\tilde{\mathbf{x}}) - \Phi(x) \tag{3.115}$$

Furthermore, the right-hand side of (3.114) *is minimised by the weights of the SK estimator of Proposition 3.5 for* $R = 0$ *and* $\tilde{R} = 0$*, which gives:*

$$|\hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \psi(\tilde{\mathbf{x}}), 0) - \psi(x)| \leq \|\psi\|_{\mathcal{H}_k} \sqrt{R_{\mathrm{SK}}(x, \tilde{\mathbf{x}})} \tag{3.116}$$

where $\sqrt{R_{\mathrm{SK}}(x, \tilde{\mathbf{x}})}$ *is known as the power function [88].*

Proof. Let $w_1, \ldots, w_{\tilde{N}}$ denote the components of W, and let:

$$\varphi \triangleq W^\top \Phi(\tilde{\mathbf{x}}) - \Phi(x) = \sum_{i=1}^{\tilde{N}} w_i \, \Phi(\tilde{x}_i) - \Phi(x) \tag{3.117}$$

The reproducing property and the Cauchy-Schwarz inequality give:

$$\begin{aligned} \left|W^\top \psi(\tilde{\mathbf{x}}) - \psi(x)\right| &= \left| \sum_{i=1}^{\tilde{N}} w_i \, \langle \psi \mid \Phi(\tilde{x}_i) \rangle_{\mathcal{H}_k} - \langle \psi \mid \Phi(x) \rangle_{\mathcal{H}_k} \right| \\ &= |\langle \psi \mid \varphi \rangle_{\mathcal{H}_k}| \\ &\leq \|\psi\|_{\mathcal{H}_k} \, \|\varphi\|_{\mathcal{H}_k} \end{aligned} \tag{3.118}$$

with equality when $\psi = \varphi$.

We also have:

$$\begin{aligned} \|\varphi\|_{\mathcal{H}_k}^2 &= \sum_{i=1}^{\tilde{N}} \sum_{j=1}^{\tilde{N}} w_i \, w_j \, \langle \Phi(\tilde{x}_i) \mid \Phi(\tilde{x}_j) \rangle_{\mathcal{H}_k} - 2 \sum_{i=1}^{\tilde{N}} w_i \, \langle \Phi(\tilde{x}_i) \mid \Phi(x) \rangle_{\mathcal{H}_k} \\ &\quad + \langle \Phi(x) \mid \Phi(x) \rangle_{\mathcal{H}_k} \\ &= \sum_{i=1}^{\tilde{N}} \sum_{j=1}^{\tilde{N}} w_i \, w_j \, k(\tilde{x}_i, \tilde{x}_j) - 2 \sum_{i=1}^{\tilde{N}} w_i \, k(\tilde{x}_i, x) + k(x, x) \\ &= W^\top k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \, W - 2 W^\top k(\tilde{\mathbf{x}}, x) + k(x, x) \end{aligned} \tag{3.119}$$

which is exactly (3.36) for $R = 0$ and $\tilde{R} = 0$, and thus the variance of $W^\top f(\tilde{\mathbf{x}}) - f(x)$. As established in the proof of Proposition 3.5, this variance is minimised by the SK weights, for which it is equal to $R_{\mathrm{SK}}(x, \tilde{\mathbf{x}})$. □

It must be noted that this interpretation only holds for noiseless observations of a function of $\mathcal{H}_k$, whose norm is unknown in practice. Furthermore, it can be shown [86] that the sample paths of a GP of covariance kernel k almost surely do not belong to $\mathcal{H}_k$ when it is infinite-dimensional.

3.5.3 Kernel Mean Embedding

Beyond evaluating functions inside $\mathcal{H}_k$, the feature map can also be used to represent the moments of a probability distribution. Definition 3.8 introduces the kernel mean embedding (KME) of a random variable [89], while Definition 3.9 generalises this concept to two random variables.

Definition 3.8 (Kernel mean embedding)**.** *Let X be a random variable on $\mathcal{U}$ with probability density p, and $\varphi \in \mathcal{H}_k$. The KME of X is:*

$$\mu_X \triangleq \mathbb{E}_p\left[\Phi(X)\right] = \int_{\mathcal{U}} \Phi(x)\, p(x)\, dx \in \mathcal{H}_k \tag{3.120}$$

By the reproducing property, the expectation of $\varphi(X)$ can then be expressed as an inner product in $\mathcal{H}_k$:

$$\begin{aligned} \mathbb{E}_p\left[\varphi(X)\right] &= \int_{\mathcal{U}} \varphi(x)\, p(x)\, dx \\ &= \int_{\mathcal{U}} \langle \varphi \mid \Phi(x) \rangle_{\mathcal{H}_k}\, p(x)\, dx \\ &= \left\langle \varphi \,\middle|\, \int_{\mathcal{U}} \Phi(x)\, p(x)\, dx \right\rangle_{\mathcal{H}_k} \\ &= \langle \varphi \mid \mu_X \rangle_{\mathcal{H}_k} \end{aligned} \tag{3.121}$$

Since (3.120) integrates a function taking its values in $\mathcal{H}_k$, its existence is not granted. Proposition 3.22 gives a sufficient condition [82], which every bounded kernel satisfies.

Proposition 3.22 (Existence of the KME)**.** *If $\mathbb{E}_p\left[\sqrt{k(X,X)}\right] < \infty$, then μ_X is well defined, as the unique element of $\mathcal{H}_k$ for which (3.121) holds for all $\varphi \in \mathcal{H}_k$. Moreover:*

$$\forall x \in \mathcal{U},\ \mu_X(x) = \mathbb{E}_p\left[k(X,x)\right] \tag{3.122}$$

Proof. Let $\varphi \in \mathcal{H}_k$. We have:

$$\begin{aligned} \left|\mathbb{E}_p\left[\varphi(X)\right]\right| &\leq \mathbb{E}_p\left[\left|\varphi(X)\right|\right] \\ &= \mathbb{E}_p\left[\left|\langle \varphi \mid \Phi(X) \rangle_{\mathcal{H}_k}\right|\right] \\ &\leq \|\varphi\|_{\mathcal{H}_k}\, \mathbb{E}_p\left[\|\Phi(X)\|_{\mathcal{H}_k}\right] \\ &= \|\varphi\|_{\mathcal{H}_k}\, \mathbb{E}_p\left[\sqrt{k(X,X)}\right] \end{aligned} \tag{3.123}$$

By linearity of the expectation, $\varphi \mapsto \mathbb{E}_p\left[\varphi(X)\right]$ is thus a continuous linear form on $\mathcal{H}_k$. By the Riesz representation theorem [90], it is the inner product with a unique element of $\mathcal{H}_k$. This element is the meaning given to the integral of (3.120), and it satisfies (3.121) by construction.

Finally, let $x \in \mathcal{U}$. The reproducing property and (3.121) applied to $\varphi = \Phi(x)$ give:

$$\mu_X(x) = \langle \mu_X \mid \Phi(x) \rangle_{\mathcal{H}_k} = \mathbb{E}_p\left[(\Phi(x))(X)\right] = \mathbb{E}_p\left[k(X,x)\right] \tag{3.124}$$

□

It must be noted that two different probability densities may have the same KME. A kernel for which the map $p \mapsto \mu_X$ is injective is called a characteristic kernel [91]. In this case, μ_X uniquely determines p. In particular, the SE and Matérn kernels of section 3.2.2 are characteristic.

Definition 3.9 (Joint KME). *Let X and Z be random variables on two sets $\mathcal{U}_X$ and $\mathcal{U}_Z$, with joint probability density p_{XZ}. Let k_X and k_Z be SPD kernels on $\mathcal{U}_X$ and $\mathcal{U}_Z$, with respective native RKHSs $(\mathcal{H}_{k_X}, \langle \cdot \mid \cdot \rangle_{\mathcal{H}_{k_X}})$ and $(\mathcal{H}_{k_Z}, \langle \cdot \mid \cdot \rangle_{\mathcal{H}_{k_Z}})$, and feature maps Φ_X and Φ_Z. The joint KME of (X, Z) is defined as:*

$$\begin{aligned} \mu_{XZ} &\triangleq \mathbb{E}_{p_{XZ}} \left[\Phi_X(X) \otimes \Phi_Z(Z) \right] \\ &= \iint_{\mathcal{U}_X \times \mathcal{U}_Z} \Phi_X(x) \otimes \Phi_Z(z)\, p_{XZ}(x, z)\, dx\, dz \\ &\in \mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z} \end{aligned} \tag{3.125}$$

where $\otimes$ denotes the tensor product in the product space.

The space $\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}$ is the completion of the set spanned by the tensors $\varphi \otimes \psi$, for $\varphi \in \mathcal{H}_{k_X}$ and $\psi \in \mathcal{H}_{k_Z}$, under the inner product defined on these generators by:

$$\langle \varphi \otimes \psi \mid \varphi' \otimes \psi' \rangle_{\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}} \triangleq \langle \varphi \mid \varphi' \rangle_{\mathcal{H}_{k_X}} \langle \psi \mid \psi' \rangle_{\mathcal{H}_{k_Z}} \tag{3.126}$$

Each of its elements T also acts as a continuous linear operator from $\mathcal{H}_{k_Z}$ to $\mathcal{H}_{k_X}$, the tensor $\varphi' \otimes \psi'$ mapping ψ to $\langle \psi' \mid \psi \rangle_{\mathcal{H}_{k_Z}} \varphi'$, so that:

$$\forall \varphi \in \mathcal{H}_{k_X},\ \forall \psi \in \mathcal{H}_{k_Z},\ \langle \varphi \mid T\psi \rangle_{\mathcal{H}_{k_X}} = \langle \varphi \otimes \psi \mid T \rangle_{\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}} \tag{3.127}$$

Similarly to Proposition 3.22, Proposition 3.23 gives a sufficient condition for the existence of μ_{XZ}. It also generalises (3.121) to expectations involving both X and Z [92].

Proposition 3.23 (Existence of the joint KME). *With the notations of Definition 3.9, if $\mathbb{E}_{p_{XZ}} \left[\sqrt{k_X(X, X)\, k_Z(Z, Z)} \right] < \infty$, then μ_{XZ} is well defined and:*

$$\forall \varphi \in \mathcal{H}_{k_X},\ \forall \psi \in \mathcal{H}_{k_Z},\ \langle \varphi \mid \mu_{XZ}\, \psi \rangle_{\mathcal{H}_{k_X}} = \mathbb{E}_{p_{XZ}} \left[\varphi(X)\, \psi(Z) \right] \tag{3.128}$$

Proof. Let $x \in \mathcal{U}_X$ and $z \in \mathcal{U}_Z$. We have:

$$\| \Phi_X(x) \otimes \Phi_Z(z) \|^2_{\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}} = k_X(x, x)\, k_Z(z, z) \tag{3.129}$$

The same arguments presented in the proof of Proposition 3.22 can be applied in $\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}$ to define μ_{XZ} as the unique element of this space such that, for all $T \in \mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}$:

$$\langle T \mid \mu_{XZ} \rangle_{\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}} = \mathbb{E}_{p_{XZ}} \left[\langle T \mid \Phi_X(X) \otimes \Phi_Z(Z) \rangle_{\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}} \right] \tag{3.130}$$

Let $\varphi \in \mathcal{H}_{k_X}$ and $\psi \in \mathcal{H}_{k_Z}$. Applying the previous results with $T = \varphi \otimes \psi$ yields:

$$\begin{aligned} \langle \varphi \mid \mu_{XZ}\, \psi \rangle_{\mathcal{H}_{k_X}} &= \langle \varphi \otimes \psi \mid \mu_{XZ} \rangle_{\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}} \\ &= \mathbb{E}_{p_{XZ}} \left[\langle \varphi \otimes \psi \mid \Phi_X(X) \otimes \Phi_Z(Z) \rangle_{\mathcal{H}_{k_X} \otimes \mathcal{H}_{k_Z}} \right] \\ &= \mathbb{E}_{p_{XZ}} \left[\langle \varphi \mid \Phi_X(X) \rangle_{\mathcal{H}_{k_X}} \langle \psi \mid \Phi_Z(Z) \rangle_{\mathcal{H}_{k_Z}} \right] \\ &= \mathbb{E}_{p_{XZ}} \left[\varphi(X)\, \psi(Z) \right] \end{aligned} \tag{3.131}$$

□

Definition 3.10 introduces the embedding of a conditional law, and Proposition 3.24 shows that it depends linearly on the feature of the value taken by the conditioning variable [93]. In the following, μ_{ZX} and μ_{XX} denote the joint KMEs of (Z, X) and of (X, X), the latter being taken under the kernel k_X alone.

Definition 3.10 (Conditional KME)**.** *With the notations of Definition 3.9, let $p_{Z|X}$ be the conditional probability density of Z given X, and $x \in \mathcal{U}_X$. The conditional KME of Z given $X = x$ is:*

$$\mu_{Z|x} \triangleq \mathbb{E}\left[\Phi_Z(Z)|X = x\right] = \int_{\mathcal{U}_Z} \Phi_Z(z)\, p_{Z|X}(z|x)\, dz \in \mathcal{H}_{k_Z} \tag{3.132}$$

As in Definition 3.8, it can be used to express conditional expectations in terms of inner products:

$$\forall \psi \in \mathcal{H}_{k_Z},\ \mathbb{E}[\psi(Z)|X = x] = \langle \psi \mid \mu_{Z|x} \rangle_{\mathcal{H}_{k_Z}} \tag{3.133}$$

Proposition 3.24 (Conditional embedding operator)**.** *With the notations of Definition 3.10, assume that:*

$$\psi \mapsto \mathbb{E}[\psi(Z)|X = \cdot] \tag{3.134}$$

is a continuous linear operator from $\mathcal{H}_{k_Z}$ to $\mathcal{H}_{k_X}$.

We also let $\mu_{Z|X}$ be its adjoint, i.e. the continuous linear operator from $\mathcal{H}_{k_X}$ to $\mathcal{H}_{k_Z}$ such that:

$$\forall \varphi \in \mathcal{H}_{k_X},\ \forall \psi \in \mathcal{H}_{k_Z},\ \langle \psi \mid \mu_{Z|X}\, \varphi \rangle_{\mathcal{H}_{k_Z}} = \langle \mathbb{E}[\psi(Z)|X = \cdot] \mid \varphi \rangle_{\mathcal{H}_{k_X}} \tag{3.135}$$

The conditional KME is then linear in the feature of the conditioning value:

$$\forall x \in \mathcal{U}_X,\ \mu_{Z|x} = \mu_{Z|X} \Phi_X(x) \tag{3.136}$$

and $\mu_{Z|X}$ maps the embeddings of X onto those of (Z, X) and Z:

$$\begin{cases} \mu_{Z|X}\, \mu_{XX} = \mu_{ZX} \\ \mu_{Z|X}\, \mu_X = \mu_Z \end{cases} \tag{3.137}$$

Proof. Let $\varphi \in \mathcal{H}_{k_X}$, $\psi \in \mathcal{H}_{k_Z}$ and $x \in \mathcal{U}_X$. We have:

$$\begin{aligned} \langle \psi \mid \mu_{Z|x} \rangle_{\mathcal{H}_{k_Z}} &= \mathbb{E}[\psi(Z)|X = x] \\ &= \langle \mathbb{E}[\psi(Z)|X = \cdot] \mid \Phi_X(x) \rangle_{\mathcal{H}_{k_X}} \\ &= \langle \psi \mid \mu_{Z|X} \Phi_X(x) \rangle_{\mathcal{H}_{k_Z}} \end{aligned} \tag{3.138}$$

Similarly, the tower property of conditional expectation yields:

$$\begin{aligned} \langle \psi \mid \mu_{Z|X}\, \mu_{XX}\, \varphi \rangle_{\mathcal{H}_{k_Z}} &= \langle \mathbb{E}[\psi(Z)|X = \cdot] \mid \mu_{XX}\, \varphi \rangle_{\mathcal{H}_{k_X}} \\ &= \mathbb{E}\left[\mathbb{E}[\psi(Z)|X]\, \varphi(X)\right] \\ &= \mathbb{E}_{p_{XZ}}\left[\psi(Z)\, \varphi(X)\right] \\ &= \langle \psi \mid \mu_{ZX}\, \varphi \rangle_{\mathcal{H}_{k_Z}} \end{aligned} \tag{3.139}$$

while the law of total expectation yields:

$$\begin{aligned} \langle \psi \mid \mu_{Z|X}\, \mu_X \rangle_{\mathcal{H}_{k_Z}} &= \langle \mathbb{E}[\psi(Z)|X = \cdot] \mid \mu_X \rangle_{\mathcal{H}_{k_X}} \\ &= \mathbb{E}\left[\mathbb{E}[\psi(Z)|X]\right] \\ &= \mathbb{E}\left[\psi(Z)\right] \\ &= \langle \psi \mid \mu_Z \rangle_{\mathcal{H}_{k_Z}} \end{aligned} \tag{3.140}$$

□

3.5.4 Empirical Embeddings

When k_X is an SE or Matérn kernel on $\mathbb{R}^{d_x}$, every element of $\mathcal{H}_{k_X}$ is square-integrable [88]. However, when X and Z are independent, $\mathbb{E}[\psi(Z)|X = \cdot]$ is the constant function $\langle \psi \mid \mu_Z \rangle_{\mathcal{H}_{k_Z}}$. Such a function belongs to $\mathcal{L}^2(\mathbb{R}^{d_x}, \mathbb{R})$ only when this constant is zero. Consequently, the assumption of Proposition 3.24 is not satisfied in general, and the conditional embedding operator may not exist.

Furthermore, (3.120) and (3.132) are no more computable than the integrals of chapter 2. Similarly to Proposition 2.4, the KMEs can be approximated by a Monte Carlo estimator, as presented by Definition 3.11 and Proposition 3.25. The estimator of the conditional KME is also regularised, so that it remains well defined even when the assumption of Proposition 3.24 is not satisfied.

Definition 3.11 (Empirical KME)**.** *Let $N \in \mathbb{N}^*$ and $X^{(1)}, \dots, X^{(N)}$ be i.i.d. random variables with probability density p. The empirical KME is:*

$$\hat{\mu}_X \triangleq \frac{1}{N} \sum_{i=1}^{N} \Phi\left(X^{(i)}\right) \tag{3.141}$$

It converges a.s. to μ_X by the strong law of large numbers. If moreover $\mathbb{E}_p[k(X,X)] < \infty$, its mean squared error can be computed as in the proof of Proposition 2.4:

$$\mathbb{E}\left[\|\hat{\mu}_X - \mu_X\|_{\mathcal{H}_k}^2\right] = \frac{1}{N}\left(\mathbb{E}_p[k(X,X)] - \|\mu_X\|_{\mathcal{H}_k}^2\right) \tag{3.142}$$

When μ_{XX} is injective, Proposition 3.24 gives on the range of μ_{XX}:

$$\mu_{Z|X} = \mu_{ZX}\, \mu_{XX}^{-1} \tag{3.143}$$

which suggests replacing μ_{ZX} and μ_{XX} with their empirical counterparts. However, the empirical counterpart of μ_{XX} has rank at most N, and is therefore not injective in general. To alleviate this issue, a regularisation term is added before the inversion, similarly to (3.105). The resulting estimator is given by Proposition 3.25.

Proposition 3.25 (Empirical conditional KME)**.** *Let $N \in \mathbb{N}^*$, $\lambda \in \mathbb{R}_+^*$, and let $\left(X^{(i)}, Z^{(i)}\right)_{i \in [\![1,N]\!]}$ be i.i.d. pairs of random variables with joint probability density p_{XZ}. We write $\mathbf{x} \triangleq \left(X^{(1)}, \dots, X^{(N)}\right)$ and $\mathbf{z} \triangleq \left(Z^{(1)}, \dots, Z^{(N)}\right)$, and we let:*

$$\begin{cases} \hat{\mu}_{XX} \triangleq \dfrac{1}{N} \displaystyle\sum_{i=1}^{N} \Phi_X\left(X^{(i)}\right) \otimes \Phi_X\left(X^{(i)}\right) \\ \hat{\mu}_{ZX} \triangleq \dfrac{1}{N} \displaystyle\sum_{i=1}^{N} \Phi_Z\left(Z^{(i)}\right) \otimes \Phi_X\left(X^{(i)}\right) \end{cases} \tag{3.144}$$

be the empirical counterparts of μ_{XX} and μ_{ZX}.

The regularised estimator of $\mu_{Z|X}$:

$$\hat{\mu}_{Z|X} \triangleq \hat{\mu}_{ZX}\left(\hat{\mu}_{XX} + \lambda I\right)^{-1} \tag{3.145}$$

where I denotes the identity of $\mathcal{H}_{k_X}$, is well defined and verifies:

$$\forall \varphi \in \mathcal{H}_{k_X},\ \hat{\mu}_{Z|X}\, \varphi = \varphi(\mathbf{x})^\top \left(k_X(\mathbf{x},\mathbf{x}) + \lambda N I_N\right)^{-1} \Phi_Z(\mathbf{z}) \tag{3.146}$$

In particular, for $x \in \mathcal{U}_X$, the conditional KME of Z given $X = x$ is estimated by:

$$\hat{\mu}_{Z|x} \triangleq \hat{\mu}_{Z|X}\Phi_X(x) = \sum_{i=1}^{N} w^{(i)}(x)\, \Phi_Z\left(Z^{(i)}\right) \tag{3.147}$$

where the weights are given by:

$$\mathbf{w}(x) \triangleq \left[w^{(1)}(x), \dots, w^{(N)}(x)\right]^\top = \left(k_X(\mathbf{x},\mathbf{x}) + \lambda N I_N\right)^{-1} k_X(\mathbf{x}, x) \tag{3.148}$$

Proof. Let $u \in \mathcal{H}_{k_X}$. By definition of the operator associated with a tensor, and by the reproducing property, we have:

$$\begin{cases} \hat{\mu}_{XX}\, u = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} u\left(X^{(i)}\right)\Phi_X\left(X^{(i)}\right) = \dfrac{1}{N}\, u(\mathbf{x})^\top \Phi_X(\mathbf{x}) \\ \hat{\mu}_{ZX}\, u = \dfrac{1}{N}\displaystyle\sum_{i=1}^{N} u\left(X^{(i)}\right)\Phi_Z\left(Z^{(i)}\right) = \dfrac{1}{N}\, u(\mathbf{x})^\top \Phi_Z(\mathbf{z}) \end{cases} \tag{3.149}$$

In particular:

$$\langle u \mid (\hat{\mu}_{XX} + \lambda I)\, u\rangle_{\mathcal{H}_{k_X}} = \frac{1}{N}\|u(\mathbf{x})\|_2^2 + \lambda\|u\|_{\mathcal{H}_{k_X}}^2 \geq \lambda\|u\|_{\mathcal{H}_{k_X}}^2 \tag{3.150}$$

and thus $\hat{\mu}_{XX} + \lambda I$ is injective.

Let $\varphi \in \mathcal{H}_{k_X}$. Since $k_X(\mathbf{x},\mathbf{x}) \succeq 0$ and $\lambda > 0$, the matrix $k_X(\mathbf{x},\mathbf{x}) + \lambda N I_N$ is SPD, hence invertible, and we can define:

$$\begin{cases} c \triangleq \left(k_X(\mathbf{x},\mathbf{x}) + \lambda N I_N\right)^{-1}\varphi(\mathbf{x}) \\ u \triangleq \dfrac{1}{\lambda}\left(\varphi - c^\top \Phi_X(\mathbf{x})\right) \end{cases} \tag{3.151}$$

By the kernel trick, it comes that:

$$u(\mathbf{x}) = \frac{1}{\lambda}\left(\varphi(\mathbf{x}) - k_X(\mathbf{x},\mathbf{x})\, c\right) = Nc \tag{3.152}$$

and thus:

$$(\hat{\mu}_{XX} + \lambda I)\, u = c^\top \Phi_X(\mathbf{x}) + \varphi - c^\top \Phi_X(\mathbf{x}) = \varphi \tag{3.153}$$

Therefore, $\hat{\mu}_{XX} + \lambda I$ is invertible, and $u = \left(\hat{\mu}_{XX} + \lambda I\right)^{-1}\varphi$, which finally gives:

$$\hat{\mu}_{Z|X}\, \varphi = \hat{\mu}_{ZX}\, u = c^\top \Phi_Z(\mathbf{z}) \tag{3.154}$$

The expression of the weights is obtained by taking $\varphi = \Phi_X(x)$, whose values at $\mathbf{x}$ are $k_X(\mathbf{x}, x)$. □

Remark 3.18 (Conditional KME as a regression)**.** *Let $\psi \in \mathcal{H}_{k_Z}$. According to Proposition 3.25, the conditional expectation $\mathbb{E}[\psi(Z)|X = x]$ is estimated by:*

$$\langle \psi \mid \hat{\mu}_{Z|x}\rangle_{\mathcal{H}_{k_Z}} = k_X(x,\mathbf{x})\left(k_X(\mathbf{x},\mathbf{x}) + \lambda N I_N\right)^{-1}\psi(\mathbf{z}) \tag{3.155}$$

which is the solution given by Proposition 3.20 for the training points $\mathbf{x}$ and the observations $\psi(\mathbf{z})$. The empirical conditional KME can thus be interpreted as a centred SK estimator, in which λN plays the role of the training noise covariance [94]. This interpretation does not require the assumption of Proposition 3.24, which is why the regularised estimator is used when it is not satisfied.

It must also be noted that, unlike the uniform weights of Definition 3.11, the weights $\mathbf{w}(x)$ depend on x, do not necessarily sum to one, and may be negative, so that $\hat{\mu}_{Z|x}$ is not the KME of a probability distribution in general.

3.5.5 Kernel Kalman Filter

The kernel Kalman filter (KKF) [95] solves the filtering problem (2.1) by propagating the KME of the PDF instead of the PDF itself. Formally, let k_x and k_z be SPD kernels on the state space $\mathbb{R}^{d_x}$ and on the observation space $\mathbb{R}^{d_z}$, with respective native RKHSs $\mathcal{H}_{k_x}$ and $\mathcal{H}_{k_z}$, and feature maps Φ_x and Φ_z. Using the index convention of Proposition 2.3, the KMEs of the predicted and corrected densities of Algorithm 2.1 are:

$$\begin{cases} \mu_{k|k-1} \triangleq \mathbb{E}\left[\Phi_x(x_k)|\mathbf{z}_{1:k-1}\right] \\ \mu_{k|k} \triangleq \mathbb{E}\left[\Phi_x(x_k)|\mathbf{z}_{1:k}\right] \end{cases} \tag{3.156}$$

When k_x is characteristic, these KMEs uniquely determine both densities. Proposition 3.26 shows that the state-space model (2.1) becomes linear when it is expressed in terms of features.

Proposition 3.26 (Embedded state-space model)**.** *Assume that the transition density $p(x_k|x_{k-1})$ and the likelihood $p(z_k|x_k)$ satisfy the assumption of Proposition 3.24, and let $\mu_{x_k|x_{k-1}}$ and $\mu_{z_k|x_k}$ denote their respective conditional embedding operators. Then:*

$$\begin{cases} \mathbb{E}\left[\Phi_x(x_k)|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}\right] = \mu_{x_k|x_{k-1}} \Phi_x(x_{k-1}) \\ \mathbb{E}\left[\Phi_z(z_k)|\mathbf{x}_{0:k}, \mathbf{z}_{1:k-1}\right] = \mu_{z_k|x_k} \Phi_x(x_k) \end{cases} \tag{3.157}$$

Furthermore, the prediction step of Algorithm 2.1 becomes:

$$\mu_{k|k-1} = \mu_{x_k|x_{k-1}} \, \mu_{k-1|k-1} \tag{3.158}$$

Proof. According to Assumptions 2.1 and 2.2, these conditional expectations only depend on x_{k-1} and x_k respectively. They are thus the conditional KMEs of Definition 3.10, and Proposition 3.24 gives the desired expressions.

Since $\mu_{x_k|x_{k-1}}$ is linear and continuous, the tower property of conditional expectation finally yields:

$$\begin{aligned} \mu_{k|k-1} &= \mathbb{E}\left[\mathbb{E}\left[\Phi_x(x_k)|\mathbf{x}_{0:k-1}, \mathbf{z}_{1:k-1}\right]|\mathbf{z}_{1:k-1}\right] \\ &= \mathbb{E}\left[\mu_{x_k|x_{k-1}} \Phi_x(x_{k-1})|\mathbf{z}_{1:k-1}\right] \\ &= \mu_{x_k|x_{k-1}} \, \mu_{k-1|k-1} \end{aligned} \tag{3.159}$$

□

According to Proposition 3.26, the features of the state and of the observation follow a linear model in conditional expectation, regardless of the non-linearity of f_k and h_k. In particular, the Chapman-Kolmogorov equation (2.30) reduces to a linear relation. However, the correction step (2.32) does not admit such a simple expression. While the kernel Bayes rule [92] gives an expression of $\mu_{k|k}$, it relies on conditional embedding operators that depend on the predicted density. These operators must therefore be estimated and inverted at every time step, which is numerically unstable.

The kernel Kalman rule circumvents this issue by treating the embedded model as the linear model (2.35) of a Kalman filter, in which $\Phi_x(x_k)$ plays the role of the state, and $\Phi_z(z_k)$ that of the observation. The predicted KME is then corrected by a linear function of the innovation:

$$\hat{\mu}_{k|k} = \mu_{k|k-1} + \mathcal{Q}_k \left(\Phi_z(z_k) - \mu_{z_k|x_k} \, \mu_{k|k-1}\right) \tag{3.160}$$

where the gain $\mathcal{Q}_k$ is the continuous linear operator from $\mathcal{H}_{k_z}$ to $\mathcal{H}_{k_x}$ that minimises the mean squared norm of the estimation error $\Phi_x(x_k) - \hat{\mu}_{k|k}$. Since the differences between the features and their conditional expectations are centred but not Gaussian, the corrected KME $\hat{\mu}_{k|k}$ is the best linear estimator of $\Phi_x(x_k)$, but does not coincide with $\mu_{k|k}$ in general.

In practice, neither $\mu_{x_k|x_{k-1}}$ nor $\mu_{z_k|x_k}$ is known. The KKF estimates them once from a training set of $\tilde{N} \in \mathbb{N}^*$ triples:

$$\left(\tilde{x}_i^-, \tilde{x}_i, \tilde{z}_i\right)_{i \in [\![1, \tilde{N}]\!]} \tag{3.161}$$

where $\tilde{x}_i$ is the successor of the state $\tilde{x}_i^-$, and $\tilde{z}_i$ is the observation of $\tilde{x}_i$. We let $\tilde{\mathbf{x}}^-$, $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{z}}$ denote the corresponding tuples. The model (2.1) is thus assumed to be stationary, but does not need to be known.

The two operators are estimated by applying Proposition 3.25 to the pairs $\left(\tilde{x}_i^-, \tilde{x}_i\right)$ and $(\tilde{x}_i, \tilde{z}_i)$ respectively, with $\lambda_x \triangleq \lambda \tilde{N}$. For $\mathbf{w} \in \mathbb{R}^{\tilde{N}}$, the function $\mathbf{w}^\top \Phi_x(\tilde{\mathbf{x}})$ takes the values $k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}\right)\mathbf{w}$ at $\tilde{\mathbf{x}}^-$ and $k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})\,\mathbf{w}$ at $\tilde{\mathbf{x}}$, which gives:

$$\begin{cases} \hat{\mu}_{x_k|x_{k-1}}\left(\mathbf{w}^\top \Phi_x(\tilde{\mathbf{x}})\right) = \left(k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}\right)\mathbf{w}\right)^\top \left(k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}^-\right) + \lambda_x I_{\tilde{N}}\right)^{-1} \Phi_x(\tilde{\mathbf{x}}) \\ \hat{\mu}_{z_k|x_k}\left(\mathbf{w}^\top \Phi_x(\tilde{\mathbf{x}})\right) = \left(k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})\,\mathbf{w}\right)^\top \left(k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda_x I_{\tilde{N}}\right)^{-1} \Phi_z(\tilde{\mathbf{z}}) \end{cases} \tag{3.162}$$

which can be rewritten as:

$$\begin{cases} \hat{\mu}_{x_k|x_{k-1}}\left(\mathbf{w}^\top \Phi_x(\tilde{\mathbf{x}})\right) = (T\mathbf{w})^\top \Phi_x(\tilde{\mathbf{x}}) \\ \hat{\mu}_{z_k|x_k}\left(\mathbf{w}^\top \Phi_x(\tilde{\mathbf{x}})\right) = (O\mathbf{w})^\top \Phi_z(\tilde{\mathbf{z}}) \end{cases} \tag{3.163}$$

with:

$$\begin{cases} T \triangleq \left(k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}^-\right) + \lambda_x I_{\tilde{N}}\right)^{-1} k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}\right) \\ O \triangleq \left(k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda_x I_{\tilde{N}}\right)^{-1} k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) \end{cases} \tag{3.164}$$

In particular, the set spanned by the features $\Phi_x(\tilde{\mathbf{x}})$, called the dictionary, is stable under the estimated transition operator. A KME of this set is then entirely described by its weights $\mathbf{w}$, on which the two operators act as the matrices T and O.

The remaining quantities of the model are obtained as follows:

- **Transition residual.** The values of $\Phi_x\left(\tilde{x}_i^-\right)$ at $\tilde{\mathbf{x}}^-$ form the i-th column of $k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}^-\right)$. According to Proposition 3.25, the image of this feature by the estimated transition operator thus has for weights the i-th column of $D + I_{\tilde{N}}$, where:

  $$D \triangleq \left(k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}^-\right) + \lambda_x I_{\tilde{N}}\right)^{-1} k_x\left(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}^-\right) - I_{\tilde{N}} \tag{3.165}$$

  Since $\Phi_x(\tilde{x}_i)$ has for weights the i-th column of $I_{\tilde{N}}$, the residual $\Phi_x(\tilde{x}_i) - \hat{\mu}_{x_k|x_{k-1}}\Phi_x\left(\tilde{x}_i^-\right)$ of the i-th training triple has for weights the i-th column of $-D$. The covariance of the transition residual is then estimated by the empirical second-order moment of these weights:

  $$V \triangleq \frac{1}{\tilde{N}} D\, D^\top \tag{3.166}$$

- **Observation residual.** The covariance of the observation residual is chosen as $\lambda_z I$, where $\lambda_z \in \mathbb{R}_+^*$ and I denotes the identity of $\mathcal{H}_{k_z}$.

- **Observation.** According to the reproducing property, the inner products of $\psi \in \mathcal{H}_{k_z}$ with the features $\Phi_z(\tilde{\mathbf{z}})$ are its values $\psi(\tilde{\mathbf{z}})$. The predicted observation $(O\mathbf{w})^\top \Phi_z(\tilde{\mathbf{z}})$ belongs to the set spanned by these features, and the covariance of the observation residual is proportional to the identity. The component of $\Phi_z(z_k)$ that is orthogonal to this set is therefore uncorrelated with the other one, and $\Phi_z(z_k)$ is only used through its values at $\tilde{\mathbf{z}}$. By the kernel trick, the observation and its prediction become:

$$\begin{cases} \left(\Phi_z(z_k)\right)(\tilde{\mathbf{z}}) = k_z(\tilde{\mathbf{z}}, z_k) \\ \left((O\mathbf{w})^\top \Phi_z(\tilde{\mathbf{z}})\right)(\tilde{\mathbf{z}}) = G^z O\mathbf{w} \end{cases} \tag{3.167}$$

  where:

$$G^z \triangleq k_z(\tilde{\mathbf{z}}, \tilde{\mathbf{z}}) \tag{3.168}$$

  Similarly, a residual r of covariance $\lambda_z I$ verifies, for $i, j \in [\![1, \tilde{N}]\!]$:

$$\mathbb{E}\left[r(\tilde{z}_i)\, r(\tilde{z}_j)\right] = \lambda_z \langle \Phi_z(\tilde{z}_i) \mid \Phi_z(\tilde{z}_j) \rangle_{\mathcal{H}_{k_z}} = \lambda_z k_z(\tilde{z}_i, \tilde{z}_j) \tag{3.169}$$

  and the covariance of the observation residual becomes $\lambda_z G^z$.

The KKF is then the Kalman filter of Algorithm 2.2 applied to the weights, under the substitutions:

$$\begin{cases} \hat{x}_{k|k} \longrightarrow \mathbf{w}_{k|k} & F_k \longrightarrow T & H_k \longrightarrow G^z O \\ P_{k|k} \longrightarrow \Sigma_{k|k} & Q_k \longrightarrow V & R_k \longrightarrow \lambda_z G^z \\ z_k \longrightarrow k_z(\tilde{\mathbf{z}}, z_k) & b_k^{(e)} \longrightarrow 0 & b_k^{(o)} \longrightarrow 0 \end{cases} \tag{3.170}$$

where $\mathbf{w}_{k|k} \in \mathbb{R}^{\tilde{N}}$ denotes the weights of the corrected KME:

$$\hat{\mu}_{k|k} = \mathbf{w}_{k|k}^\top \Phi_x(\tilde{\mathbf{x}}) \tag{3.171}$$

and $\Sigma_{k|k} \in \mathcal{S}_{\tilde{N}}^+(\mathbb{R})$ is the covariance of their estimation error. The resulting recursion is given by Proposition 3.27.

Proposition 3.27 (Kernel Kalman filter)**.** *Assume that the training observations $\tilde{\mathbf{z}}$ are pairwise distinct. Under the substitutions (3.170), Algorithm 2.2 reduces to Algorithm 3.1.*

Proof. The prediction and correction steps of Algorithm 2.2 are obtained by direct substitution. Since k_z is SPD and the training observations are pairwise distinct, we have $G^z \succ 0$. The innovation covariance can thus be factorised as:

$$S_k = G^z O\, \Sigma_{k|k-1}\, O^\top G^z + \lambda_z G^z = \left(G^z O\, \Sigma_{k|k-1}\, O^\top + \lambda_z I_{\tilde{N}}\right) G^z \tag{3.172}$$

where both factors are invertible since $S_k \succ 0$. The Kalman gain is therefore:

$$K_k = \Sigma_{k|k-1}\, O^\top G^z\, S_k^{-1} = \Sigma_{k|k-1}\, O^\top \left(G^z O\, \Sigma_{k|k-1}\, O^\top + \lambda_z I_{\tilde{N}}\right)^{-1} \tag{3.173}$$

□

Algorithm 3.1 Kernel Kalman Filter

$\mathbf{w}_{k|k-1} = T\,\mathbf{w}_{k-1|k-1}$ ▷ Weight prediction

$\Sigma_{k|k-1} = T\,\Sigma_{k-1|k-1}\,T^\top + V$ ▷ Covariance prediction

$K_k = \Sigma_{k|k-1}\,O^\top \left(G^z O\,\Sigma_{k|k-1}\,O^\top + \lambda_z I_{\tilde{N}}\right)^{-1}$ ▷ Kernel Kalman gain

$\mathbf{w}_{k|k} = \mathbf{w}_{k|k-1} + K_k \left(k_z\left(\tilde{\mathbf{z}}, z_k\right) - G^z O\,\mathbf{w}_{k|k-1}\right)$ ▷ Weight correction

$\Sigma_{k|k} = \Sigma_{k|k-1} - K_k\,G^z O\,\Sigma_{k|k-1}$ ▷ Covariance correction

The KKF is initialised using samples of the initial density p_0. By Proposition 3.25 applied to the pairs $(\tilde{x}_i, \tilde{x}_i)$, a sample $x \in \mathbb{R}^{d_x}$ is represented in the dictionary by the weights $\left(k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda_x I_{\tilde{N}}\right)^{-1} k_x(\tilde{\mathbf{x}}, x)$, and $\mathbf{w}_{0|0}$ and $\Sigma_{0|0}$ are the empirical mean and covariance of these weights over the samples.

In particular, the weight correction of Algorithm 3.1 shows that the gain $\mathcal{Q}_k$ of the kernel Kalman rule is estimated by the operator:

$$\psi \in \mathcal{H}_{k_z} \mapsto \left(K_k\,\psi(\tilde{\mathbf{z}})\right)^\top \Phi_x(\tilde{\mathbf{x}}) \tag{3.174}$$

The state is estimated from the corrected KME using (3.121). Since the coordinate functions of the state do not belong to $\mathcal{H}_{k_x}$ in general, they are replaced by their regularised least-squares approximations in the dictionary. Proposition 3.20 applied to the training points $\tilde{\mathbf{x}}$ and to the j-th coordinates $[\tilde{\mathbf{x}}]_j \in \mathbb{R}^{\tilde{N}}$ of the training states, with the same λ, approximates the j-th coordinate function by $a^\top \Phi_x(\tilde{\mathbf{x}})$, with:

$$a \triangleq \left(k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \lambda_x I_{\tilde{N}}\right)^{-1} [\tilde{\mathbf{x}}]_j \tag{3.175}$$

By the kernel trick, its inner product with $\hat{\mu}_{k|k}$ is:

$$a^\top k_x(\tilde{\mathbf{x}}, \tilde{\mathbf{x}})\,\mathbf{w}_{k|k} = [\tilde{\mathbf{x}}]_j^\top O\,\mathbf{w}_{k|k} \tag{3.176}$$

which is the j-th coordinate of:

$$\hat{x}_k = \sum_{i=1}^{\tilde{N}} \left[O\,\mathbf{w}_{k|k}\right]_i \tilde{x}_i \tag{3.177}$$

Remark 3.19 (Cost of the KKF)**.** *As for the Kalman filter, the covariances and gains of Algorithm 3.1 do not depend on the observations, and can be computed beforehand. Otherwise, the inversion required by the gain costs* $\mathcal{O}\left(\tilde{N}^3\right)$ *operations per time step.*

3.5.6 Adaptive Kernel Kalman Filter

Since its operators are estimated from a fixed training set, the KKF does not require any knowledge of the model. However, a KME can only be represented in the regions where training states are available, and the filter fails when the state leaves these regions [96]. In the navigation problem (2.1), the functions f_k and h_k are known. The adaptive kernel Kalman filter (AKKF) uses them to generate a new training set at every time step, located around the current state estimate.

Formally, at time k, the triple $(\tilde{\mathbf{x}}^-, \tilde{\mathbf{x}}, \tilde{\mathbf{z}})$ of the KKF is replaced by three sets of $N \in \mathbb{N}^*$ particles:

- The resampled particles $\mathbf{x}_{k-1}^+$, drawn around the previous state estimate.
- The propagated particles $\mathbf{x}_k$, obtained by propagating $\mathbf{x}_{k-1}^+$ through the evolution model.
- The simulated observations $\mathbf{z}_k$, obtained by applying the observation model to $\mathbf{x}_k$.

The dictionary thus changes at every time step. In particular, the corrected KME of time $k-1$ is represented by the particles $\mathbf{x}_{k-1}$, with weights $\mathbf{w}_{k-1}$ and covariance $\Sigma_{k-1|k-1}$. On the other hand, the transition operator is estimated from the pairs $\left(x_{k-1}^{+(i)}, x_k^{(i)}\right)$, and returns KMEs represented by the particles $\mathbf{x}_k$. Proposition 3.28 gives the corresponding change of weights.

Proposition 3.28 (Prediction over a moving dictionary)**.** *Let $\hat{\mu}_{x_k|x_{k-1}}$ be the transition operator estimated by Proposition 3.25 from the pairs $\left(x_{k-1}^{+(i)}, x_k^{(i)}\right)_{i\in[\![1,N]\!]}$, with $\lambda_x \triangleq \lambda N$, and let $G_{k-1}^x \triangleq k_x\left(\mathbf{x}_{k-1}^+, \mathbf{x}_{k-1}^+\right)$. Then, for all $\mathbf{w} \in \mathbb{R}^N$:*

$$\hat{\mu}_{x_k|x_{k-1}}\left(\mathbf{w}^\top \Phi_x(\mathbf{x}_{k-1})\right) = (T_{k-1}\mathbf{w})^\top \Phi_x(\mathbf{x}_k) \tag{3.178}$$

with:

$$T_{k-1} \triangleq \left(G_{k-1}^x + \lambda_x I_N\right)^{-1} k_x\left(\mathbf{x}_{k-1}^+, \mathbf{x}_{k-1}\right) \tag{3.179}$$

Furthermore, the residual $\Phi_x\left(x_k^{(i)}\right) - \hat{\mu}_{x_k|x_{k-1}}\Phi_x\left(x_{k-1}^{+(i)}\right)$ of the i-th pair has for weights in $\Phi_x(\mathbf{x}_k)$ the i-th column of $-D_k$, where:

$$D_k \triangleq \left(G_{k-1}^x + \lambda_x I_N\right)^{-1} G_{k-1}^x - I_N \tag{3.180}$$

Proof. Let $\mathbf{w} \in \mathbb{R}^N$. The function $\mathbf{w}^\top \Phi_x(\mathbf{x}_{k-1})$ takes the values $k_x\left(\mathbf{x}_{k-1}^+, \mathbf{x}_{k-1}\right)\mathbf{w}$ at $\mathbf{x}_{k-1}^+$, and Proposition 3.25 thus gives:

$$\begin{aligned}\hat{\mu}_{x_k|x_{k-1}}\left(\mathbf{w}^\top \Phi_x(\mathbf{x}_{k-1})\right) &= \left(k_x\left(\mathbf{x}_{k-1}^+, \mathbf{x}_{k-1}\right)\mathbf{w}\right)^\top \left(G_{k-1}^x + \lambda_x I_N\right)^{-1} \Phi_x(\mathbf{x}_k) \\ &= (T_{k-1}\mathbf{w})^\top \Phi_x(\mathbf{x}_k)\end{aligned} \tag{3.181}$$

Similarly, the values of $\Phi_x\left(x_{k-1}^{+(i)}\right)$ at $\mathbf{x}_{k-1}^+$ form the i-th column of G_{k-1}^x. Its image thus has for weights the i-th column of:

$$\left(G_{k-1}^x + \lambda_x I_N\right)^{-1} G_{k-1}^x = D_k + I_N \tag{3.182}$$

while $\Phi_x\left(x_k^{(i)}\right)$ has for weights the i-th column of I_N. The weights of their difference are therefore the i-th column of $-D_k$. □

Each iteration of the AKKF is then computed in four steps:

- **Embedded prediction.** According to Proposition 3.28, the prediction step of Algorithm 3.1 is performed with T_{k-1} instead of T, and $\frac{1}{N} D_k D_k^\top$ instead of V. Since T_{k-1} only depends on the particles of time $k-1$, it is applied as soon as $\mathbf{x}_{k-1}^+$ is drawn. This operation is called embedded resampling, and gives:

$$\begin{cases} \mathbf{w}_{k-1}^+ \triangleq T_{k-1}\,\mathbf{w}_{k-1} \\ \Sigma_{k-1|k-1}^+ \triangleq T_{k-1}\,\Sigma_{k-1|k-1}\,T_{k-1}^\top \end{cases} \tag{3.183}$$

The particle propagation then leaves the weights unchanged, and only adds the covariance of the residuals:

$$\begin{cases} \mathbf{w}_{k|k-1} = \mathbf{w}^+_{k-1} \\ \Sigma_{k|k-1} = \Sigma^+_{k-1|k-1} + \dfrac{1}{N} D_k \, D_k^\top \end{cases} \tag{3.184}$$

The predicted weights are associated with the propagated particles $\mathbf{x}_k$.

- **Embedded correction.** The correction step is that of Algorithm 3.1, where the observation operator is estimated from the pairs $\left(x_k^{(i)}, z_k^{(i)}\right)$, with the Gram matrix:

$$G_k^z \triangleq k_z(\mathbf{z}_k, \mathbf{z}_k) \tag{3.185}$$

Moreover, the matrix O is replaced by the identity. This corresponds to an observation operator estimated without regularisation, which maps the feature of each particle to the feature of its simulated observation. The correction thus becomes:

$$\begin{cases} K_k = \Sigma_{k|k-1} \left(G_k^z \, \Sigma_{k|k-1} + \lambda_z I_N\right)^{-1} \\ \mathbf{w}_k = \mathbf{w}_{k|k-1} + K_k \left(k_z\left(\mathbf{z}_k, z_k\right) - G_k^z \, \mathbf{w}_{k|k-1}\right) \\ \Sigma_{k|k} = \Sigma_{k|k-1} - K_k \, G_k^z \, \Sigma_{k|k-1} \end{cases} \tag{3.186}$$

- **State estimation.** The estimator (3.177) then reduces to the weighted mean of the particles:

$$\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)} \, x_k^{(i)} \tag{3.187}$$

Since this estimator is linear in the weights, the covariance $\Sigma_{k|k}$ of their estimation error gives that of the state [95]:

$$\hat{P}_k = \sum_{i=1}^{N} \sum_{j=1}^{N} \left[\Sigma_{k|k}\right]_{ij} \, x_k^{(i)} \left(x_k^{(j)}\right)^\top \tag{3.188}$$

Both are exact when the coordinate functions belong to $\mathcal{H}_{k_x}$, which is the case for a polynomial kernel, and are approximations otherwise.

- **Resampling.** The new particles $\mathbf{x}_k^+$ are drawn from $\mathcal{N}\left(\hat{x}_k, \hat{P}_k\right)$.

The AKKF is initialised with particles $\mathbf{x}_0^+$ drawn independently from p_0, with:

$$\mathbf{w}_0^+ = \frac{1}{N} \mathbf{1}_N \quad \text{and} \quad \Sigma^+_{0|0} = \frac{1}{N} I_N \tag{3.189}$$

It is presented by Algorithm 3.2, where $\mathbf{w}_k$ denotes the corrected weights $\mathbf{w}_{k|k}$. Its pseudo-code includes the state estimation step, since the resampling step depends on it.

It must be noted that the AKKF only requires samples of the likelihood $p(z_k|x_k)$, and never evaluates it.

Algorithm 3.2 Adaptive Kernel Kalman Filter

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim p\left(\cdot | x_{k-1}^{+(i)}\right)$ ▷ Particle propagation

$D_k = \left(G_{k-1}^x + \lambda_x I_N\right)^{-1} G_{k-1}^x - I_N$ ▷ Embedded prediction

$\Sigma_{k|k-1} = \Sigma_{k-1|k-1}^+ + \dfrac{1}{N} D_k\, D_k^\top$

$\forall i \in [\![1, N]\!],\ z_k^{(i)} = h_k\left(x_k^{(i)}, n_k^{(i)}\right),\ n_k^{(i)} \sim p_k^{(o)}$ ▷ Simulated observations

$K_k = \Sigma_{k|k-1}\left(G_k^z\, \Sigma_{k|k-1} + \lambda_z I_N\right)^{-1}$ ▷ Embedded correction

$\mathbf{w}_k = \mathbf{w}_{k-1}^+ + K_k\left(k_z\left(\mathbf{z}_k, z_k\right) - G_k^z\, \mathbf{w}_{k-1}^+\right)$

$\Sigma_{k|k} = \Sigma_{k|k-1} - K_k\, G_k^z\, \Sigma_{k|k-1}$

$\hat{x}_k = \sum_{i=1}^N w_k^{(i)}\, x_k^{(i)}$ ▷ State estimation

$\hat{P}_k = \sum_{i=1}^N \sum_{j=1}^N \left[\Sigma_{k|k}\right]_{ij}\, x_k^{(i)} \left(x_k^{(j)}\right)^\top$

$\forall i \in [\![1, N]\!],\ x_k^{+(i)} \sim \mathcal{N}\left(\hat{x}_k, \hat{P}_k\right)$ ▷ Resampling

$T_k = \left(G_k^x + \lambda_x I_N\right)^{-1} k_x\left(\mathbf{x}_k^+, \mathbf{x}_k\right)$ ▷ Embedded resampling

$\mathbf{w}_k^+ = T_k\, \mathbf{w}_k$

$\Sigma_{k|k}^+ = T_k\, \Sigma_{k|k}\, T_k^\top$

Remark 3.20 (Sign of the weights)**.** *The embedded correction solves a regularised least-squares problem in the feature space, rather than a Bayesian update of a discrete measure. Its weights are thus not constrained to be positive, and the KME they represent is not that of a probability measure in general. While the estimator $\hat{x}_k$ remains well defined, the effective sample size of section 2.5 does not, and neither does the resampling criterion based on it. This is why the AKKF resamples at every time step, by drawing a new particle set from a Gaussian distribution.*

Nonetheless, $\hat{P}_k$ is SPSD as soon as $\Sigma_{k|k}$ is, up to the numerical errors addressed in section 2.4.3.

Remark 3.21 (Cost of the AKKF)**.** *Each iteration of Algorithm 3.2 solves three linear systems of size N, respectively for D_k, K_k and T_k. Its cost is thus $\mathcal{O}\left(N^3\right)$ operations per time step.*

Part II

Navigation with Gaussian Processes

CHAPTER 4

KRIGING-BASED NAVIGATION

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4.1 Introduction

Assume a navigation scenario similar to that of chapter 2:

$$\begin{cases} x_k = f_k(x_{k-1}, w_k) \\ z_k = h_k(x_k, n_k) \end{cases} \tag{4.1}$$

with the exception that h_k is unknown, and the observation noise is additive for the sake of simplicity. Since the likelihood density cannot be evaluated without an observation model, Bayesian data fusion is no longer applicable. The state x_k must then be estimated from inertial measurements alone. However, inertial navigation is subject to drift and therefore cannot provide a reliable state estimate for longer missions.

In our scenarios, the observation function describes the measurement of a physical field, whose spatial distribution does not change over the course of the mission, by an embedded sensor at each time step. Furthermore, we assume that the environment has been sampled beforehand, whether during a previous reconnaissance mission, or using a known but deteriorated map. This is formally modelled by assumption 4.1.

Assumption 4.1 (Observation model)**.** *The observation function is a time-invariant GP, and its noise is an additive Gaussian white noise of covariance matrix $R_k \succeq 0$:*

$$z_k = h(x_k) + n_k, \qquad n_k \sim \mathcal{N}(0, R_k) \tag{4.2}$$

where $m : \mathbb{R}^{d_x} \to \mathbb{R}^{d_z}$ is the potentially unknown mean of the GP, and $k : \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \to \mathbb{R}^{d_z}$ is its chosen covariance kernel. Here, n_k plays the role of the query noise ε of section 3.3, and R_k is its associated covariance R.

We also have access to $\tilde{N} \in \mathbb{N}^*$ distinct training inputs $\tilde{\mathbf{x}}$ and their noisy observations $\tilde{\mathbf{z}}$, degraded by a noise of covariance $\tilde{R} \succeq 0$, in the sense of section 3.3.

Remark 4.1 (Time-varying observation function)**.** *In full generality, the observed physical field could vary over time, and so the underlying GP would need to include a temporal input. While the kriging theory perfectly works with this additional parameter, it would require the training data to be sampled over time as well as space.*

This chapter is organised as follows:

- Section 4.2 combines chapters 2 and 3 to create a kriging-based framework for autonomous navigation in a poorly known environment.
- Section 4.3 discusses non-stationary observation fields, and the necessity of training data selection for kriging.
- Section 4.4 improves upon this framework by adapting the shape of the subsampling window to that of the particle cloud to better capture the non-stationarities of the field.
- Finally, section 4.5 quantifies the estimation error committed by subsampling the training data used for kriging.

4.2 Kriging the Observation Model

Using assumption 4.1, the likelihood can be expressed as:

$$z_k|x_k, \tilde{\mathbf{z}} \sim \mathcal{N}\left(\hat{z}_{\mathrm{K}}(x_k, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g), R_{\mathrm{K}}(x_k, \tilde{\mathbf{x}}; g)\right) \tag{4.3}$$

where $\hat{z}_{\mathrm{K}}$ denotes any of the three kriging estimators, R_{K} is the corresponding covariance matrix, and m and g are optional arguments depending on the chosen kriging method. In particular, R_{K} already carries the query noise R_k, so it must not be added a second time.

The innovation associated with (4.3) combines three contributions:

$$\begin{aligned} z_k - \hat{z}_{\mathrm{K}}(x_k, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) &= h(x_k) - \hat{z}_{\mathrm{K}}(x_k, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) + n_k \\ &= \zeta_h(x_k) + \zeta_\varepsilon(x_k) + n_k \end{aligned} \tag{4.4}$$

where $\zeta_h(x_k)$ is the error that the kriging estimator would commit from a noise-free training set:

$$\zeta_h : x \in \mathbb{R}^{d_x} \mapsto h(x) - \hat{z}_{\mathrm{K}}\left(x, \tilde{\mathbf{x}}, h(\tilde{\mathbf{x}}); m, g\right) \tag{4.5}$$

and $\zeta_\varepsilon(x_k)$ corresponds to the impact of the training noise:

$$\zeta_\varepsilon : x \in \mathbb{R}^{d_x} \mapsto \hat{z}_{\mathrm{K}}\left(x, \tilde{\mathbf{x}}, h(\tilde{\mathbf{x}}); m, g\right) - \hat{z}_{\mathrm{K}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) \tag{4.6}$$

which is linear in $\tilde{\mathbf{z}} - h(\tilde{\mathbf{x}})$, every kriging estimator being affine in its training observations.

While n_k is drawn at each time step, ζ_h and ζ_ε are functions of the position, drawn once with the training data and the field, and fixed throughout the mission.

Given the trajectory, the observations are jointly Gaussian:

$$\mathbf{z}_{1:k}|\mathbf{x}_{1:k}, \tilde{\mathbf{z}} \sim \mathcal{N}\left(\hat{z}_{\mathrm{K}}(\mathbf{x}_{1:k}, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g), \mathbb{V}\left[\zeta_h(\mathbf{x}_{1:k}) + \zeta_\varepsilon(\mathbf{x}_{1:k})\right] + \mathbb{V}\left[\mathbf{n}_{1:k}\right]\right) \tag{4.7}$$

where $\mathbf{n}_{1:k} \triangleq (n_1, \ldots, n_k)$. Since the observation noises are white, $\mathbb{V}\left[\mathbf{n}_{1:k}\right]$ is block diagonal.

However, the off-diagonal blocks of $\mathbb{V}\left[\zeta_h(\mathbf{x}_{1:k}) + \zeta_\varepsilon(\mathbf{x}_{1:k})\right]$ correspond to the cross-covariances of the map error between the states at different time steps. Since they are not zero in general, the observations remain dependent even when the states are known, and Assumption 2.2 no longer holds.

In a particle filter, evaluating (4.3) separately for each particle introduces no approximation, but Applying it independently at each time step, only keeps the diagonal blocks $R_{\mathrm{K}}(x_1, \tilde{\mathbf{x}}; g), \ldots, R_{\mathrm{K}}(x_k, \tilde{\mathbf{x}}; g)$ of the joint covariance along the path of each particle, which amounts to drawing a new map error at every step.

This approximation is accurate when the distance travelled during a single time step exceeds the correlation length of the map error. Otherwise, the same error is counted at several successive steps, which can introduce a systematic bias in the state estimation and make the filter overconfident.

To account for these correlations, the map error can either be modelled as a Gauss-Markov process along the trajectory and appended to the state [97], or the map reconstruction itself can be updated using observations collected in flight. Both approaches yield conditionally linear-Gaussian models given the trajectory, and can thus be estimated by a Rao-Blackwellised filter, as in Algorithm 2.10.

4.2.1 Choosing the Kriging Method

Section 3.3 leaves the choice between the three kriging methods open. Table 4.1 presents a $\mathbb{R} \to \mathbb{R}$ kriging scenario, from which every kriging estimator attempts to reconstruct the true function. A numerical comparison of the estimators is presented at the end of this section.

True function	$h : x \in \mathbb{R} \mapsto x\ \sin(x) + 10 \in \mathbb{R}$
GP mean	$m : x \mapsto 10$ or $m : x \mapsto 0$
GP covariance	$k : x, x' \in \mathbb{R} \mapsto 5^2 \exp\left(-\frac{\|x - x'\|_2^2}{2 \times 1.2^2}\right)$
Training data	$\tilde{\mathbf{x}} = \{0.5, 1.5\} \cup \{5.5, 6.5, \ldots, 9.5\}$
Training noise	$\tilde{R} = 1$
Query noise	$R = 0$
Trend functions	$g : x \mapsto \begin{bmatrix} 1 & x \end{bmatrix}$ or $g : x \mapsto \begin{bmatrix} 1 & x & \ldots & x^5 \end{bmatrix}$
Prediction domain	$[-3, 13]$

Table 4.1: An example kriging scenario. The SK and UK estimators are given either mean function and trend bases to illustrate their reliance on prior knowledge, as shown in their respective figures.

Five kriging estimators, two simple, one ordinary, and two universal, are evaluated on this scenario, as illustrated by Figures 4.1 to 4.3 respectively. The continuous blue line draws the true function, while the training points are represented by the black crosses. The kriged function is in dashed red, and its associated confidence interval is the pale red envelope. Since h is scalar, the confidence interval is computed as:

$$\left[\hat{z}_{\mathrm{K}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) - 3\sqrt{R_{\mathrm{K}}(x, \tilde{\mathbf{x}}; g)}, \hat{z}_{\mathrm{K}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}; m, g) + 3\sqrt{R_{\mathrm{K}}(x, \tilde{\mathbf{x}}; g)}\right] \tag{4.8}$$

and contains $h(x)$ with probability $\mathrm{erf}(3/\sqrt{2}) \approx 99.73\%$, where erf is the Gauss error function.

While it is the computationally simplest estimator, the SK estimator relies heavily on the prior knowledge of the mean function m, and poorly represents the true function in regions with a lower density of training points when that knowledge is wrong. This is particularly apparent in the gap within the training points distribution, and outside the sampled region.

As motivated by section 3.3, ordinary kriging tackles this issue by directly estimating a constant mean function instead of relying on a given prior. Its behaviour is similar to that of the SK estimator with the correct mean function, at the cost of a slightly increased confidence interval. This is due to the additional uncertainty associated with estimating the mean, and is quantified later in this section.

For universal kriging, we considered polynomial means of degree 1 and 5. The affine basis behaves much like ordinary kriging, which it extends by a single coefficient. The affine UK envelope is barely wider than that of OK near the samples but grows faster outside the interpolation domain.

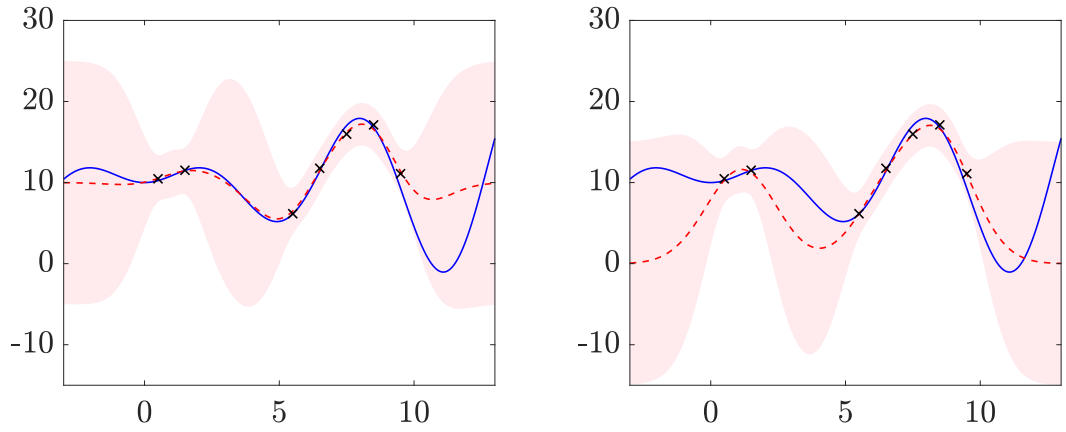


Figure 4.1: Simple kriging under a correct mean function, on the left, and under a wrong one, on the right. Since R_{SK} does not depend on m in (3.35), both shaded areas have the same width along the y-axis.

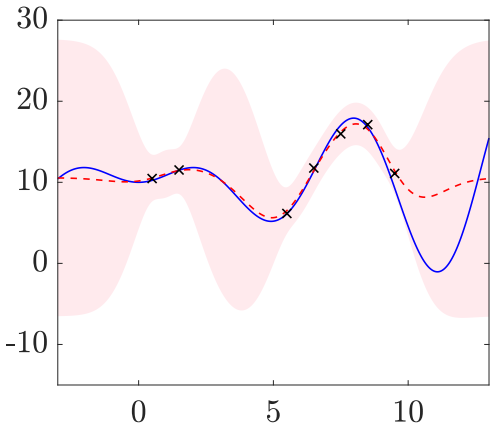


Figure 4.2: Simultaneous kriging of the training data and mean estimation by ordinary kriging. The constant mean is reconstructed here at $\hat{m}_{\mathrm{OK}} = 10.56$.

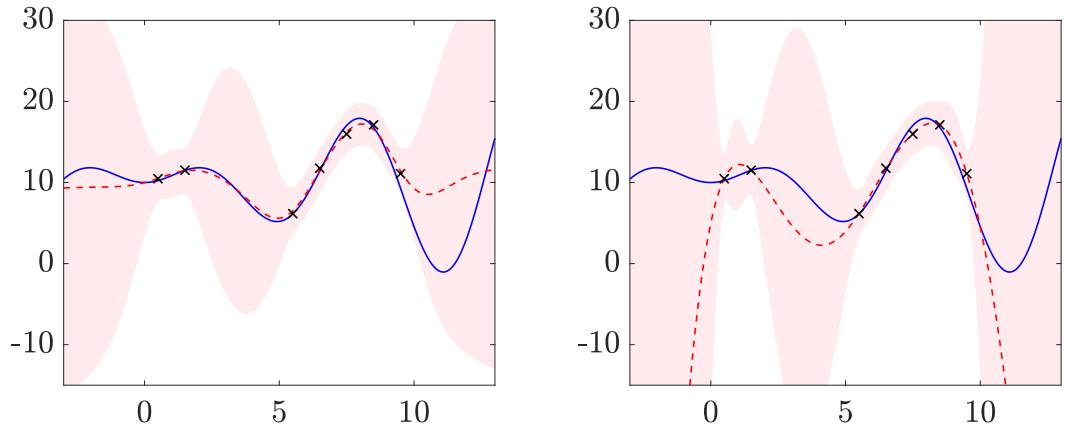


Figure 4.3: Universal kriging over an affine trend basis, on the left, and over a degree five monomial basis, on the right. The trend coefficients are estimated here at $\hat{m}_{\mathrm{UK}} = [9.78, 0.15]^\top$ and $\hat{m}_{\mathrm{UK}} = [4.95, 15.85, -10.82, 2.40, -0.20, 0.01]^\top$ respectively.

On the other hand, the degree 5 model overfits the seven training points with six coefficients, which causes the corresponding estimator to diverge from the true function farther away from the training data. This is reflected in the UK covariance, which indicates high uncertainty in poorly sampled regions.

Since the training observations are noisy, a single draw of $\tilde{\mathbf{z}}$ cannot accurately represent the performance of the estimators above. To alleviate this issue, the training observations are redrawn over $n_{\mathrm{MC}} = 1000$ Monte Carlo simulations, in order to compute the root mean squared error (RMSE):

$$\mathrm{RMSE}_{\hat{z}_{\mathrm{K}}} : x \mapsto \sqrt{\frac{1}{n_{\mathrm{MC}}} \sum_{i=1}^{n_{\mathrm{MC}}} \left\| \hat{z}_{\mathrm{K}}\left(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}^{(i)}; m, g\right) - h(x) \right\|_2^2} \tag{4.9}$$

of the estimators, where $\tilde{\mathbf{z}}^{(i)}$ denotes the i-th draw of the training observations.

Their average RMSE (ARMSE):

$$\mathrm{ARMSE}_{\hat{z}_{\mathrm{K}}}(E) \triangleq \frac{\displaystyle\int_E \mathrm{RMSE}_{\hat{z}_{\mathrm{K}}}(x)\, dx}{\displaystyle\int_E dx} \tag{4.10}$$

is then computed over three subsets:

$$\begin{cases} E_{\mathrm{in}} \triangleq [0.5, 1.5] \cup [5.5, 9.5] \\ E_{\mathrm{gap}} \triangleq [1.5, 5.5] \\ E_{\mathrm{out}} \triangleq [-3, 0.5] \cup [9.5, 13] \end{cases} \tag{4.11}$$

of the prediction domain, as presented in Table 4.2.

Estimator	ARMSE		
	in	gap	out
Simple, correct mean	0.85	1.15	3.23
Simple, wrong mean	0.87	3.69	5.88
Ordinary	0.85	1.26	3.25
Universal, affine	0.85	1.25	3.54
Universal, degree 5	0.98	4.02	70.88

Table 4.2: ARMSE of the five estimators over the three subsets E_{in}, E_{gap}, and E_{out} of the prediction domain.

All five estimators perform similarly in the densely sampled region, their ARMSE values ranging from 0.85 to 0.87, with the exception of the overfitted UK estimator. In this region, the high density of training data limits the influence of the mean on the estimation, and is thus not informative for assessing the individual impact of each estimator.

In the gap region, the choice of mean becomes more influential due to the sparsity of training points. The SK estimator with the correct mean achieves an ARMSE

of 1.15, while using an incorrect one increases the error to 3.69. In contrast, both the OK and affine UK estimators remain close to the correctly informed SK estimator, with an increase of less than 10% in ARMSE. The degree 5 UK estimator performs worst, its ARMSE representing a 250% increase over that of the correct SK estimator.

This tendency suggests that estimating the mean from the available observations generally makes the estimator more robust, provided that the mean model does not overfit the data. It is even more pronounced in the extrapolation region, where the SK estimator with an incorrect mean, and especially the high-order UK estimator, show a substantial increase in ARMSE compared with the three other estimators. However, this robustness comes at a cost, as neither the OK nor the affine UK estimator ever outperforms a correctly informed simple kriging.

Although the previous results highlight the differences in point-estimation accuracy between the five estimators, a key advantage of kriging is its ability to quantify the corresponding uncertainty. Table 4.3 compares the confidence intervals and the contribution of the mean estimation to the overall covariance for the five estimators.

Estimator	Confidence half-width			$R_{\hat{m}}/R_{\mathrm{K}}$		
	in	gap	out	in	gap	out
Simple, correct mean	2.71	9.05	11.58	0.0%	0.0%	0.0%
Simple, wrong mean	2.71	9.05	11.58	0.0%	0.0%	0.0%
Ordinary	2.71	9.55	12.73	0.4%	8.9%	15.0%
Universal, affine	2.72	9.74	16.55	0.8%	11.9%	43.0%
Universal, degree 5	3.22	15.70	320.03	20.5%	59.6%	94.5%

Table 4.3: Average confidence interval half-width and average share of the covariance caused by the estimation of the mean for the five estimators over each region. The share is computed as the pointwise ratio of $R_{\hat{m}}$ over $R_{\mathrm{K}} = R_{\hat{m}} + R_{\mathrm{SK}}$.

This table confirms the trend observed in the ARMSE results, as the average confidence intervals also increase with distance from the training points. As shown by the first row, the kriging covariance grows on its own, since the lack of close samples causes the estimator to rely more on its mean. In such regions, the estimated mean becomes more influential, and thus accounts for an increasing share of the total uncertainty.

Furthermore, it must be noted that both SK confidence intervals are the same, showing that simple kriging is unable to account for an incorrectly specified prior mean. Both present a half-width of 9.05 in the gap, well inside the sampled range, although their ARMSE differ by a factor of three. This issue is particularly relevant in autonomous navigation scenarios, as a poorly informed SK estimator will be overconfident, resulting in skewed likelihood peaks. This, in turn, may lead to relevant particles being discarded while incorrect ones are retained, and so simple kriging is set aside.

Conversely, the degree 5 UK estimator fails by trying to fit six trend coefficients to only seven training points. Outside the sampled range, the mean becomes highly uncertain, and accounts for an average of 94.5% of the total covariance. Unlike simple kriging, the UK estimator propagates this uncertainty to its confidence interval,

whose half-width reaches 320.03. From a computational point of view, universal kriging requires $\tilde{N} \geq r_{\mathrm{UK}}$, where we recall that $r_{\mathrm{UK}} \in \mathbb{N}^*$ is the number of trend functions used for kriging. This causes the matrix $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})$ to become ill-conditioned when the training set is too small relative to the number of trend functions.

Furthermore, every estimator requires the $\mathcal{O}(\tilde{N}^3)$ inversion of $G(\tilde{\mathbf{x}})$ once, after which, solving any related system costs $\mathcal{O}(\tilde{N}^2)$. Simple kriging thus requires an additional $\mathcal{O}(\tilde{N}^2)$ at each query point. Estimating the mean amounts to solving $r_{\mathrm{UK}} d_z$ equations, which costs $\mathcal{O}(r_{\mathrm{UK}} \tilde{N}^2)$ once and $\mathcal{O}(r_{\mathrm{UK}} \tilde{N})$ per query. The inversion of $g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}) \in \mathcal{S}^{++}_{r_{\mathrm{UK}} d_z}(\mathbb{R})$ does not depend on $\tilde{N}$, and is negligible as soon as $r_{\mathrm{UK}} d_z \ll \tilde{N}$. Table 4.4 gives the empirical cost of each estimator in floating-point operations (flop), for $\tilde{N} \in \{7, 100, 1600\}$.

Estimator	Setup (flops)			Per query (flops)		
	$\tilde{N} = 7$	100	1600	$\tilde{N} = 7$	100	1600
Simple	114	333 k	1.37 G	49	10 k	2.56 M
Ordinary	+112	+20.2 k	+5.12 M	+14	+200	+3.2 k
Universal, affine	+255	+40.8 k	+10.3 M	+28	+400	+6.4 k
Universal, degree 5	+1.16 k	+127 k	+30.8 M	+84	+1.2 k	+19.2 k

Table 4.4: Computational cost of the five estimators, split into the setup and the query costs. The simple kriging row is the cost of reconstructing the field, paid by every estimator, and the following ones are what estimating the mean adds to it. Since their cost is independent of the mean they were given, the two simple estimators share the same row.

From 7 to 1600 training points, the setup cost of simple kriging rises from 114 flops to 1.37 Gflop, and its per-query cost from 49 to 2.56 Mflop. In our scenarios, every particle is a query point at each time step, so real-time constraints rule out a training set of that size. The other end of the table is no more comfortable: on seven points, the affine basis costs twice the reconstruction of the field, 255 flops against 114, and the degree 5 basis ten times.

Ordinary kriging is thus retained for the remainder of this chapter. Estimating a constant mean from the observations is robust to misinformation, its ARMSE staying within 10% of a correctly informed simple kriging everywhere, when a wrong mean triples the error in the gap. It also reports that estimation, the term $R_{\hat{m}_{\mathrm{OK}}}$ of remark 3.8 rising from 0.4% of the covariance near the samples to 8.9% in the gap, so its confidence interval widens exactly where its accuracy degrades. Finally, the additional $d_z \times d_z$ inversion it requires is the smallest trend a kriging estimator can carry, and remains affordable at both ends of table 4.4.

4.2.2 Hyperparameter Estimation

Recall that section 3.2.2 considered every kernel subsequently used for kriging to be fully known, with both their form and hyperparameters specified. This assumption is relaxed here by considering only the form to be known, while the hyperparameters θ must be estimated from the available data.

Unlike the mean, which is estimated in closed form by ordinary kriging, the hyperparameters do not admit an analytical estimator in general. Indeed, while the mean appears linearly in the kriging equations, θ only appears through the inverse of the Gram matrix. The standard approach [98] is to select the hyperparameters maximising the likelihood of the training observations, a procedure known as maximum likelihood estimation.

Formally, let $d_\theta \in \mathbb{N}^*$ be the number of hyperparameters, let k_θ denote the kernel of hyperparameters θ, and let $\Theta \subseteq \mathbb{R}^{d_\theta}$ be the set of hyperparameters for which k_θ is SPD and $\theta \mapsto k_\theta$ is differentiable. The Gram matrix associated with k_θ is written as:

$$G_\theta(\tilde{\mathbf{x}}) \triangleq k_\theta(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R} \tag{4.12}$$

Since ordinary kriging is retained, the mean of the GP is an unknown constant $m \in \mathbb{R}^{d_z}$, which must be estimated alongside θ. By Proposition 3.1, and since the training noises are Gaussian and independent of the field, the training observations follow:

$$\tilde{\mathbf{z}} \sim \mathcal{N}\left(C\, m, G_\theta(\tilde{\mathbf{x}})\right) \tag{4.13}$$

where C is defined by (3.49). Let:

$$\begin{aligned} L(m, \theta) \triangleq &-\frac{1}{2}\left(\tilde{\mathbf{z}} - C\, m\right)^\top G_\theta(\tilde{\mathbf{x}})^{-1}\left(\tilde{\mathbf{z}} - C\, m\right) \\ &-\frac{1}{2}\log\det\left(G_\theta(\tilde{\mathbf{x}})\right) - \frac{\tilde{N} d_z}{2}\log(2\pi) \end{aligned} \tag{4.14}$$

be the corresponding log-likelihood.

Remark 4.2 (Model complexity)**.** *The first two terms of (4.14) have opposite effects. The first one measures the fit to the training observations, and increases as the field is allowed to vary more freely, while the second one decreases. Maximising the log-likelihood therefore retains the least flexible field that remains consistent with the training data [99].*

Since the logarithm is increasing, the maximum likelihood estimators of m and θ are:

$$\left(\hat{m}_{\mathrm{ML}}, \hat{\theta}_{\mathrm{ML}}\right) \triangleq \underset{m \in \mathbb{R}^{d_z},\, \theta \in \Theta}{\operatorname{argmax}} \left(L(m, \theta)\right) \tag{4.15}$$

Proposition 4.1 gives the derivatives of L, and shows that the mean can be eliminated from (4.15).

Proposition 4.1 (Maximum likelihood hyperparameters)**.** *Let $m \in \mathbb{R}^{d_z}$ and $\theta \in \Theta$. The derivatives of the log-likelihood are:*

$$\begin{cases} \dfrac{\partial L}{\partial m}(m, \theta) = C^\top a \\ \dfrac{\partial L}{\partial \theta_i}(m, \theta) = \dfrac{1}{2}\operatorname{tr}\left(\left(a\, a^\top - G_\theta(\tilde{\mathbf{x}})^{-1}\right)\dfrac{\partial G_\theta(\tilde{\mathbf{x}})}{\partial \theta_i}\right), \quad i \in [\![1, d_\theta]\!] \end{cases} \tag{4.16}$$

where:

$$a \triangleq G_\theta(\tilde{\mathbf{x}})^{-1}\left(\tilde{\mathbf{z}} - C\, m\right) \tag{4.17}$$

and θ_i denotes the i-th component of θ.

Furthermore, for a fixed θ, the log-likelihood is maximised by the OK mean estimator of Proposition 3.8, computed with the kernel k_θ:

$$\hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}; \theta) = \left(C^\top G_\theta(\tilde{\mathbf{x}})^{-1} C\right)^{-1} C^\top G_\theta(\tilde{\mathbf{x}})^{-1} \tilde{\mathbf{z}} \tag{4.18}$$

Proof. Writing G_θ for $G_\theta(\tilde{\mathbf{x}})$, differentiating (4.14) with respect to m yields:

$$\frac{\partial L}{\partial m}(m,\theta) = C^\top G_\theta^{-1}\left(\tilde{\mathbf{z}} - C\,m\right) = C^\top a \tag{4.19}$$

Let $i \in [\![1, d_\theta]\!]$. The derivatives of the inverse and of the determinant of a matrix [100] give:

$$\begin{cases} \dfrac{\partial G_\theta^{-1}}{\partial \theta_i} = -G_\theta^{-1}\,\dfrac{\partial G_\theta}{\partial \theta_i}\,G_\theta^{-1} \\ \dfrac{\partial \log\det(G_\theta)}{\partial \theta_i} = \operatorname{tr}\left(G_\theta^{-1}\,\dfrac{\partial G_\theta}{\partial \theta_i}\right) \end{cases} \tag{4.20}$$

and thus, by the cyclic property of the trace:

$$\begin{aligned} \frac{\partial L}{\partial \theta_i}(m,\theta) &= \frac{1}{2}\,a^\top \frac{\partial G_\theta}{\partial \theta_i}\,a - \frac{1}{2}\operatorname{tr}\left(G_\theta^{-1}\,\frac{\partial G_\theta}{\partial \theta_i}\right) \\ &= \frac{1}{2}\operatorname{tr}\left(\left(a\,a^\top - G_\theta^{-1}\right)\frac{\partial G_\theta}{\partial \theta_i}\right) \end{aligned} \tag{4.21}$$

Finally, since $G_\theta \succ 0$ and C has full column rank, the function $L(\cdot,\theta)$ is strictly concave. It is therefore maximised where its derivative vanishes:

$$\begin{aligned} & \frac{\partial L}{\partial m}(m,\theta) = 0 \\ \iff\quad & C^\top G_\theta^{-1}\tilde{\mathbf{z}} - C^\top G_\theta^{-1} C\,m = 0 \\ \iff\quad & m = \left(C^\top G_\theta^{-1} C\right)^{-1} C^\top G_\theta^{-1}\tilde{\mathbf{z}} \end{aligned} \tag{4.22}$$

which is exactly (4.18). □

According to Proposition 4.1, the maximum likelihood estimator of the mean is the OK mean estimator computed with $\hat{\theta}_{\mathrm{ML}}$:

$$\hat{m}_{\mathrm{ML}} = \hat{m}_{\mathrm{OK}}\left(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}; \hat{\theta}_{\mathrm{ML}}\right) \tag{4.23}$$

and (4.15) reduces to a maximisation over θ alone. However, the derivatives of L with respect to the hyperparameters cannot be cancelled analytically in general, and so $\hat{\theta}_{\mathrm{ML}}$ is computed by a gradient ascent.

To ensure that the hyperparameters remain strictly positive, as the Matérn kernels of definition 3.4 require, this ascent is performed on their logarithm, with a step size $\tau > 0$. Since the derivative of L with respect to $\log\theta_i$ is θ_i times its derivative with respect to θ_i, each hyperparameter is multiplied by a positive factor at each iteration. One iteration of the resulting algorithm is presented by Algorithm 4.1.

Algorithm 4.1 Maximum Likelihood Estimation

Compute $\hat{m} = \hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}; \theta)$ with (4.18) ▷ Mean estimation

$\forall i \in [\![1, d_\theta]\!]$, compute $\dfrac{\partial L}{\partial \theta_i}(\hat{m}, \theta)$ with (4.16) ▷ Log-likelihood gradient

$\forall i \in [\![1, d_\theta]\!]$, $\theta_i \leftarrow \theta_i \exp\left(\tau\,\theta_i\,\dfrac{\partial L}{\partial \theta_i}(\hat{m}, \theta)\right)$ ▷ Gradient ascent

This iteration is repeated from an initial value θ_{init} until the norm of the gradient falls below a given threshold, or a maximum number of iterations is reached. The resulting estimate is written as:

$$\hat{\theta} = \text{MLE}\left(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, \theta_{\text{init}}\right) \tag{4.24}$$

In practice, the gradient ascent can be replaced with any other gradient-based method, such as a quasi-Newton method [101], in order to reduce the number of iterations.

Remark 4.3 (Initialisation)**.** *The log-likelihood is not concave in θ in general, and may admit several local maxima [99]. The estimate returned by Algorithm 4.1 therefore depends on θ_{init}. In the absence of a previous estimate, the initial value can be chosen from the empirical statistics of the training observations, such as their variance.*

Remark 4.4 (Uncertainty of the hyperparameters)**.** *The kriging estimators of section 3.3 are then computed with $\hat{\theta}$ in place of the true hyperparameters. While the uncertainty associated with the estimation of the mean is captured by $R_{\hat{m}_{\text{OK}}}$ (see Remark 3.8), that of the hyperparameters is not included in the OK covariance, which thus tends to underestimate the covariance of the estimation error [102]. However, this effect decreases as the number of training points increases [103], and is therefore mostly noticeable when few training points are available.*

Remark 4.5 (Cost of the hyperparameter estimation)**.** *Each iteration of Algorithm 4.1 requires the inversion of $G_\theta(\tilde{\mathbf{x}})$, which costs $\mathcal{O}\left(\tilde{N}^3\right)$, followed by $\mathcal{O}\left(\tilde{N}^2\right)$ operations for each component of the gradient. This inversion is the one required by kriging itself, and corresponds to the setup cost of Table 4.4. However, it is repeated at each iteration, whereas kriging with fixed hyperparameters only requires it once.*

4.2.3 Simulation Scenario

Every navigation result of this chapter is produced by the same scenario, as described in this section. The observation field represented by h is a magnetic anomaly map of a 200 km×200 km region above Mexico, and extracted from [104]. It is sampled on a regular grid of 201×201 points with a resolution of 1 km×1 km, and is read by a bilinear interpolation between the points.

From this map are extracted $\tilde{N}$ training samples $(\tilde{\mathbf{x}}, \tilde{\mathbf{z}})$, where $\tilde{\mathbf{x}}$ is sampled from a regular grid, each node being independently jittered by up to 30% of the grid resolution along both axes. The grid resolution Δ_g is varied from 2.5 km to 7.5 km, with increments of 1 km. This corresponds to $\tilde{N} \in \{79^2, 56^2, 44^2, 36^2, 30^2, 26^2\}$. The corresponding observations $\tilde{\mathbf{z}}$ are corrupted by a Gaussian white noise of variance $\tilde{R} = 10^2$ $(\text{nT})^2$. The irregular grid models navigation errors during the prior survey mission, while the white noise represents the imperfect measurements of a real sensor.

The vehicle flies over the map at a constant speed of 500 m/s along a mower pattern consisting of three parallel branches, each 80 km long, and connected by half-turns with a radius of 25.5 km. The trajectory is sampled every $\Delta_t = 5$ s, and lasts $n_{\text{steps}} = 150$ steps. A single time step thus covers 2.50 km.

The vehicle state consists of its horizontal position and velocity:

$$x_k = \begin{bmatrix} p_k^{(x)} & p_k^{(y)} & v_k^{(x)} & v_k^{(y)} \end{bmatrix}^\top, \qquad d_x = 4 \tag{4.25}$$

and its propagation is degraded with a Gaussian white noise of covariance matrix:

$$Q_k = \begin{bmatrix} 100^2 & 0 & 0 & 0 \\ 0 & 100^2 & 0 & 0 \\ 0 & 0 & 1^2 & 0 \\ 0 & 0 & 0 & 1^2 \end{bmatrix} \tag{4.26}$$

whose position and velocity entries are respectively expressed in m^2 and $\mathrm{m}^2\mathrm{s}^{-2}$.

The observation function is the scalar ($d_z = 1$) ordinary kriging of the magnetic anomaly field, conditioned on the $\tilde{N}$ training points, and degraded by a Gaussian white noise of covariance $R_k = 5^2\ (\mathrm{nT})^2$. The chosen kernel is a Matérn kernel with smoothness $\nu = 3/2$, which corresponds to the smallest half-integer for which the field and the kriging mean are differentiable, and its hyperparameters are estimated as described in section 4.2.2. A bilinear observation function on the same samples is also used as comparison.

The filter is an RPF, initialised on a cloud of $N = 5000$ particles, drawn around a point itself drawn from $\mathcal{N}(x_0, P_0)$, with:

$$P_0 = \begin{bmatrix} 8000^2 & 0 & 0 & 0 \\ 0 & 8000^2 & 0 & 0 \\ 0 & 0 & 2^2 & 0 \\ 0 & 0 & 0 & 2^2 \end{bmatrix} \tag{4.27}$$

in the same units, to prevent the particle cloud from being centred on the true initial state. The resampling threshold is fixed to $N_{\mathrm{th}} = 0.5\,N$, and the regularisation coefficient is $\alpha_{\mathrm{reg}} = 0.3$. This scenario is illustrated by Figure 4.4 and summarised by Table 4.5.

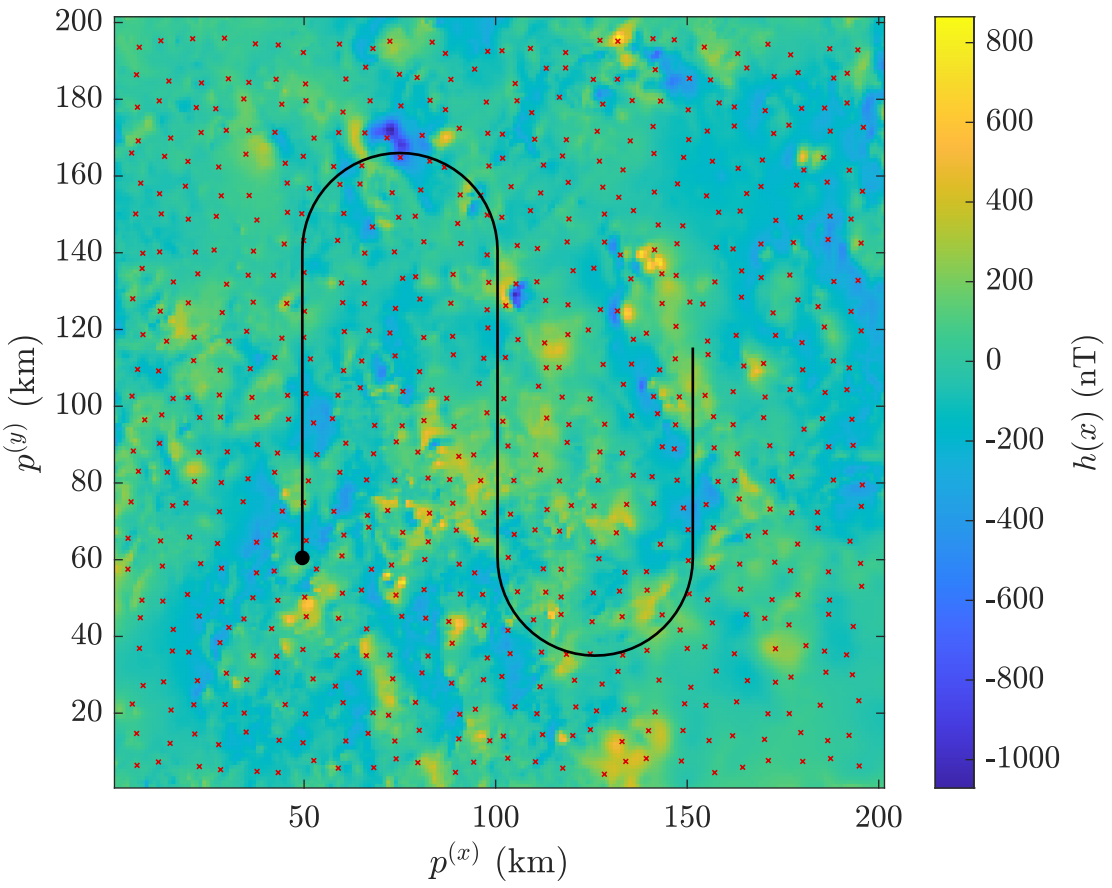


Figure 4.4: The magnetic field anomaly used throughout this chapter. The black line represents the overflown trajectory, and the red crosses correspond to the training points on a 7.5 km-resolution grid. The black dot signals the beginning of the trajectory.

Each configuration is run over $n_{\mathrm{MC}} = 1000$ Monte Carlo realisations, redrawing the initial state and white noises. The training data are also redrawn every 10 runs so that the metrics are averaged over the reconstruction as well as the trajectory. The chosen estimator is the MMSE estimator from (2.83). The performance of each compared approach is read through two quantities:

- The estimation RMSE, defined as:

$$\mathrm{RMSE}_{\hat{x}} : k \mapsto \sqrt{\frac{1}{n_{\mathrm{MC}}} \sum_{i=1}^{n_{\mathrm{MC}}} \left\| \Pi_{\mathrm{pos}}\, \hat{x}_k^{(i)} - \Pi_{\mathrm{pos}}\, x_k \right\|_2^2} \tag{4.28}$$

  where $\hat{x}_k^{(i)}$ is the state estimated by the i-th filter at time k, x_k is the corresponding true state, and Π_{pos} is the projection matrix onto the position subspace:

$$\Pi_{\mathrm{pos}} \triangleq \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{4.29}$$

  This metric gives an absolute value for the average error made by the filter.

Map	Surface	200 km×200 km
	Resolution	1 km×1 km
Kriging	Samples	Jittered grid
	Jitter	$0.3\,\Delta_g$
	Δ_g	$\in \{2.5, 3.5, \ldots, 7.5\}$ km
	$\tilde{N}$	$\in \{79^2, 56^2, 44^2, 36^2, 30^2, 26^2\}$
	$\tilde{R}$	10^2 (nT)^2
Trajectory	Form	3-branch mower
	Branch length	80 km
	Turn radius	25.5 km
	Speed	500 m/s
	Δ_t	5 s
	n_{steps}	150
State	d_x	4
	x_k	$\left[p_k^{(x)}, p_k^{(y)}, v_k^{(x)}, v_k^{(y)}\right]^\top$
	Q_k	100^2 m^2, 1^2 m^{2}s^{-2}
Observation	d_z	1
	Function	OK or bilinear
	Kriging kernel	Matérn, $\nu = 3/2$
	R_k	5^2 (nT)^2
Filter	Filter	RPF
	N	5000
	N_{th}	$0.5\,N$
	α_{reg}	0.3
	P_0	8000^2 m^2, 2^2 m^{2}s^{-2}

Table 4.5: Simulation parameters for the navigation scenario. For improved readability, covariance matrices are represented by their diagonal coefficients only.

- The normalised estimation error squared (NEES):

$$\text{NEES}_{\hat{x}} : k \mapsto \frac{1}{n_{\text{MC}}\, d_x} \sum_{i=1}^{n_{\text{MC}}} \left\| \hat{x}_k^{(i)} - x_k \right\|_{\hat{P}_k^{(i)}}^2 \tag{4.30}$$

where $\| \cdot \|_{\hat{P}_k^{(i)}}$ is the Mahalanobis norm associated with the error covariance:

$$\left\| \hat{x}_k^{(i)} - x_k \right\|_{\hat{P}_k^{(i)}}^2 \triangleq \left(\hat{x}_k^{(i)} - x_k \right)^\top \left(\hat{P}_k^{(i)} \right)^{-1} \left(\hat{x}_k^{(i)} - x_k \right) \tag{4.31}$$

This metric detects overconfident filters by using the Mahalanobis norm associated with the error covariance matrix rather than the standard Euclidean distance.

Similarly, a run is declared convergent under two different criteria, as each describes a different behaviour:

- A fixed threshold of 2000 m, under which the median of the error over the last ten steps of a filter must be. This metric is simple to read and ensures that the estimation is relatively close to the true state. However, the threshold value is arbitrary.
- A Mahalanobis criterion, where the median of (4.31) over the last ten steps must remain under the 99% quantile of a χ^2 law with d_x degrees of freedom, written $\chi^2_{d_x}(99\%)$. While this criterion follows the property of a Gaussian error, it primarily assesses the consistency between the estimation error and the uncertainty reported by the filter. Consequently, a filter with a large estimation error may still satisfy this criterion if it reports a sufficiently large uncertainty.

4.2.4 Numerical Results

In this section, we compare the performance of the filter using 13 different observation functions:

- The first configuration corresponds to the known map, and serves as a reference case.
- The six next use ordinary kriging at different grid resolutions. This method is called static kriging for the purpose of distinguishing it from the adaptive methods presented in the subsequent sections.
- The final six instead use bilinear interpolation on the same grid.

Before analysing the filters themselves, it is important to assess the quality of the reconstructed maps. Figure 4.5 presents the reconstruction ARMSE along the true trajectory for ordinary kriging and bilinear interpolation across the different grid resolutions, in the sense of (4.10).

As one could have expected, denser grids lead to better reconstructions, the ARMSE evolving as the logarithm of Δ_g. However, kriging and bilinear interpolation exhibit similar errors, within 1 nT of each other at every resolution but the finest. Thus, the advantage of kriging cannot be primarily attributed to the reconstruction accuracy alone.

Focusing now on the filter performance, Figures 4.6 and 4.7 respectively compare the RMSE and NEES of the filter using the true map, kriging and bilinear interpolation on each of the six grid resolutions. For the sake of readability, both metrics are only computed on the converged runs with respect to the threshold criterion.

Three main results can be drawn from the RMSE curves. First, filters with access to more training points from a denser grid achieve better performance, with the limit case of the filter using the true map outperforming every other configuration. Second, kriging-based filters consistently achieve lower RMSEs than their bilinear counterparts despite having similar map reconstruction accuracy. This result, combined with Figure 4.5, indicates that the advantage of kriging lies in its ability to quantify reconstruction uncertainty through its covariance matrix. Finally, the time

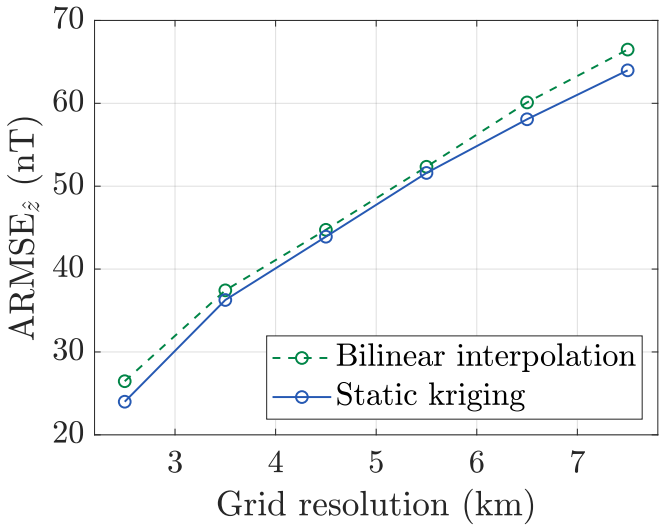


Figure 4.5: Reconstruction ARMSE of the field over the trajectory corridor for ordinary kriging and bilinear interpolation as a function of the training grid resolution. The domain E considered is a rectangular corridor around the true trajectory, with a 28 km margin. It covers 74% of the map.

required to reach a stable estimation error increases from 13 to 25 steps as the grid resolution decreases from 2.5 to 7.5 km. In particular, the vehicle begins its first turn at $k = 33$, so any potential information gained from this turn does not affect this result.

The NEES curves further highlight the superiority of kriging methods by taking the estimated error covariance matrix into account. Kriging-based filters stay within the consistency band at every resolution, whereas bilinear interpolation lies above it even on the densest map. This shows that kriging not only leads to more accurate state estimates than bilinear interpolation, but also provides a better characterisation of their associated uncertainty.

In terms of convergence rates, Figure 4.8 starts by extracting the relevant median errors from the previous curves, and Figure 4.9 computes the associated convergence rates using the two metrics introduced at the beginning of this section.

The first two figures show that the bilinear to kriging RMSE ratio ranges from 1.4 to 1.9, and increases to 9 to 56 for the NEES. Furthermore, while the RMSE ratio tends to decrease at lower grid resolutions, the NEES ratio follows the opposite trend. Bilinear interpolation thus completely fails to measure its accuracy at lower resolutions, while kriging degrades in step with its own accuracy.

The convergence rates follow the same pattern. Under the threshold criterion, the rate for kriging decreases from 100% to 60% as the grid resolution decreases, while that of bilinear interpolation decreases from 99 to 42%. The two thus remain comparable. For the Mahalanobis criterion, however, kriging only falls to 81% while bilinear interpolation collapses to 1%, having lost every run whose reported uncertainty is consistent with its error.

However, the advantages of kriging come at an increased computational and memory costs, as described by Table 4.6.

While the setup cost can be paid before the mission, the vehicle must still be able to store between 3.67 and 312 MB of data, and sustain an average computational

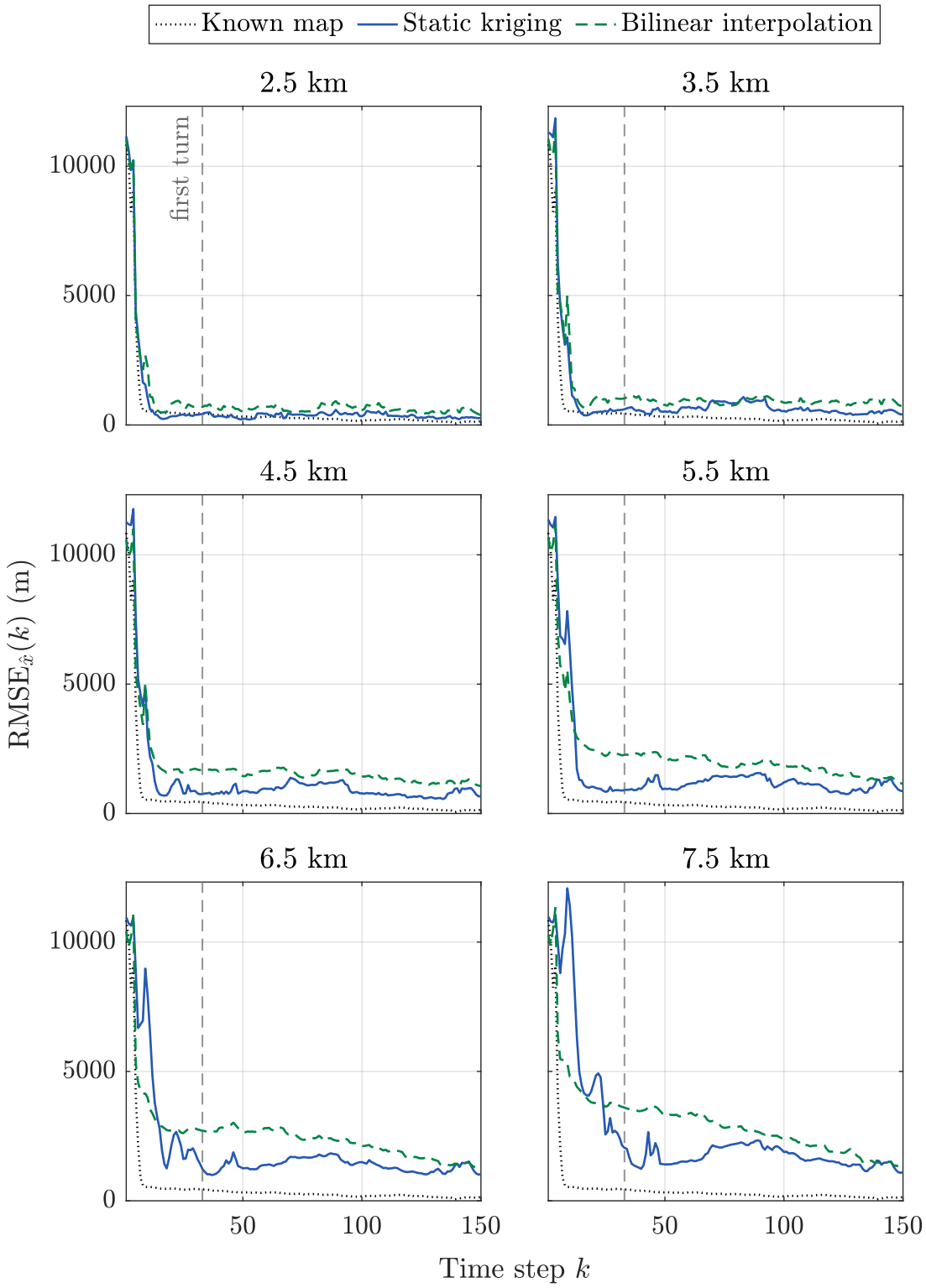


Figure 4.6: Estimation RMSE of the navigation algorithms using the known map, bilinear interpolation and static kriging. Each panel corresponds to a different grid resolution, and the true map result is repeated for comparison.

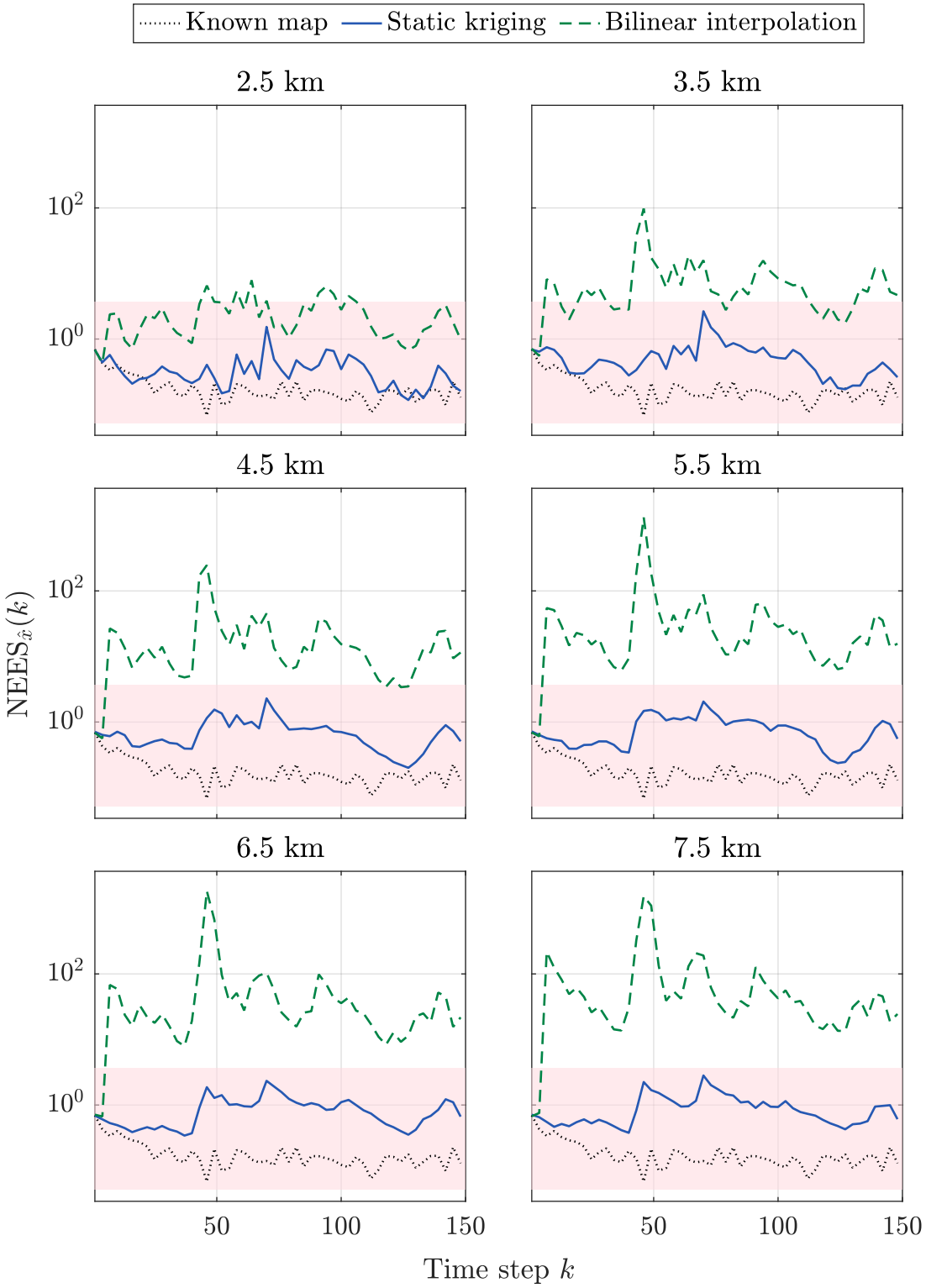


Figure 4.7: Estimation NEES of the navigation algorithms using the known map, bilinear interpolation and static kriging. The shaded area delimits the quantiles of a χ^2 law with d_x degrees of freedom, within which the NEES of a consistent filter falls with probability 0.99.

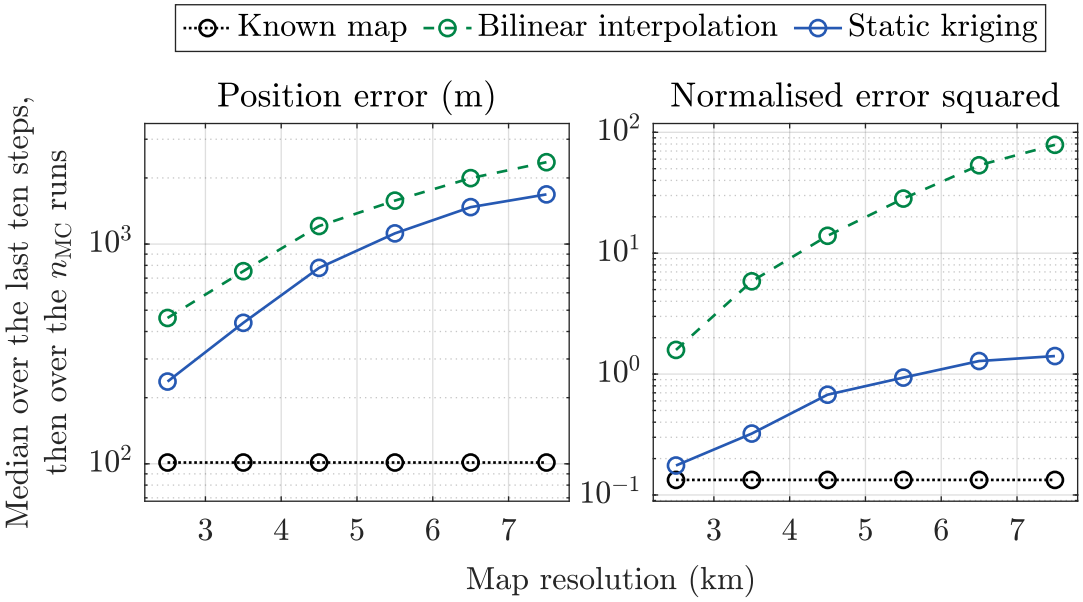


Figure 4.8: Average median RMSE and NEES at the end of the flight using the known map, bilinear interpolation and static kriging as a function of grid resolution.

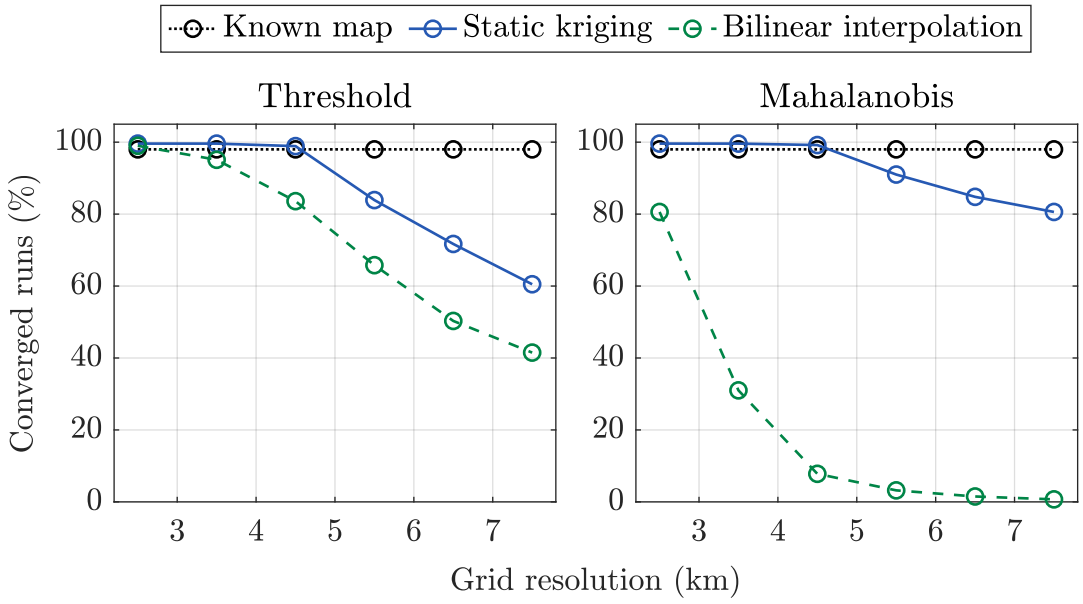


Figure 4.9: Percentage of converged runs for the known map, bilinear interpolation and static kriging against the grid resolution for the two metrics. The true map does not depend on the resolution, and has a convergence rate of 98% under both criteria.

rate of 0.46 to 38.9 Gflops/s over the navigation period. These requirements remain affordable at low grid resolutions, but these configurations result in poor navigation performance, as shown above. Increasing the grid resolution improves navigation accuracy, but quickly leads to prohibitive storage and computational requirements.

Observation model	Cost (flops)			Memory (bytes)
	Setup	Queries	Total	
Known map	0	6 M	6 M	323 k
Bilinear interpolation	0	6 M	6 M	150 k
Static kriging, 2.5 km	81 G	29.2 T	29.3 T	312 M
Static kriging, 3.5 km	10.3 G	7.38 T	7.39 T	78.8 M
Static kriging, 4.5 km	2.42 G	2.81 T	2.81 T	30 M
Static kriging, 5.5 km	726 M	1.26 T	1.26 T	13.5 M
Static kriging, 6.5 km	243 M	608 G	608 G	6.5 M
Static kriging, 7.5 km	103 M	343 G	343 G	3.67 M

Table 4.6: Memory and computational costs of the navigation algorithm using the known map, bilinear interpolation, or static kriging. Since the cost of kriging depends on the number of training data, one row is assigned per grid resolution.

4.3 Adaptive Kriging

In this section, we propose to use training data subsampling, *i.e.* selecting a subset $\tilde{\mathbf{x}}^r$ of $\tilde{N}^r$ training points and using only these points for regression, as a way to alleviate the computational issues raised above. Since the computational and memory costs of kriging scale quadratically with the number of training points, discarding half the data reduces the computational load by a factor of 4 and requires 4 times less storage.

However, this method comes at a cost. In the following, we describe two subsampling strategies, each requiring to perform a subsampling step at each time step. This means that no precomputation can be done before the mission, and in particular that inverting $G(\tilde{\mathbf{x}}^r) \in \mathcal{S}^{++}_{\tilde{N}^r\, d_z}(\mathbb{R})$ and $C^\top G(\tilde{\mathbf{x}}^r)^{-1} C \in \mathcal{S}^{++}_{d_z}(\mathbb{R})$ must be done online.

4.3.1 Random Subsampling

The simplest idea to reduce the number of training points is to let $p \in [0, 1]$, and draw independently $s_{i,k}$ from a Bernoulli distribution of success probability p, for each $i \in [\![1, \tilde{N}]\!]$. Any training sample $\tilde{x}_i$ for which the corresponding $s_{i,k}$ is a success is retained to form $\tilde{\mathbf{x}}^r_{\mathrm{RS},k}$, while the others are discarded. This operation is performed at the beginning of each time step to avoid skewing the reconstruction. Since the training points are randomly selected across the map rather than localised to a specific region of space, the hyperparameters are estimated before the mission on the entire training set, similarly to the static kriging version.

By linearity of the expectation, the number of retained data is, on average:

$$\mathbb{E}\left[\tilde{N}^r_{\mathrm{RS},k}\right] = p\, \tilde{N} \tag{4.32}$$

and corresponds, on average, to the density of a regular grid of resolution $\Delta_g/\sqrt{p}$.

Similarly to the case of bilinear interpolation and static kriging presented above,

the simulations described in section 4.2.3 were run using random subsampling with:

$$p = 0.5 \tag{4.33}$$

The results are compared here with those obtained with static kriging. Figures 4.10 and 4.11 respectively compare the RMSE and NEES of the two approaches on each of the six grid resolutions, while Figure 4.12 presents the corresponding convergence rates. Finally, Table 4.7 gives the comparative computational cost of each method across the six resolutions.

For the RMSE curves, random subsampling is systematically outperformed by static kriging, especially before the first turn. From 4.5 km onward, random subsampling only reaches its error plateau during this turn, whereas static kriging stabilises before it. This indicates that, although the subsampling (RS) is randomised at each time step, and each training point is therefore available to the filter on average 50% of the time, the resulting map is too degraded to provide an accurate navigation solution. The late convergence of the filters with access to the fewest training points is likely a consequence of the additional information provided by the first turn.

Nevertheless, the similarity between random subsampling at $\Delta_g = 2.5$ and 5.5 km and static kriging at $\Delta_g = 3.5 \approx 2.5/\sqrt{0.5}$ and $7.5 \approx 5.5/\sqrt{0.5}$ km respectively, is noteworthy. This suggests that random subsampling effectively behaves as a map with a degraded resolution. Although the full training set is likely to have been sampled at least once over the course of the mission, only half of the points are available to the filter at each time step.

Regarding the NEES, the curves are consistently similar between both methods, further highlighting the ability of kriging to accurately quantify its uncertainty. As already established in section 4.2.1, the kriging covariance increases in regions where less data are available, and thus the predicted uncertainty naturally follows the degradation of the map. Despite this, random subsampling achieves lower convergence rates at every grid resolution, showing that this method is unable to match the navigation performance of static kriging.

Finally, Table 4.7 illustrates the only advantage of random subsampling over static kriging. By reducing the number of available training points by half on average, this method reduces the computational cost by approximately 75% at every grid resolution.

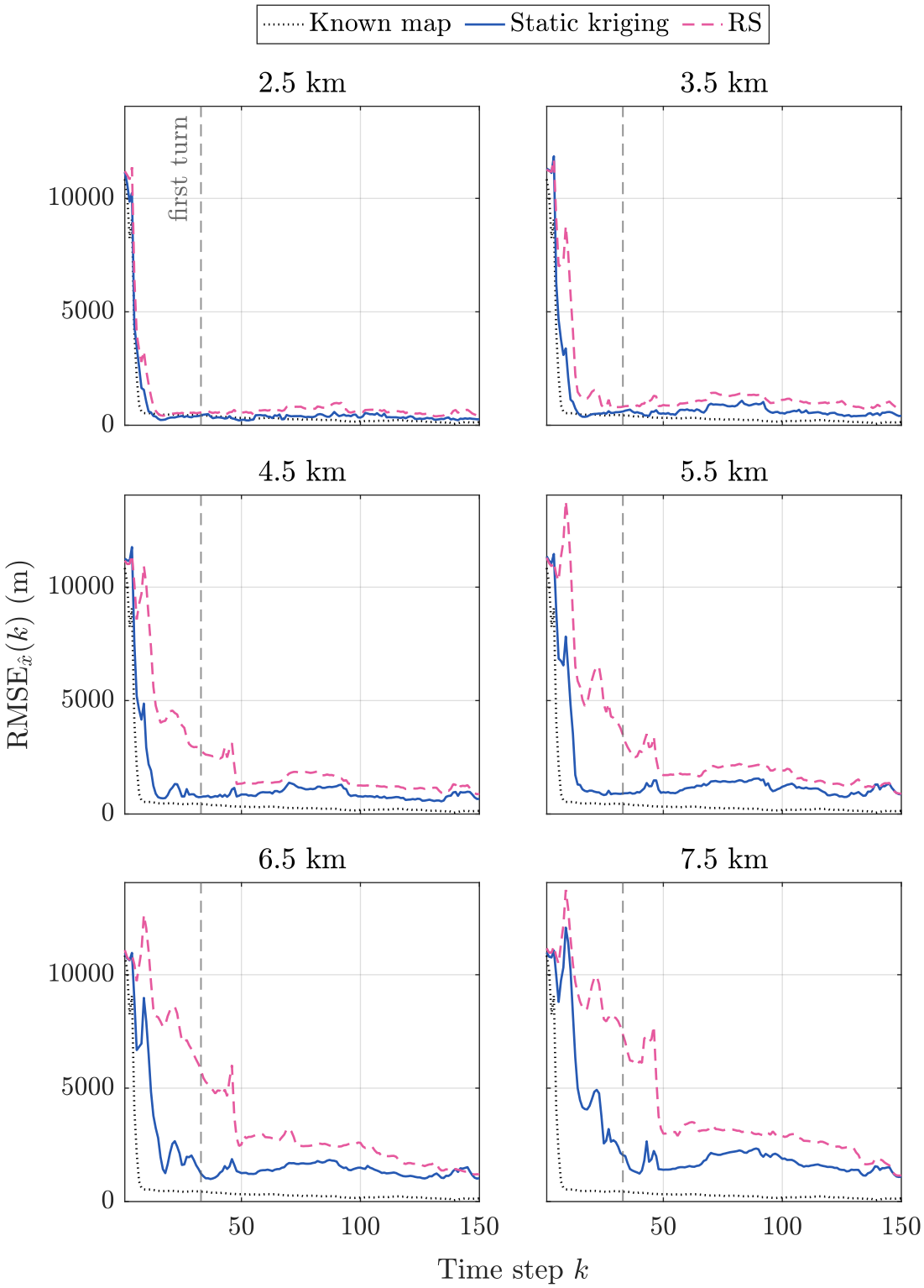


Figure 4.10: Comparative RMSE of the navigation algorithms using static kriging or random subsampling across the six grid resolutions. The performance of the filter with access to the true map is given as a reference.

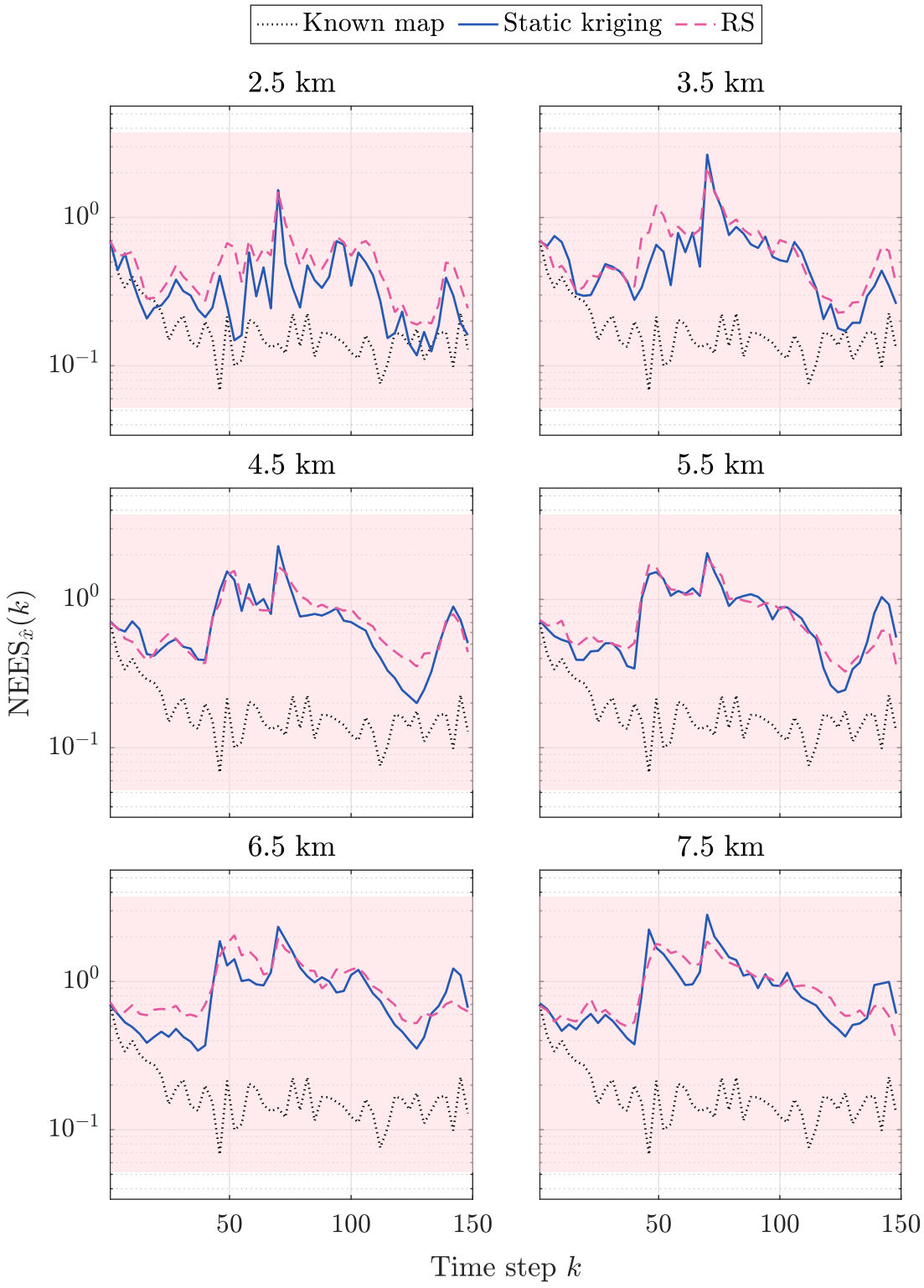


Figure 4.11: Comparative NEES of the navigation algorithms using static kriging or random subsampling across the six grid resolutions. The shaded area corresponds to that of the previous NEES figure.

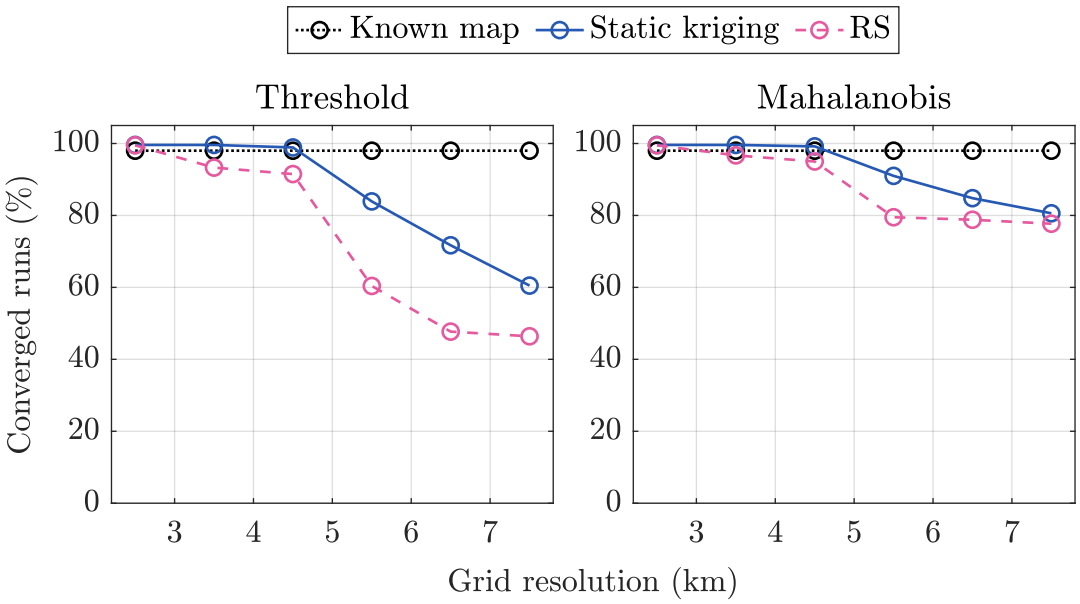


Figure 4.12: Convergence rates for the navigation algorithms using static kriging and random subsampling as a function of grid resolution.

Δ_g	Static kriging	RS
2.5 km	29.3 T	7.3 T
3.5 km	7.39 T	1.84 T
4.5 km	2.81 T	704 G
5.5 km	1.26 T	314 G
6.5 km	608 G	152 G
7.5 km	343 G	85.9 G

Table 4.7: Total computational cost, in flops, of using static kriging or random subsampling throughout the mission, for each grid resolution.

4.3.2 Adaptive Subsampling Window

The above results show that, while subsampling does reduce the computational cost of the algorithm, indiscriminately discarding training points is detrimental to the navigation performance. A more suitable strategy should therefore retain only a limited number of training points while selecting those that are most relevant to the current situation.

Since, at each time step, the filter only queries the map where its particles lie, this suggests retaining the points surrounding the particle set. Beyond reducing the computational cost, a local selection also relaxes a second limitation of static kriging. Indeed, the estimators introduced in section 3.3 assume the use of a single kernel over the entire domain, and assumption 4.1 inherits that hypothesis. In section 4.2.2, we estimated the hyperparameters of the kernel once before the mission, meaning that

they are assumed to be independent of the inputs:

$$k_{3/2} : x, x' \mapsto \sigma_f^2 \left(1 + \frac{\sqrt{3}\,\|x - x'\|_2}{l}\right) \exp\left(-\frac{\sqrt{3}\,\|x - x'\|_2}{l}\right) \tag{4.34}$$

The considered kernel is thus radial, and the resulting GP is therefore stationary. This is unrealistic for a physical field, especially at large spatial scales. However, stationarity cannot be simply dropped, as the hyperparameters of such a kernel would have to be fitted at every point of the domain, where a single sample is available. We therefore assume the field to be non-stationary, but its hyperparameters to vary over a scale much larger than the grid resolution of the training data. Under this hypothesis, the hyperparameters must be estimated over a region large enough to provide sufficient data for accurate estimation, while remaining small enough to avoid averaging them over distances on which they vary significantly.

In this section, we propose an adaptive subsampling strategy to optimise the kriging performance of the filter according to four criteria:

- The retained training points must cover an area large enough to avoid extrapolation at the particle locations.
- A sufficient number of training points must be considered to ensure that the hyperparameters are accurately estimated.
- The retained training points must be confined to a sufficiently small region for the hyperparameters to be approximately constant within this region.
- Finally, their number must remain small enough to ensure that the filter can be computed within real-time constraints.

The first constraint naturally gives the shape of the retained subset. Let $\hat{x}_{k|k-1}$ be the predicted estimate of the filter at time k:

$$\hat{x}_{k|k-1} \triangleq \sum_{i=1}^{N} w_{k-1}^{(i)}\, x_k^{(i)} \approx \mathbb{E}_{p(\cdot|\mathbf{z}_{1:k-1})}[x_k] \tag{4.35}$$

and $\hat{P}_{k|k-1}$ denote the corresponding covariance matrix:

$$\hat{P}_{k|k-1} \triangleq \sum_{i=1}^{N} w_{k-1}^{(i)} \left(x_k^{(i)} - \hat{x}_{k|k-1}\right)\left(x_k^{(i)} - \hat{x}_{k|k-1}\right)^\top \tag{4.36}$$

These two quantities can be computed after the prediction step at time k, but before the following weight update, and represent a Gaussian approximation of the particle set at time k before kriging the environment. It must be noted that, while the selection itself should only depend on the particle locations, both sums are weighted by the $w_{k-1}^{(i)}$ to limit the effect of resampling on the resulting moments. Although this may result in some particles being left outside the retained region, this is not detrimental when their associated weights are low.

In this approximation, each particle has a 99% chance to verify:

$$\left\|\Pi_{\text{pos}}\, \hat{x}_{k|k-1} - \Pi_{\text{pos}}\, x_k^{(i)}\right\|^2_{\hat{P}_{\text{pos},k|k-1}} \leq \chi^2_{d_{\text{pos}}}(99\%) \tag{4.37}$$

where $d_{\text{pos}} \in \mathbb{N}^*$ is the dimension of the position subspace, $\hat{P}_{\text{pos},k|k-1}$ is the covariance matrix associated with the projection of the predicted state onto this subspace:

$$\hat{P}_{\text{pos},k|k-1} \triangleq \Pi_{\text{pos}}\, \hat{P}_{k|k-1}\, \Pi_{\text{pos}}^{\top} \tag{4.38}$$

and $\chi^2_{d_{\text{pos}}}(99\%)$ is the 99% quantile of a χ^2 distribution with d_{pos} degrees of freedom. This means that the particles are likely to be contained in an ellipsoid centred on $\Pi_{\text{pos}}\, \hat{x}_{k|k-1}$, and whose shape is given by the eigenvectors of $\hat{P}_{\text{pos},k|k-1}$.

However, this geometry is tied to the particle set, and does not directly translate into the physical position space. Furthermore, when $\hat{P}_{\text{pos},k|k-1}$ is highly anisotropic, the resulting ellipsoid may provide insufficient coverage along directions of lower uncertainty, leading to extrapolation. To solve this issue, we use the Rayleigh quotient [100] to convert the Mahalanobis inequality into a Euclidean one:

$$\begin{aligned}\left\|\Pi_{\text{pos}}\, \hat{x}_{k|k-1} - \Pi_{\text{pos}}\, x_k^{(i)}\right\|_2^2 &\leq \lambda_{\max}\left(\hat{P}_{\text{pos},k|k-1}\right) \left\|\Pi_{\text{pos}}\, \hat{x}_{k|k-1} - \Pi_{\text{pos}}\, x_k^{(i)}\right\|^2_{\hat{P}_{\text{pos},k|k-1}} \\ &\leq \chi^2_{d_{\text{pos}}}(99\%)\, \lambda_{\max}\left(\hat{P}_{\text{pos},k|k-1}\right)\end{aligned} \tag{4.39}$$

The first subsampling criterion is thus satisfied by retaining training points within a ball of radius:

$$r_{\text{cov},k} \triangleq \alpha_{\text{cov}} \sqrt{\chi^2_{d_{\text{pos}}}(99\%)} \sqrt{\lambda_{\max}\left(\hat{P}_{\text{pos},k|k-1}\right)} \tag{4.40}$$

and centred on $\Pi_{\text{pos}}\hat{x}_{k|k-1}$, where $\alpha_{\text{cov}} \geq 1$ is a dilation factor, introduced to provide a margin around the particle set and reduce the risk of extrapolation at its boundaries.

However, when the particle set collapses on a small region of space, the ball may not contain enough training points to satisfy the second criterion. We thus introduce a second radius to ensure that at least $\tilde{N}_{\min}$ training points are retained:

$$r_{\min,k} \triangleq \min_{r \in \mathbb{R}^+} \left(\text{Card}\left\{i \in [\![1, \tilde{N}]\!] \;\middle|\; \left\|\Pi_{\text{pos}}\, \hat{x}_{k|k-1} - \tilde{x}_i\right\|_2 \leq r\right\} \geq \tilde{N}_{\min}\right) \tag{4.41}$$

The retained training set is thus defined at each time step k as:

$$\tilde{\mathbf{x}}^r_{\text{AK},k} \triangleq \left\{\tilde{x}_i \in \tilde{\mathbf{x}} \;\middle|\; \left\|\Pi_{\text{pos}}\, \hat{x}_{k|k-1} - \tilde{x}_i\right\|_2 \leq \max\left(r_{\text{cov},k}, r_{\min,k}\right)\right\} \tag{4.42}$$

and:

$$\tilde{\mathbf{z}}^r_{\text{AK},k} \triangleq \left\{\tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}^r_{\text{AK},k}\right\} \tag{4.43}$$

At each time step, the hyperparameters $\hat{\theta}_k$ of the kernel are then re-estimated by maximum likelihood on the retained training set, as described in section 4.2.2. In particular, since the navigation estimate varies slowly between two consecutive time steps, the latter is likely to remain similar. The hyperparameters estimated at time $k-1$ can thus be used as initial values for the estimation at time k, reducing the number of iterations required for convergence of the hyperparameter estimation algorithm.

In particular, the number of retained points is, on average, the d_{pos}-volume of the window divided by that of a grid cell:

$$\mathbb{E}\left[\tilde{N}^r_{\text{AK},k}\right] = \frac{\pi^{d_{\text{pos}}/2}}{\Gamma\left(\frac{d_{\text{pos}}}{2}+1\right)} \left(\frac{\max\left(r_{\text{cov},k}, r_{\min,k}\right)}{\Delta_g}\right)^{d_{\text{pos}}} \tag{4.44}$$

The likelihood (4.3) is thus computed on the retained subset alone and using the updated hyperparameters:

$$z_k|x_k, \tilde{\mathbf{z}}^r_{\mathrm{AK},k} \sim \mathcal{N}\left(\hat{z}_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}^r_{\mathrm{AK},k}, \tilde{\mathbf{z}}^r_{\mathrm{AK},k}; \hat{\theta}_k\right), R_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}^r_{\mathrm{AK},k}; \hat{\theta}_k\right)\right) \tag{4.45}$$

This method, referred to as Adaptive Kriging (AK) and originally proposed in [105], can be integrated into any navigation algorithm presented in chapter 2. In this section, we focus on its integration into the RPF of Algorithm 2.8 using the transition density as proposal, as described in (2.97). The resulting algorithm is the Adaptive Kriging Regularised Particle Filter (AK-RPF) given in Algorithm 4.2.

Algorithm 4.2 Adaptive Kriging Regularised Particle Filter

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim p\left(\cdot|x_{k-1}^{(i)}\right)$ ▷ Particle propagation

$\hat{x}_{k|k-1} = \sum_{i=1}^{N} w_{k-1}^{(i)} x_k^{(i)}$ ▷ **Predicted estimation**

$\hat{P}_{k|k-1} = \sum_{i=1}^{N} w_{k-1}^{(i)} \left(x_k^{(i)} - \hat{x}_{k|k-1}\right)\left(x_k^{(i)} - \hat{x}_{k|k-1}\right)^\top$

Compute $\tilde{\mathbf{x}}^r_{\mathrm{AK},k}$ with (4.42) ▷ **Adaptive subsampling**

Compute $\tilde{\mathbf{z}}^r_{\mathrm{AK},k}$ with (4.43)

$\hat{\theta}_k = \mathrm{MLE}\left(\tilde{\mathbf{x}}^r_{\mathrm{AK},k}, \tilde{\mathbf{z}}^r_{\mathrm{AK},k}, \hat{\theta}_{k-1}\right)$ ▷ **Hyperparameter estimation**

$\forall i \in [\![1, N]\!]$, compute $\tilde{p}\left(z_k|x_k^{(i)}\right)$ from (4.45) ▷ **Kriging**

$\forall i \in [\![1, N]\!],\ \tilde{w}_k^{(i)} = w_{k-1}^{(i)}\, \tilde{p}\left(z_k|x_k^{(i)}\right)$ ▷ Weight update

$\forall i \in [\![1, N]\!],\ w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

$\hat{N}_{\mathrm{eff}} = 1 \Big/ \sum_{i=1}^{N} \left(w_k^{(i)}\right)^2$ ▷ Resampling criterion

if $\hat{N}_{\mathrm{eff}} < N_{\mathrm{th}}$ **then**

  $\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)}$ ▷ Whitening

  $\hat{P}_k = \sum_{i=1}^{N} w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right)\left(x_k^{(i)} - \hat{x}_k\right)^\top$

  A_k s.t. $\hat{P}_k = A_k\, A_k^\top$

  $\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^{N} \leftarrow \mathrm{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^{N}\right)$ ▷ Resampling

  $\forall j \in [\![1, N]\!],\ \varepsilon^{(j)} \sim K$ ▷ Regularisation

  $\forall j \in [\![1, N]\!],\ x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\mathrm{reg}} h A_k\, \varepsilon^{(j)}$

end if

Remark 4.6 (Dependence on the particle set)**.** *While Bayesian estimation allows the observation function to change at each time step, (4.45) depends on the realisation of the particle set, through both $\tilde{\mathbf{x}}^r_{\mathrm{AK},k}$ and $\hat{\theta}_k$. Two runs on the same data may therefore assign different likelihoods to their particles.*

However, AK only replaces the fixed likelihood (4.3) on the whole training set by a subsampled approximation, whose error is quantified in section 4.5. On the other hand, the hyperparameters are assumed to vary slowly in space, and depend on the region covered by the window rather than on the particles themselves. At a given time step, the likelihood is the same for every particle, so that their weights remain comparable.

Unfortunately, the third and fourth criteria may conflict with the first one in situations where the particle set spans over a large region of space. These two criteria are addressed in section 4.4.

4.3.3 Numerical Results

As for static kriging, this section evaluates the performance of the AK-RPF across the six grid resolutions, with:

$$\alpha_{\text{cov}} = 2 \quad \text{and} \quad \tilde{N}_{\min} = 25 \tag{4.46}$$

The performance of the static kriging RPF is also given as a reference. Alongside the RMSE and the NEES, another metric is introduced to help explain the variations in computational cost between each configuration. This metric is the number of retained training points, *i.e.* the value of $\tilde{N}^r$, at each time step. Figures 4.13 and 4.14 present the comparative RMSE and NEES curves across the six grid resolutions, and Figure 4.15 gives the associated convergence rates. Finally, Table 4.8 presents the average total computational cost of static kriging and AK during the entire mission for each of the six grid resolutions, and Figure 4.16 shows the evolution of the average number of retained training points throughout the mission.

The first point to note is the strong similarity between the RMSE curves obtained with static kriging and AK for all grid resolutions. Apart from a slight reduction in the AK error between $k = 60$ and $k = 90$, and a slight increase during the first turn, particularly for the 7.5 km grid resolution, the curves almost completely overlap. The absence of any noticeable accuracy loss despite discarding most of the training set can be attributed to the local nature of the AK window.

Since discarded points are far from the particle cloud, they have negligible contributions to the map reconstruction at the particle locations. Their removal thus has little effect on the relative particle likelihoods. While this is only shown numerically here, its theoretical analysis is the subject of section 4.5.

On the other hand, the NEES curves of both algorithms are significantly different. Although both remain within the χ^2 confidence bounds throughout the trajectory, the NEES of static kriging rises above 2 around $k = 70$ for every resolution but the finest, before decreasing again around $k = 125$. In contrast, the NEES of AK remains below 1 throughout the mission and for all grid resolutions. This may be explained by the non-stationarity of the field. The global kernel would provide a poor description of this portion of the field, whereas the local estimation of the hyperparameters would allow AK to adapt to it. It must also be noted that the RMSE of AK is lower precisely around $k = 70$, which further illustrates this explanation.

Regarding convergence, the two filters exhibit the same overall trend, with a drop in convergence for both criteria from $\Delta_g = 5.5$ km onward. Since its NEES

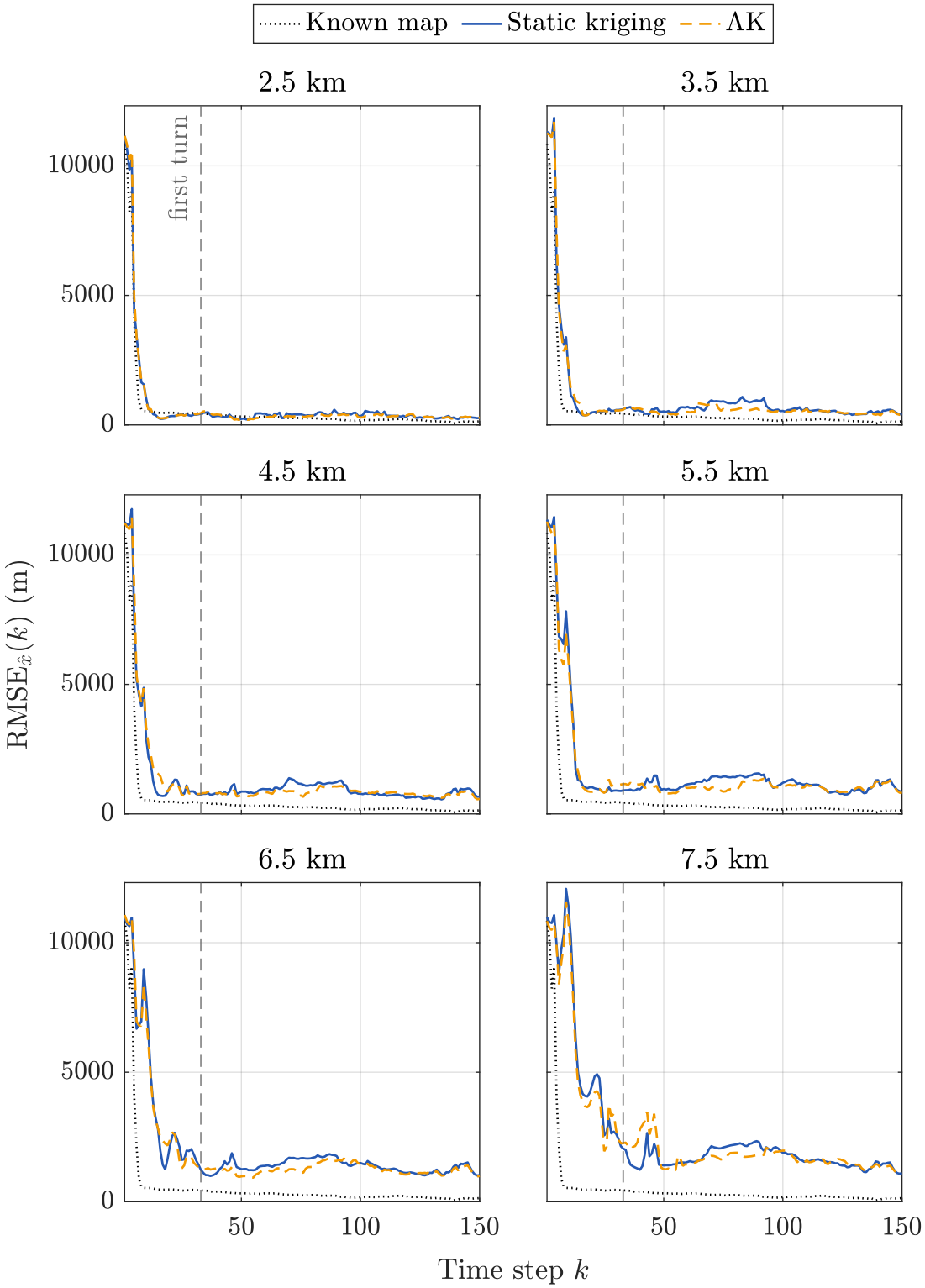


Figure 4.13: RMSE curves for the AK-RPF compared with that of an RPF using static kriging across the six grid resolutions.

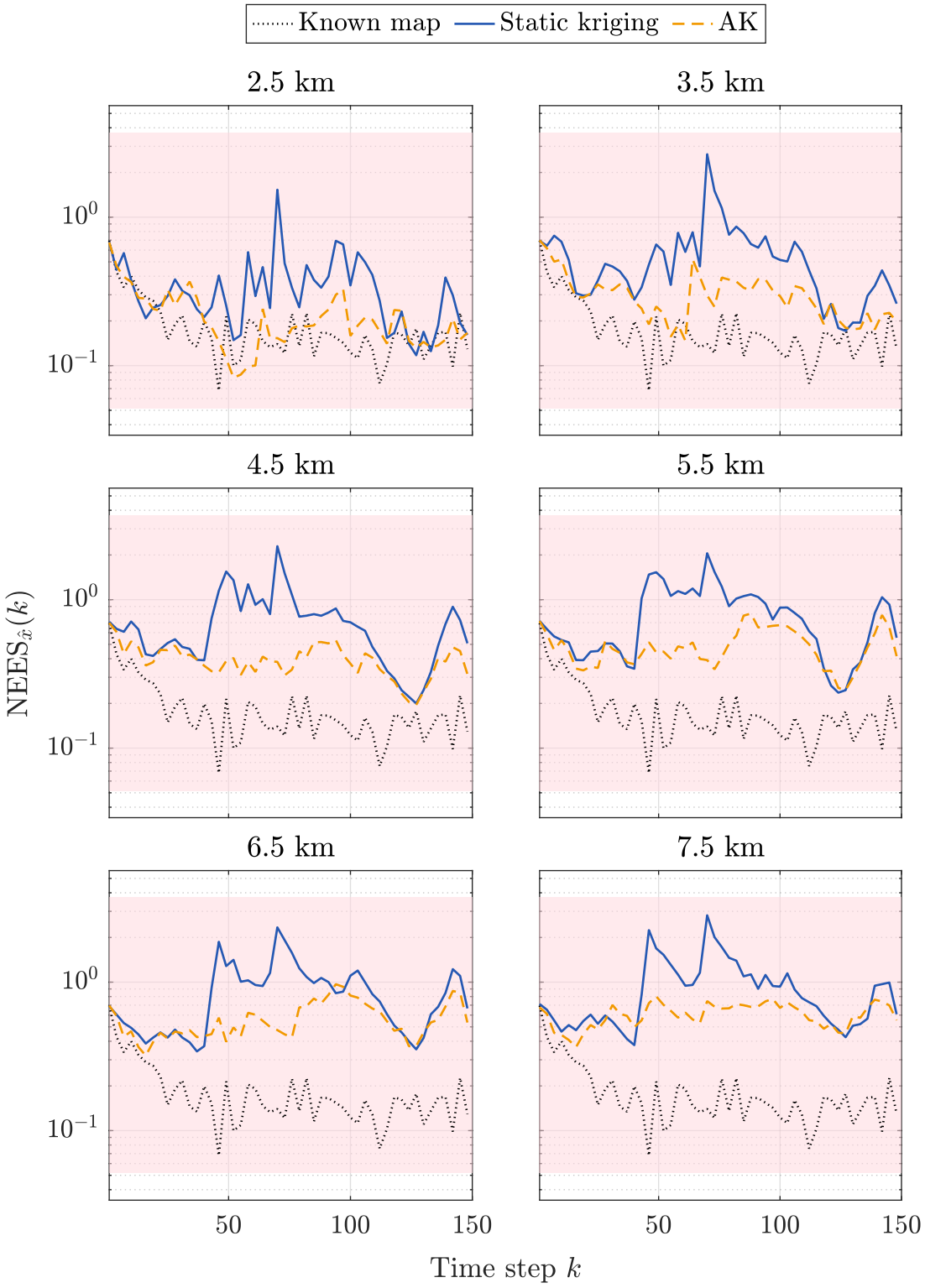


Figure 4.14: NEES curves for the AK-RPF compared with that of an RPF using static kriging across the six grid resolutions.

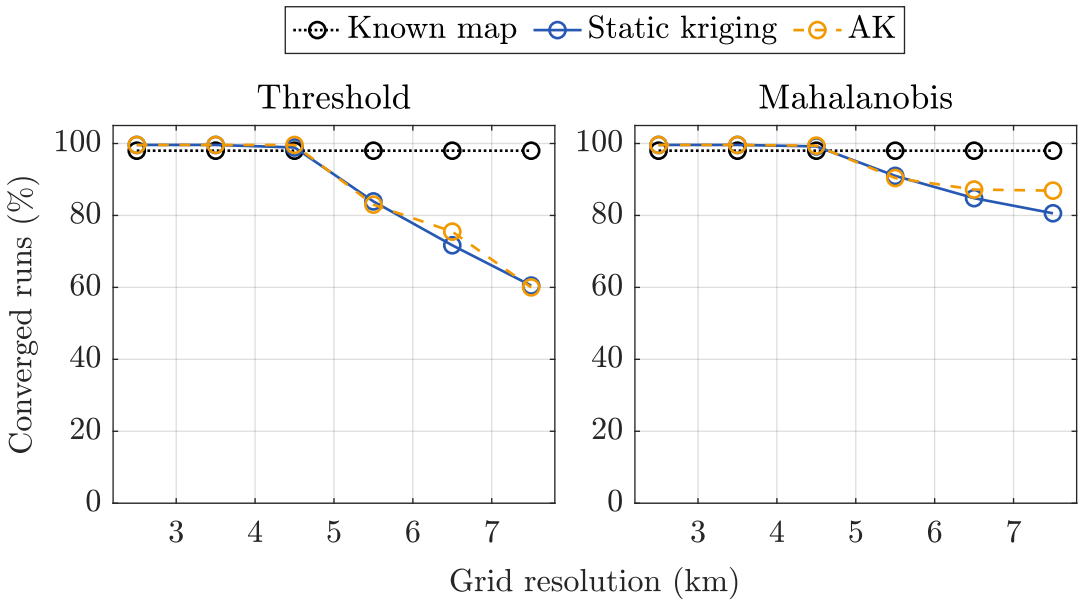


Figure 4.15: Convergence rates of the AK-RPF and the RPF using static kriging using the threshold and the Mahalanobis criteria.

Δ_g	Static kriging	AK	
		Kriging	Hyperparameters
2.5 km	29.3 T	46 G	564 M
3.5 km	7.39 T	14.4 G	224 M
4.5 km	2.81 T	5.75 G	121 M
5.5 km	1.26 T	3.09 G	89.8 M
6.5 km	608 G	1.93 G	70.1 M
7.5 km	343 G	1.54 G	66.7 M

Table 4.8: Total computational cost, in flops, of using static kriging or AK throughout the mission, for each grid resolution. The cost of AK is split between the kriging itself and the local estimation of the hyperparameters, which static kriging only performs once before the mission.

is generally more conservative, AK achieves convergence for 87% of the runs at 7.5 km according to the Mahalanobis criterion, compared with 81% for static kriging. Their convergence rates remain similar for the threshold criterion, with the same drop at grid resolutions greater than 5.5 km.

Turning finally to the number of retained data, two regimes emerge from Figure 4.16. The first one, during which the size of the adaptive window is determined by the spatial dispersion of the particles, gives way to the second around $k = 25$, after which the number of retained points remains of the same order of magnitude as $\tilde{N}_{\min} = 25$. In the first regime, using (4.44) with $d_{\text{pos}} = 2$ establishes that the average number of retained training points is:

$$\tilde{N}^r_{\text{AK},k} \approx \frac{\pi\, r^2_{\text{cov},k}}{\Delta_g^2} \tag{4.47}$$

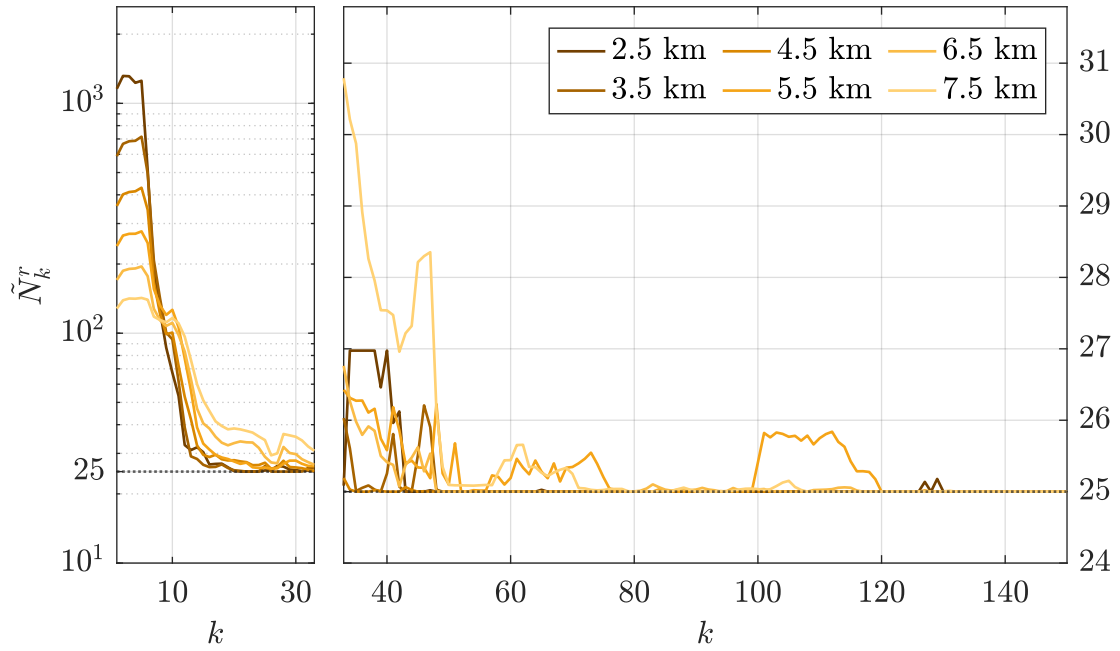


Figure 4.16: Evolution of the average number of training points retained by the AK-RPF during the mission for each grid resolution. A logarithmic scale is used before the first turn at $k = 33$ to capture the large variations in $\tilde{N}^r_{\mathrm{AK},k}$, while a linear scale is used afterwards, where these variations are smaller.

which explains the difference between the six curves at the beginning of the mission.

The rapid reduction in the number of retained training points also explains the computational cost of AK. Using the approximation that every filter uses 25 points after $k = 50$, the last hundred steps require approximately $100 \times 5000 \times 25^2 \approx 0.31$ Gflop of kriging at every grid resolution. This represents 0.7% of the AK kriging cost at $\Delta_g = 2.5$ km, up to 20% at 7.5 km. The local hyperparameter estimation only adds between 1 and 5% to these costs, so most of the computational load is incurred at the beginning of the mission, when the window is at its largest.

While AK is between 213 and 629 times less costly than static kriging for grid resolutions of 7.5 and 2.5 km respectively, this gap decreases as Δ_g increases. This happens because the cost of static kriging scales as $\tilde{N}^2$, decreasing by a factor of 85 between the finest and coarsest grids, whereas the cost of AK-RPF, bounded below by the minimum window size, decreases by only a factor of 29.

Taken together, these results show that AK removes the accuracy/cost compromise observed with static kriging, and that random subsampling failed to alleviate. Despite retaining only a small fraction of the available training set, it achieves comparable or better performance, while substantially reducing the overall cost of the filter. At $\Delta_g = 2.5$ km, its most expensive configuration, the entire AK mission only needs 46.6 Gflop, less than the 81 Gflop required by static kriging for the initial factorisation alone at the same resolution, and even less than the 343 Gflop of the least expensive static kriging configuration, at $\Delta_g = 7.5$ km.

4.4 Non-Spherical Subsampling Windows

While it outperforms static kriging, the AK window can still fail to satisfy the bottom two requirements established at the beginning of section 4.3.2. When a sufficiently large number of particles uniformly covers a wide region of the position subspace, any window that contains the particle set necessarily retains a large number of training points.

In other cases however, the limitation stems from the shape of the subsampling window. In particular, a single spherical window is poorly suited to non-Gaussian particle distributions, especially multimodal PDFs, for which large regions within the window may contain few or no particles. This section introduces two methods to address this issue, either by adapting the shape of a single window to that of the particle distribution, or by using multiple spherical windows around clusters of particles.

4.4.1 Nearest Neighbour Selection

As established above, choosing the retained training points within a sphere centred on the predicted state provides a simple subsampling heuristic, but may include large regions without particles. A more refined selection should therefore follow the shape of the particle set more closely, avoiding the unnecessary computational cost of reconstructing the map in regions where it will not be queried.

A simple way to do this is to define, for each particle, a spherical selection window centred on it, similarly to AK but with a smaller radius, and to retain the training points contained within the union of these spheres. However, unlike the entire particle set, a single particle has no individual covariance matrix to compute (4.40). Instead, the windows are defined using only the $r_{\min,k}$ criterion, applied to a single particle:

$$\forall i \in [\![1, N]\!],\ \tilde{\mathbf{x}}^{r,(i)}_{\mathrm{NNAK},k} \triangleq \left\{ \tilde{x}_j \in \tilde{\mathbf{x}} \;\middle|\; \left\| \Pi_{\mathrm{pos}}\, x^{(i)}_k - \tilde{x}_j \right\|_2 \leq r^{(i)}_{\min,k} \right\} \tag{4.48}$$

where:

$$r^{(i)}_{\min,k} \triangleq \min_{r \in \mathbb{R}^+} \left(\mathrm{Card} \left\{ j \in [\![1, \tilde{N}]\!] \;\middle|\; \left\| \Pi_{\mathrm{pos}}\, x^{(i)}_k - \tilde{x}_j \right\|_2 \leq r \right\} \geq \tilde{N}_{\min} \right) \tag{4.49}$$

Each individual selection window thus contains at least the $\tilde{N}_{\min}$ training points closest to the corresponding particle, with exactly $\tilde{N}_{\min}$ points unless several of them are equidistant from the considered particle. The global subsampling window is then defined as the union of the individual windows:

$$\tilde{\mathbf{x}}^{r}_{\mathrm{NNAK},k} \triangleq \bigcup_{i=1}^{N} \tilde{\mathbf{x}}^{r,(i)}_{\mathrm{NNAK},k} \tag{4.50}$$

and:

$$\tilde{\mathbf{z}}^{r}_{\mathrm{NNAK},k} \triangleq \{ \tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}^{r}_{\mathrm{NNAK},k} \} \tag{4.51}$$

Due to the nature of the selection, we call this method Nearest Neighbour Adaptive Kriging (NNAK). Similarly to AK, NNAK re-estimates the hyperparameters of

the kernel by maximum likelihood estimation on $\tilde{\mathbf{x}}^r_{\mathrm{NNAK},k}$ and $\tilde{\mathbf{z}}^r_{\mathrm{NNAK},k}$, and computes the likelihood (4.3) using the entire retained data:

$$z_k|x_k, \tilde{\mathbf{z}}^r_{\mathrm{NNAK},k} \sim \mathcal{N}\left(\hat{z}_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}^r_{\mathrm{NNAK},k}, \tilde{\mathbf{z}}^r_{\mathrm{NNAK},k}; \hat{\theta}_k\right), R_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}^r_{\mathrm{NNAK},k}; \hat{\theta}_k\right)\right) \tag{4.52}$$

Regarding the parameters of this method, NNAK does not rely on the covariance-based criterion of AK, and has therefore no α_{cov} parameter to tune. In addition, $\tilde{N}_{\min}$ is deliberately chosen smaller than for AK, as NNAK aims to follow the shape of the particle set without unnecessarily enlarging or distorting it. It must be noted that, unlike AK, no floor is applied to the size of the entire NNAK subsampling window. The pseudo-code of a NNAK-based RPF is given by Algorithm 4.3.

Algorithm 4.3 Nearest Neighbour Adaptive Kriging RPF

$\forall i \in [\![1, N]\!]$, $x_k^{(i)} \sim p\left(\cdot|x_{k-1}^{(i)}\right)$ ▷ Particle propagation

Compute $\tilde{\mathbf{x}}^r_{\mathrm{NNAK},k}$ with (4.50) ▷ **Adaptive subsampling**

Compute $\tilde{\mathbf{z}}^r_{\mathrm{NNAK},k}$ with (4.51)

$\hat{\theta}_k = \mathrm{MLE}\left(\tilde{\mathbf{x}}^r_{\mathrm{NNAK},k}, \tilde{\mathbf{z}}^r_{\mathrm{NNAK},k}, \hat{\theta}_{k-1}\right)$ ▷ **Hyperparameter estimation**

$\forall i \in [\![1, N]\!]$, compute $\tilde{p}\left(z_k|x_k^{(i)}\right)$ from (4.52) ▷ **Kriging**

$\forall i \in [\![1, N]\!]$, $\tilde{w}_k^{(i)} = w_{k-1}^{(i)}\, \tilde{p}\left(z_k|x_k^{(i)}\right)$ ▷ Weight update

$\forall i \in [\![1, N]\!]$, $w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

$\hat{N}_{\mathrm{eff}} = 1 \Big/ \sum_{i=1}^{N} \left(w_k^{(i)}\right)^2$ ▷ Resampling criterion

if $\hat{N}_{\mathrm{eff}} < N_{\mathrm{th}}$ **then**

$\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)}$ ▷ Whitening

$\hat{P}_k = \sum_{i=1}^{N} w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right)\left(x_k^{(i)} - \hat{x}_k\right)^\top$

A_k s.t. $\hat{P}_k = A_k\, A_k^\top$

$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^{N} \leftarrow \mathrm{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^{N}\right)$ ▷ Resampling

$\forall j \in [\![1, N]\!]$, $\varepsilon^{(j)} \sim K$ ▷ Regularisation

$\forall j \in [\![1, N]\!]$, $x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\mathrm{reg}} h A_k\, \varepsilon^{(j)}$

end if

4.4.2 Clustering

The main difference between AK and NNAK lies in the scale at which the particle set is considered. While AK views the particle set as a whole from a macroscopic perspective, NNAK adopts a microscopic point of view by treating each particle individually. In this section, we introduce an intermediate, mesoscopic perspective by partitioning the particle set into clusters.

Since this partition is recomputed at each time step, and determines the subsampling windows, the clustering method must satisfy four criteria:

- The number of clusters is unknown prior to the mission, and may vary from one time step to the next. It therefore cannot be an input of the method.
- The particles and their weights approximate the PDF. Clustering should thus account for particle weights, so that modes represented by few high-weight particles are properly identified.
- Every particle must belong to a cluster. Otherwise, its likelihood could not be computed by kriging.
- Finally, the computational cost of clustering must remain small compared with that of kriging.

For the sake of simplicity, the time index is omitted throughout this section, and only the spatial positions of the particles are considered. We thus write:

$$\forall i \in [\![1, N]\!],\ x^{(i)} \triangleq \Pi_{\text{pos}}\, x_k^{(i)} \quad \text{and} \quad w^{(i)} \triangleq w_{k-1}^{(i)} \tag{4.53}$$

for the position and the weight of the i-th particle. A clustering method partitions the particle set into $C \in \mathbb{N}^*$ clusters:

$$\left\{\mathcal{C}^{(c)}\right\}_{c=1}^{C} = \text{Clustering}\left(\left\{x^{(i)}, w^{(i)}\right\}_{i=1}^{N}\right) \tag{4.54}$$

where $\mathcal{C}^{(c)}$ denotes the indices of the particles belonging to the c-th cluster.

Several clustering methods can be used in (4.54). In this section, we present four classical methods.

Hierarchical Clustering Analysis

Hierarchical clustering analysis (HCA) [106, 107] does not compute a single partition, but a hierarchy of nested partitions. Starting from the partition in which each particle forms its own cluster, it repeatedly merges the two closest clusters, in the sense of a given dissimilarity, until a single cluster remains. For instance, Ward's criterion merges the two clusters whose union least increases the weighted dispersion of the particles around the barycentre of their cluster. The $N-1$ successive merges, and the dissimilarities, or heights, at which they occur, form the dendrogram of the particle set.

A partition is then read from the dendrogram in one of two ways. Stopping after $N - C$ merges imposes a partition into exactly C clusters, whereas cutting the dendrogram at a given height determines the number of clusters according to a prescribed dissimilarity threshold. This method is illustrated by Figure 4.17.

However, HCA requires the dissimilarities between all pairs of clusters. This represents $N(N-1)/2$ values at the first merge, *i.e.* more than 12 million for the $N = 5000$ particles considered in section 4.2.3, and updating them over the $N-1$ merges costs $\mathcal{O}\left(N^2 \log N\right)$ operations at each time step.

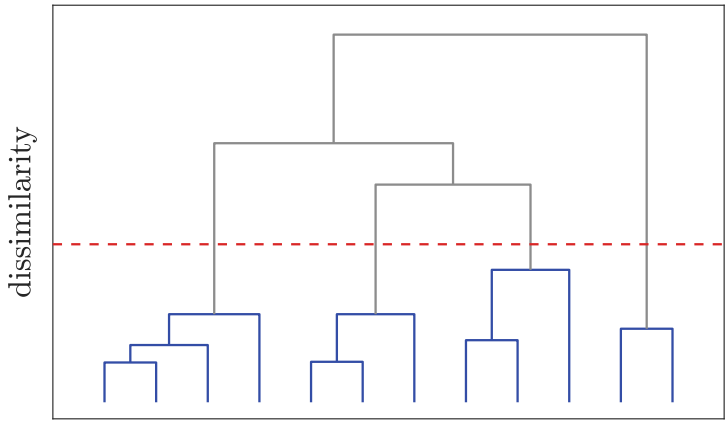


Figure 4.17: Dendrogram of a particle set, cut at a certain dissimilarity threshold. The particles connected below the cut, highlighted in blue, delimit four distinct clusters.

k-means

Instead of building a hierarchy, the k-means algorithm [108] searches for a partition into a given number C of clusters. Each cluster is represented by a centre $\breve{x}^{(c)} \in \mathbb{R}^{d_{\text{pos}}}$, and both the partition and the centres are chosen to minimise the weighted dispersion of the particles around the centre of their cluster:

$$\underset{\mathcal{C}^{(1)},\dots,\mathcal{C}^{(C)},\, \breve{x}^{(1)},\dots,\breve{x}^{(C)}}{\operatorname{argmin}} \left(\sum_{c=1}^{C} \sum_{i \in \mathcal{C}^{(c)}} w^{(i)} \left\| x^{(i)} - \breve{x}^{(c)} \right\|_2^2 \right) \tag{4.55}$$

In practice, this problem is addressed by Lloyd's algorithm [109], which starts from C initial centres, and alternates between two steps:

- **Assignment.** Each particle is assigned to the cluster of the closest centre:

$$\forall c \in [\![1, C]\!],\ \mathcal{C}^{(c)} = \left\{ i \in [\![1, N]\!] \;\middle|\; c = \underset{c' \in [\![1,C]\!]}{\operatorname{argmin}} \left\| x^{(i)} - \breve{x}^{(c')} \right\|_2 \right\} \tag{4.56}$$

- **Update.** Each centre is replaced by the weighted barycentre of its cluster:

$$\forall c \in [\![1, C]\!],\ \breve{x}^{(c)} = \sum_{i \in \mathcal{C}^{(c)}} w^{(i)}\, x^{(i)} \Bigg/ \sum_{i \in \mathcal{C}^{(c)}} w^{(i)} \tag{4.57}$$

Each of these steps does not increase the criterion (4.55). Since the number of partitions is finite, the algorithm stops after a finite number of iterations, at a local minimum depending on the initial centres. One iteration costs $\mathcal{O}(N\,C)$ operations, which makes the method affordable in real time. However, the number of clusters must be known beforehand.

Mean Shift

Rather than minimising a distance criterion, mean shift [110, 111] defines the clusters from the modes of the PDF. Since the particle set is a weighted sample of this density, the latter can be estimated by regularising the particle set similarly to the RPF. For a regularisation kernel K on $\mathbb{R}^{d_{\text{pos}}}$, in the sense of Definition 2.5, and a bandwidth $h_{\text{MS}} > 0$, the estimated density is:

$$p_{\text{pos}} : x \in \mathbb{R}^{d_{\text{pos}}} \mapsto \sum_{i=1}^{N} w^{(i)} \, K_{h_{\text{MS}}}\left(x - x^{(i)}\right) \tag{4.58}$$

When K is the Epanechnikov kernel K_{Epa} of section 2.5 taken in dimension d_{pos}, the gradient of this density is given, wherever it is defined, by:

$$\frac{\partial p_{\text{pos}}}{\partial x}(x) = \frac{d_{\text{pos}} + 2}{c_{d_{\text{pos}}} \, h_{\text{MS}}^{d_{\text{pos}}+2}} \left(\sum_{i \in \mathcal{B}(x, h_{\text{MS}})} w^{(i)} \right) \left(\frac{\displaystyle\sum_{i \in \mathcal{B}(x, h_{\text{MS}})} w^{(i)} \, x^{(i)}}{\displaystyle\sum_{i \in \mathcal{B}(x, h_{\text{MS}})} w^{(i)}} - x \right) \tag{4.59}$$

where $c_{d_{\text{pos}}}$ is the volume of the unit ball of $\mathbb{R}^{d_{\text{pos}}}$ and, for $r > 0$:

$$\mathcal{B}(x, r) \triangleq \left\{ i \in [\![1, N]\!] \;\middle|\; \left\| x^{(i)} - x \right\|_2 \leq r \right\} \tag{4.60}$$

contains the indices of the particles within a distance r of x. This set is assumed to be non-empty in (4.59).

The second factor of (4.59) points from x towards the weighted barycentre of its neighbourhood, and the first one is a non-negative scalar. The mean-shift vector therefore points in the direction of the density gradient, leading to the following fixed-point iteration, run from each particle:

$$\forall i \in [\![1, N]\!], \; \forall n \in \mathbb{N}, \; x_{n+1}^{(i)} \triangleq \sum_{j \in \mathcal{B}\left(x_n^{(i)}, h_{\text{MS}}\right)} w^{(j)} \, x^{(j)} \Bigg/ \sum_{j \in \mathcal{B}\left(x_n^{(i)}, h_{\text{MS}}\right)} w^{(j)} \tag{4.61}$$

where the sequence of the i-th particle starts at the particle itself:

$$x_0^{(i)} \triangleq x^{(i)} \tag{4.62}$$

Since the Epanechnikov kernel has bounded support, each sequence takes its values, after its first iteration, among the finitely many weighted barycentres of the subsets of the particle set. As the estimated density increases strictly as long as the barycentre moves [111], a sequence cannot return to a previous value, and therefore becomes stationary after a finite number of iterations.

Formally, let $x_\infty^{(i)}$ denote the limit of the sequence generated by the i-th particle, and let $\breve{x}^{(1)}, \ldots, \breve{x}^{(C)}$ be the distinct limits reached by the particle set. The relation defined by two particles sharing the same limit is a binary equivalence relation, whose equivalence classes define the clusters:

$$\forall c \in [\![1, C]\!], \; \mathcal{C}^{(c)} = \left\{ i \in [\![1, N]\!] \;\middle|\; x_\infty^{(i)} = \breve{x}^{(c)} \right\} \tag{4.63}$$

Unlike k-means, mean shift does not require the number of clusters, which is determined by the bandwidth h_{MS} instead. Its computational cost is $\mathcal{O}(N^2 T)$, where T is the number of iterations required for (4.61) to converge.

DBSCAN

TODO: à revoir

Density-based spatial clustering of applications with noise (DBSCAN) [112] also relies on the density of the particle set, but compares it with a threshold instead of searching for its modes. Formally, let $r_{\mathrm{DB}} > 0$ and $w_{\min} \in [0, 1]$. The i-th particle is a core particle if the total weight of the particles lying within a distance r_{DB} of it reaches $w_{\min}$:

$$\sum_{j \in \mathcal{B}\left(x^{(i)}, r_{\mathrm{DB}}\right)} w^{(j)} \geq w_{\min} \tag{4.64}$$

Two core particles are connected if they can be joined by a chain of core particles, each lying within a distance r_{DB} of the next. This defines an equivalence relation over the core particles, whose classes are the clusters. Every remaining particle lying within a distance r_{DB} of a core particle then joins the cluster of the latter, while the others are labelled as noise and do not belong to any cluster.

Similarly to mean shift, DBSCAN does not require the number of clusters. Its clusters are the connected components of the dense regions of the particle set, and can therefore have an arbitrary shape and extent. Its cost is $\mathcal{O}(N \log N)$ when the neighbourhoods are computed using a spatial index.

4.4.3 Clustered Adaptive Kriging

AK and NNAK can thus be seen as limiting cases of clustering, corresponding respectively to a single cluster containing the entire particle set and to one cluster per particle. The framework introduced in section 4.4.2 can be used to construct the intermediate, mesoscopic view. However, the question of which clustering algorithm is best suited for this view must be addressed.

This choice is constrained by both computational and structural considerations. First, the heavy computational cost of HCA makes it unsuitable for real-time applications. Furthermore, the number of clusters at each time step is unknown prior to the mission, which prevents the use of k-means. Finally, neither mean shift nor DBSCAN require the number of clusters to be specified in advance, and both are computationally affordable at each time step.

Mean shift introduces a characteristic scale through its bandwidth, which may artificially limit the size of each cluster and split a single, spatially extended cluster into several ones, whose windows may then overlap. However, this bound helps satisfy the last two requirements of section 4.3.2. On the other hand, DBSCAN identifies clusters based on the connected components of the neighbourhood graph of the particle set. A chain of dense regions is thus returned as a single cluster, which is precisely the situation that the clustering step was introduced to prevent. The clustering step is thus performed using mean shift on the weighted particle set, with the weights from the previous time step carrying the particle density between two resamplings.

While substantial work on clustered particle filters exists [113], this work focuses on the use of clustering for adaptive kriging rather than for particle management and mode preservation. Nonetheless, the two approaches are not mutually exclusive and can be applied jointly.

Similarly to AK and NNAK, the clustering step must be performed after the particle propagation, but before the correction step. Let C_k be the number of clusters at time step k, and $\mathcal{C}_k^{(c)}$ denote the indices of the particles belonging to the c-th. For $k \geq 0$ and $c \in [\![1, C_k]\!]$, the barycentre and empirical covariance matrix of the c-th cluster are respectively defined as:

$$\hat{x}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} \, x_k^{(i)} \tag{4.65}$$

and:

$$\hat{P}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)}\right) \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)}\right)^\top \tag{4.66}$$

where:

$$\forall i \in \mathcal{C}_k^{(c)}, \; w_{k-1}^{(i,c)} \triangleq w_{k-1}^{(i)} \Big/ \sum_{j \in \mathcal{C}_k^{(c)}} w_{k-1}^{(j)} \tag{4.67}$$

is the weight of the i-th particle, renormalised within its associated cluster.

Applying (4.40) and (4.41) to these moments respectively gives the two radii of the cluster:

$$r_{\mathrm{cov},k}^{(c)} \triangleq \alpha_{\mathrm{cov}} \sqrt{\chi^2_{d_{\mathrm{pos}}}(99\%)} \sqrt{\lambda_{\max}\left(\Pi_{\mathrm{pos}} \, \hat{P}_{k|k-1}^{(c)} \, \Pi_{\mathrm{pos}}^\top\right)} \tag{4.68}$$

and:

$$r_{\mathrm{min},k}^{(c)} \triangleq \min_{r \in \mathbb{R}^+} \left(\mathrm{Card}\left\{i \in [\![1, \tilde{N}]\!] \;\middle|\; \left\|\Pi_{\mathrm{pos}} \, \hat{x}_{k|k-1}^{(c)} - \tilde{x}_i\right\|_2 \leq r\right\} \geq \tilde{N}_{\min}\right) \tag{4.69}$$

Each cluster thus defines its own subsampling window:

$$\tilde{\mathbf{x}}_{\mathrm{ICAK},k}^{r,(c)} \triangleq \left\{\tilde{x}_i \in \tilde{\mathbf{x}} \;\middle|\; \left\|\Pi_{\mathrm{pos}} \, \hat{x}_{k|k-1}^{(c)} - \tilde{x}_i\right\|_2 \leq \max\left(r_{\mathrm{cov},k}^{(c)}, r_{\mathrm{min},k}^{(c)}\right)\right\} \tag{4.70}$$

and:

$$\tilde{\mathbf{z}}_{\mathrm{ICAK},k}^{r,(c)} \triangleq \left\{\tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}_{\mathrm{ICAK},k}^{r,(c)}\right\} \tag{4.71}$$

The C_k windows can be handled in two ways. The first one, called Independent Clustering Adaptive Kriging (ICAK) and originally presented in [114], keeps them separate in order to preserve the locality induced by the clustering step. Every cluster estimates its hyperparameters $\hat{\theta}_k^{(c)}$ on its own window, and the likelihood of a particle of $\mathcal{C}_k^{(c)}$ is computed using that window alone:

$$z_k | x_k, \tilde{\mathbf{z}}_{\mathrm{ICAK},k}^{r,(c)} \sim \mathcal{N}\left(\hat{z}_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}_{\mathrm{ICAK},k}^{r,(c)}, \tilde{\mathbf{z}}_{\mathrm{ICAK},k}^{r,(c)}; \hat{\theta}_k^{(c)}\right), R_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}_{\mathrm{ICAK},k}^{r,(c)}; \hat{\theta}_k^{(c)}\right)\right) \tag{4.72}$$

Unlike AK, the hyperparameters of a cluster cannot be directly inherited from a unique cluster at the previous time step. Instead, for each current cluster, they are initialised from those estimated for the parent cluster associated with the largest number of particles in the current cluster. A weighted average of the hyperparameters could also be considered, but may yield values that are not representative of any individual region. Additionally, each new particle inherits the cluster label of its parent during resampling. The pseudo-code for the ICAK-RPF is given by Algorithm 4.4.

Algorithm 4.4 Independent Clustering Adaptive Kriging RPF

$\forall i \in [\![1, N]\!]$, $x_k^{(i)} \sim p\left(\cdot | x_{k-1}^{(i)}\right)$ ▷ Particle propagation

$\left\{\mathcal{C}_k^{(c)}\right\}_{c=1}^{C_k} = \text{Clustering}\left(\left\{\Pi_{\text{pos}}\, x_k^{(i)}, w_{k-1}^{(i)}\right\}_{i=1}^{N}\right)$ ▷ **Clustering**

for $c \in [\![1, C_k]\!]$ **do**

$\hat{x}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)}\, x_k^{(i)}$ ▷ **Predicted estimation**

$\hat{P}_{k|k-1}^{(c)} = \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)}\right)\left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)}\right)^\top$

Compute $\tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}$ with (4.70) ▷ **Adaptive subsampling**

Compute $\tilde{\mathbf{z}}_{\text{ICAK},k}^{r,(c)}$ with (4.71)

$\hat{\theta}_k^{(c)} = \text{MLE}\left(\tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}, \tilde{\mathbf{z}}_{\text{ICAK},k}^{r,(c)}, \hat{\theta}_{k-1}^{(c)}\right)$ ▷ **Hyperparameter estimation**

$\forall i \in \mathcal{C}_k^{(c)}$, compute $\tilde{p}\left(z_k | x_k^{(i)}\right)$ from (4.72) ▷ **Kriging**

end for

$\forall i \in [\![1, N]\!]$, $\tilde{w}_k^{(i)} = w_{k-1}^{(i)}\, \tilde{p}\left(z_k | x_k^{(i)}\right)$ ▷ Weight update

$\forall i \in [\![1, N]\!]$, $w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

$\hat{N}_{\text{eff}} = 1 \Big/ \sum_{i=1}^{N} \left(w_k^{(i)}\right)^2$ ▷ Resampling criterion

if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**

$\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)}$ ▷ Whitening

$\hat{P}_k = \sum_{i=1}^{N} w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right)\left(x_k^{(i)} - \hat{x}_k\right)^\top$

A_k s.t. $\hat{P}_k = A_k\, A_k^\top$

$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^{N} \leftarrow \text{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^{N}\right)$ ▷ Resampling

$\forall j \in [\![1, N]\!]$, $\varepsilon^{(j)} \sim K$ ▷ Regularisation

$\forall j \in [\![1, N]\!]$, $x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k\, \varepsilon^{(j)}$

end if

On the other hand, the clustering step was originally introduced to bridge the gap between AK and NNAK. While AK is the limit case for a single cluster, ICAK does not reduce to NNAK when each particle is given its own cluster. Indeed, ICAK treats each cluster independently, while NNAK considers the union of all the windows jointly. This motivates the introduction of the Merged Clustering Adaptive Kriging (MCAK), which performs the regression on the union of the clustered windows.

Formally, MCAK still computes (4.70) as its independent counterpart, but also

defines:

$$\tilde{\mathbf{x}}^r_{\mathrm{MCAK},k} \triangleq \bigcup_{c=1}^{C_k} \tilde{\mathbf{x}}^{r,(c)}_{\mathrm{ICAK},k} \tag{4.73}$$

and:

$$\tilde{\mathbf{z}}^r_{\mathrm{MCAK},k} \triangleq \left\{ \tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}^r_{\mathrm{MCAK},k} \right\} \tag{4.74}$$

Since a single window is considered in this method, the hyperparameters can be estimated as for AK, and the likelihood (4.3) becomes:

$$z_k | x_k, \tilde{\mathbf{z}}^r_{\mathrm{MCAK},k} \sim \mathcal{N}\left(\hat{z}_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}^r_{\mathrm{MCAK},k}, \tilde{\mathbf{z}}^r_{\mathrm{MCAK},k}; \hat{\theta}_k \right), R_{\mathrm{OK}}\left(x_k, \tilde{\mathbf{x}}^r_{\mathrm{MCAK},k}; \hat{\theta}_k \right) \right) \tag{4.75}$$

Figure 4.18 illustrates the shapes of the subsampling windows produced by AK, NNAK and MCAK for a bimodal particle set, and Algorithm 4.5 presents the pseudo-code for the MCAK-RPF.

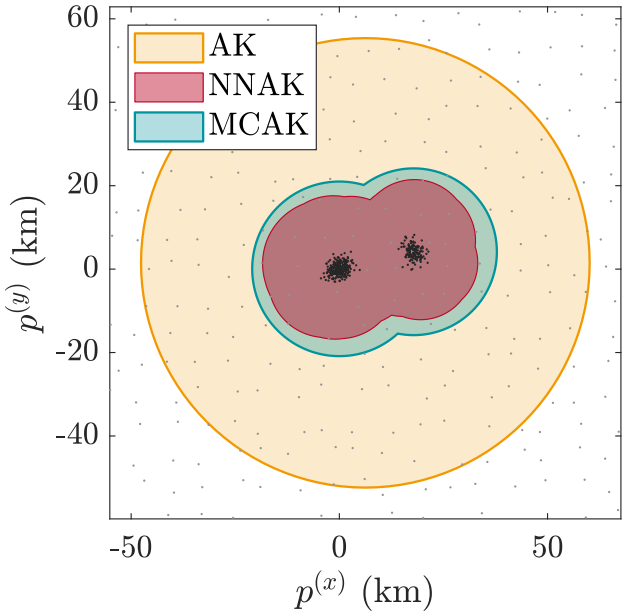


Figure 4.18: Subsampling windows retained by AK, NNAK and MCAK for the same particle set, drawn in black over the training grid, in grey. The shape of the window for ICAK is identical to that of MCAK, up to union.

Remark 4.7 (Independent clusters)**.** *The weight comparability of Remark 4.6 rests on the assumption that every particle is kriged on the same training set and the same hyperparameters, which holds for MCAK, but not for ICAK.*

When the clusters are far apart, section 4.5 bounds the influence of the data of one cluster on the kriging at the particles of another, leaving only the estimated mean and the hyperparameters to distinguish the two variants. The former contributes little wherever training data are available, as measured in section 4.2.1, so that the difference mainly lies in the latter.

However, mean shift does not guarantee any minimum distance between clusters, as mentioned at the beginning of this section, and the difference must then be assessed numerically.

Algorithm 4.5 Merged Clustering Adaptive Kriging RPF

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim p\left(\cdot | x_{k-1}^{(i)}\right)$ ▷ Particle propagation

$\left\{\mathcal{C}_k^{(c)}\right\}_{c=1}^{C_k} = \text{Clustering}\left(\left\{\Pi_{\text{pos}}\, x_k^{(i)}, w_{k-1}^{(i)}\right\}_{i=1}^{N}\right)$ ▷ **Clustering**

for $c \in [\![1, C_k]\!]$ **do**

$\hat{x}_{k|k-1}^{(c)} \triangleq \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)}\, x_k^{(i)}$ ▷ **Predicted estimation**

$\hat{P}_{k|k-1}^{(c)} = \sum_{i \in \mathcal{C}_k^{(c)}} w_{k-1}^{(i,c)} \left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)}\right)\left(x_k^{(i)} - \hat{x}_{k|k-1}^{(c)}\right)^\top$

Compute $\tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}$ with (4.70) ▷ **Adaptive subsampling**

end for

$\tilde{\mathbf{x}}_{\text{MCAK},k}^{r} = \bigcup_{c=1}^{C_k} \tilde{\mathbf{x}}_{\text{ICAK},k}^{r,(c)}$ ▷ **Cluster merging**

$\tilde{\mathbf{z}}_{\text{MCAK},k}^{r} \triangleq \left\{\tilde{z}_i \mid \tilde{x}_i \in \tilde{\mathbf{x}}_{\text{MCAK},k}^{r}\right\}$

$\hat{\theta}_k = \text{MLE}\left(\tilde{\mathbf{x}}_{\text{MCAK},k}^{r}, \tilde{\mathbf{z}}_{\text{MCAK},k}^{r}, \hat{\theta}_{k-1}\right)$ ▷ **Hyperparameter estimation**

$\forall i \in [\![1, N]\!]$, compute $\tilde{p}\left(z_k | x_k^{(i)}\right)$ from (4.75) ▷ **Kriging**

$\forall i \in [\![1, N]\!],\ \tilde{w}_k^{(i)} = w_{k-1}^{(i)}\, \tilde{p}\left(z_k | x_k^{(i)}\right)$ ▷ Weight update

$\forall i \in [\![1, N]\!],\ w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

$\hat{N}_{\text{eff}} = 1 \Big/ \sum_{i=1}^{N} \left(w_k^{(i)}\right)^2$ ▷ Resampling criterion

if $\hat{N}_{\text{eff}} < N_{\text{th}}$ **then**

$\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)} x_k^{(i)}$ ▷ Whitening

$\hat{P}_k = \sum_{i=1}^{N} w_k^{(i)} \left(x_k^{(i)} - \hat{x}_k\right)\left(x_k^{(i)} - \hat{x}_k\right)^\top$

A_k s.t. $\hat{P}_k = A_k\, A_k^\top$

$\left\{x_k^{(j)}, w_k^{(j)}, -\right\}_{j=1}^{N} \leftarrow \text{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^{N}\right)$ ▷ Resampling

$\forall j \in [\![1, N]\!],\ \varepsilon^{(j)} \sim K$ ▷ Regularisation

$\forall j \in [\![1, N]\!],\ x_k^{(j)} \leftarrow x_k^{(j)} + \alpha_{\text{reg}} h A_k\, \varepsilon^{(j)}$

end if

4.4.4 Numerical Results

The four strategies are evaluated across the six grid resolutions, using the filter with access to the true map as a reference. As in section 4.3, AK, ICAK and MCAK use:

$$\alpha_{\text{cov}} = 2 \quad \text{and} \quad \tilde{N}_{\min} = 25 \tag{4.76}$$

while NNAK retains the nearest:

$$\tilde{N}_{\min} = 10 \tag{4.77}$$

training points for each particle. Finally, mean shift is performed with a bandwidth of:

$$h_{\mathrm{MS}} = 1 \text{ km} \tag{4.78}$$

Figures 4.19 and 4.20 compare the RMSE and NEES curves for the four methods. The associated convergence rates are given by Figure 4.21, while Figure 4.22 shows the number of training points retained by each method throughout the mission. Finally, Table 4.9 presents their average computational cost.

The RMSE curves of AK, NNAK and MCAK are indistinguishable at every grid resolution. Following the shape of the particle set, whether through one window per particle or one window per mode, therefore costs nothing in accuracy compared with the single window of AK. ICAK, on the other hand, achives significantly higher RMSEs on grid resolutions $\Delta_g = 6.5$ and $\Delta_g = 7.5$ km during the first 50 iterations, before rejoining them once the particle set has collapsed to a single mode.

Similarly, all four strategies exhibit almost identical NEES curves and convergence rates. Adapting the shape of the window thus preserves both the consistency and the robustness of AK.

TODO: à reprendre Figure 4.22 explains where the differences lie. Once the particle set has concentrated, AK sits exactly at its floor $\tilde{N}_{\min}$, while MCAK and ICAK sit slightly above it, their window being the union of one disc per mode. NNAK remains below this floor, as announced in section 4.4.1, since the floor is applied to each individual ball rather than to their union. The ordering is reversed during acquisition, where the spread of the particle set drives the window of AK and those of the clusters to a thousand points at $\Delta_g = 2.5$ km, against a few hundred for the union of the balls.

These selections are reflected in the computational costs. At $\Delta_g = 2.5$ km, the kriging of AK costs 46 Gflop per run, against 4.3 G for NNAK, 2.26 G for ICAK and 2.47 G for MCAK, so that adapting the shape of the window divides the dominant term by more than ten, for a clustering step or a neighbour search of about 0.1 G. As the grid coarsens, this term collapses faster than the partition that produces it: at $\Delta_g = 7.5$ km, MCAK pays 150 M of kriging for 274 M of clustering. ICAK is also the only strategy whose hyperparameter estimation grows with the grid resolution, from 87.3 M to 131 M, as it fits one set of hyperparameters per cluster, whereas that of MCAK falls from 25.3 M to 3.4 M.

Merging the clustered windows therefore retains the accuracy, the consistency and the convergence of AK for a tenth of its cost, and is the cheapest of the four strategies at every grid resolution but the finest, where ICAK is marginally below it. Resolving each cluster separately, on the other hand, buys nothing: it degrades the acquisition, and costs more than the merged variant as soon as the clusters multiply.

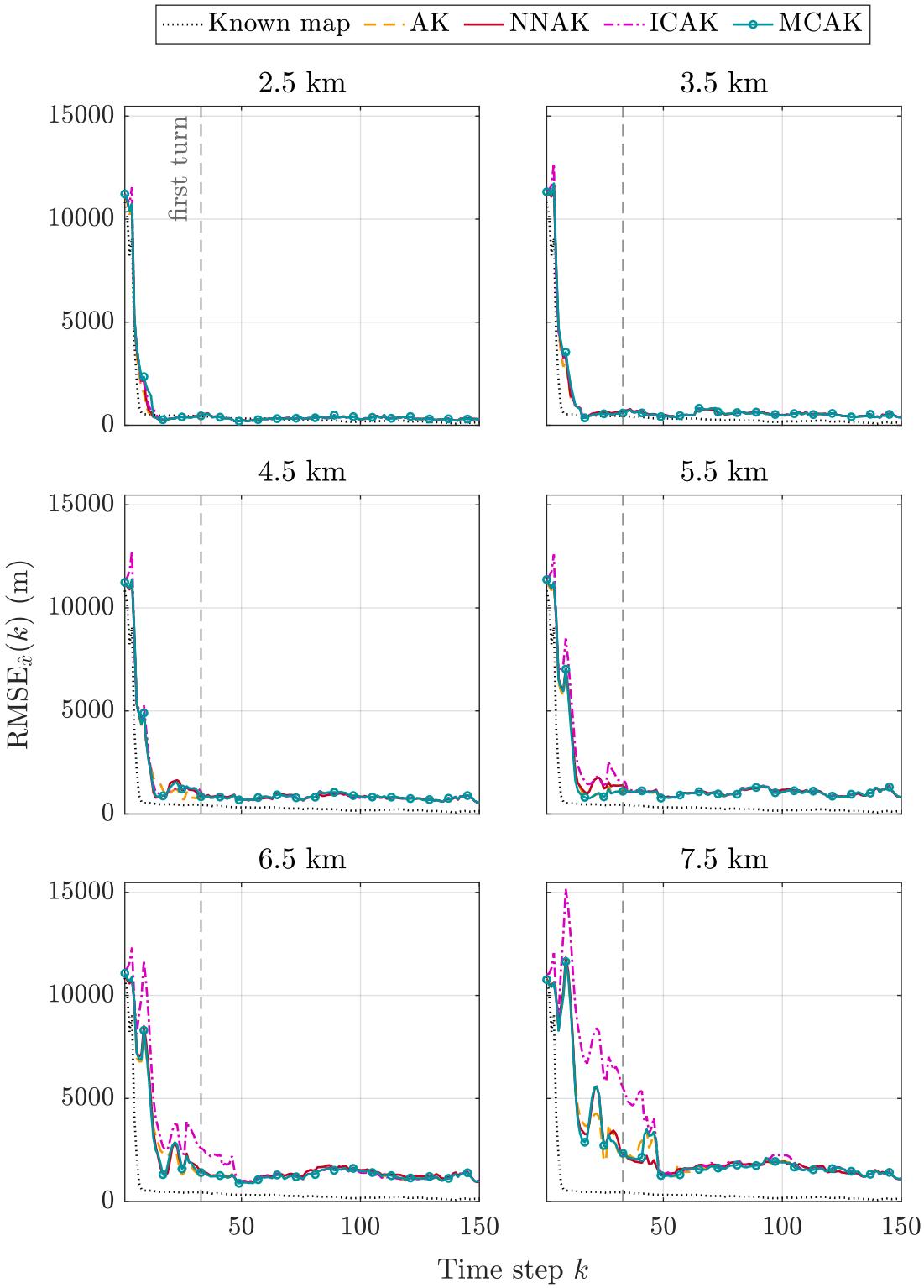


Figure 4.19: Comparative RMSE of the navigation algorithms using AK, NNAK, MCAK and ICAK across the six grid resolutions. The performance of the filter with access to the true map is given as a reference.

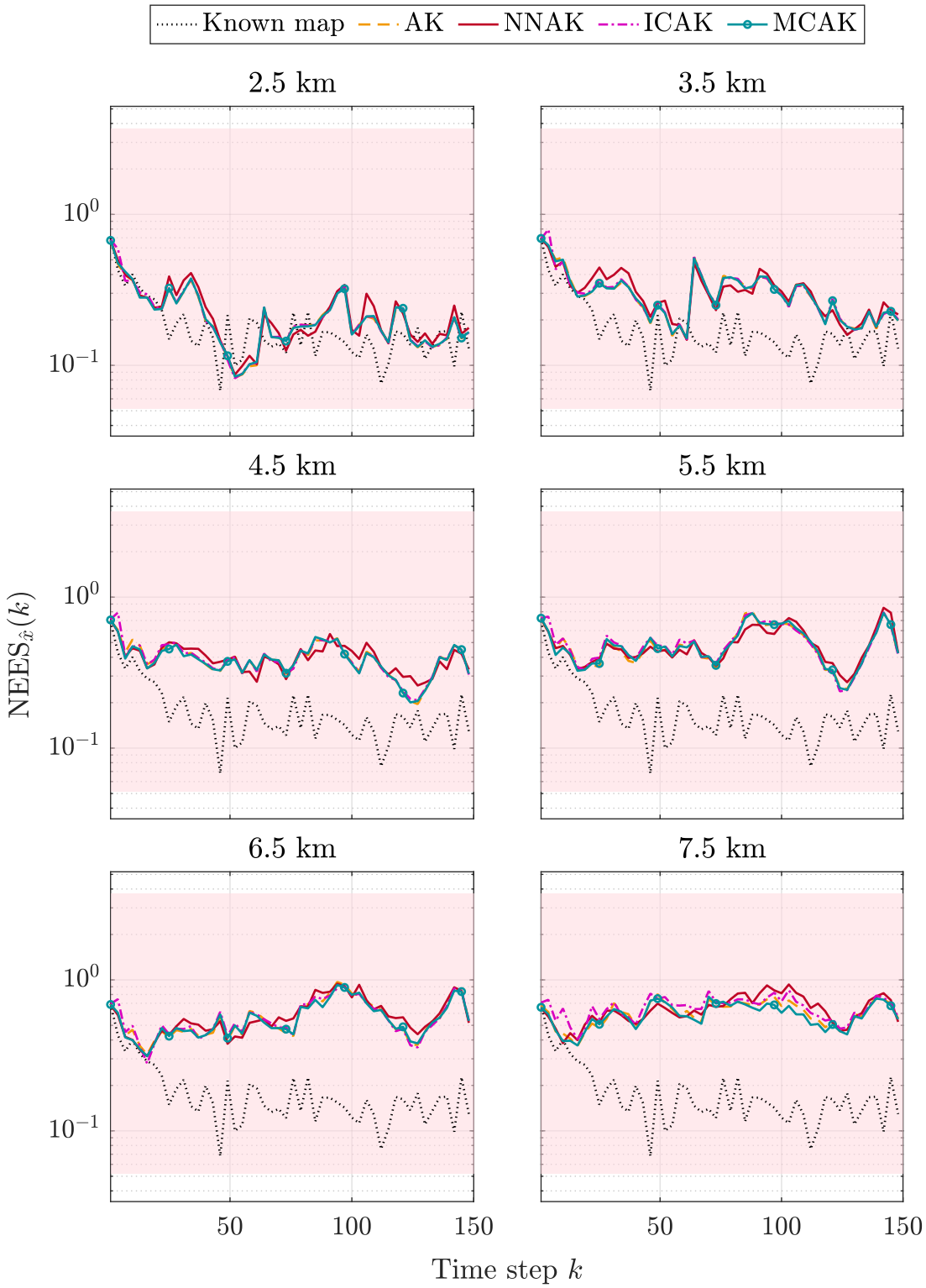


Figure 4.20: Comparative NEES of the navigation algorithms using AK, NNAK, MCAK and ICAK across the six grid resolutions. The shaded area corresponds to that of the previous NEES figures.

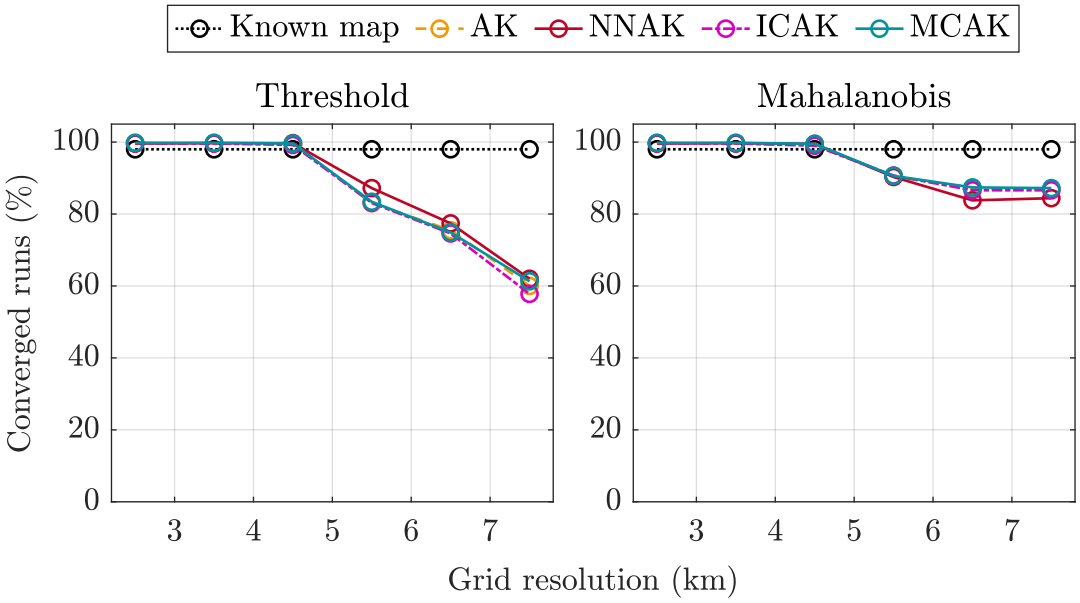


Figure 4.21: Convergence rates of the AK-RPF, NNAK-RPF, MCAK-RPF and ICAK-RPF as a function of grid resolution, using the threshold and the Mahalanobis criteria.

Δ_g	AK		NNAK		
	Kriging	HP	Kriging	HP	NN
2.5 km	46 G	564 M	4.3 G	57 M	47.3 M
3.5 km	14.4 G	224 M	2.05 G	36.1 M	43.6 M
4.5 km	5.75 G	121 M	1.23 G	25.9 M	40.9 M
5.5 km	3.09 G	89.8 M	885 M	21.7 M	38.8 M
6.5 km	1.93 G	70.1 M	705 M	19.4 M	36.8 M
7.5 km	1.54 G	66.7 M	603 M	18.1 M	35.3 M

Δ_g	ICAK			MCAK		
	Kriging	HP	Clustering	Kriging	HP	Clustering
2.5 km	2.26 G	87.3 M	114 M	2.47 G	25.3 M	114 M
3.5 km	901 M	83.2 M	158 M	811 M	21.5 M	158 M
4.5 km	616 M	83.3 M	194 M	496 M	17.2 M	195 M
5.5 km	547 M	91.4 M	223 M	352 M	11.9 M	224 M
6.5 km	513 M	107 M	251 M	241 M	7.12 M	253 M
7.5 km	499 M	131 M	270 M	150 M	3.4 M	274 M

Table 4.9: Total computational cost, in flops, of the four windowing strategies throughout the mission, for each grid resolution. Each cost is split between kriging, hyperparameters estimation, and clustering step when applicable

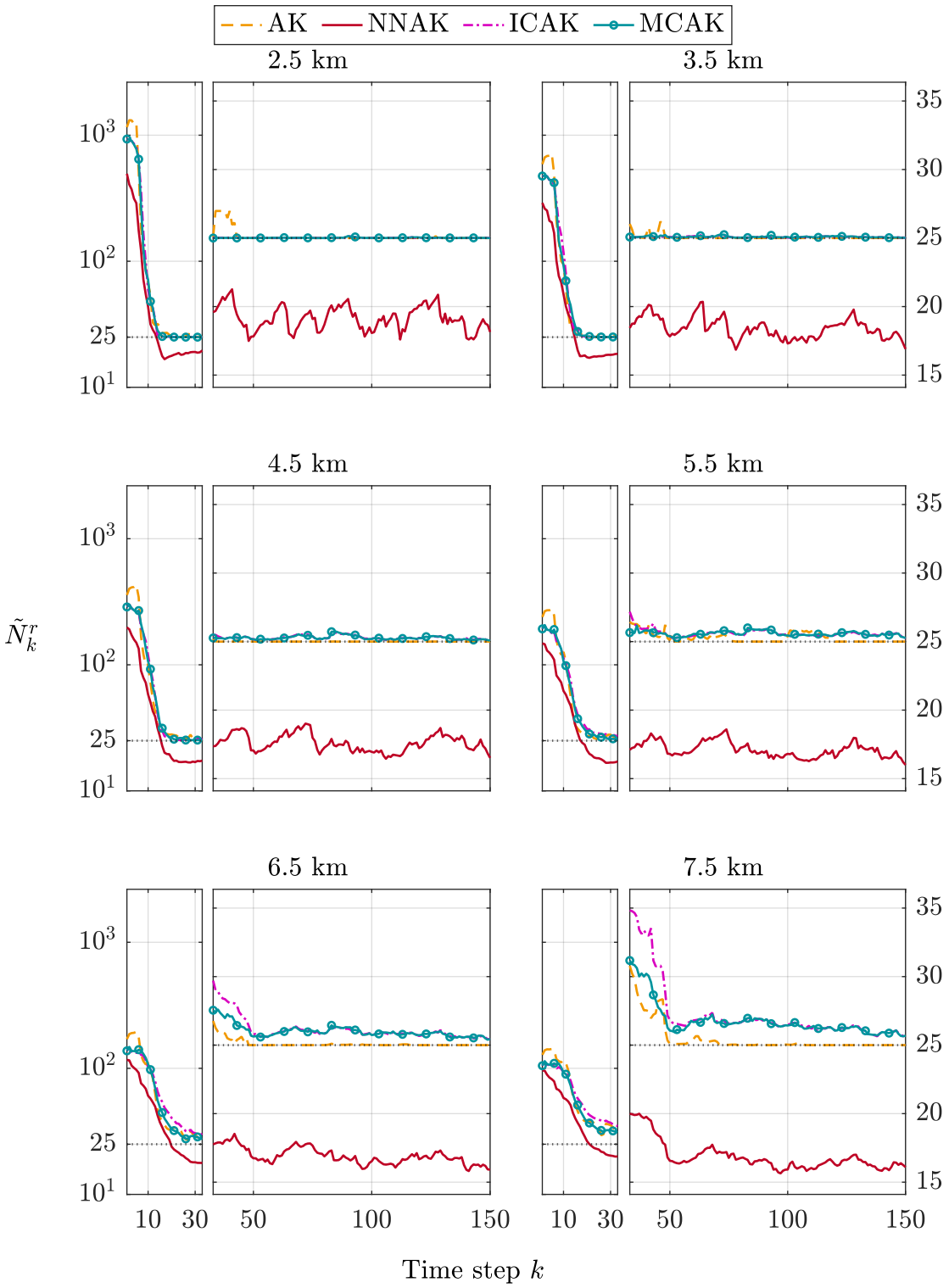


Figure 4.22: Evolution of the average number of training points retained by each windowing strategy during the mission, for each grid resolution. The two scales are those of Figure 4.16.

4.5 Subsampling Error Bounds

In this section, we use the setting and notations of Proposition 3.9, and we recall that ordinary kriging retained in this chapter corresponds to the special case $r_{\mathrm{UK}} = 1$ and $g_1 \equiv 1$. In particular, we write:

$$M(\tilde{\mathbf{x}}) \triangleq \big(g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}})\big)^{-1} \tag{4.79}$$

for the covariance of the estimated trend coefficients $\hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)$.

The kriging equations can be written in a block form, in which the unbiasedness constraints are gathered into the Gram matrix. Formally, let:

$$\mathcal{G}(\tilde{\mathbf{x}}) \triangleq \begin{bmatrix} G(\tilde{\mathbf{x}}) & g(\tilde{\mathbf{x}}) \\ g(\tilde{\mathbf{x}})^\top & 0 \end{bmatrix}, \qquad \bar{k}(\tilde{\mathbf{x}}, x) \triangleq \begin{bmatrix} k(\tilde{\mathbf{x}}, x) \\ g(x)^\top \end{bmatrix}, \qquad \bar{\mathbf{z}} \triangleq \begin{bmatrix} \tilde{\mathbf{z}} \\ 0 \end{bmatrix} \tag{4.80}$$

be obtained by appending the trend to $G(\tilde{\mathbf{x}})$, $k(\tilde{\mathbf{x}}, x)$ and $\tilde{\mathbf{z}}$, with:

$$\bar{k}(x, \tilde{\mathbf{x}}) = \bar{k}(\tilde{\mathbf{x}}, x)^\top \tag{4.81}$$

Proposition 4.2 shows that these blocks give universal kriging a form similar to that of simple kriging.

Proposition 4.2 (Block form of universal kriging)**.** *The matrix $\mathcal{G}(\tilde{\mathbf{x}})$ is invertible, the UK estimator and covariance matrix verify:*

$$\begin{cases} \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{\mathbf{z}} \\ R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, g) = k(x, x) + R - \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{k}(\tilde{\mathbf{x}}, x) \end{cases} \tag{4.82}$$

Proof. For the sake of brevity, the arguments of G, g and M are omitted here, all three being taken at $\tilde{\mathbf{x}}$. The Schur complement of $\mathcal{G}(\tilde{\mathbf{x}})$ with respect to G is:

$$0 - g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}) = -M^{-1} \tag{4.83}$$

which is invertible. Proposition A.2 thus applies, and gives:

$$\mathcal{G}(\tilde{\mathbf{x}})^{-1} = \begin{bmatrix} G^{-1} - G^{-1} g\, M\, g^\top G^{-1} & G^{-1} g\, M \\ M\, g^\top G^{-1} & -M \end{bmatrix} \tag{4.84}$$

Using:

$$\hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) = M\, g^\top G^{-1} \tilde{\mathbf{z}} \tag{4.85}$$

it comes that:

$$\begin{aligned} \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{\mathbf{z}} &= \begin{bmatrix} k(x, \tilde{\mathbf{x}}) & g(x) \end{bmatrix} \begin{bmatrix} G^{-1} - G^{-1} g\, M\, g^\top G^{-1} & G^{-1} g\, M \\ M\, g^\top G^{-1} & -M \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{z}} \\ 0 \end{bmatrix} \\ &= \begin{bmatrix} k(x, \tilde{\mathbf{x}}) & g(x) \end{bmatrix} \begin{bmatrix} G^{-1} \left(\tilde{\mathbf{z}} - g\, M\, g^\top G^{-1} \tilde{\mathbf{z}}\right) \\ M\, g^\top G^{-1} \tilde{\mathbf{z}} \end{bmatrix} \\ &= \begin{bmatrix} k(x, \tilde{\mathbf{x}}) & g(x) \end{bmatrix} \begin{bmatrix} G^{-1} (\tilde{\mathbf{z}} - g\, \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)) \\ \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) \end{bmatrix} \\ &= k(x, \tilde{\mathbf{x}})\, G^{-1} (\tilde{\mathbf{z}} - g\, \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)) + g(x)\, \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) \\ &= \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) \end{aligned} \tag{4.86}$$

Similarly, writing:**TODO: reprendre ici**

$$V(x) \triangleq g(x) - k(x, \tilde{\mathbf{x}})\, G^{-1} g \tag{4.87}$$

for the weights given to the estimated trend coefficients gives:

$$\mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{k}(\tilde{\mathbf{x}}, x) = \begin{bmatrix} G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} g\, M\, V(x)^\top \\ -M\, V(x)^\top \end{bmatrix} \tag{4.88}$$

and therefore:

$$\bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{k}(\tilde{\mathbf{x}}, x) = k(x, \tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}}, x) - V(x)\, M\, V(x)^\top \tag{4.89}$$

The subtracted term is $R_{\hat{m}_{\mathrm{UK}}}(x, \tilde{\mathbf{x}}, g)$, as given by (3.59), which gives the second line of (3.58). □

A major limitation of this method is the $\mathcal{O}(\tilde{N}^3)$ computational cost, stemming from the inversion of the Gram matrix $G(\tilde{\mathbf{x}})$, or equivalently of $\mathcal{G}(\tilde{\mathbf{x}})$, in Proposition 4.2. One common approach to mitigate this is to omit certain elements of $\tilde{\mathbf{x}}$ to reduce the size of $G(\tilde{\mathbf{x}})$. This is the principle of the methods presented in sections 4.3 and 4.4. However, this approximation introduces a loss of accuracy in (**??**). This section quantifies the error induced by such subsampling approximations. We first define two metrics for this error, and express them as functions of residual quantities that describe the information carried by the discarded data alone. We then derive bounds that do not require inverting any matrix of the size of these data.

4.5.1 Subsampling Error Metrics

Formally, subsampling is achieved by splitting $\tilde{\mathbf{x}}$ into $\begin{bmatrix} \tilde{\mathbf{x}}^r & \tilde{\mathbf{x}}^d \end{bmatrix}^\top$, with $\tilde{\mathbf{x}}^r$ being the $\tilde{N}^r$ retained data, usually close to the considered kriging location $x \in \mathcal{U}$, and $\tilde{\mathbf{x}}^d$ being the $\tilde{N}^d$ discarded data, usually far from x. A similar operation is performed on $\tilde{\mathbf{z}}$. This decomposition provides the foundation for all subsequent error bounds, as it explicitly isolates the predictive contribution of the discarded points from that of the retained training set. We assume that $g(\tilde{\mathbf{x}}^r)$ has full column rank, which requires $\tilde{N}^r \geq r_{\mathrm{UK}}$, so that the subsampled estimator is well defined.

Two error metrics between $\hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g)$ and $\hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$ are considered here, each capturing a distinct aspect of the kriging approximation. The first metric corresponds to the covariance of their difference, and is introduced by Definition 4.1 together with the associated standard deviation. In contrast, since the kriging estimator is a deterministic function conditionally to $\tilde{\mathbf{z}}$, a natural alternative is to measure the distance between the two estimators, as proposed by Definition 4.2.

Definition 4.1 (Subsampling error covariance)**.** *The subsampling error covariance E_r is the covariance of the difference between the full and subsampled estimators:*

$$E_r : x \in \mathcal{U} \mapsto \mathbb{V}\left[\hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)\right] \tag{4.90}$$

The associated error e_r is the largest standard deviation of this difference over the unit directions of $\mathbb{R}^{d_z}$:

$$e_r : x \in \mathcal{U} \mapsto \max_{\|a\|_2 = 1} \sqrt{a^\top E_r(x)\, a} = \sqrt{\lambda_{\max}\left(E_r(x)\right)} \tag{4.91}$$

Definition 4.2 (Interpolation error)**.** *The interpolation error e_f between the full and subsampled estimators is defined as:*

$$e_f : x \in \mathcal{U} \mapsto \|\hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)\|_2 \tag{4.92}$$

It must be noted that neither metric depends on the unknown trend coefficients β. Indeed, since both estimators are unbiased, their difference is a linear combination of the training observations whose weights cancel the trend matrix $g(\tilde{\mathbf{x}})$.

4.5.2 Importance of the Gram Matrix

As stated above, most of the computational cost comes from inverting the Gram matrix. Up to a reordering of its rows and columns, in which the retained data and the unbiasedness constraints come first, the decomposition introduced in section 4.5.1 leads to the following block partitioning:

$$\mathcal{G}(\tilde{\mathbf{x}}) = \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} = \begin{bmatrix} \mathcal{G}(\tilde{\mathbf{x}}^r) & \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)^\top & G(\tilde{\mathbf{x}}^d) \end{bmatrix} \text{ with } \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \triangleq \begin{bmatrix} k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ g(\tilde{\mathbf{x}}^d)^\top \end{bmatrix} \tag{4.93}$$

The vectors of Proposition 4.2 are reordered accordingly:

$$\bar{k}(\tilde{\mathbf{x}}, x) = \begin{bmatrix} \bar{k}(\tilde{\mathbf{x}}^r, x) \\ k(\tilde{\mathbf{x}}^d, x) \end{bmatrix}, \qquad \bar{\mathbf{z}} = \begin{bmatrix} \bar{\mathbf{z}}^r \\ \tilde{\mathbf{z}}^d \end{bmatrix}, \qquad \bar{\mathbf{z}}^r \triangleq \begin{bmatrix} \tilde{\mathbf{z}}^r \\ 0 \end{bmatrix} \tag{4.94}$$

Let $\mathcal{G}_h(\tilde{\mathbf{x}}^r)$ be the hollowed-out Gram matrix, of the same size $(\tilde{N} + r_{\mathrm{UK}})d_z$ as $\mathcal{G}(\tilde{\mathbf{x}})$:

$$\mathcal{G}_h(\tilde{\mathbf{x}}^r) \triangleq \begin{bmatrix} \mathcal{G}(\tilde{\mathbf{x}}^r) & 0_{(\tilde{N}^r + r_{\mathrm{UK}})d_z, \tilde{N}^d d_z} \\ 0_{\tilde{N}^d d_z, (\tilde{N}^r + r_{\mathrm{UK}})d_z} & 0_{\tilde{N}^d d_z, \tilde{N}^d d_z} \end{bmatrix} \tag{4.95}$$

Proposition 4.3 states that $\mathcal{G}_h(\tilde{\mathbf{x}}^r)$ acts as a substitute for $\mathcal{G}(\tilde{\mathbf{x}})$ in the subsampled case with respect to Proposition 4.2.

Proposition 4.3 (Subsampled kriging estimator)**.** *The subsampled kriging estimator verifies:*

$$\begin{cases} \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) = \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger\, \bar{\mathbf{z}} \\ R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, g) = k(x, x) + R - \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger\, \bar{k}(\tilde{\mathbf{x}}, x) \end{cases} \tag{4.96}$$

where $\dagger$ denotes the Moore-Penrose pseudo-inverse symbol [7].

Proof. The result comes from applying Proposition 4.2 to $\tilde{\mathbf{x}}^r$, and noting that:

$$\mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger = \begin{bmatrix} \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} & 0 \\ 0 & 0 \end{bmatrix} \tag{4.97}$$

□

Although it is not directly used in computations, Proposition 4.3 further highlights the importance of the Gram matrix in the subsampling problem. In particular, Definition 4.3 serves as the foundation for the key reformulations of the studied errors established by Proposition 4.4.

Definition 4.3 (Gram inverse difference matrix)**.** *The Gram inverse difference matrix $\mathfrak{D}_G$ is the quantity defined as:*

$$\mathfrak{D}_G \triangleq \mathcal{G}(\tilde{\mathbf{x}})^{-1} - \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger \tag{4.98}$$

Proposition 4.4 (Errors and Gram inverse difference)**.** *The two subsampling errors can be respectively expressed as:*

$$\begin{cases} E_r(x) = \bar{k}(x, \tilde{\mathbf{x}})\, \mathfrak{D}_G \bar{k}(\tilde{\mathbf{x}}, x) \\ e_f(x) = \left\| \bar{k}(x, \tilde{\mathbf{x}})\, \mathfrak{D}_G \bar{\mathbf{z}} \right\|_2 \end{cases} \tag{4.99}$$

Proof. See section B.14. □

Remark 4.8 (Covariance inflation)**.** *An alternative reformulation of $E_r(x)$ is obtained in the proof of Proposition 4.4, and is worth comparing with the original definition of $e_f(x)$ given in Definition 4.2:*

$$E_r(x) = R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) \tag{4.100}$$

In particular, $R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) \succeq R_{\text{UK}}(x, \tilde{\mathbf{x}}, g)$. Subsampling thus increases the covariance reported by the estimator by exactly the covariance of the error it introduces, and so the subsampled estimator is not overconfident. ***alors oui mais surtout on en déduit que l'erreur du sous échantillonnage est bornée par la covariance du krigeage sur tout le jeu de données***

4.5.3 Block Inversion and Residual Quantities

To isolate the impact of discarding $\tilde{\mathbf{x}}^d$ on the kriging estimator, $\mathfrak{D}_G$ is decomposed using the Schur inversion formula (see Proposition A.2):

$$\begin{aligned} \mathfrak{D}_G &= \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix}^{-1} - \begin{bmatrix} A^{-1} & 0 \\ 0 & 0 \end{bmatrix} \\ &= \begin{bmatrix} A^{-1}BS_{d|r}^{-1}B^\top A^{-1} & -A^{-1}BS_{d|r}^{-1} \\ -S_{d|r}^{-1}B^\top A^{-1} & S_{d|r}^{-1} \end{bmatrix} \\ &= \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^d d_z} \end{bmatrix} S_{d|r}^{-1} \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^d d_z} \end{bmatrix}^\top \end{aligned} \tag{4.101}$$

where $S_{d|r}$ is the Schur complement of $\mathcal{G}(\tilde{\mathbf{x}})$ with respect to $A = \mathcal{G}(\tilde{\mathbf{x}}^r)$, which is invertible since both $\mathcal{G}(\tilde{\mathbf{x}})$ and A are:

$$S_{d|r} \triangleq D - B^\top A^{-1} B \tag{4.102}$$

This decomposition, combined with Proposition 4.4, reveals two residual quantities that contribute to the error introduced by subsampling. The first one, called residual covariance function, quantifies the portion of the covariance between x and the discarded training set $\tilde{\mathbf{x}}^d$ not accounted for by the retained set $\tilde{\mathbf{x}}^r$. It is the object of Definition 4.4. Similarly, the residual observation vector captures the information contained within the discarded observations $\tilde{\mathbf{z}}^d$ and unexplained by the retained observations $\tilde{\mathbf{z}}^r$. Its expression is given by Definition 4.5.

Definition 4.4 (Residual covariance function)**.** *The residual covariance function* $k_{d|r}$ *is given by:*

$$k_{d|r} : x \in \mathcal{U} \mapsto k(\tilde{\mathbf{x}}^d, x) - \bar{k}(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\, \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} \bar{k}(\tilde{\mathbf{x}}^r, x) \tag{4.103}$$

with $\bar{k}(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r) \triangleq \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)^\top$.

Definition 4.5 (Residual observation vector)**.** *The residual observation vector* $\tilde{\mathbf{z}}_{d|r}$ *is defined as:*

$$\tilde{\mathbf{z}}_{d|r} \triangleq \tilde{\mathbf{z}}^d - \bar{k}(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\, \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} \bar{\mathbf{z}}^r \tag{4.104}$$

Proposition 4.5 gives relevant identities that link (4.101) with $k_{d|r}$ and $\tilde{\mathbf{z}}_{d|r}$. Proposition 4.6 and Proposition 4.8 push this further by showing that the studied metrics can be fully expressed as functions of those residual quantities. In between, Proposition 4.7 relates these quantities to their simple kriging counterparts.

Proposition 4.5 (Residual identities)**.** *The following residual identities hold:*

$$\begin{cases} k_{d|r}(x) = \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^d d_z} \end{bmatrix}^\top \bar{k}(\tilde{\mathbf{x}}, x) \\ k_{d|r}(\tilde{\mathbf{x}}^d) = S_{d|r} - I_{\tilde{N}^d} \otimes \tilde{R} \\ \tilde{\mathbf{z}}_{d|r} = \begin{bmatrix} -A^{-1}B \\ I_{\tilde{N}^d d_z} \end{bmatrix}^\top \bar{\mathbf{z}} \end{cases} \tag{4.105}$$

Proof. Recall that $A = \mathcal{G}(\tilde{\mathbf{x}}^r)$ and $B = \bar{k}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)$. Direct calculations give the desired results. □

Proposition 4.6 (Residual forms of the errors)**.** *The residual forms of the two errors are given below:*

$$\begin{cases} E_r(x) = k_{d|r}(x)^\top \left(k_{d|r}(\tilde{\mathbf{x}}^d) + I_{\tilde{N}^d} \otimes \tilde{R} \right)^{-1} k_{d|r}(x) \\ e_f(x) = \left\| k_{d|r}(x)^\top \left(k_{d|r}(\tilde{\mathbf{x}}^d) + I_{\tilde{N}^d} \otimes \tilde{R} \right)^{-1} \tilde{\mathbf{z}}_{d|r} \right\|_2 \end{cases} \tag{4.106}$$

Proof. These expressions are obtained by applying (4.101) and Proposition 4.5 to Proposition 4.4. □

Proposition 4.7 (Link with simple kriging)**.** *Let* $k^0_{d|r}$*,* $S^0_{d|r}$ *and* $\tilde{\mathbf{z}}^0_{d|r}$ *be the residual quantities of simple kriging, obtained by replacing* $\mathcal{G}(\tilde{\mathbf{x}}^r)$ *with the Gram matrix alone:*

$$\begin{cases} k^0_{d|r} : x \in \mathcal{U} \mapsto k(\tilde{\mathbf{x}}^d, x) - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\, G(\tilde{\mathbf{x}}^r)^{-1} k(\tilde{\mathbf{x}}^r, x) \\ S^0_{d|r} \triangleq G(\tilde{\mathbf{x}}^d) - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\, G(\tilde{\mathbf{x}}^r)^{-1} k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ \tilde{\mathbf{z}}^0_{d|r} \triangleq \tilde{\mathbf{z}}^d - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\, G(\tilde{\mathbf{x}}^r)^{-1} \tilde{\mathbf{z}}^r \end{cases} \tag{4.107}$$

and let:

$$U : x \in \mathcal{U} \mapsto g(x) - k(x, \tilde{\mathbf{x}}^r)\, G(\tilde{\mathbf{x}}^r)^{-1} g(\tilde{\mathbf{x}}^r) \tag{4.108}$$

be the weights given to the estimated trend coefficients by the subsampled estimator. Then:

$$\begin{cases} k_{d|r}(x) = k^0_{d|r}(x) + U(\tilde{\mathbf{x}}^d)\, M(\tilde{\mathbf{x}}^r)\, U(x)^\top \\ S_{d|r} = S^0_{d|r} + U(\tilde{\mathbf{x}}^d)\, M(\tilde{\mathbf{x}}^r)\, U(\tilde{\mathbf{x}}^d)^\top \\ \tilde{\mathbf{z}}_{d|r} = \tilde{\mathbf{z}}^0_{d|r} - U(\tilde{\mathbf{x}}^d)\, \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) \end{cases} \tag{4.109}$$

Furthermore, the covariances of the trend coefficients estimated from $\tilde{\mathbf{x}}$ *and from* $\tilde{\mathbf{x}}^r$ *are linked by:*

$$M(\tilde{\mathbf{x}})^{-1} = M(\tilde{\mathbf{x}}^r)^{-1} + U(\tilde{\mathbf{x}}^d)^\top \left(S^0_{d|r}\right)^{-1} U(\tilde{\mathbf{x}}^d) \tag{4.110}$$

In particular, $S_{d|r}$ *is SPD, and* $\tilde{\mathbf{z}}_{d|r}$ *is unchanged when a trend* $g(\tilde{\mathbf{x}})\, \beta_0$*, with* $\beta_0 \in \mathbb{R}^{r_{\mathrm{UK}} d_z}$*, is added to the observations.*

Proof. See section B.15. □

Proposition 4.8 (Errors as Mahalanobis norms)**.** *Let* $\|\cdot\|_{\mathrm{op}}$ *denote the operator norm associated with the Euclidean norm* $\|\cdot\|_2$*, and let* $\langle\cdot|\cdot\rangle_{S_{d|r}}$ *be the Mahalanobis inner product weighted by the inverse of* $S_{d|r} = k_{d|r}(\tilde{\mathbf{x}}^d) + I_{\tilde{N}^d} \otimes \tilde{R}$*, which is SPD by Proposition 4.7:*

$$\forall n \in \mathbb{N}^*,\ \forall Y, Y' \in \mathcal{M}_{\tilde{N}^d d_z, n}(\mathbb{R}),\ \begin{cases} \langle Y | Y' \rangle_{S_{d|r}} \triangleq Y^\top S^{-1}_{d|r} Y' \\ \|Y\|_{S_{d|r}} \triangleq \left\| S^{-\frac{1}{2}}_{d|r} Y \right\|_{\mathrm{op}} \end{cases} \tag{4.111}$$

The kriging errors can be interpreted as metrics in the Euclidean space to which the columns of $k_{d|r}(x)$ *and* $\tilde{\mathbf{z}}_{d|r}$ *belong, namely* $(\mathbb{R}^{\tilde{N}^d d_z}, \langle\cdot|\cdot\rangle_{S_{d|r}})$*:*

$$\begin{cases} E_r(x) = \langle k_{d|r}(x) | k_{d|r}(x) \rangle_{S_{d|r}}, & e_r(x) = \|k_{d|r}(x)\|_{S_{d|r}} \\ e_f(x) = \left\| \langle k_{d|r}(x) | \tilde{\mathbf{z}}_{d|r} \rangle_{S_{d|r}} \right\|_2 \end{cases} \tag{4.112}$$

Proof. These expressions are direct from Propositions 4.5 and 4.6, since $e_r(x)^2 = \lambda_{\max}\left(E_r(x)\right)$. □

Remark 4.9 (Uncorrelated discarded data)**.** *Contrary to simple kriging, discarding data that are uncorrelated with both* $\tilde{\mathbf{x}}^r$ *and* x *still introduces an error. Indeed, if* $k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r) = 0$ *and* $k(\tilde{\mathbf{x}}^d, x) = 0$*, then* $k^0_{d|r}(x) = 0$ *and* $U(\tilde{\mathbf{x}}^d) = g(\tilde{\mathbf{x}}^d)$*, and Propositions 4.7 and 4.8 yield:*

$$\begin{cases} E_r(x) = U(x)\left(M(\tilde{\mathbf{x}}^r) - M(\tilde{\mathbf{x}})\right) U(x)^\top \\ e_f(x) = \left\| U(x)\left(\hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)\right) \right\|_2 \end{cases} \tag{4.113}$$

In this case, the discarded data still reduce the covariance $M(\tilde{\mathbf{x}})$ *of the estimated trend coefficients, and both errors correspond to the loss of accuracy of this estimation. For ordinary kriging,* $E_r(x)$ *is thus the increase in covariance of the estimated mean, weighted by* $U(x)$*.*

4.5.4 Relations Between the Errors

This section introduces two relations between $e_r(x)$ and $e_f(x)$. The first relation is the concentration inequality described by Proposition 4.9, while the second comes from algebraic considerations, and is given by Proposition 4.10.

Proposition 4.9 (Concentration inequality)**.** *Let $\chi^2_{d_z}(1-\delta)$ denote the quantile of order $1-\delta$ of the χ^2 distribution with d_z degrees of freedom. For $\delta \in (0,1]$:*

$$e_f(x) \leq \sqrt{\chi^2_{d_z}(1-\delta)}\, e_r(x) \tag{4.114}$$

with a probability of at least $1-\delta$.

Proof. Since both estimators are unbiased linear functions of the Gaussian vector $\tilde{\mathbf{z}}$, their difference follows a centred Gaussian distribution of covariance $E_r(x)$. It can thus be written as $E_r(x)^{\frac{1}{2}}\xi$ with $\xi \sim \mathcal{N}(0, I_{d_z})$, which gives:

$$e_f(x)^2 = \xi^\top E_r(x)\, \xi \leq \lambda_{\max}\left(E_r(x)\right) \|\xi\|_2^2 = e_r(x)^2\, \|\xi\|_2^2 \tag{4.115}$$

Since $\|\xi\|_2^2$ follows a χ^2 distribution with d_z degrees of freedom, we have $\|\xi\|_2^2 \leq \chi^2_{d_z}(1-\delta)$ with a probability of $1-\delta$, which concludes the proof. □

Remark 4.10 (Scalar field)**.** *For a scalar field, a simpler, albeit more pessimistic, version of Proposition 4.9 can be derived from the inverse form of Mill's inequality [115]:*

$$\forall \delta \in (0,1],\ \mathbb{P}\left(e_f(x) \leq \sqrt{2\log\left(\frac{2}{\delta}\right)}\, e_r(x)\right) \geq 1-\delta \tag{4.116}$$

Proposition 4.10 (Cauchy-Schwarz inequality)**.** *$e_r(x)$ and $e_f(x)$ are linked by the following inequality:*

$$e_f(x) \leq e_r(x)\, \|\tilde{\mathbf{z}}_{d|r}\|_{S_{d|r}} \tag{4.117}$$

Proof. Applying the Cauchy-Schwarz inequality [7] to Proposition 4.8 proves the relation. □

4.5.5 Computable Error Bounds

The residual forms of Proposition 4.8 still involve $S^{-1}_{d|r}$, whose size $\tilde{N}^d d_z$ is that of the discarded observations. Computing $E_r(x)$ exactly is thus as costly as the full kriging estimator that subsampling aims to avoid. This section derives upper bounds that do not require this inversion. We first separate the contributions of simple kriging and of the trend estimation. We then nest the retained data into a larger set, whose own subsampling error is small. Finally, we derive bounds from entrywise bounds on the covariances between the data.

Simple Kriging and Trend Contributions

Proposition 4.7 splits the residual covariance function into its simple kriging counterpart $k^0_{d|r}(x)$ and a trend contribution. Proposition 4.11 bounds these two parts separately.

Proposition 4.11 (Simple kriging and trend contributions)**.** *Let* $\|\cdot\|_{S^0_{d|r}}$ *be defined as in Proposition 4.8, with* $S^0_{d|r}$ *in place of* $S_{d|r}$*, and let* $e_{r,0}(x) \triangleq \|k^0_{d|r}(x)\|_{S^0_{d|r}}$ *be the error that simple kriging would commit if the trend were known. Then:*

$$e_r(x) \leq e_{r,0}(x) + \sqrt{\|U(x)\,(M(\tilde{\mathbf{x}}^r) - M(\tilde{\mathbf{x}}))\,U(x)^\top\|_{\mathrm{op}}} \tag{4.118}$$

with equality when $k^0_{d|r}(x) = 0$*.*

Proof. See section B.16. □

The second term of this bound is the contribution of the trend estimation, as described by Remark 4.9. However, $e_{r,0}(x)$ still requires the inversion of $S^0_{d|r}$, and $M(\tilde{\mathbf{x}})$ that of $G(\tilde{\mathbf{x}})$. When the training observations are noisy, Proposition 4.12 removes both inversions.

Proposition 4.12 (Computable bound)**.** *Assume that* $\tilde{R} \succ 0$*, and let* $\tilde{\lambda} \triangleq \lambda_{\min}(\tilde{R})$*. Then* $e_r(x) \leq e_r^{\max}(x)$*, where:*

$$\begin{aligned} e_r^{\max}(x) &\triangleq \frac{\|k^0_{d|r}(x)\|_{\mathrm{op}}}{\sqrt{\tilde{\lambda}}} \\ &+ \sqrt{\left\|U(x)\left(M(\tilde{\mathbf{x}}^r) - \left(M(\tilde{\mathbf{x}}^r)^{-1} + \frac{U(\tilde{\mathbf{x}}^d)^\top U(\tilde{\mathbf{x}}^d)}{\tilde{\lambda}}\right)^{-1}\right)U(x)^\top\right\|_{\mathrm{op}}} \end{aligned} \tag{4.119}$$

Proof. See section B.16. □

Remark 4.11 (Cost of the computable bound)**.** *Unlike the residual forms,* $e_r^{\max}(x)$ *only requires the inversion of* $G(\tilde{\mathbf{x}}^r)$*, which is already performed by the subsampled estimator. Its cost is thus* $\mathcal{O}\left(d_z^3\left(\left(\tilde{N}^r\right)^3 + \tilde{N}^r\tilde{N}^d\right)\right)$*, and grows linearly with the number of discarded data. Moreover, by Proposition 4.9, it also bounds the interpolation error with a probability of at least* $1-\delta$*:*

$$e_f(x) \leq \sqrt{\chi^2_{d_z}(1-\delta)}\, e_r^{\max}(x) \tag{4.120}$$

Remark 4.12 (Noise-dominated regime)**.** *The two terms of Proposition 4.11 behave differently when the training noise increases. Formally, let* $\tilde{R} = \sigma^2\tilde{R}_0$*, with* $\tilde{R}_0 \succ 0$ *fixed and* $\sigma \to +\infty$*. The simple kriging term* $e_{r,0}(x)$ *then vanishes, since the discarded and retained data become equally uninformative about the field. On the other hand, the trend term does not. In particular, for ordinary kriging, it comes that:*

$$E_r(x) \underset{\sigma\to+\infty}{\sim} \left(\frac{1}{\tilde{N}^r} - \frac{1}{\tilde{N}}\right)\tilde{R} \tag{4.121}$$

In this regime, both estimators reduce to averages of the observations, and discarding data increases the covariance of the estimated mean.

Nested Subsampling

The bound of Proposition 4.12 replaces $\left(S^0_{d|r}\right)^{-1}$ with $\left(I_{\tilde{N}^d} \otimes \tilde{R}\right)^{-1}$, which is pessimistic when the discarded data are strongly correlated with the retained ones. This is typically the case of the discarded data lying just outside a spatial window. Proposition 4.13 handles these data exactly, and only applies the bound to the remaining ones.

Proposition 4.13 (Nested subsampling)**.** *Let $\tilde{\mathbf{x}}^e$ be a set of training data such that $\tilde{\mathbf{x}}^r \subset \tilde{\mathbf{x}}^e \subset \tilde{\mathbf{x}}$. Let $E_e(x)$ denote the error covariance committed by subsampling $\tilde{\mathbf{x}}$ into $\tilde{\mathbf{x}}^e$, and $E_{r|e}(x)$ the one committed by subsampling $\tilde{\mathbf{x}}^e$ into $\tilde{\mathbf{x}}^r$. Then:*

$$E_r(x) = E_{r|e}(x) + E_e(x) \tag{4.122}$$

Furthermore, if $\tilde{R} \succ 0$ and $e_e^{\max}(x)$ denotes the bound of Proposition 4.12 for the retained set $\tilde{\mathbf{x}}^e$:

$$\left\|E_{r|e}(x)\right\|_{\text{op}} \leq e_r(x)^2 \leq \left\|E_{r|e}(x)\right\|_{\text{op}} + e_e^{\max}(x)^2 \tag{4.123}$$

Proof. By Remark 4.8, applied to each of the three subsamplings, we have:

$$\begin{cases} E_r(x) = R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) \\ E_e(x) = R_{\text{UK}}(x, \tilde{\mathbf{x}}^e, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}, g) \\ E_{r|e}(x) = R_{\text{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\text{UK}}(x, \tilde{\mathbf{x}}^e, g) \end{cases} \tag{4.124}$$

which gives (4.122). Since $E_{r|e}(x)$ and $E_e(x)$ are SPSD, it follows that:

$$\lambda_{\max}\left(E_{r|e}(x)\right) \leq \lambda_{\max}\left(E_r(x)\right) \leq \lambda_{\max}\left(E_{r|e}(x)\right) + \lambda_{\max}\left(E_e(x)\right) \tag{4.125}$$

and Proposition 4.12 bounds the last eigenvalue by $e_e^{\max}(x)^2$. □

Remark 4.13 (Choice of the intermediate set)**.** *The term $E_{r|e}(x)$ only involves the $\tilde{N}^e$ data of $\tilde{\mathbf{x}}^e$, and is computed exactly at the $\mathcal{O}\left(\left(\tilde{N}^e\right)^3 d_z^3\right)$ cost of two kriging covariances. It also provides a lower bound on the error, which Proposition 4.12 does not. Moreover, the bracket tightens as $\tilde{\mathbf{x}}^e$ grows, since fewer data are left to Proposition 4.12. In particular, when $\tilde{\mathbf{x}}^e$ gathers $\tilde{\mathbf{x}}^r$ and the discarded data lying within a few correlation lengths of x, the remaining data are screened by the closer ones for most kernels used in practice [116]. In this case, $e_e^{\max}(x)$ is small, while $\tilde{N}^e$ remains far below $\tilde{N}$.*

Entrywise Bounds

The previous bounds are evaluated on the data at hand. Bounds on the residual quantities can also be derived from bounds on the covariances between the data, which make the dependence of the errors on these covariances explicit. Formally, let $\|K\|_{\max}$ denote the largest operator norm among the blocks of a matrix K, of size $d_z \times d_z$ for a covariance matrix and $d_z \times 1$ for an observation vector. It is thus the

largest entry of K in absolute value when $d_z = 1$. For $j \in \{r, d\}$, we write $\tilde{x}_i^j$ for the i-th element of $\tilde{\mathbf{x}}^j$, and we assume that:

$$\begin{cases} \left\|k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\right\|_{\max} \leq \alpha, \qquad \left\|k(\tilde{\mathbf{x}}^r, x)\right\|_{\max} \leq \kappa, \qquad \left\|k(\tilde{\mathbf{x}}^d, x)\right\|_{\max} \leq \eta \\ \left\|k(\tilde{x}_i^j, \tilde{x}_l^j)\right\|_{\text{op}} \leq \kappa \text{ for } i \neq l, \qquad \left\|k(\tilde{x}_i^j, \tilde{x}_i^j)\right\|_{\text{op}} \leq k_{\max} \\ \left\|\tilde{\mathbf{z}}^j - g(\tilde{\mathbf{x}}^j)\,\beta_0\right\|_{\max} \leq \tilde{z}_{\max} \end{cases} \tag{4.126}$$

for all the indices involved, where $\beta_0 \in \mathbb{R}^{r_{\text{UK}} d_z}$ defines an arbitrary reference trend. The trend functions are also assumed to verify $\|g(x')\|_{\text{op}} \leq g_{\max}$ for $x' = x$ and for every element x' of $\tilde{\mathbf{x}}$.

In (4.126), κ bounds the covariances within $\tilde{\mathbf{x}}^r$, within $\tilde{\mathbf{x}}^d$, and between $\tilde{\mathbf{x}}^r$ and x, while α and η respectively bound the covariances between $\tilde{\mathbf{x}}^r$ and $\tilde{\mathbf{x}}^d$, and between $\tilde{\mathbf{x}}^d$ and x. The constants $k_{\max}$ and $\tilde{z}_{\max}$ respectively bound the prior covariances of the training data and the deviation of the observations from the reference trend. Since $\tilde{\mathbf{z}}_{d|r}$ is unchanged when a trend is added to the observations, β_0 can be chosen to minimise $\tilde{z}_{\max}$. The subsampling strategy described in section 4.5.1, which retains the data close to x and discards the farther ones, aims at the regime:

$$0 < \eta, \alpha \ll \kappa \leq k_{\max} \tag{4.127}$$

We further define:

$$\begin{cases} \lambda_j^- \triangleq \lambda_{\min}\left(G(\tilde{\mathbf{x}}^j)\right), & j \in \{r, d\} \\ \lambda_j^+ \triangleq k_{\max} + (\tilde{N}^j - 1)\,\kappa + \lambda_{\max}(\tilde{R}), & j \in \{r, d\} \\ \lambda_r^g \triangleq \lambda_{\min}\left(g(\tilde{\mathbf{x}}^r)^\top g(\tilde{\mathbf{x}}^r)\right) \end{cases} \tag{4.128}$$

where λ_j^+ bounds $\lambda_{\max}\left(G(\tilde{\mathbf{x}}^j)\right)$ by the blockwise Gershgorin circle theorem [7], recalled in section B.17, and λ_r^g measures the conditioning of the trend on the retained data. For ordinary kriging, $g_{\max} = 1$ and $\lambda_r^g = \tilde{N}^r$. Proposition 4.14 bounds the residual quantities, from which Proposition 4.15 deduces bounds on the two errors.

Proposition 4.14 (Entrywise bounds on the residual quantities)**.** *Assume that* (4.126) *holds and that* $\lambda_r^- \lambda_d^- > \alpha^2 \tilde{N}^r \tilde{N}^d$*, and let* $\rho_r \triangleq (\lambda_r^- + \alpha \tilde{N}^r)/\lambda_r^-$*. Then:*

$$\begin{cases} \|k_{d|r}(x)\|_{\text{op}} \leq k_{d|r}^{\max} \triangleq \sqrt{\tilde{N}^d}\left(\dfrac{\lambda_r^- \eta + \alpha \tilde{N}^r \kappa}{\lambda_r^-} + \dfrac{g_{\max}^2\, \lambda_r^+ \rho_r \left(\lambda_r^- + \tilde{N}^r \kappa\right)}{\lambda_r^g\, \lambda_r^-}\right) \\ \|S_{d|r}^{-1}\|_{\text{op}} \leq s_{\max} \triangleq \dfrac{\lambda_r^-}{\lambda_r^- \lambda_d^- - \alpha^2 \tilde{N}^r \tilde{N}^d} \\ \|\tilde{\mathbf{z}}_{d|r}\|_2 \leq \tilde{\mathbf{z}}_{d|r}^{\max} \triangleq \sqrt{\tilde{N}^d}\, \rho_r \left(1 + \dfrac{\tilde{N}^r g_{\max}^2\, \lambda_r^+}{\lambda_r^g\, \lambda_r^-}\right) \tilde{z}_{\max} \end{cases} \tag{4.129}$$

Proof. See section B.17. □

Proposition 4.15 (Entrywise bounds on the errors)**.** *Under the assumptions of Proposition 4.14, the two errors verify:*

$$\begin{cases} e_r(x) \leq k_{d|r}^{\max} \sqrt{s_{\max}} \\ e_f(x) \leq k_{d|r}^{\max}\, \tilde{\mathbf{z}}_{d|r}^{\max}\, s_{\max} \end{cases} \tag{4.130}$$

and $\|\tilde{\mathbf{z}}_{d|r}\|_{S_{d|r}} \leq \tilde{\mathbf{z}}_{d|r}^{\max} \sqrt{s_{\max}}$.

Proof. Let Y and Y' be two matrices with $\tilde{N}^d d_z$ rows. We have:

$$\left\| \langle Y | Y' \rangle_{S_{d|r}} \right\|_{\text{op}} \leq \|Y\|_{\text{op}} \, \|Y'\|_{\text{op}} \, \|S_{d|r}^{-1}\|_{\text{op}} \tag{4.131}$$

Applying this inequality and Proposition 4.14 to Proposition 4.8 concludes the proof. □

As in Proposition 4.10, the bound on $e_f(x)$ is the product of the bounds on $e_r(x)$ and $\|\tilde{\mathbf{z}}_{d|r}\|_{S_{d|r}}$. Unlike their simple kriging counterparts, these bounds do not vanish when α and η tend to 0, since their second term accounts for the estimation of the trend described by Remark 4.9. They also decrease with λ_r^- and λ_d^-, and thus remain valid when these eigenvalues are replaced with lower bounds such as $\lambda_{\min}(\tilde{R})$. Moreover, it is shown in [117] that the smallest eigenvalue of a covariance matrix remains bounded away from 0 as the number of data grows, provided that the training inputs are separated by a minimal distance.

Remark 4.14 (Scope of the entrywise bounds)**.** *The condition $\lambda_r^- \lambda_d^- > \alpha^2 \tilde{N}^r \tilde{N}^d$ requires the retained and discarded data to be separated by a gap wide enough for their cross covariances to be small compared with the eigenvalues of the Gram matrices. When $\tilde{\mathbf{x}}^r$ is a window cut out of a continuous set of training data, α is reached by the closest data on either side of its boundary, and is thus close to κ. In this case, Proposition 4.15 does not apply, and Propositions 4.12 and 4.13 must be used instead. Furthermore, even when it applies, the successive triangle and Cauchy-Schwarz inequalities used in its proof make it loose. Its interest thus lies in the dependence on α, η and κ that it exhibits, rather than in its numerical value.*

CHAPTER 5
SIMULTANEOUS LOCALISATION AND MAPPING

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5.1 Introduction

Throughout chapter 4, we assumed to have access to at least some prior knowledge of the environment in the form of a sparse and noisy set of training data. Such training points formed the basis on which we were able to reconstruct the observation function using kriging, without which no Bayesian correction could be performed.

In this chapter, we address the navigation problem described by (2.1) without any prior training data. This situation corresponds to Simultaneous Localisation and Mapping (SLAM), where the environment must be learnt while simultaneously estimating the vehicle's position and kinematic state. SLAM is a well-established problem in autonomous navigation. A standard solution is to use exteroceptive sensors, such as a stereo camera, LiDAR, or sonar, which provide information about the surrounding environment beyond the vehicle's immediate vicinity.

In our case however, we focus on local sensors, such as magnetometers or gravimeters, whose measurements are restricted to the field at the vehicle's current location. They consequently provide no direct information about the neighbouring environment, preventing the evaluation of the likelihood at any location other than the true one, which a Bayesian filter is precisely trying to recover.

TODO: il faut dire qu'on peut utiliser une base de fonctions quelconques aussi, cf Camille et son polynôme + partir d'une zon krigée ce serait chouette et ça ferait le lien avec ce qui précède (en ouverture plutôt)

Similarly to chapter 4, additional assumptions on the environment must be made in order to be able to do something. In this chapter, we assume the field to be stationary, such that it can be fully represented by the same GP on the whole of $\mathbb{R}^{d_x}$ and across time. This is formally modelled by assumption 5.1.

Assumption 5.1 (Stationary observation model)**.** *The observation function is a time-invariant scalar GP, and its noise is an additive Gaussian white noise of variance $R_k > 0$:*

$$z_k = h(x_k) + n_k, \qquad n_k \sim \mathcal{N}(0, R_k) \tag{5.1}$$

where the mean of the GP is known, and thus assumed to be zero without loss of generality, and its covariance kernel is a Matérn kernel k_ν with known hyperparameters, in the sense of Definition 3.4.

Unlike assumption 4.1, no training data are available.

This chapter is organised as follows:

- Section 5.2 presents an existing SLAM method based on an RBPF formulation to estimate the coefficients of the Karhunen-Loève decomposition associated with the environment.
- Section 5.3 uses the AKKF to embed the field into an RKHS, and solve for its coefficients in this space.
- Finally, section 5.4 compares the two methods on a common simulation scenario.

5.2 Rao-Blackwellised Mapping

Without training data, the kriging estimators of section 3.3 cannot be computed, and the observation function must be estimated alongside the state. However, a GP is an infinite-dimensional object, and cannot be added as such to the state of a filter.

The method proposed in [118] addresses this issue by representing the field with a finite number of coefficients, which are then estimated jointly with the state by the RBPF of section 2.5. This method is not a contribution of this thesis. It is presented in this section as the state-of-the-art approach to which the method proposed in section 5.3 is compared.

In its original form, this method is applied to the three components of the magnetic field, modelled as the gradient of a scalar potential [119, 120], and the domain is divided into hexagonal tiles, each associated with its own set of coefficients. For the sake of simplicity, it is presented here for a scalar field and a single domain.

5.2.1 Reduced-Rank Field Representation

According to Proposition 3.15, a GP defined on a compact set can be decomposed over the eigenfunctions of its Mercer operator. Truncating this decomposition at rank $r_{\mathrm{KL}} \in \mathbb{N}^*$ gives the finite-dimensional parametrisation of Remark 3.15, in which the field is described by r_{KL} coefficients. Formally, we assume that the state of the vehicle remains within a bounded domain:

$$\mathcal{U} \triangleq \prod_{d=1}^{d_x} [a_d, b_d] \tag{5.2}$$

and we let $L_d \triangleq b_d - a_d$ denote its length along the d-th axis.

However, as noted in Remark 3.13, the eigenpairs of the Mercer operator are not available in closed form in general. For a radial kernel, they can be approximated using the eigenfunctions of the Laplace operator on $\mathcal{U}$ with Dirichlet boundary conditions, which are known explicitly [121]. Let $\omega_1, \dots, \omega_{r_{\mathrm{KL}}} \in \mathbb{R}^{d_x}$ denote the r_{KL} vectors of smallest norm among the frequencies:

$$\pi \begin{bmatrix} \frac{n_1}{L_1} & \cdots & \frac{n_{d_x}}{L_{d_x}} \end{bmatrix}^\top, \qquad n_1, \dots, n_{d_x} \in \mathbb{N}^* \tag{5.3}$$

The eigenpairs of the Mercer operator are then approximated by:

$$\begin{cases} \psi_j : x \in \mathcal{U} \mapsto \displaystyle\prod_{d=1}^{d_x} \sqrt{\frac{2}{L_d}} \sin\left(\omega_j^{(d)} \left(x^{(d)} - a_d\right)\right) \\ \lambda_j = S_\nu(\omega_j) \end{cases} \tag{5.4}$$

where $\omega_j^{(d)}$ and $x^{(d)}$ denote the d-th components of ω_j and x, and S_ν is the Fourier transform of k_ν given in section B.4, also known as its spectral density.

Let Ψ and Λ be defined as in section 3.4.2 from these approximated eigenpairs. Under assumption 5.1, Remark 3.15 then gives:

$$\forall x \in \mathcal{U},\ h(x) \approx \Psi(x)\,\alpha, \qquad \alpha \sim \mathcal{N}(0, \Lambda) \tag{5.5}$$

Furthermore, the basis functions only depend on the domain, and can thus be computed before the mission.

Remark 5.1 (Choice of the domain)**.** *The basis functions of* (5.4) *are equal to zero on the boundary of* $\mathcal{U}$*, and so is the field reconstructed from* (5.5)*. The domain must therefore be chosen larger than the region covered by the vehicle. However, the frequencies are spaced by* π/L_d *along the* d*-th axis. The rank* r_{KL} *required to reach a given frequency thus increases with the size of the domain, which is why the original method divides it into tiles.*

5.2.2 Rao-Blackwellised Estimation

Under assumption 5.1 and the approximation (5.5), the observation equation becomes:

$$z_k = \Psi(x_k)\,\alpha + n_k \tag{5.6}$$

and is thus linear in the coefficients for a given state. Assuming that the evolution noise is additive and Gaussian, the navigation problem (2.1) is then the system (2.139) with:

$$\begin{cases} x_k^n = x_k & F_k^n = 0 & f_k^n = f_k & Q_k^n = Q_k \\ x_k^l = \alpha & F_k^l = I_{r_{\mathrm{KL}}} & f_k^l = 0 & Q_k^l = 0 \\ H_k = \Psi & h_k = 0 & Q_k^{nl} = 0 & \end{cases} \tag{5.7}$$

Since $F_k^n = 0$, the propagation of the state does not depend on the coefficients, and the Kalman pre-correction (2.149) through (2.154) is omitted. Moreover, since the field does not vary over time, the coefficients are constant, and the Kalman prediction (2.155) through (2.160) leaves their estimate and its covariance unchanged. For the sake of simplicity, we thus write $\hat{\alpha}_k^{(i)}$ and $P_k^{\alpha(i)}$ for the estimate $\hat{x}_{k|k}^{l(i)}$ of the coefficients associated with the i-th particle and for its covariance $P_{k|k}^{l(i)}$. They are initialised with the prior distribution of (5.5):

$$\hat{\alpha}_0^{(i)} = 0 \quad \text{and} \quad P_0^{\alpha(i)} = \Lambda \tag{5.8}$$

Using the transition density as proposal, as described in (2.97), each iteration of the RBPF is then computed in three steps:

- **Particle propagation.** The particles are propagated according to the transition density:

$$x_k^{(i)} \sim p\left(\cdot | x_{k-1}^{(i)}\right) \tag{5.9}$$

- **Weight update.** Let $\Psi_k^{(i)} \triangleq \Psi\left(x_k^{(i)}\right)$. The innovation and its covariance are:

$$\begin{cases} y_k^{(i)} = z_k - \Psi_k^{(i)}\,\hat{\alpha}_{k-1}^{(i)} \\ S_k^{(i)} = \Psi_k^{(i)}\,P_{k-1}^{\alpha(i)}\left(\Psi_k^{(i)}\right)^\top + R_k \end{cases} \tag{5.10}$$

and the weights are updated as in (2.98), with the likelihood:

$$z_k | \mathbf{x}_{0:k}^{(i)}, \mathbf{z}_{1:k-1} \sim \mathcal{N}\left(\Psi_k^{(i)}\,\hat{\alpha}_{k-1}^{(i)}, S_k^{(i)}\right) \tag{5.11}$$

Similarly to (4.3), its covariance accounts for both the uncertainty of the estimated field and the observation noise.

- **Kalman correction.** The coefficients associated with each particle are corrected as:

$$\begin{cases} K_k^{\alpha(i)} = P_{k-1}^{\alpha(i)} \left(\Psi_k^{(i)}\right)^\top \left(S_k^{(i)}\right)^{-1} \\ \hat{\alpha}_k^{(i)} = \hat{\alpha}_{k-1}^{(i)} + K_k^{\alpha(i)}\, y_k^{(i)} \\ P_k^{\alpha(i)} = \left(I_{r_{\mathrm{KL}}} - K_k^{\alpha(i)}\, \Psi_k^{(i)}\right) P_{k-1}^{\alpha(i)} \end{cases} \tag{5.12}$$

The state and the coefficients are then estimated as in section 2.5, by the weighted means:

$$\hat{x}_k = \sum_{i=1}^{N} w_k^{(i)}\, x_k^{(i)} \quad \text{and} \quad \hat{\alpha}_k = \sum_{i=1}^{N} w_k^{(i)}\, \hat{\alpha}_k^{(i)} \tag{5.13}$$

and the field is reconstructed as $x \mapsto \Psi(x)\, \hat{\alpha}_k$.

As for the generic PF, the particles are only resampled when the effective sample size falls below N_{th}. In this case, the coefficients and their covariance are duplicated along with their particle before the Kalman correction. The resulting algorithm is given by Algorithm 5.1, where $i^{(j)}$ denotes the parent index of the j-th resampled particle.

Algorithm 5.1 Rao-Blackwellised SLAM

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim p\left(\cdot | x_{k-1}^{(i)}\right)$ ▷ Particle propagation

$\forall i \in [\![1, N]\!]$, compute $y_k^{(i)}$ and $S_k^{(i)}$ with (5.10) ▷ **Innovation**

$\forall i \in [\![1, N]\!]$, compute $\tilde{p}\left(z_k | x_k^{(i)}\right)$ from (5.11)

$\forall i \in [\![1, N]\!],\ \tilde{w}_k^{(i)} = w_{k-1}^{(i)}\, \tilde{p}\left(z_k | x_k^{(i)}\right)$ ▷ Weight update

$\forall i \in [\![1, N]\!],\ w_k^{(i)} = \tilde{w}_k^{(i)} \Big/ \sum_{j=1}^{N} \tilde{w}_k^{(j)}$ ▷ Normalisation

$\hat{N}_{\mathrm{eff}} = 1 \Big/ \sum_{i=1}^{N} \left(w_k^{(i)}\right)^2$ ▷ Resampling criterion

if $\hat{N}_{\mathrm{eff}} < N_{\mathrm{th}}$ **then**

$\left\{x_k^{(j)}, w_k^{(j)}, i^{(j)}\right\}_{j=1}^{N} \leftarrow \mathrm{Resampling}\left(\left\{x_k^{(i)}, w_k^{(i)}\right\}_{i=1}^{N}\right)$ ▷ Resampling

$\forall j \in [\![1, N]\!],\ \left(\hat{\alpha}_{k-1}^{(j)}, P_{k-1}^{\alpha(j)}\right) \leftarrow \left(\hat{\alpha}_{k-1}^{\left(i^{(j)}\right)}, P_{k-1}^{\alpha\left(i^{(j)}\right)}\right)$ ▷ **Map duplication**

end if

$\forall i \in [\![1, N]\!]$, compute $\hat{\alpha}_k^{(i)}$ and $P_k^{\alpha(i)}$ with (5.12) ▷ **Kalman correction**

Remark 5.2 (Cost of the Rao-Blackwellised SLAM)**.** *Each particle is associated with its own coefficients and covariance matrix. Each iteration of Algorithm 5.1 thus requires* $\mathcal{O}\left(N\, r_{\mathrm{KL}}^2\right)$ *operations, and the storage of* $\mathcal{O}\left(N\, r_{\mathrm{KL}}^2\right)$ *values. Unlike kriging, these costs do not depend on the number of observations collected since the beginning of the mission.*

5.3 The RKHS Approach

As described by Remark 5.2, the RBPF associates each particle with its own coefficients and covariance matrix, which becomes costly when the number of particles or of basis functions is large. In this section, we propose to estimate a single set of coefficients instead, shared by all the particles. The proposed method relies on the AKKF of Algorithm 3.2.

The AKKF never evaluates the likelihood, and only requires samples of $p(z_k|x_k)$. However, under assumption 5.1, drawing these samples requires the observation function h, which is unknown. Since the filtering problem is already formulated in an RKHS, it is natural to estimate h in such a space as well. This section introduces random Fourier features (RFF) [122, 123] to obtain a finite-dimensional approximation of this space, and recursive least squares (RLS) [124] to estimate the corresponding coefficients.

5.3.1 Random Fourier Features

Let $\mathcal{H}_{k_\nu}$ be the native RKHS of the kernel k_ν. According to Proposition 3.20, the regularised least-squares estimate of h in $\mathcal{H}_{k_\nu}$ is a linear combination of the features of its training points. As for kriging, its number of coefficients is thus equal to the number of training points, which increases at each time step when the observations are collected along the trajectory. To obtain a fixed number of coefficients, the kernel is approximated by a finite number of known functions instead.

Let p_ν be the probability density obtained by normalising the spectral density of k_ν:

$$p_\nu : \omega \in \mathbb{R}^{d_x} \mapsto S_\nu(\omega) \Big/ \int_{\mathbb{R}^{d_x}} S_\nu(\omega')\, d\omega' \tag{5.14}$$

According to Bochner's theorem, already used in section B.4, a continuous SPD radial kernel is the Fourier transform of a positive measure. For k_ν, the density of this measure is proportional to S_ν, and thus to p_ν. Since k_ν is real-valued and $k_\nu(x,x) = \sigma_f^2$, this gives:

$$\forall x, x' \in \mathbb{R}^{d_x},\ k_\nu(x,x') = \sigma_f^2 \int_{\mathbb{R}^{d_x}} \cos\left(\omega^\top (x - x')\right) p_\nu(\omega)\, d\omega \tag{5.15}$$

Let ω be a random variable of probability density p_ν, and $b \sim \mathcal{U}([0, 2\pi])$ be independent of ω. We have:

$$\begin{aligned} \mathbb{E}\left[2\cos\left(\omega^\top x + b\right)\cos\left(\omega^\top x' + b\right)\right] &= \mathbb{E}\left[\cos\left(\omega^\top (x - x')\right)\right] \\ &\quad + \mathbb{E}\left[\cos\left(\omega^\top (x + x') + 2b\right)\right] \\ &= \mathbb{E}\left[\cos\left(\omega^\top (x - x')\right)\right] \end{aligned} \tag{5.16}$$

since the expectation of the second term with respect to b is zero. The kernel can thus be written as an expectation:

$$k_\nu(x,x') = \sigma_f^2\, \mathbb{E}\left[2\cos\left(\omega^\top x + b\right)\cos\left(\omega^\top x' + b\right)\right] \tag{5.17}$$

Similarly to Proposition 2.4, this expectation can be approximated by a Monte Carlo estimator. Let $r_{\mathrm{RFF}} \in \mathbb{N}^*$, and let $(\omega_j, b_j)_{j \in [\![1, r_{\mathrm{RFF}}]\!]}$ be i.i.d. draws of (ω, b). In

this section, ψ_j and Ψ denote the corresponding features:

$$\psi_j : x \in \mathbb{R}^{d_x} \mapsto \sqrt{\frac{2}{r_{\text{RFF}}}} \cos\left(\omega_j^\top x + b_j\right) \tag{5.18}$$

and:

$$\Psi(x) \triangleq \begin{bmatrix} \psi_1(x) & \cdots & \psi_{r_{\text{RFF}}}(x) \end{bmatrix} \tag{5.19}$$

The kernel is then approximated by:

$$k_\nu(x, x') \approx \sigma_f^2 \, \Psi(x) \, \Psi(x')^\top \tag{5.20}$$

Every function of the native RKHS of this approximated kernel is a linear combination of $\psi_1, \ldots, \psi_{r_{\text{RFF}}}$. The observation function is thus estimated as:

$$\forall x \in \mathbb{R}^{d_x}, \; h(x) \approx \Psi(x) \, \alpha \tag{5.21}$$

with $\alpha \in \mathbb{R}^{r_{\text{RFF}}}$. This representation has the same form as (5.5). However, its basis functions are defined on the whole of $\mathbb{R}^{d_x}$, and do not depend on any domain.

5.3.2 Recursive Least Squares

The coefficients of (5.21) are estimated from the observations collected by the vehicle. Since the true states are unknown, each observation z_l is associated with the state estimate $\hat{x}_l$ of the filter at time l. Formally, the estimate $\hat{\alpha}_k$ of the coefficients at time k is defined as the minimiser of the regularised least-squares criterion:

$$J_k : \alpha \in \mathbb{R}^{r_{\text{RFF}}} \mapsto \sum_{l=1}^{k} \beta^{k-l} \left(z_l - \Psi\left(\hat{x}_l\right) \alpha\right)^2 + \beta^k \lambda_{\text{RLS}} \|\alpha\|_2^2 \tag{5.22}$$

where $\lambda_{\text{RLS}} \in \mathbb{R}_+^*$ is a regularisation coefficient, and $\beta \in (0, 1]$ is a forgetting factor, which reduces the influence of the oldest observations when $\beta < 1$.

Minimising (5.22) at each time step would require storing every pair $(\hat{x}_l, z_l)$. Proposition 5.1 shows that $\hat{\alpha}_k$ can instead be computed from $\hat{\alpha}_{k-1}$ and the last pair alone, which is known as the RLS algorithm.

Proposition 5.1 (Recursive least squares)**.** *Let $k \in \mathbb{N}^*$. The criterion (5.22) admits a unique minimiser $\hat{\alpha}_k$, which verifies:*

$$\begin{cases} K_k^\alpha = P_{k-1}^\alpha \, \Psi\left(\hat{x}_k\right)^\top \left(\beta + \Psi\left(\hat{x}_k\right) P_{k-1}^\alpha \, \Psi\left(\hat{x}_k\right)^\top\right)^{-1} \\ \hat{\alpha}_k = \hat{\alpha}_{k-1} + K_k^\alpha \left(z_k - \Psi\left(\hat{x}_k\right) \hat{\alpha}_{k-1}\right) \\ P_k^\alpha = \dfrac{1}{\beta} \left(I_{r_{\text{RFF}}} - K_k^\alpha \, \Psi\left(\hat{x}_k\right)\right) P_{k-1}^\alpha \end{cases} \tag{5.23}$$

with:

$$\hat{\alpha}_0 = 0 \quad \textit{and} \quad P_0^\alpha = \frac{1}{\lambda_{\text{RLS}}} I_{r_{\text{RFF}}} \tag{5.24}$$

Proof. Writing Ψ_l for $\Psi\left(\hat{x}_l\right)$, we let:

$$P_k^\alpha \triangleq \left(\sum_{l=1}^{k} \beta^{k-l} \, \Psi_l^\top \Psi_l + \beta^k \lambda_{\text{RLS}} I_{r_{\text{RFF}}}\right)^{-1} \tag{5.25}$$

which is well defined as the inverse of the sum of an SPSD and an SPD matrix. In particular, P_0^α is the matrix given above. The function J_k is strictly convex, and differentiating it with respect to α yields:

$$\begin{aligned} & \frac{\partial J_k}{\partial \alpha} = 0 \\ \iff \quad & \sum_{l=1}^{k} \beta^{k-l} \, \Psi_l^\top \left(\Psi_l \, \alpha - z_l \right) + \beta^k \lambda_{\mathrm{RLS}} \, \alpha = 0 \\ \iff \quad & \alpha = P_k^\alpha \sum_{l=1}^{k} \beta^{k-l} \, \Psi_l^\top z_l \end{aligned} \tag{5.26}$$

which gives the unique minimiser $\hat{\alpha}_k$. This expression also holds for $k = 0$ with $\hat{\alpha}_0 = 0$.

By definition, we have:

$$\begin{cases} \left(P_k^\alpha\right)^{-1} = \beta \left(P_{k-1}^\alpha\right)^{-1} + \Psi_k^\top \Psi_k \\ \left(P_k^\alpha\right)^{-1} \hat{\alpha}_k = \beta \left(P_{k-1}^\alpha\right)^{-1} \hat{\alpha}_{k-1} + \Psi_k^\top z_k \end{cases} \tag{5.27}$$

The Woodbury inversion formula (see Proposition A.5) applied to the first equation gives:

$$\begin{aligned} P_k^\alpha &= \frac{1}{\beta} \left(P_{k-1}^\alpha - P_{k-1}^\alpha \, \Psi_k^\top \left(\beta + \Psi_k \, P_{k-1}^\alpha \, \Psi_k^\top \right)^{-1} \Psi_k \, P_{k-1}^\alpha \right) \\ &= \frac{1}{\beta} \left(I_{r_{\mathrm{RFF}}} - K_k^\alpha \, \Psi_k \right) P_{k-1}^\alpha \end{aligned} \tag{5.28}$$

and thus:

$$\begin{aligned} P_k^\alpha \, \Psi_k^\top &= \frac{1}{\beta} \left(K_k^\alpha \left(\beta + \Psi_k \, P_{k-1}^\alpha \, \Psi_k^\top \right) - K_k^\alpha \, \Psi_k \, P_{k-1}^\alpha \, \Psi_k^\top \right) \\ &= K_k^\alpha \end{aligned} \tag{5.29}$$

Combining these results finally yields:

$$\begin{aligned} \hat{\alpha}_k &= P_k^\alpha \left(\beta \left(P_{k-1}^\alpha\right)^{-1} \hat{\alpha}_{k-1} + \Psi_k^\top z_k \right) \\ &= P_k^\alpha \left(\left(\left(P_k^\alpha\right)^{-1} - \Psi_k^\top \Psi_k \right) \hat{\alpha}_{k-1} + \Psi_k^\top z_k \right) \\ &= \hat{\alpha}_{k-1} + K_k^\alpha \left(z_k - \Psi_k \, \hat{\alpha}_{k-1} \right) \end{aligned} \tag{5.30}$$

□

5.3.3 AKKF-SLAM

Since (5.23) only involves the pair $(\hat{x}_k, z_k)$, the coefficients can be updated at each time step, within the recursion of the AKKF. Each iteration of the resulting filter is then computed in four steps:

- **Prediction.** The particles are propagated through the evolution model, and the embedded prediction is performed as in Algorithm 3.2.

- **Correction.** Since h is unknown, the simulated observations are generated from the current estimate of the field:

$$\forall i \in [\![1, N]\!],\ z_k^{(i)} = \Psi\left(x_k^{(i)}\right)\hat{\alpha}_{k-1} + n_k^{(i)}, \qquad n_k^{(i)} \sim p_k^{(o)} \tag{5.31}$$

  The embedded correction and the state estimation are then those of Algorithm 3.2.

- **Map update.** The coefficients $\hat{\alpha}_k$ are computed with (5.23), from the real observation z_k and the state estimate $\hat{x}_k$.

- **Resampling.** The new particles are drawn from $\mathcal{N}\left(\hat{x}_k, \hat{P}_k\right)$, and the embedded resampling is performed as in Algorithm 3.2.

It must be noted that the map update only relies on the real observation z_k, whereas the correction step uses the observations simulated from the estimated field. The resulting algorithm, referred to as AKKF-SLAM and originally proposed in [125], is given by Algorithm 5.2.

Algorithm 5.2 AKKF-SLAM

$\forall i \in [\![1, N]\!],\ x_k^{(i)} \sim p\left(\cdot | x_{k-1}^{+(i)}\right)$ ▷ Particle propagation

$D_k = \left(G_{k-1}^x + \lambda_x I_N\right)^{-1} G_{k-1}^x - I_N$ ▷ Embedded prediction

$\Sigma_{k|k-1} = \Sigma_{k-1|k-1}^+ + \dfrac{1}{N} D_k\, D_k^\top$

$\forall i \in [\![1, N]\!],\ z_k^{(i)} = \Psi\left(x_k^{(i)}\right)\hat{\alpha}_{k-1} + n_k^{(i)},\ n_k^{(i)} \sim p_k^{(o)}$ ▷ **Simulated observations**

$K_k = \Sigma_{k|k-1}\left(G_k^z\, \Sigma_{k|k-1} + \lambda_z I_N\right)^{-1}$ ▷ Embedded correction

$\mathbf{w}_k = \mathbf{w}_{k-1}^+ + K_k\left(k_z\left(\mathbf{z}_k, z_k\right) - G_k^z\, \mathbf{w}_{k-1}^+\right)$

$\Sigma_{k|k} = \Sigma_{k|k-1} - K_k\, G_k^z\, \Sigma_{k|k-1}$

$\hat{x}_k = \displaystyle\sum_{i=1}^{N} w_k^{(i)}\, x_k^{(i)}$ ▷ State estimation

$\hat{P}_k = \displaystyle\sum_{i=1}^{N}\sum_{j=1}^{N} \left[\Sigma_{k|k}\right]_{ij} x_k^{(i)}\left(x_k^{(j)}\right)^\top$

Compute K_k^α, $\hat{\alpha}_k$ and P_k^α from (5.23) ▷ **Map update**

$\forall i \in [\![1, N]\!],\ x_k^{+(i)} \sim \mathcal{N}\left(\hat{x}_k, \hat{P}_k\right)$ ▷ Resampling

$T_k = \left(G_k^x + \lambda_x I_N\right)^{-1} k_x\left(\mathbf{x}_k^+, \mathbf{x}_k\right)$ ▷ Embedded resampling

$\mathbf{w}_k^+ = T_k \mathbf{w}_k$

$\Sigma_{k|k}^+ = T_k\, \Sigma_{k|k}\, T_k^\top$

Remark 5.3 (Dependence on the origin)**.** *The weights $\mathbf{w}_k$ solve a regularised least-squares problem, and are not constrained to sum to one. Since k_x is radial, they are unchanged when the origin is translated by $t \in \mathbb{R}^{d_x}$, whereas $\hat{x}_k$ is moved by $\left(\mathbf{1}_N^\top \mathbf{w}_k\right) t$ instead of t. The state estimate thus carries an error proportional to the distance between the particles and the origin, and $\hat{P}_k$ a term growing with its square.*

Both dominate as soon as this distance is large compared with the dispersion of the particles.

This dependence is removed by computing the two moments with respect to the empirical mean $\bar{x}_k$ of the particles $\mathbf{x}_k$:

$$\begin{cases} \hat{x}_k = \bar{x}_k + \sum_{i=1}^{N} w_k^{(i)} \left(x_k^{(i)} - \bar{x}_k \right) \\ \hat{P}_k = \sum_{i=1}^{N} \sum_{j=1}^{N} \left[\Sigma_{k|k} \right]_{ij} \left(x_k^{(i)} - \bar{x}_k \right) \left(x_k^{(j)} - \bar{x}_k \right)^\top \end{cases} \tag{5.32}$$

Remark 5.4 (Estimated training inputs)**.** *The criterion (5.22) is computed with the state estimates $\hat{x}_l$ in place of the unknown true states. The estimation errors of the filter are thus transferred to the estimated field, from which the following observations are simulated. Unlike Algorithm 5.1, in which each set of coefficients is associated with the trajectory of its particle, Algorithm 5.2 does not account for the uncertainty of these training inputs.*

Remark 5.5 (Cost of AKKF-SLAM)**.** *In addition to the $\mathcal{O}(N^3)$ operations of Algorithm 3.2, each iteration of Algorithm 5.2 evaluates the r_{RFF} features at the N particles, and updates the coefficients in $\mathcal{O}(r_{\mathrm{RFF}}^2)$ operations. Its cost is thus $\mathcal{O}(N^3 + N\, r_{\mathrm{RFF}} + r_{\mathrm{RFF}}^2)$ operations per time step. Since a single set of coefficients is estimated, it only requires the storage of $\mathcal{O}(N^2 + r_{\mathrm{RFF}}^2)$ values.*

5.4 Numerical Simulations

5.4.1 Simulation Scenario

The two methods are compared on a scenario derived from that of section 4.2.3. The state of the vehicle, its evolution model, its speed and the time step are unchanged, as well as the covariances Q_k and R_k. However, three elements differ.

First, the magnetic anomaly map is replaced by the analytical field of [125], scaled to the dimensions of the scenario:

$$\begin{aligned} h : x \mapsto 151.5 \Bigg(& \sin\left(\frac{2\pi\, p^{(x)}}{25} \right) + 0.5 \cos\left(\frac{2\pi\, p^{(y)}}{14.58} \right) \\ & + 0.3 \sin\left(\frac{2\pi \left(p^{(x)} + p^{(y)} \right)}{43.75} \right) \Bigg) \end{aligned} \tag{5.33}$$

where $h(x)$ is expressed in nT, and the positions $p^{(x)}$ and $p^{(y)}$ in km. Its standard deviation is 124 nT. Unlike the magnetic anomaly map, whose variations cover a wide range of spatial scales, this field only contains three spatial frequencies, and can thus be represented by a basis of moderate rank.

Second, the mower pattern is replaced by a closed circuit, which the vehicle follows several times. This circuit is a square of side 60 km, whose corners are rounded with the turn radius of 25.5 km, and has a length of 196 km. The mission lasts $n_{\mathrm{steps}} = 300$ steps, which corresponds to 3.8 laps. Unlike the branches of the mower pattern, which are 51 km apart, this trajectory repeatedly crosses regions

that have already been observed. This is required for the field estimated during a lap to be used during the following ones.

Finally, the initial uncertainty is reduced to:

$$P_0 = \begin{bmatrix} 1000^2 & 0 & 0 & 0 \\ 0 & 1000^2 & 0 & 0 \\ 0 & 0 & 2^2 & 0 \\ 0 & 0 & 0 & 2^2 \end{bmatrix} \tag{5.34}$$

in the units of section 4.2.3. Indeed, since the kernel is radial, translating both the trajectory and the field does not change the distribution of the observations. Without training data, the error on the initial position can therefore not be corrected, and a SLAM method can only limit the growth of the estimation error over time.

Five filters are compared, each using $N = 500$ particles:

- The RPF of section 4.2.3 using the known field, which serves as a reference case.
- The same RPF using static kriging on a training set of resolution $\Delta_g = 3.5$ km, built as in section 4.2.3. It corresponds to the method of section 4.2, when a prior survey of the field is available.
- The inertial navigation, in which the particles are propagated without any correction.
- The RBPF of Algorithm 5.1.
- AKKF-SLAM, as given by Algorithm 5.2.

Both SLAM methods are given the same kernel, which is a Matérn kernel of smoothness $\nu = 3/2$, variance $\sigma_f^2 = 124^2$ (nT)^2 and correlation length $l = 4$ km. Since the field only depends on the position of the vehicle, their basis functions are defined on the position subspace, of dimension $d_{\text{pos}} = 2$, and evaluated at $\Pi_{\text{pos}}\, x$.

For the RBPF, the domain $\mathcal{U}$ is the smallest rectangle containing the trajectory, extended by a margin of 25 km on each side, and $r_{\text{KL}} = 300$ basis functions are used. Its particles are regularised as in Algorithm 2.8, with the parameters of section 4.2.3. Moreover, the variance of the field that is not represented by the truncated basis is added to R_k in (5.10).

For AKKF-SLAM, $r_{\text{RFF}} = 500$ features are used, with $\lambda_x = \lambda_z = 0.1$, $\lambda_{\text{RLS}} = 0.01$ and $\beta = 0.995$. The kernel k_x is an SE kernel of unit variance and correlation length, using the Mahalanobis norm associated with P_0. The kernel k_z is an SE kernel of unit variance, whose correlation length is set at each time step to the median of the distances between the simulated observations and z_k, with a minimum of $\sqrt{R_k}$. Since the positions are large compared with the dispersion of the particles, the state estimate and its covariance are computed with respect to the empirical mean of the particles, as described in Remark 5.3.

This scenario is illustrated by Figure 5.1 and summarised by Table 5.1.

Each filter is run over $n_{\text{MC}} = 100$ Monte Carlo realisations. As in section 4.2.3, their performance is read through the estimation RMSE (4.28) and the NEES. The

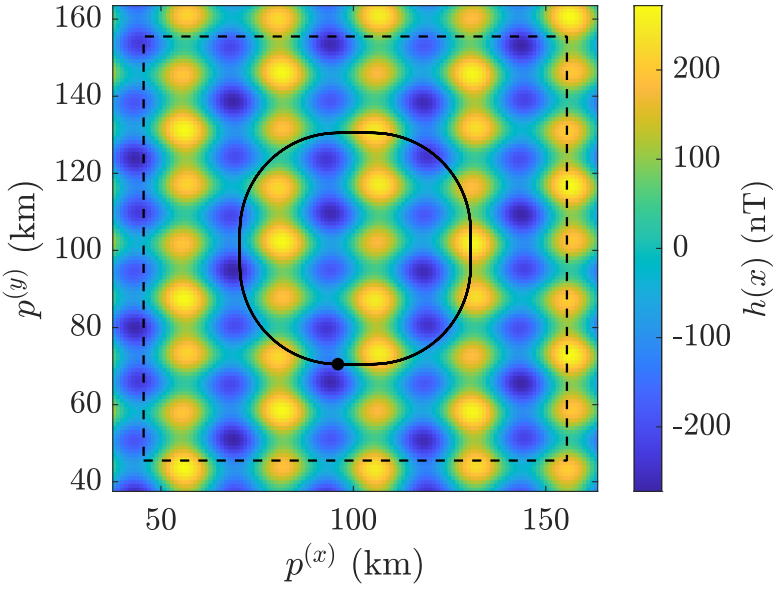


Figure 5.1: The analytical field used in this chapter. The black line represents the circuit followed by the vehicle, and the black dot signals the beginning of the trajectory. The dashed rectangle delimits the domain $\mathcal{U}$ of the RBPF.

Field	Function	Analytical
	Standard deviation	124 nT
Trajectory	Form	Closed circuit
	Dimensions	60 km×60 km
	Turn radius	25.5 km
	n_{steps}	300
Filters	N	500
	P_0	1000^2 m^2, 2^2 m^{2}s^{-2}
Kernel	ν	3/2
	σ_f^2	124^2 (nT)^2
	l	4 km
RBPF	r_{KL}	300
	$\mathcal{U}$	110 km×110 km
AKKF-SLAM	r_{RFF}	500
	λ_x, λ_z	0.1
	λ_{RLS}	0.01
	β	0.995

Table 5.1: Simulation parameters of the SLAM scenario. The parameters that are not listed are those of Table 4.5.

RMSE is also averaged over a set E of time steps, which gives the $\mathrm{ARMSE}_{\hat{x}}(E)$ in the sense of (4.10).

For the two SLAM methods, the field reconstructed at the end of the mission is also compared with h. Its RMSE is defined as in (4.9), and its ARMSE is computed over the points of $\mathcal{U}$ located within 8 km of the trajectory.

5.4.2 Numerical Results

Figure 5.2 presents the RMSE and NEES of the five filters throughout the mission, and Table 5.2 gives their average values, along with the reconstruction error of the field.

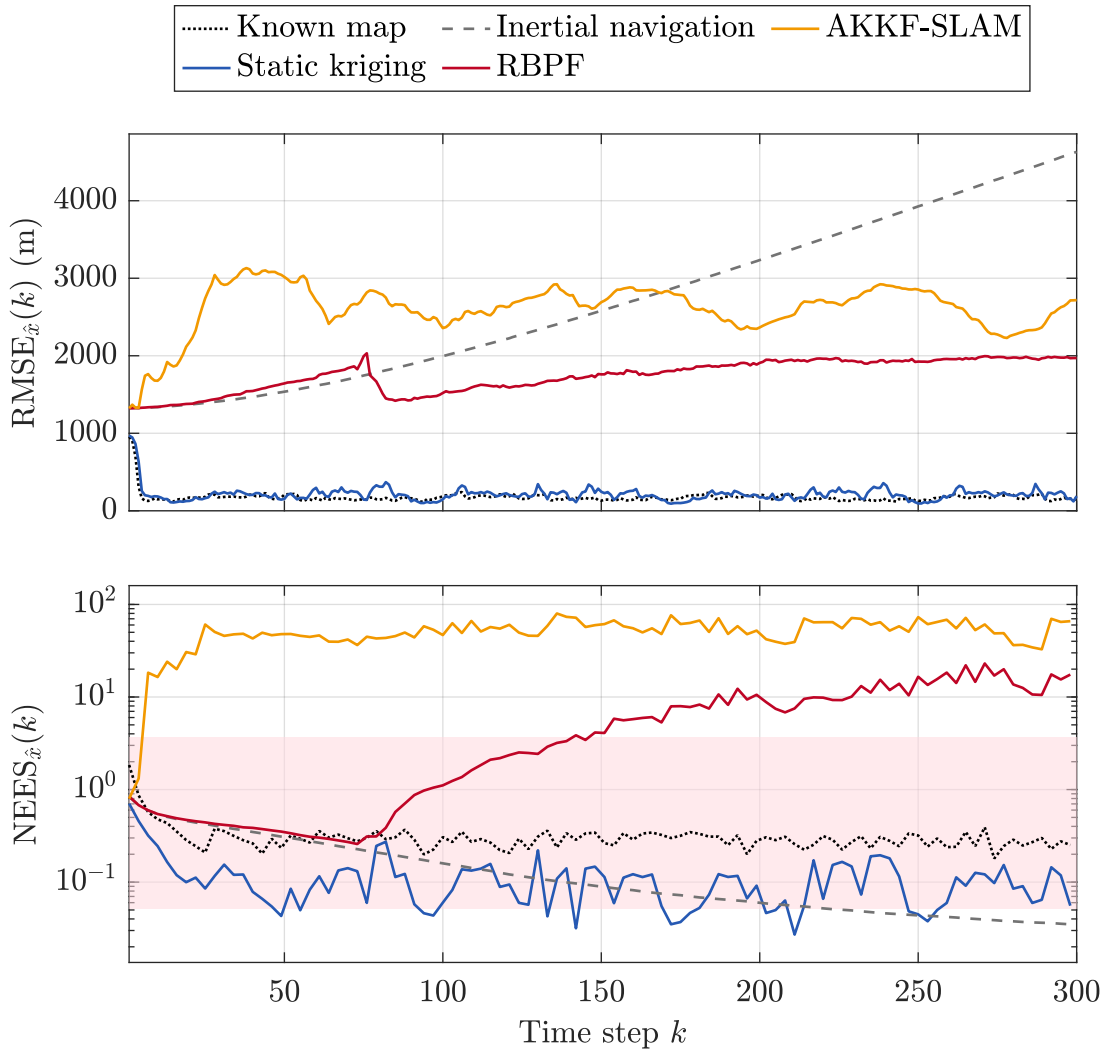


Figure 5.2: Estimation RMSE and NEES of the five filters on the SLAM scenario. The shaded area corresponds to that of the NEES figures of chapter 4.

When a map is available, whether known or kriged from a prior survey, the RPF corrects the initial error within the first ten time steps, and its RMSE then remains around 200 m. Conversely, the error of the inertial navigation increases throughout the mission, from 1.3 km to 4.6 km.

Filter	$\mathrm{ARMSE}_{\hat{x}}$ (m)			NEES	Field ARMSE (nT)
	$[\![1, 100]\!]$	$[\![101, 200]\!]$	$[\![201, 300]\!]$		
Known map	184	173	163	0.28	–
Static kriging	223	206	208	0.10	–
Inertial navigation	1581	2598	3935	0.04	103.7
RBPF	1543	1733	1952	13	78.4
AKKF-SLAM	2564	2660	2610	56	89.4

Table 5.2: Estimation ARMSE of the five filters over three successive parts of the mission, NEES averaged over the last one, and reconstruction ARMSE of the field along the trajectory. For the inertial navigation, the field is reconstructed as zero.

As expected, neither SLAM method reduces the initial error. During the first lap, the RMSE of the RBPF increases slightly faster than that of the inertial navigation, and reaches 2.0 km at $k = 76$. It then decreases to 1.4 km at $k = 85$, when the vehicle crosses the beginning of the circuit for the second time. From this point onward, it increases much more slowly than the error of the inertial navigation, and is equal to 2.0 km at the end of the mission. However, its NEES exceeds 1 from $k = 95$, and leaves the consistency band from $k = 142$. The RBPF thus becomes overconfident as the mission progresses.

On the other hand, the RMSE of AKKF-SLAM increases faster than that of the inertial navigation at the beginning of the mission, and reaches 3.1 km at $k = 38$. It then remains between 2.2 and 2.9 km until the end of the mission, without any noticeable growth, and becomes lower than that of the inertial navigation from $k = 169$. However, it remains higher than that of the RBPF, and the NEES of AKKF-SLAM lies around 50, far above the consistency band.

Regarding the field, both methods reduce the reconstruction error along the trajectory, from 103.7 nT without any map to 78.4 nT for the RBPF, and 89.4 nT for AKKF-SLAM. These errors remain large compared with the standard deviation of the field. This may be explained by the position error. Indeed, the field is estimated at the estimated positions of the vehicle, so that an error of about 2 km on these positions is transferred to the reconstructed field, whose shortest wavelength is 14.58 km. Outside this region, the field is not observed, and neither method improves on the absence of map.

Finally, the two methods differ in their memory requirements. The RBPF stores $N\, r_{\mathrm{KL}}^2 = 4.5 \times 10^7$ values for the covariance matrices of its coefficients, whereas AKKF-SLAM stores $r_{\mathrm{RFF}}^2 = 2.5 \times 10^5$ values for its single set of coefficients, in addition to its matrices of size $N \times N$.

Taken together, these results show that both methods limit the growth of the estimation error in the absence of training data. On this scenario, the RBPF is the more accurate of the two, while AKKF-SLAM bounds its error with a single set of coefficients, at a lower memory cost. However, neither method provides a consistent estimate of its own uncertainty, and both remain far from the accuracy reached when a prior survey is available.

CHAPTER 6

CONCLUSION

This thesis set out to navigate autonomously on physical fields when neither satellite positioning nor a reliable embedded map can be taken for granted. The guiding idea has been to reconstruct the observation function of the field with Gaussian processes and to embed this reconstruction within a Bayesian, particle-based navigation filter, so that a single probabilistic language carries both the uncertainty of the state and the uncertainty of the map.

The foundations laid in the first part recalled the Bayesian filters that cope with the non-linear and multimodal densities of terrain-aided navigation – from the Kalman filter to the regularised and Rao-Blackwellised particle filters – and the Gaussian-process machinery – kriging, its reduced-rank approximations and the reproducing-kernel-Hilbert-space viewpoint – that turns field data into an observation function endowed with its own uncertainty. The second part brought the two together. Kriging was integrated into the particle filter, yielding a navigation scheme that no longer requires a dense and exact embedded map but reconstructs the field from whatever data are available, and that feeds the uncertainty of this reconstruction back into the state estimate. To keep the reconstruction affordable and to respect the non-stationarity of real fields, an adaptive kriging scheme restricts it to the data relevant to the current region, the resulting subsampling error being bounded theoretically. Finally, for the case of an altogether unknown field, localisation and mapping were carried out jointly – through a Rao-Blackwellised filter and a reproducing-kernel-Hilbert-space reconstruction from a dictionary of basis functions – and the learning bias inherent to feeding the map with the vehicle's own estimated positions was identified and characterised.

TODO: bilan quantitatif – résumer ici les résultats des campagnes (précision, robustesse, coût AK/statique), une fois les chiffres figés.

6.1 Limitations

Several limitations temper these results. Adaptive kriging trades exactness for tractability: restricting the reconstruction to a local window rests on a local-stationarity assumption that has the status of a heuristic rather than of a guarantee, and the cost of the reconstruction, although reduced, remains higher than the mere reading of a stored map. In the map-free case, the learning bias was identified

and characterised but not corrected; in the absence of a joint law over the state and the map, the joint estimate is moreover left without a reference against which its consistency could be certified. The observation function, finally, is treated as time-invariant, and the results rest on simulation rather than on data from a field campaign.

6.2 Perspectives

Several directions extend this work. On the side of the models, a spatially varying set of hyperparameters, fitted offline, would relax the local-stationarity heuristic on which adaptive kriging rests, and richer covariance families would widen the range of fields the method can handle. On the side of the methodology, the learning bias of the map-free case calls for a correction rather than a mere characterisation, and for a way of certifying the consistency of a joint estimate that no joint law underwrites.

The most natural opening, however, is the fusion of several complementary physical fields – magnetic, gravimetric, bathymetric. Their joint informativeness could be quantified through the Fisher information, and their combination could resolve the multimodal likelihoods that a single field leaves ambiguous, extending the reconstruction pursued throughout this thesis from one field to several.

Part III

Appendices

APPENDIX A
USEFUL MATRIX IDENTITIES

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A.1 Gaussian Conditioning Formula

Proposition A.1 (Gaussian conditioning [126]). *For $n, p \in \mathbb{N}$, let $U \in \mathbb{R}^n$ and $V \in \mathbb{R}^p$ be jointly Gaussian random variables, of respective means μ_U, μ_V and covariance matrices $A \succ 0$ and $D \succ 0$. There exists $B \in \mathcal{M}_{n,p}(\mathbb{R})$ such that:*

$$\begin{bmatrix} U \\ V \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mu_U \\ \mu_V \end{bmatrix}, \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \right) \tag{A.1}$$

We assume B to be such that this joint covariance is SPD in the sense of Definition 2.1. The conditional distribution of U given $V = v$ is then:

$$U|V = v \sim \mathcal{N}\left(\mu_U + BD^{-1}(v - \mu_V), P_{U|V}\right) \tag{A.2}$$

with:

$$P_{U|V} \triangleq A - BD^{-1}B^\top \succ 0 \tag{A.3}$$

Proof. The existence of B is direct by definition of jointly Gaussian random variables. As $D \succ 0$ is invertible, $S \triangleq A - BD^{-1}B^\top$ the Schur complement with respect to D is well-defined.

Let $u \neq 0$:

$$\begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix}^\top \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix} \begin{bmatrix} u \\ -D^{-1}B^\top u \end{bmatrix} = u^\top \left(A - BD^{-1}B^\top\right) u > 0 \tag{A.4}$$

by SPDness of the joint covariance, the left-hand vector being non-zero. S is thus also invertible, and we can apply the inferior right Schur inversion formula (see Proposition A.2) to obtain:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \triangleq \begin{bmatrix} A & B \\ B^\top & D \end{bmatrix}^{-1} \tag{A.5}$$

For some realisation u of U and v of V, we define $\tilde{u} \triangleq u - \mu_U$ and $\tilde{v} \triangleq v - \mu_V$, and we compute:

$$\begin{aligned} \alpha &\triangleq \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix}^\top \begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} \begin{bmatrix} \tilde{u} \\ \tilde{v} \end{bmatrix} \\ &= \tilde{u}^\top \bar{A} \tilde{u} + \tilde{u}^\top \bar{B} \tilde{v} + \tilde{v}^\top \bar{C} \tilde{u} + \tilde{v}^\top \bar{D} \tilde{v} \\ &= \tilde{u}^\top S^{-1} \tilde{u} + \tilde{u}^\top S^{-1} \left(-BD^{-1}\tilde{v}\right) + \left(-BD^{-1}\tilde{v}\right)^\top S^{-1} \tilde{u} + \\ &\quad \left(-BD^{-1}\tilde{v}\right)^\top S^{-1} \left(-BD^{-1}\tilde{v}\right) + \tilde{v}^\top D^{-1} \tilde{v} \\ &= (\tilde{u} - BD^{-1}\tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1}\tilde{v}) + \tilde{v}^\top D^{-1} \tilde{v} \end{aligned} \tag{A.6}$$

The conditional density is thus:

$$\begin{aligned} p\left(\tilde{u}|\tilde{v}\right) &\propto \frac{p\left(\tilde{u}, \tilde{v}\right)}{p\left(\tilde{v}\right)} \\ &\propto \frac{\exp\left(-\frac{1}{2}\alpha\right)}{\exp\left(-\frac{1}{2}\tilde{v}^\top D^{-1} \tilde{v}\right)} \\ &\propto \exp\left(-\frac{1}{2}(\tilde{u} - BD^{-1}\tilde{v})^\top S^{-1} (\tilde{u} - BD^{-1}\tilde{v})\right) \end{aligned} \tag{A.7}$$

With this formulation, it comes that $U|V = v$ is Gaussian with mean:

$$\mu_{U|V} \triangleq \mu_U + BD^{-1}(v - \mu_V) \tag{A.8}$$

and covariance:

$$P_{U|V} \triangleq S = A - BD^{-1}B^\top \succ 0 \tag{A.9}$$

□

A.2 Schur Inversion Formula

Proposition A.2 (Schur inversion formula [127]). *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$, with A invertible. We define the matrix M and its Schur complement with respect to A as:*

$$M \triangleq \begin{bmatrix} A & B \\ C & D \end{bmatrix}, \qquad M/A \triangleq D - CA^{-1}B \tag{A.10}$$

If M/A is invertible, then M is invertible and:

$$M^{-1} = \begin{bmatrix} A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} & -A^{-1}B(M/A)^{-1} \\ -(M/A)^{-1}CA^{-1} & (M/A)^{-1} \end{bmatrix} \tag{A.11}$$

Proof. Let $\bar{M}$ be the matrix defined by the right-hand side of the formula, and write its blocks as:

$$\begin{bmatrix} \bar{A} & \bar{B} \\ \bar{C} & \bar{D} \end{bmatrix} = \bar{M} \tag{A.12}$$

The identity $M\bar{M} = I$ yields:

$$\begin{cases} A\bar{A} + B\bar{C} = I \\ A\bar{B} + B\bar{D} = 0 \\ C\bar{A} + D\bar{C} = 0 \\ C\bar{B} + D\bar{D} = I \end{cases} \tag{A.13}$$

In particular, the second equation can be rewritten as:

$$\bar{B} = -A^{-1}B\bar{D} \tag{A.14}$$

which can be plugged into the fourth equation to give:

$$\begin{aligned} & C\bar{B} + D\bar{D} = I \\ \iff \quad & -CA^{-1}B\bar{D} + D\bar{D} = I \\ \iff \quad & (D - CA^{-1}B)\bar{D} = I \\ \iff \quad & (M/A)\bar{D} = I \\ \iff \quad & \bar{D} = (M/A)^{-1} \end{aligned} \tag{A.15}$$

(A.14) can then be solved:

$$\bar{B} = -A^{-1}B(M/A)^{-1} \tag{A.16}$$

A similar operation can be performed on the two remaining equations to obtain:

$$\bar{A} = A^{-1}(I - B\bar{C}) \tag{A.17}$$

and:

$$\begin{aligned} & C\bar{A} + D\bar{C} = 0 \\ \iff \quad & CA^{-1}(I - B\bar{C}) + D\bar{C} = 0 \\ \iff \quad & CA^{-1} + (D - CA^{-1}B)\bar{C} = 0 \\ \iff \quad & CA^{-1} + (M/A)\bar{C} = 0 \\ \iff \quad & \bar{C} = -(M/A)^{-1}CA^{-1} \end{aligned} \tag{A.18}$$

Finally:

$$\bar{A} = A^{-1}(I - B\bar{C}) = A^{-1} + A^{-1}B(M/A)^{-1}CA^{-1} \tag{A.19}$$

It can be verified by direct calculations that:

$$M\bar{M} = \bar{M}M = I \tag{A.20}$$

which proves that M is indeed invertible with inverse $\bar{M}$. □

Remark A.1 (Inversion with respect to D)**.** *If D is invertible, one can instead introduce the Schur complement with respect to D:*

$$M/D \triangleq A - BD^{-1}C \tag{A.21}$$

When M/D is also invertible, an analogous block inversion formula holds:

$$M^{-1} = \begin{bmatrix} (M/D)^{-1} & -(M/D)^{-1}BD^{-1} \\ -D^{-1}C(M/D)^{-1} & D^{-1} + D^{-1}C(M/D)^{-1}BD^{-1} \end{bmatrix} \tag{A.22}$$

Remark A.2 (Combining both formulations)**.** *When both A and D are invertible, the Woodbury lemma proves that the two formulations give the same matrices for each block. In particular, they can be combined block by block at will.*

A.3 Woodbury Inversion Formula

Proposition A.3 (Inversion identity)**.** *For $n \in \mathbb{N}$, let $P \in \mathcal{M}_{n,n}(\mathbb{R})$ such that $I+P$ is invertible. Then:*

$$(I + P)^{-1} = I - (I + P)^{-1}P \tag{A.23}$$

Proof. This result is direct from:

$$I = (I + P)^{-1}(I + P) = (I + P)^{-1} + (I + P)^{-1}P \tag{A.24}$$

□

Proposition A.4 (Push-through identity [100])**.** *For $n, p \in \mathbb{N}$, let $P \in \mathcal{M}_{n,p}(\mathbb{R})$ and $Q \in \mathcal{M}_{p,n}(\mathbb{R})$ such that $I_n + PQ$ is invertible. Then $I_p + QP$ is also invertible and:*

$$(I_n + PQ)^{-1}P = P(I_p + QP)^{-1} \tag{A.25}$$

Proof. We start by using the identity:

$$P + PQP = P(I_p + QP) = (I_n + PQ)P \tag{A.26}$$

Thus:

$$\begin{aligned} & P(I_p + QP) = (I_n + PQ)P \\ \iff & P = (I_n + PQ)P(I_p + QP)^{-1} \\ \iff & (I_n + PQ)^{-1}P = P(I_p + QP)^{-1} \end{aligned} \tag{A.27}$$

with $I_p + QP$ admitting a right inverse, hence being invertible. □

Proposition A.5 (Generalised Woodbury formula [128])**.** *For $n, p \in \mathbb{N}$, let $A \in \mathcal{M}_{n,n}(\mathbb{R})$, $B \in \mathcal{M}_{n,p}(\mathbb{R})$, $C \in \mathcal{M}_{p,n}(\mathbb{R})$, and $D \in \mathcal{M}_{p,p}(\mathbb{R})$. We assume that A, D, and $D + CA^{-1}B$ are invertible. Then $A + BD^{-1}C$ is also invertible, and:*

$$\left(A + BD^{-1}C\right)^{-1} = A^{-1} - A^{-1}B\left(D + CA^{-1}B\right)^{-1}CA^{-1} \tag{A.28}$$

Proof. We first factorise:

$$\begin{aligned} \left(A + BD^{-1}C\right)^{-1} &= \left(A(I + A^{-1}BD^{-1}C)\right)^{-1} \\ &= \left(I + A^{-1}BD^{-1}C\right)^{-1}A^{-1} \end{aligned} \tag{A.29}$$

Applying Proposition A.3 to $A^{-1}BD^{-1}C$ yields:

$$\begin{aligned} \left(A + BD^{-1}C\right)^{-1} &= \left(I - \left(I + A^{-1}BD^{-1}C\right)^{-1}A^{-1}BD^{-1}C\right)A^{-1} \\ &= A^{-1} - \left(I + A^{-1}BD^{-1}C\right)^{-1}A^{-1}BD^{-1}CA^{-1} \end{aligned} \tag{A.30}$$

Using Proposition A.4 with $P = A^{-1}B$ and $Q = D^{-1}C$ further gives:

$$\left(I + A^{-1}BD^{-1}C\right)^{-1}A^{-1}B = A^{-1}B\left(I + D^{-1}CA^{-1}B\right)^{-1} \tag{A.31}$$

Therefore:

$$\begin{aligned} \left(A + BD^{-1}C\right)^{-1} &= A^{-1} - A^{-1}B\left(I + D^{-1}CA^{-1}B\right)^{-1}D^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B\left(D(I + D^{-1}CA^{-1}B)\right)^{-1}CA^{-1} \\ &= A^{-1} - A^{-1}B\left(D + CA^{-1}B\right)^{-1}CA^{-1} \end{aligned} \tag{A.32}$$

□

APPENDIX B
DEFERRED PROOFS

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B.1 Proof of Proposition 2.3

Prediction. Let $\mathbb{V}[X]$ denote the variance of the random variable X:

$$\mathbb{V}[X] \triangleq \mathbb{E}\left[(X - \mathbb{E}[X])(X - \mathbb{E}[X])^\top\right] \tag{B.1}$$

Since $x_k = F_k\, x_{k-1} + b_k^{(e)} + w_k$ is an affine transformation of two independent Gaussian random variables $x_{k-1}|\mathbf{z}_{1:k-1} \sim \mathcal{N}\left(\hat{x}_{k-1|k-1}, P_{k-1|k-1}\right)$ and $w_k \sim \mathcal{N}(0, Q_k)$, the predicted state $x_k|\mathbf{z}_{1:k-1}$ is also Gaussian, with:

$$\begin{aligned}\hat{x}_{k|k-1} &\triangleq \mathbb{E}[x_k|\mathbf{z}_{1:k-1}] \\ &= F_k\, \mathbb{E}[x_{k-1}|\mathbf{z}_{1:k-1}] + b_k^{(e)} + \mathbb{E}[w_k|\mathbf{z}_{1:k-1}] \\ &= F_k\, \hat{x}_{k-1|k-1} + b_k^{(e)}\end{aligned} \tag{B.2}$$

and:

$$\begin{aligned}P_{k|k-1} &\triangleq \mathbb{V}[x_k|\mathbf{z}_{1:k-1}] \\ &= F_k\, \mathbb{V}[x_{k-1}|\mathbf{z}_{1:k-1}]\, F_k^\top + \mathbb{V}[w_k|\mathbf{z}_{1:k-1}] \\ &= F_k\, P_{k-1|k-1}\, F_k^\top + Q_k\end{aligned} \tag{B.3}$$

The predicted covariance is thus SPD as the sum of an SPSD matrix and the SPD matrix Q_k.

Correction. The joint distribution of (x_k, z_k) given $\mathbf{z}_{1:k-1}$ is also Gaussian, since $z_k = H_k\, x_k + b_k^{(o)} + n_k$ is an affine transformation of the independent Gaussian variables $x_k|\mathbf{z}_{1:k-1} \sim \mathcal{N}\left(\hat{x}_{k|k-1}, P_{k|k-1}\right)$ and $n_k \sim \mathcal{N}(0, R_k)$. It comes that:

$$\begin{bmatrix} x_k \\ z_k \end{bmatrix} \middle| \mathbf{z}_{1:k-1} \sim \mathcal{N}\left(\begin{bmatrix} \hat{x}_{k|k-1} \\ H_k \hat{x}_{k|k-1} + b_k^{(o)} \end{bmatrix}, \begin{bmatrix} P_{k|k-1} & P_{k|k-1} H_k^\top \\ H_k P_{k|k-1} & S_k \end{bmatrix}\right) \tag{B.4}$$

where:

$$S_k \triangleq H_k\, P_{k|k-1}\, H_k^\top + R_k \tag{B.5}$$

is called the innovation covariance, and is the covariance associated with the innovation vector:

$$y_k \triangleq z_k - \left(H_k\, \hat{x}_{k|k-1} + b_k^{(o)}\right) \tag{B.6}$$

Since $R_k \succ 0$, the independent pair (x_k, n_k) has the covariance:

$$\begin{bmatrix} P_{k|k-1} & 0 \\ 0 & R_k \end{bmatrix} \succ 0 \tag{B.7}$$

Its image under the invertible linear map $(x, n) \mapsto (x, H_k\, x + n)$ therefore has an SPD covariance as well. The Gaussian conditioning formula (see Proposition A.1) can thus be applied to give the posterior distribution $x_k|\mathbf{z}_{1:k}$ is Gaussian, with:

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k\, y_k \tag{B.8}$$

$$P_{k|k} = (I - K_k\, H_k)\, P_{k|k-1} \tag{B.9}$$

where:

$$K_k \triangleq P_{k|k-1}\, H_k^\top\, S_k^{-1} \tag{B.10}$$

is the Kalman gain.

Finally, Proposition A.1 proves that the Schur complement of the SPD joint covariance:

$$P_{k|k-1} - \left(P_{k|k-1}\, H_k^\top\right) S_k^{-1} \left(H_k\, P_{k|k-1}\right) = P_{k|k-1} - K_k\, H_k\, P_{k|k-1} = P_{k|k} \tag{B.11}$$

is itself SPD, so $P_{k|k} \succ 0$.

B.2 Proof of Proposition 2.10

As a linear combination of the centred noises w_k^l and w_k^n, $\overline{w}_k^l$ is centred. Its covariance develops as:

$$\begin{aligned}
\overline{Q}_k^l &= \mathbb{E}\left[\overline{w}_k^l\, \overline{w}_k^{l\,\top}\right] \\
&= \mathbb{E}\left[(w_k^l - Q_k^{nl}\,(Q_k^n)^{-1}\, w_k^n)\,(w_k^l - Q_k^{nl}\,(Q_k^n)^{-1}\, w_k^n)^\top\right] \\
&= \mathbb{E}\left[w_k^l\,(w_k^l)^\top\right] - \mathbb{E}\left[w_k^l\,(Q_k^{nl}\,(Q_k^n)^{-1}\, w_k^n)^\top\right] - \mathbb{E}\left[(Q_k^{nl}\,(Q_k^n)^{-1}\, w_k^n)\,(w_k^l)^\top\right] \\
&\quad + \mathbb{E}\left[(Q_k^{nl}\,(Q_k^n)^{-1}\, w_k^n)\,(Q_k^{nl}\,(Q_k^n)^{-1}\, w_k^n)^\top\right] \\
&= \mathbb{E}\left[w_k^l\,(w_k^l)^\top\right] - \mathbb{E}\left[w_k^l\,(w_k^n)^\top\right](Q_k^n)^{-\top}\,(Q_k^{nl})^\top - Q_k^{nl}\,(Q_k^n)^{-1}\,\mathbb{E}\left[w_k^n\,(w_k^l)^\top\right] \\
&\quad + Q_k^{nl}\,(Q_k^n)^{-1}\,\mathbb{E}\left[w_k^n\,(w_k^n)^\top\right]\,(Q_k^n)^{-\top}\,(Q_k^{nl})^\top \\
&= Q_k^l - Q_k^{nl}\,(Q_k^n)^{-\top}\,(Q_k^{nl})^\top - Q_k^{nl}\,(Q_k^n)^{-1}\,(Q_k^{nl})^\top + Q_k^{nl}\,(Q_k^n)^{-1}\,Q_k^n\,(Q_k^n)^{-\top}\,(Q_k^{nl})^\top \\
&= Q_k^l - Q_k^{nl}\,(Q_k^n)^{-1}\,(Q_k^{nl})^\top - Q_k^{nl}\,(Q_k^n)^{-1}\,(Q_k^{nl})^\top + Q_k^{nl}\,(Q_k^n)^{-1}\,(Q_k^{nl})^\top \\
&= Q_k^l - Q_k^{nl}\,(Q_k^n)^{-1}\,(Q_k^{nl})^\top
\end{aligned} \tag{B.12}$$

Finally, computing:

$$\begin{aligned}
\mathbb{E}\left[\overline{w}_k^l\,(w_k^n)^\top\right] &= \mathbb{E}\left[(w_k^l - Q_k^{nl}\,(Q_k^n)^{-1}\, w_k^n)\,(w_k^n)^\top\right] \\
&= \mathbb{E}\left[w_k^l\,(w_k^n)^\top\right] - Q_k^{nl}\,(Q_k^n)^{-1}\,\mathbb{E}\left[w_k^n\,(w_k^n)^\top\right] \\
&= Q_k^{nl} - Q_k^{nl}\,(Q_k^n)^{-1}\,Q_k^n \\
&= 0
\end{aligned} \tag{B.13}$$

shows that $\overline{w}_k^l$ and w_k^n are uncorrelated.

B.3 Proof of Proposition 3.2

Forward implication. Let $n \in \mathbb{N}^*$ and $\mathbf{x} \in \mathcal{U}^n$. It comes from the symmetry of k that:

$$\begin{aligned} k(\mathbf{x},\mathbf{x}) &= \begin{bmatrix} k(x_1,x_1) & \cdots & k(x_1,x_n) \\ \vdots & \ddots & \vdots \\ k(x_n,x_1) & \cdots & k(x_n,x_n) \end{bmatrix} \\ &= \begin{bmatrix} k(x_1,x_1)^\top & \cdots & k(x_n,x_1)^\top \\ \vdots & \ddots & \vdots \\ k(x_1,x_n)^\top & \cdots & k(x_n,x_n)^\top \end{bmatrix} \\ &= \begin{bmatrix} k(x_1,x_1) & \cdots & k(x_1,x_n) \\ \vdots & \ddots & \vdots \\ k(x_n,x_1) & \cdots & k(x_n,x_n) \end{bmatrix}^\top \\ &= k(\mathbf{x},\mathbf{x})^\top \end{aligned} \tag{B.14}$$

Let $a \in \mathbb{R}^{nd_z}$. We decompose $a = [a_1^\top, \ldots, a_n^\top]^\top$ with $a_i \in \mathbb{R}^{d_z}$, $i \in [\![1,n]\!]$ to obtain:

$$\begin{aligned} a^\top k(\mathbf{x},\mathbf{x})\, a &= \begin{bmatrix} a_1^\top & \cdots & a_n^\top \end{bmatrix} \begin{bmatrix} k(x_1,x_1) & \cdots & k(x_1,x_n) \\ \vdots & \ddots & \vdots \\ k(x_n,x_1) & \cdots & k(x_n,x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\ &= \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i,x_j)\, a_j \\ &\geq 0 \end{aligned} \tag{B.15}$$

Reverse implication. Let $x, x' \in \mathcal{U}$. By symmetry of $k((x,x'),(x,x'))$, it comes that:

$$\begin{aligned} \begin{bmatrix} k(x,x) & k(x,x') \\ k(x',x) & k(x',x') \end{bmatrix} &= \begin{bmatrix} k(x,x) & k(x,x') \\ k(x',x) & k(x',x') \end{bmatrix}^\top \\ &= \begin{bmatrix} k(x,x)^\top & k(x',x)^\top \\ k(x,x')^\top & k(x',x')^\top \end{bmatrix} \end{aligned} \tag{B.16}$$

In particular, the upper-right block gives:

$$k(x,x') = k(x',x)^\top \tag{B.17}$$

Let $n \in \mathbb{N}^*$, $\mathbf{x} \in \mathcal{U}^n$, and $a_1, \ldots, a_n \in \mathbb{R}^{d_z}$, and set $a \triangleq [a_1^\top, \ldots, a_n^\top]^\top \in \mathbb{R}^{nd_z}$. Then:

$$\begin{aligned} \sum_{i=1}^n \sum_{j=1}^n a_i^\top k(x_i,x_j)\, a_j &= \begin{bmatrix} a_1^\top & \cdots & a_n^\top \end{bmatrix} \begin{bmatrix} k(x_1,x_1) & \cdots & k(x_1,x_n) \\ \vdots & \ddots & \vdots \\ k(x_n,x_1) & \cdots & k(x_n,x_n) \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} \\ &= a^\top k(\mathbf{x},\mathbf{x})\, a \\ &\geq 0 \end{aligned} \tag{B.18}$$

Strict version. The same computations hold with strict inequalities, provided the inputs are pairwise distinct and the coefficients not all zero.

B.4 Proof of Proposition 3.3

Assume that $\nu \in \mathbb{R}_+^*$ and let k_ν denote such a function. We also let:

$$g : \begin{cases} \mathbb{R}_+ \to \mathbb{R}_+ \\ \quad r \mapsto \sigma_f^2 \dfrac{2^{1-\nu}}{\Gamma(\nu)} r^\nu K_\nu(r) \end{cases} \quad \text{and} \quad \tilde{r} : \begin{cases} \mathbb{R}^{d_x} \times \mathbb{R}^{d_x} \to \mathbb{R}_+ \\ \qquad x, x' \mapsto \dfrac{\sqrt{2\nu}}{l} \|x - x'\| \end{cases} \tag{B.19}$$

$\tilde{r}$ and K_ν are continuous on $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$ and $\mathbb{R}_+^*$ respectively, and the latter admits the following asymptotic expansion as $r \to 0^+$:

$$K_\nu(r) = 2^{\nu-1}\Gamma(\nu)\, r^{-\nu} + o\left(r^{-\nu}\right) \tag{B.20}$$

Therefore, f can be continuously extended to 0 by setting $f(0) = \sigma_f^2$, and k_ν is thus continuous on the entirety of $\mathbb{R}^{d_x} \times \mathbb{R}^{d_x}$.

Let $x, x' \in \mathbb{R}^{d_x}$. By symmetry of the norm, $\tilde{r}(x, x') = \tilde{r}(x', x)$, and thus:

$$k_\nu = g \circ \tilde{r} \tag{B.21}$$

is symmetric, with $\circ$ being the composition operator.

Furthermore, Bochner's theorem [129] states that a continuous integrable function with a strictly positive Fourier transform is SPD. It can be shown [130] that the Fourier transform of k_ν on $\mathbb{R}^{d_x}$ is:

$$S_\nu : \begin{cases} \mathbb{R}^{d_x} \to \mathbb{R} \\ \quad \omega \mapsto C \left(\dfrac{2\nu}{l^2} + \|\omega\|^2\right)^{-(\nu + d_x/2)} \end{cases} \tag{B.22}$$

where $C > 0$ is a normalising constant which only depends on d_x and the parameters of k_ν. Since $\nu > 0$, $l > 0$, it comes that:

$$\forall \omega \in \mathbb{R}^{d_x},\ S_\nu(\omega) > 0 \tag{B.23}$$

and thus k_ν is a SPD kernel. By definition, it is also a radial kernel.

The case $\nu = +\infty$ is more direct. The squared exponential kernel is continuous on the whole of $\mathbb{R}_+$ without any extension, and symmetric by the same composition argument. Its Fourier transform on $\mathbb{R}^{d_x}$ is:

$$S_\nu : \begin{cases} \mathbb{R}^{d_x} \to \mathbb{R} \\ \quad \omega \mapsto C \exp\left(-\dfrac{l^2\|\omega\|^2}{2}\right) \end{cases} \tag{B.24}$$

which is again strictly positive, so Bochner's theorem applies as above.

B.5 Proof of Proposition 3.8

Let:

$$\mathcal{L}(W, \xi) \triangleq \operatorname{tr}\left(\mathbb{V}\left[W^\top \tilde{\mathbf{z}} - z\right]\right) + 2 \operatorname{tr}\left(\xi^\top \left(C^\top W - I_{d_z}\right)\right) \tag{B.25}$$

be the Lagrangian associated with the optimisation problem (3.32) under the constraint (3.48), and $\xi \in \mathcal{M}_{d_z,d_z}(\mathbb{R})$ the corresponding Lagrange multiplier.

This problem is solved by finding the minimum of $\mathcal{L}$, *i.e.* finding a zero of its derivative. Writing G for $G(\tilde{\mathbf{x}})$, the same properties used in the proof of Proposition 3.5 give:

$$\begin{aligned}
& \frac{\partial \mathcal{L}}{\partial W}(W, \xi) = 0 \\
\iff \quad & 2G\,W - 2k(\tilde{\mathbf{x}}, x) + 2\frac{\partial}{\partial W}\left(\operatorname{tr}\left(\xi^\top \left(C^\top W - I_{d_z}\right)\right)\right) = 0 \\
\iff \quad & 2G\,W - 2k(\tilde{\mathbf{x}}, x) + 2C\xi = 0 \\
\iff \quad & G\,W + C\xi = k(\tilde{\mathbf{x}}, x)
\end{aligned} \tag{B.26}$$

Together with the constraint $C^\top W = I_{d_z}$, this yields the block system:

$$\begin{bmatrix} G & C \\ C^\top & 0 \end{bmatrix} \begin{bmatrix} W \\ \xi \end{bmatrix} = \begin{bmatrix} k(\tilde{\mathbf{x}}, x) \\ I_{d_z} \end{bmatrix} \tag{B.27}$$

The left matrix block can be inverted using the upper left Schur inversion formula from Proposition A.2. Its Schur complement,

$$S \triangleq 0 - C^\top G^{-1} C = -C^\top G^{-1} C, \tag{B.28}$$

is invertible since $G \succ 0$ and C has full column rank. The convention of Proposition A.2 gives the four blocks of the inverse:

$$\begin{cases} \bar{A} = G^{-1} + G^{-1} C\, S^{-1} C^\top G^{-1} \\ \bar{B} = -G^{-1} C\, S^{-1} \\ \bar{C} = -S^{-1} C^\top G^{-1} \\ \bar{D} = S^{-1} \end{cases} \tag{B.29}$$

Let $M \triangleq -S^{-1} = \left(C^\top G^{-1} C\right)^{-1}$ and $U \triangleq I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)$. The weight matrix can be derived from the upper blocks of the inverse:

$$\begin{aligned}
W &= \bar{A}\, k(\tilde{\mathbf{x}}, x) + \bar{B}\, I_{d_z} \\
&= \left(G^{-1} + G^{-1} C\, S^{-1} C^\top G^{-1}\right) k(\tilde{\mathbf{x}}, x) - G^{-1} C\, S^{-1} \\
&= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C\, S^{-1} \left(C^\top G^{-1} k(\tilde{\mathbf{x}}, x) - I_{d_z}\right) \\
&= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C\, M \left(I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)\right) \\
&= G^{-1} k(\tilde{\mathbf{x}}, x) + G^{-1} C\, M\, U
\end{aligned} \tag{B.30}$$

and ξ from the lower blocks:

$$\begin{aligned}
\xi &= \bar{C}\, k(\tilde{\mathbf{x}}, x) + \bar{D}\, I_{d_z} \\
&= -S^{-1} C^\top G^{-1} k(\tilde{\mathbf{x}}, x) + S^{-1} \\
&= S^{-1} \left(I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)\right) \\
&= -M \left(I_{d_z} - C^\top G^{-1} k(\tilde{\mathbf{x}}, x)\right) \\
&= -M\, U
\end{aligned} \tag{B.31}$$

Since M is symmetric, the OK estimator is:

$$\begin{aligned}\hat{z}_{\mathrm{OK}} &= W^\top \tilde{\mathbf{z}} \\ &= k(x,\tilde{\mathbf{x}})\, G^{-1}\tilde{\mathbf{z}} + U^\top M\, C^\top G^{-1}\tilde{\mathbf{z}} \\ &= k(x,\tilde{\mathbf{x}})\, G^{-1}\tilde{\mathbf{z}} + \left(I_{d_z} - k(x,\tilde{\mathbf{x}})\, G^{-1}C\right) M\, C^\top G^{-1}\tilde{\mathbf{z}} \\ &= M\, C^\top G^{-1}\tilde{\mathbf{z}} + k(x,\tilde{\mathbf{x}})\, G^{-1}\left(\tilde{\mathbf{z}} - C\, M\, C^\top G^{-1}\tilde{\mathbf{z}}\right)\end{aligned} \tag{B.32}$$

which is exactly (3.50), the estimated mean being $\hat{m}_{\mathrm{OK}}(\tilde{\mathbf{x}},\tilde{\mathbf{z}}) = M\, C^\top G^{-1}\tilde{\mathbf{z}}$.

Since R_{OK} is a centred moment of the estimation error, it is insensitive to the unknown mean and (3.36) can be applied. Using (B.26) and the constraint $W^\top C = I_{d_z}$ gives the identity:

$$\begin{aligned}W^\top G\, W &= W^\top\left(k(\tilde{\mathbf{x}},x) - C\,\xi\right) \\ &= W^\top k(\tilde{\mathbf{x}},x) - W^\top C\,\xi \\ &= W^\top k(\tilde{\mathbf{x}},x) - \xi\end{aligned} \tag{B.33}$$

Combining these results yields:

$$\begin{aligned}R_{\mathrm{OK}} &= W^\top G\, W - W^\top k(\tilde{\mathbf{x}},x) - k(x,\tilde{\mathbf{x}})\, W + k(x,x) + R \\ &= W^\top k(\tilde{\mathbf{x}},x) - \xi - W^\top k(\tilde{\mathbf{x}},x) - k(x,\tilde{\mathbf{x}})\, W + k(x,x) + R \\ &= -\xi - k(x,\tilde{\mathbf{x}})\, W + k(x,x) + R \\ &= M\, U - k(x,\tilde{\mathbf{x}})\, G^{-1}k(\tilde{\mathbf{x}},x) - k(x,\tilde{\mathbf{x}})\, G^{-1}C\, M\, U + k(x,x) + R \\ &= \left(I_{d_z} - k(x,\tilde{\mathbf{x}})\, G^{-1}C\right) M\, U + k(x,x) + R - k(x,\tilde{\mathbf{x}})\, G^{-1}k(\tilde{\mathbf{x}},x) \\ &= U^\top M\, U + k(x,x) + R - k(x,\tilde{\mathbf{x}})\, G^{-1}k(\tilde{\mathbf{x}},x)\end{aligned} \tag{B.34}$$

Finally, recognising $R_{\hat{m}_{\mathrm{OK}}}(x,\tilde{\mathbf{x}}) = U^\top M\, U \succeq 0$, since $M \succ 0$, gives the desired expression.

B.6 Proof of Proposition 3.9

The unbiased assumption now reads:

$$g(\tilde{\mathbf{x}})^\top W - g(x)^\top = 0 \tag{B.35}$$

and the Lagrange multiplier method of Proposition 3.8 can be transposed to universal kriging under the substitutions:

$$\begin{cases} C \longrightarrow g(\tilde{\mathbf{x}}) \\ I_{d_z} \longrightarrow g(x)^\top \\ \xi \in \mathcal{M}_{r_{\mathrm{UK}}d_z,d_z}(\mathbb{R}) \end{cases} \tag{B.36}$$

The full column rank of $g(\tilde{\mathbf{x}})$ ensures that S is invertible, and that M is SPD. Writing G for $G(\tilde{\mathbf{x}})$, the shorthands of that proof thus become:

$$\begin{cases} M \triangleq -S^{-1} = \left(g(\tilde{\mathbf{x}})^\top G^{-1} g(\tilde{\mathbf{x}})\right)^{-1} \\ U \triangleq g(x)^\top - g(\tilde{\mathbf{x}})^\top G^{-1} k(\tilde{\mathbf{x}},x) \end{cases} \tag{B.37}$$

so that the weight matrix and Lagrange multiplier are:

$$\begin{cases} W = G^{-1}k(\tilde{\mathbf{x}}, x) + G^{-1}g(\tilde{\mathbf{x}})\, M\, U \\ \xi = -M\, U \end{cases} \tag{B.38}$$

Finally, the bilinear identity (B.33) becomes:

$$W^\top G\, W = W^\top k(\tilde{\mathbf{x}}, x) - g(x)\, \xi \tag{B.39}$$

The UK estimator and covariance matrices are derived exactly like for Proposition 3.8, using the corrected results:

$$\begin{aligned} \hat{z}_{\text{UK}} &= k(x, \tilde{\mathbf{x}})\, G^{-1}\tilde{\mathbf{z}} + U^\top M\, g(\tilde{\mathbf{x}})^\top G^{-1}\tilde{\mathbf{z}} \\ &= g(x)\, M\, g(\tilde{\mathbf{x}})^\top G^{-1}\tilde{\mathbf{z}} + k(x, \tilde{\mathbf{x}})\, G^{-1}\left(\tilde{\mathbf{z}} - g(\tilde{\mathbf{x}})\, M\, g(\tilde{\mathbf{x}})^\top G^{-1}\tilde{\mathbf{z}}\right) \end{aligned} \tag{B.40}$$

and:

$$R_{\text{UK}} = U^\top M\, U + k(x, x) + R - k(x, \tilde{\mathbf{x}})\, G^{-1}k(\tilde{\mathbf{x}}, x) \tag{B.41}$$

B.7 Proof of Proposition 3.10

Using (3.54), it comes that:

$$f \mid \beta \sim \mathcal{GP}(g\, \beta, k) \tag{B.42}$$

and thus:

$$f - g\, \beta \mid \beta \sim \mathcal{GP}(0, k) \tag{B.43}$$

Since this law is free of β, $f - g\, \beta$ is independent of β, which gives:

$$f - g\, \beta \sim \mathcal{GP}(0, k) \tag{B.44}$$

Let $x, x' \in \mathcal{U}$. The mean of f can thus be computed by decomposing:

$$\mathbb{E}[f(x)] = g(x)\, \mathbb{E}[\beta] + \mathbb{E}[f(x) - g(x)\, \beta] = 0 \tag{B.45}$$

By independence of $f - g\, \beta$ and β, it follows that:

$$\begin{aligned} \mathbb{E}\left[f(x)\, f(x')^\top\right] &= \mathbb{E}\left[(f(x) - g(x)\, \beta)(f(x') - g(x')\, \beta)^\top\right] + g(x)^\top\, \mathbb{E}\left[\beta\, \beta^\top\right] g(x') + \\ &\quad \mathbb{E}\left[(f(x) - g(x)\, \beta)\, \beta^\top\right] g(x')^\top + g(x')\, \mathbb{E}\left[\beta\, (f(x) - g(x)\, \beta)^\top\right] \\ &= k(x, x') + g(x)\, B\, g(x')^\top + 0 + 0 \\ &= k_B(x, x') \end{aligned} \tag{B.46}$$

As a linear combiantion of the jointly Gaussian distributions $f - g\, \beta$ and β, we finally have:

$$f \sim \mathcal{GP}(0, k_B) \tag{B.47}$$

In this form, the mean function of f is known, and Proposition 3.7 states that simple kriging using k_B gives the conditional distribution of $z \mid \tilde{\mathbf{z}}$:

$$z \mid \tilde{\mathbf{z}} \sim \mathcal{N}\left(\hat{z}_{\text{SK},B}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0), R_{\text{SK},B}(x, \tilde{\mathbf{x}})\right) \tag{B.48}$$

The SK estimator and error covariance matrix can be expressed as follows:

$$\begin{cases} \hat{z}_{\mathrm{SK},B}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) = k_B(x, \tilde{\mathbf{x}})\, G_B(\tilde{\mathbf{x}})^{-1}\, \tilde{\mathbf{z}} \\ \quad R_{\mathrm{SK},B}(x, \tilde{\mathbf{x}}) = k_B(x, x) + R - k_B(x, \tilde{\mathbf{x}})\, G_B(\tilde{\mathbf{x}})^{-1} k_B\left(\tilde{\mathbf{x}}, x\right) \end{cases} \tag{B.49}$$

where:

$$G_B \triangleq k_B(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + I_{\tilde{N}} \otimes \tilde{R} = G + g(\tilde{\mathbf{x}})\, B\, g(\tilde{\mathbf{x}})^\top \tag{B.50}$$

For the sake of clarity, we let G and g respectively denote $G(\tilde{\mathbf{x}})$ and $g(\tilde{\mathbf{x}})$, and we reuse:

$$\begin{cases} M \triangleq \left(g^\top G^{-1} g\right)^{-1} \\ U \triangleq g(x)^\top - g^\top G^{-1} k(\tilde{\mathbf{x}}, x) \end{cases} \tag{B.51}$$

from section B.6.

Using Proposition A.5, it comes that:

$$G_B^{-1} = G^{-1} - G^{-1} g\, M_B\, g^\top G^{-1} \tag{B.52}$$

with:

$$M_B \triangleq \left(B^{-1} + g^\top G^{-1} g\right)^{-1} = \left(B^{-1} + M^{-1}\right)^{-1} \tag{B.53}$$

By definition of M_B, we have:

$$\begin{aligned} & \left(B^{-1} + M^{-1}\right) M_B = I \\ \iff \quad & M^{-1} M_B = I - B^{-1} M_B \\ \iff \quad & B\, M^{-1} M_B = B - M_B \end{aligned} \tag{B.54}$$

We also precompute:

$$\begin{aligned} g(x)\, B\, g^\top\, G_B^{-1} &= g(x)\, B\, g^\top \left(G^{-1} - G^{-1} g\, M_B\, g^\top G^{-1}\right) \\ &= g(x) \left(B\, g^\top G^{-1} - B\, g^\top G^{-1} g\, M_B\, g^\top G^{-1}\right) \\ &= g(x) \left(B\, g^\top G^{-1} - B\, M^{-1}\, M_B\, g^\top G^{-1}\right) \\ &= g(x) \left(B\, g^\top G^{-1} - (B - M_B)\, g^\top G^{-1}\right) \\ &= g(x)\, M_B\, g^\top G^{-1} \end{aligned} \tag{B.55}$$

It comes that:

$$\begin{aligned} k_B(x, \tilde{\mathbf{x}})\, G_B^{-1} &= \left(k(x, \tilde{\mathbf{x}}) + g(x)\, B\, g^\top\right) G_B^{-1} \\ &= k(x, \tilde{\mathbf{x}})\, G_B^{-1} + g(x)\, B\, g^\top G_B^{-1} \\ &= k(x, \tilde{\mathbf{x}}) \left(G^{-1} - G^{-1} g\, M_B\, g^\top G^{-1}\right) + g(x)\, M_B\, g^\top G^{-1} \\ &= k(x, \tilde{\mathbf{x}})\, G^{-1} + \left(-k(x, \tilde{\mathbf{x}})\, G^{-1} g + g(x)\right) M_B\, g^\top G^{-1} \\ &= k(x, \tilde{\mathbf{x}})\, G^{-1} + U^\top M_B\, g^\top G^{-1} \end{aligned} \tag{B.56}$$

and thus:

$$\hat{z}_{\mathrm{SK},B} = k_B(x, \tilde{\mathbf{x}})\, G_B^{-1} \tilde{\mathbf{z}} = k(x, \tilde{\mathbf{x}})\, G^{-1} \tilde{\mathbf{z}} + U^\top M_B\, g^\top G^{-1} \tilde{\mathbf{z}} \tag{B.57}$$

For the covariance, we compute:

$$\begin{aligned} k_B(x,\tilde{\mathbf{x}})\, G_B^{-1} k_B(\tilde{\mathbf{x}},x) &= \left(k(x,\tilde{\mathbf{x}})\, G^{-1} + U^\top M_B\, g^\top G^{-1}\right)\left(k(\tilde{\mathbf{x}},x) + g\, B\, g(x)^\top\right) \\ &= k(x,\tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}},x) + U^\top M_B\, g^\top G^{-1} k(\tilde{\mathbf{x}},x) + \\ &\quad k(x,\tilde{\mathbf{x}})\, G^{-1} g\, B\, g(x)^\top + U^\top M_B\, g^\top G^{-1} g\, B\, g(x)^\top \\ &= k(x,\tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}},x) + U^\top M_B\, g^\top G^{-1} k(\tilde{\mathbf{x}},x) + \\ &\quad k(x,\tilde{\mathbf{x}})\, G^{-1} g\, B\, g(x)^\top + U^\top M_B\, M^{-1}\, B\, g(x)^\top \\ &= k(x,\tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}},x) + U^\top M_B\, g^\top G^{-1} k(\tilde{\mathbf{x}},x) + \\ &\quad k(x,\tilde{\mathbf{x}})\, G^{-1} g\, B\, g(x)^\top + U^\top (B - M_B)\, g(x)^\top \end{aligned} \tag{B.58}$$

and:

$$g(x)\, B\, g(x)^\top - k(x,\tilde{\mathbf{x}})\, G^{-1} g\, B\, g(x)^\top = U^\top g\, B\, g(x)^\top \tag{B.59}$$

to obtain:

$$\begin{aligned} R_{\mathrm{SK},B} &= k_B(x,x) + R - k_B(x,\tilde{\mathbf{x}})\, G_B(\tilde{\mathbf{x}})^{-1} k_B\left(\tilde{\mathbf{x}},x\right) \\ &= k(x,x) + g(x)\, B\, g(x)^\top + R - k(x,\tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}},x) - U^\top M_B\, g^\top G^{-1} k(\tilde{\mathbf{x}},x) - \\ &\quad k(x,\tilde{\mathbf{x}})\, G^{-1} g\, B\, g(x)^\top - U^\top (B - M_B)\, g(x)^\top \\ &= k(x,x) + R - k(x,\tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}},x) - U^\top M_B\, g^\top G^{-1} k(\tilde{\mathbf{x}},x) + U^\top M_B\, g(x)^\top \\ &= k(x,x) + R - k(x,\tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}},x) + U^\top M_B\, U \end{aligned} \tag{B.60}$$

Since $M_B \to M$ as $B^{-1} \to 0$, we finally have:

$$\begin{cases} \hat{z}_{\mathrm{SK},B}(x,\tilde{\mathbf{x}},\tilde{\mathbf{z}},0) \underset{B^{-1}\to 0}{\longrightarrow} k(x,\tilde{\mathbf{x}})\, G^{-1}\tilde{\mathbf{z}} + U^\top M\, g^\top G^{-1}\tilde{\mathbf{z}} \\ R_{\mathrm{SK},B}(x,\tilde{\mathbf{x}}) \underset{B^{-1}\to 0}{\longrightarrow} k(x,x) + R - k(x,\tilde{\mathbf{x}})\, G^{-1} k(\tilde{\mathbf{x}},x) + U^\top M\, U \end{cases} \tag{B.61}$$

the limit of which are exactly (B.40) and (B.41).

B.8 Proof of Proposition 3.11

As $\mathcal{U}$ is compact and k continuous, k is bounded on $\mathcal{U}\times\mathcal{U}$, and $\mathcal{K}$ is compact.

Let $\phi, \phi' \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. Cauchy-Schwarz inequality ensures that there exists a constant $c \in \mathbb{R}_+$ such that:

$$\iint_{\mathcal{U}\times\mathcal{U}} \left|\phi(y)^\top k(x,y)^\top \phi'(x)\right|\, dx\, dy \leq c\, \sqrt{\langle\phi|\phi\rangle}\, \sqrt{\langle\phi'|\phi'\rangle} < +\infty \tag{B.62}$$

Fubini's integration theorem [131] can thus be used to give:

$$\begin{aligned}
\langle \mathcal{K}\phi | \phi' \rangle &= \int_{\mathcal{U}} (\mathcal{K}\phi)(x)^\top \phi'(x)\, dx \\
&= \int_{\mathcal{U}} \left(\int_{\mathcal{U}} k(x,y)\, \phi(y)\, dy \right)^\top \phi'(x)\, dx \\
&= \int_{\mathcal{U}} \int_{\mathcal{U}} \phi(y)^\top k(x,y)^\top \phi'(x)\, dy\, dx \\
&= \int_{\mathcal{U}} \int_{\mathcal{U}} \phi(y)^\top k(y,x)\, \phi'(x)\, dx\, dy \\
&= \int_{\mathcal{U}} \phi(y)^\top \left(\int_{\mathcal{U}} k(y,x)\, \phi'(x)\, dx \right) dy \\
&= \int_{\mathcal{U}} \phi(y)^\top (\mathcal{K}\phi')(y)\, dy \\
&= \langle \phi | \mathcal{K}\phi' \rangle
\end{aligned} \tag{B.63}$$

Let $\phi : \mathcal{U} \to \mathbb{R}^{d_z}$ be a continuous function, and let $n \in \mathbb{N}^*$. We define $\left(\mathcal{U}_i^{(n)}\right)_{i \in [\![1,n]\!]}$ a partition of $\mathcal{U}$ such that the diameter of each $\mathcal{U}_i^{(n)}$, $i \in [\![1,n]\!]$ tends to 0 when $n \to +\infty$. Here, the diameter of a subset of the Euclidean space $\mathbb{R}^{d_x}$ is defined as the maximum distance between two points of this subset. For $i \in [\![1,n]\!]$, we define $x_i^{(n)} \in \mathcal{U}_i^{(n)}$, and we let:

$$a_i^{(n)} \triangleq \left(\int_{\mathcal{U}_i^{(n)}} dx \right) \phi\left(x_i^{(n)}\right) \tag{B.64}$$

By continuity of ϕ, the function:

$$(x,y) \mapsto \phi(x)^\top k(x,y)\, \phi(y) \tag{B.65}$$

is continuous on the compact $\mathcal{U} \times \mathcal{U}$, so its Riemann sums converge:

$$\sum_{i=1}^{n} \sum_{j=1}^{n} \left(a_i^{(n)}\right)^\top k\left(x_i^{(n)}, x_j^{(n)}\right) a_j^{(n)} \underset{n \to +\infty}{\longrightarrow} \iint_{\mathcal{U}\times\mathcal{U}} \phi(x)^\top k(x,y)\, \phi(y)\, dx\, dy = \langle \phi | \mathcal{K}\phi \rangle \tag{B.66}$$

The SPSDness of k ensures that each Riemann sum is non-negative. By passing to the limit, it follows that:

$$\langle \phi | \mathcal{K}\phi \rangle \geq 0 \tag{B.67}$$

for every continuous function ϕ.

Finally, recall that the set of continuous functions of $\mathcal{U}$ is dense in $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. By continuity of:

$$\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z}) \mapsto \langle \phi | \mathcal{K}\phi \rangle \tag{B.68}$$

the inequality (B.67) extends to all $\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$.

B.9 Proof of Proposition 3.12

By Proposition 3.11, $\mathcal{K}$ is a compact and self-adjoint operator on $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. The spectral theorem [132] thus states that there exists an orthonormal basis $(\psi_i)_{i \in \mathbb{N}^*}$

of $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ consisting of eigenfunctions of $\mathcal{K}$, with each corresponding eigenvalue $(\lambda_i)_{i \in \mathbb{N}^*}$ being real. Up to reordering, the sequence of eigenvalues can be considered to be non-increasing.

Since k is uniformly continuous on the compact $\mathcal{U} \times \mathcal{U}$, $\mathcal{K}\phi$ is continuous for every $\phi \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$. In particular, for each $i \in \mathbb{N}^*$ such that $\lambda_i \neq 0$, the eigenfunction:

$$\psi_i = \lambda_i^{-1} \mathcal{K} \psi_i \tag{B.69}$$

is continuous. Without loss of generality, it must be noted that the ψ_i associated with a null eigenvalue do not contribute to (3.70) and may thus be dismissed.

Since $(\psi_i)_{i \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$, the family $(\psi_i \, \psi_j^\top)_{i,j \in \mathbb{N}^*}$ is an orthonormal basis of $\mathcal{F}$. Let $\langle \cdot | \cdot \rangle_{\mathcal{U} \times \mathcal{U}}$ denote:

$$(\phi, \phi') \in \mathcal{F}^2 \mapsto \langle \phi | \phi' \rangle_{\mathcal{U} \times \mathcal{U}} \triangleq \iint_{\mathcal{U} \times \mathcal{U}} \operatorname{tr}\left(\phi(x,y)^\top \phi'(x,y) \right) \, dx \, dy \tag{B.70}$$

Let $i, j \in \mathbb{N}^*$. We have:

$$\begin{aligned}
\langle k | \psi_i \, \psi_j^\top \rangle_{\mathcal{U} \times \mathcal{U}} &= \iint_{\mathcal{U} \times \mathcal{U}} \operatorname{tr}\left(k(x,y)^\top \psi_i(x) \, \psi_j(y)^\top \right) dx \, dy \\
&= \iint_{\mathcal{U} \times \mathcal{U}} \psi_j(y)^\top k(x,y)^\top \psi_i(x) \, dx \, dy \\
&= \int_{\mathcal{U}} \psi_j(y)^\top \left(\int_{\mathcal{U}} k(y,x) \, \psi_i(x) \, dx \right) dy \\
&= \int_{\mathcal{U}} \psi_j(y)^\top \mathcal{K} \psi_i(y) \, dy \\
&= \lambda_i \int_{\mathcal{U}} \psi_j(y)^\top \, \psi_i(y) \, dy \\
&= \lambda_i \, \delta_{i,j}
\end{aligned} \tag{B.71}$$

where:

$$\delta_{i,j} \triangleq \begin{cases} 1 & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \tag{B.72}$$

is the Kronecker symbol.

Because $k \in \mathcal{F}$, it follows from the completeness of $(\psi_i \, \psi_j^\top)_{i,j \in \mathbb{N}^*}$ that:

$$k_{r_{\mathrm{KL}}}(x, x') \triangleq \sum_{i=1}^{r_{\mathrm{KL}}} \lambda_i \, \psi_i(x) \, \psi_i(x')^\top \tag{B.73}$$

converges to k in the $\mathcal{F}$ sense.

It must be noted that $\mathcal{L}^2$ convergence only identifies k up to a Lebesgue-null subset of $\mathcal{U} \times \mathcal{U}$, and gives no control at any specific point of the domain. In particular, it is not enough to justify substituting $k_{r_{\mathrm{KL}}}$ for k in kriging scenarios. However, since k is continuous on $\mathcal{U} \times \mathcal{U}$, the convergence can be extended to absolute and uniform convergence on the entire domain [80]. The argument applies Dini's theorem [133] to the non-decreasing sequence $(x \mapsto \operatorname{tr} \, k_{r_{\mathrm{KL}}}(x,x))_{r_{\mathrm{KL}} \in \mathbb{N}^*}$, whose limit is continuous on the compact $\mathcal{U}$, and extends the result off the diagonal through the Cauchy-Schwarz inequality.

Since $\mathcal{K}$ is SPSD, every eigenvalue is non-negative. Assume that there exists a rank $r_0 \in \mathbb{N}^*$ such that $\lambda_i = 0$ as soon as $i > r_0$. Then the decomposition (3.70) would reduce to the finite sum:

$$k = \sum_{i=1}^{r_0} \lambda_i \, \psi_i \, \psi_i^\top \tag{B.74}$$

whose Gram matrices have rank at most r_0 and are thus singular at any n distinct points with $nd_z > r_0$, contradicting the SPDness of k by Proposition 3.2. Consequently:

$$\lambda_1 \geq \lambda_2 \geq \cdots > 0 \tag{B.75}$$

B.10 Proof of Proposition 3.14

As (3.80) is a rank r_{KL} perturbation of $A \triangleq I_{\tilde{N}} \otimes \tilde{R}$, the Woodbury inversion formula (see Proposition A.5) can be used to reduce the $\tilde{N}d_z \times \tilde{N}d_z$ inversion of $G(\tilde{\mathbf{x}})$ to an $r_{\mathrm{KL}} \times r_{\mathrm{KL}}$ one:

$$G(\tilde{\mathbf{x}})^{-1} \approx A^{-1} - A^{-1}\Psi(\tilde{\mathbf{x}})\, \tilde{G}(\tilde{\mathbf{x}})^{-1}\Psi(\tilde{\mathbf{x}})^\top A^{-1} \tag{B.76}$$

Computing the centred SK estimator from Proposition 3.5 using this approximation and the push-through identity (see Proposition A.4) yields:

$$\begin{aligned}
\hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) &= k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1}\tilde{\mathbf{z}} \\
&\approx \Psi(x)\, \Lambda\, \Psi(\tilde{\mathbf{x}})^\top \left(I - A^{-1}\Psi(\tilde{\mathbf{x}})\, \tilde{G}(\tilde{\mathbf{x}})^{-1}\Psi(\tilde{\mathbf{x}})^\top\right) A^{-1}\tilde{\mathbf{z}} \\
&= \Psi(x)\, \Lambda \left(I - \Psi(\tilde{\mathbf{x}})^\top A^{-1}\Psi(\tilde{\mathbf{x}})\, \tilde{G}(\tilde{\mathbf{x}})^{-1}\right) \Psi(\tilde{\mathbf{x}})^\top A^{-1}\tilde{\mathbf{z}}
\end{aligned} \tag{B.77}$$

Recall that, by definition of $\tilde{G}(\tilde{\mathbf{x}})$:

$$\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1} = \Psi(\tilde{\mathbf{x}})^\top A^{-1}\Psi(\tilde{\mathbf{x}}) \tag{B.78}$$

which gives:

$$\begin{aligned}
I - \Psi(\tilde{\mathbf{x}})^\top A^{-1}\Psi(\tilde{\mathbf{x}})\, \tilde{G}(\tilde{\mathbf{x}})^{-1} &= I - \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1}\right) \tilde{G}(\tilde{\mathbf{x}})^{-1} \\
&= I - I + \Lambda^{-1}\tilde{G}(\tilde{\mathbf{x}})^{-1} \\
&= \Lambda^{-1}\tilde{G}(\tilde{\mathbf{x}})^{-1}
\end{aligned} \tag{B.79}$$

Plugging this result into (B.77) and using $A^{-1} = I_{\tilde{N}} \otimes \tilde{R}^{-1}$ finally gives:

$$\hat{z}_{\mathrm{SK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, 0) \approx \Psi(x)\, \tilde{G}(\tilde{\mathbf{x}})^{-1}\Psi(\tilde{\mathbf{x}})^\top A^{-1}\tilde{\mathbf{z}} \tag{B.80}$$

For R_{SK}, reusing:

$$k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} \approx \Psi(x)\, \tilde{G}(\tilde{\mathbf{x}})^{-1}\Psi(\tilde{\mathbf{x}})^\top A^{-1} \tag{B.81}$$

together with (B.79) yields:

$$\begin{aligned}
R_{\mathrm{SK}} &= k(x, x) + R - k(x, \tilde{\mathbf{x}})\, G(\tilde{\mathbf{x}})^{-1} k(\tilde{\mathbf{x}}, x) \\
&\approx \Psi(x)\, \Lambda\, \Psi(x)^\top + R - \Psi(x)\, \tilde{G}(\tilde{\mathbf{x}})^{-1}\Psi(\tilde{\mathbf{x}})^\top A^{-1} \times \Psi(\tilde{\mathbf{x}})\, \Lambda\, \Psi(x)^\top \\
&= \Psi(x)\, \Lambda\, \Psi(x)^\top + R - \Psi(x)\, \tilde{G}(\tilde{\mathbf{x}})^{-1} \left(\tilde{G}(\tilde{\mathbf{x}}) - \Lambda^{-1}\right) \Lambda\, \Psi(x)^\top \\
&= \Psi(x)\, \Lambda\, \Psi(x)^\top + R - \Psi(x) \left(\Lambda - \tilde{G}(\tilde{\mathbf{x}})^{-1}\right) \Psi(x)^\top \\
&= R + \Psi(x)\, \tilde{G}(\tilde{\mathbf{x}})^{-1}\Psi(x)^\top
\end{aligned} \tag{B.82}$$

Finally, when $\tilde{R} = \sigma^2 I_{d_z}$ with $\sigma > 0$, substituting $A^{-1} = \sigma^{-2} I_{\tilde{N} d_z}$ yields the announced simplifications. For such $\sigma > 0$, the matrix:

$$\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1} \succ 0 \tag{B.83}$$

is SPD, as the sum of the SPSD matrix $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}})$ and the SPD matrix $\sigma^2 \Lambda^{-1}$. The map:

$$\sigma \in \mathbb{R}_+^* \mapsto \left(\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) + \sigma^2 \Lambda^{-1}\right)^{-1} \tag{B.84}$$

is thus well-defined and continuous, and extends to $\sigma = 0$ when $\Psi(\tilde{\mathbf{x}})^\top \Psi(\tilde{\mathbf{x}}) \succ 0$, *i.e.* when $\Psi(\tilde{\mathbf{x}})$ has full column rank. As $\hat{z}_{\mathrm{SK}}$ and R_{SK} depend on σ only through this inverse and an explicit factor σ^2, both expressions remain valid at $\sigma = 0$ under this condition.

B.11 Proof of Proposition 3.15

Let $i \in \mathbb{N}^*$. Since ψ_i is bounded on the compact $\mathcal{U}$, Tonelli's theorem [131] and the continuity of k give:

$$\mathbb{E}\left[\int_{\mathcal{U}} \|f(x) - m(x)\|^2 \, dx\right] = \int_{\mathcal{U}} \operatorname{tr} k(x,x) \, dx < \infty \tag{B.85}$$

so that $f - m \in \mathcal{L}^2(\mathcal{U}, \mathbb{R}^{d_z})$ a.s.. Each $\langle f - m | \psi_i \rangle$ is thus well-defined and finite. This integrability also justifies the exchanges of expectation and integration below.

Since any finite linear combination of $(\xi_i)_{i \in \mathbb{N}^*}$ is itself a linear combination of the values of f, they are jointly Gaussian. Let $j \in \mathbb{N}^*$. It comes that:

$$\mathbb{E}[\xi_i] = \frac{1}{\sqrt{\lambda_i}} \int_{\mathcal{U}} \mathbb{E}[f(x) - m(x)]^\top \psi_i(x) \, dx = 0 \tag{B.86}$$

and:

$$\begin{aligned}
\mathbb{E}[\xi_i \, \xi_j] &= \frac{1}{\sqrt{\lambda_i \, \lambda_j}} \iint_{\mathcal{U}\times\mathcal{U}} \psi_i(x)^\top \mathbb{E}[(f(x) - m(x))(f(x') - m(x'))^\top] \, \psi_j(x') \, dx \, dx' \\
&= \frac{1}{\sqrt{\lambda_i \, \lambda_j}} \iint_{\mathcal{U}\times\mathcal{U}} \psi_i(x)^\top k(x,x') \, \psi_j(x') \, dx \, dx' \\
&= \frac{1}{\sqrt{\lambda_i \, \lambda_j}} \int_{\mathcal{U}} \psi_i(x)^\top \left(\int_{\mathcal{U}} k(x,x') \, \psi_j(x') \, dx' \right) dx \\
&= \frac{\lambda_j}{\sqrt{\lambda_i \, \lambda_j}} \int_{\mathcal{U}} \psi_i(x)^\top \psi_j(x) \, dx \\
&= \frac{\lambda_j}{\sqrt{\lambda_i \, \lambda_j}} \langle \psi_i | \psi_j \rangle \\
&= \delta_{i,j}
\end{aligned} \tag{B.87}$$

using the fact that ψ_i is an eigenfunction of the Mercer operator $\mathcal{K}$. Together with the jointly Gaussian nature of $(\xi_i)_{i \in \mathbb{N}^*}$, this proves that they are i.i.d standard Gaussian variables.

Finally, for $r \in \mathbb{N}^*$, let:

$$f_r : \begin{cases} \mathcal{U} \to \mathbb{R}^{d_z} \\ x \mapsto m(x) + \sum_{i=1}^{r} \sqrt{\lambda_i}\, \xi_i\, \psi_i(x) \end{cases} \tag{B.88}$$

Fubini's theorem can be used to compute the following cross-covariance:

$$\begin{aligned} \mathbb{E}\left[(f(x) - m(x))\, \xi_i\right] &= \frac{1}{\sqrt{\lambda_i}} \mathbb{E}[(f(x) - m(x))\langle f - m | \psi_i \rangle] \\ &= \frac{1}{\sqrt{\lambda_i}} \mathbb{E}\left[(f(x) - m(x)) \int_{\mathcal{U}} (f(y) - m(y))^\top \psi_i(y)\, dy\right] \\ &= \frac{1}{\sqrt{\lambda_i}} \int_{\mathcal{U}} \mathbb{E}\left[(f(x) - m(x))(f(y) - m(y))^\top\right] \psi_i(y)\, dy \\ &= \frac{1}{\sqrt{\lambda_i}} \int_{\mathcal{U}} k(x, y)\, \psi_i(y)\, dy \\ &= \sqrt{\lambda_i}\, \psi_i(x) \end{aligned} \tag{B.89}$$

which can be used to obtain:

$$\begin{aligned} \mathbb{E}\left[\|f(x) - f_r(x)\|^2\right] &= \mathbb{E}\left[\|f(x) - m(x)\|^2\right] - 2 \sum_{i=1}^{r} \sqrt{\lambda_i}\, \psi_i(x)^\top \mathbb{E}\left[(f(x) - m(x))\, \xi_i\right] \\ &\quad + \sum_{i=1}^{r} \sum_{j=1}^{r} \sqrt{\lambda_i\, \lambda_j}\, \mathbb{E}[\xi_i\, \xi_j]\, \psi_i(x)^\top \psi_j(x) \\ &= \operatorname{tr} k(x, x) - 2 \sum_{i=1}^{r} \lambda_i \operatorname{tr}\left(\psi_i(x)\, \psi_i(x)^\top\right) + \sum_{i=1}^{r} \lambda_i \operatorname{tr}\left(\psi_i(x)\, \psi_i(x)^\top\right) \\ &= \operatorname{tr} k(x, x) - \sum_{i=1}^{r} \lambda_i \operatorname{tr}\left(\psi_i(x)\, \psi_i(x)^\top\right) \\ &= \sum_{i=r+1}^{+\infty} \lambda_i \operatorname{tr}\left(\psi_i(x)\, \psi_i(x)^\top\right) \end{aligned} \tag{B.90}$$

By Mercer's theorem, this last sum is the tail of:

$$\operatorname{tr} k(x, x) = \sum_{i=1}^{+\infty} \lambda_i \operatorname{tr}\left(\psi_i(x)\, \psi_i(x)^\top\right) \tag{B.91}$$

and therefore converges to 0. The partial sums thus converge to f in mean square, uniformly over $\mathcal{U}$.

B.12 Proof of Proposition 3.16

For $r \in \mathbb{N}^*$, let:

$$f_r : \begin{cases} \mathcal{U} \to \mathbb{R}^{d_z} \\ x \mapsto m(x) + \sum_{i=1}^{r} \sqrt{\lambda_i}\, \xi_i\, \psi_i(x) \end{cases} \tag{B.92}$$

be the partial sum of the series.

Let $r' \in \mathbb{N}^*$ such that $r' > r$ and $x \in \mathcal{U}$. Since the $(\xi_i)_{i\in\mathbb{N}^*}$ are i.i.d standard Gaussian variables, it follows that:

$$\begin{aligned}\mathbb{E}\left[\|f_r(x) - f_{r'}(x)\|^2\right] &= \sum_{i=r+1}^{r'} \lambda_i \, \|\psi_i(x)\|^2 \\ &= \sum_{i=r+1}^{r'} \lambda_i \, \operatorname{tr}\left(\psi_i(x)\, \psi_i(x)^\top\right)\end{aligned} \tag{B.93}$$

As for Proposition 3.15, this sum is the tail of the convergent series:

$$\operatorname{tr} k(x,x) = \sum_{i=1}^{+\infty} \lambda_i \, \operatorname{tr}\left(\psi_i(x)\, \psi_i(x)^\top\right) \tag{B.94}$$

and tends to 0. The sequence $(f_r(x))_{r\in\mathbb{N}^*}$ is thus a Cauchy sequence in the space of square-integrable $\mathbb{R}^{d_z}$-valued random variables, which is complete. The sequence thus converges, and $f(x)$ is well-defined as its limit.

Let $n \in \mathbb{N}^*$, $x_1, \ldots, x_n \in \mathcal{U}$ and $a_1, \ldots, a_n \in \mathbb{R}^{d_z}$. For every $r \in \mathbb{N}^*$, the sum:

$$\sum_{i=1}^{n} a_i^\top f_r(x_i) \tag{B.95}$$

is affine in $\xi_1, \ldots, \xi_r$, and is thus Gaussian. Passing to the limit, this proves that:

$$\sum_{i=1}^{n} a_i^\top f(x_i) \tag{B.96}$$

is also Gaussian, and so f is a GP by Definition 3.1.

Since mean-square convergence carries the first two moments to the limit, the mean of f is:

$$\begin{aligned}\mathbb{E}[f(x)] &= \lim_{r\to+\infty} \mathbb{E}[f_r(x)] \\ &= m(x) + \lim_{r\to+\infty} \sum_{i=1}^{r} \sqrt{\lambda_i}\, \mathbb{E}[\xi_i]\, \psi_i(x) \\ &= m(x)\end{aligned} \tag{B.97}$$

and its covariance is:

$$\begin{aligned}\mathbb{E}\left[(f(x) - m(x))(f(x') - m(x'))^\top\right] &= \lim_{r\to+\infty} \sum_{i=1}^{r}\sum_{j=1}^{r} \sqrt{\lambda_i\, \lambda_j}\, \mathbb{E}[\xi_i\, \xi_j]\, \psi_i(x)\, \psi_j(x')^\top \\ &= \lim_{r\to+\infty} \sum_{i=1}^{r}\sum_{j=1}^{r} \sqrt{\lambda_i\, \lambda_j}\, \delta_{i,j}\, \psi_i(x)\, \psi_j(x')^\top \\ &= \sum_{i=1}^{+\infty} \lambda_i\, \psi_i(x)\, \psi_i(x')^\top \\ &= k(x,x')\end{aligned} \tag{B.98}$$

B.13 Proof of Proposition 3.18

Hilbert space. Let:

$$\mathcal{E} \triangleq \operatorname{span}\{k(x,\cdot),\ x \in \mathcal{U}\} \tag{B.99}$$

and let $\psi, \varphi \in \mathcal{E}$. There exist $n, n' \in \mathbb{N}^*$, $x_1, \dots, x_n, x'_1, \dots, x'_{n'} \in \mathcal{U}$ and $a_1, \dots, a_n, a'_1, \dots, a'_{n'} \in \mathbb{R}$ such that:

$$\psi = \sum_{i=1}^{n} a_i\, k(x_i, \cdot) \quad \text{and} \quad \varphi = \sum_{j=1}^{n'} a'_j\, k(x'_j, \cdot) \tag{B.100}$$

Since k is symmetric, extending $\langle \cdot \mid \cdot \rangle_{\mathcal{H}_k}$ to $\mathcal{E}$ by bilinearity gives:

$$\begin{aligned} \langle \psi \mid \varphi \rangle_{\mathcal{H}_k} &= \sum_{i=1}^{n} \sum_{j=1}^{n'} a_i\, a'_j\, k(x_i, x'_j) \\ &= \sum_{j=1}^{n'} a'_j\, \psi(x'_j) \\ &= \sum_{i=1}^{n} a_i\, \varphi(x_i) \end{aligned} \tag{B.101}$$

The second expression only depends on ψ through its values, and the third one only depends on φ through its values. The result thus does not depend on the decompositions of ψ and φ, which are not unique in general, and $\langle \cdot \mid \cdot \rangle_{\mathcal{H}_k}$ is well-defined on $\mathcal{E}$. It is symmetric and bilinear by construction. Since k is SPSD, it is also positive semi-definite:

$$\langle \psi \mid \psi \rangle_{\mathcal{H}_k} = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i\, a_j\, k(x_i, x_j) \geq 0 \tag{B.102}$$

Let $x \in \mathcal{U}$. In particular, for $\varphi = k(x, \cdot)$, it comes that:

$$\langle \psi \mid k(x,\cdot) \rangle_{\mathcal{H}_k} = \psi(x) \tag{B.103}$$

The Cauchy-Schwarz inequality, which holds for any symmetric positive semi-definite bilinear form, thus gives:

$$\begin{aligned} \psi(x)^2 &= \langle \psi \mid k(x,\cdot) \rangle_{\mathcal{H}_k}^2 \\ &\leq \langle \psi \mid \psi \rangle_{\mathcal{H}_k}\, \langle k(x,\cdot) \mid k(x,\cdot) \rangle_{\mathcal{H}_k} \\ &= \langle \psi \mid \psi \rangle_{\mathcal{H}_k}\, k(x,x) \end{aligned} \tag{B.104}$$

Consequently, $\langle \psi \mid \psi \rangle_{\mathcal{H}_k} = 0$ implies $\psi = 0$, and $\langle \cdot \mid \cdot \rangle_{\mathcal{H}_k}$ is thus an inner product on $\mathcal{E}$.

The completion $\mathcal{H}_k$ of $\mathcal{E}$ is a Hilbert space in which $\mathcal{E}$ is dense by construction, and its inner product is the extension of $\langle \cdot \mid \cdot \rangle_{\mathcal{H}_k}$ by continuity.

Reproducing kernel. We first identify the elements of $\mathcal{H}_k$ with functions from $\mathcal{U}$ to $\mathbb{R}$. Let $x \in \mathcal{U}$ and $\psi \in \mathcal{H}_k$, and let $(\psi_r)_{r \in \mathbb{N}^*}$ be a sequence of $\mathcal{E}$ that converges to ψ in $\mathcal{H}_k$. By continuity of the inner product, it comes that:

$$\lim_{r \to +\infty} \psi_r(x) = \lim_{r \to +\infty} \langle \psi_r \mid k(x,\cdot) \rangle_{\mathcal{H}_k} = \langle \psi \mid k(x,\cdot) \rangle_{\mathcal{H}_k} \tag{B.105}$$

The sequence $(\psi_r)_{r\in\mathbb{N}^*}$ thus converges pointwise to the function:

$$x \in \mathcal{U} \mapsto \langle \psi \mid k(x,\cdot)\rangle_{\mathcal{H}_k} \tag{B.106}$$

which only depends on ψ, and coincides with it when $\psi \in \mathcal{E}$. Assume that this function is null. Then ψ is orthogonal to $\mathcal{E}$, and thus:

$$\langle \psi \mid \psi\rangle_{\mathcal{H}_k} = \lim_{r\to+\infty} \langle \psi \mid \psi_r\rangle_{\mathcal{H}_k} = 0 \tag{B.107}$$

so $\psi = 0$. By linearity, two elements of $\mathcal{H}_k$ associated with the same function are therefore equal.

Each element of $\mathcal{H}_k$ can thus be identified with its associated function, and $\mathcal{H}_k$ is a Hilbert space of functions from $\mathcal{U}$ to $\mathbb{R}$ which verifies:

$$\forall x \in \mathcal{U},\ \forall \psi \in \mathcal{H}_k,\ \langle \psi \mid k(x,\cdot)\rangle_{\mathcal{H}_k} = \psi(x) \tag{B.108}$$

Since $k(x,\cdot) \in \mathcal{H}_k$ for every $x \in \mathcal{U}$, Definition 3.6 is satisfied, and $\mathcal{H}_k$ is an RKHS whose reproducing kernel is k.

Uniqueness. Let $(\mathcal{H}, \langle\cdot \mid \cdot\rangle_{\mathcal{H}})$ be an RKHS whose reproducing kernel is k. By Definition 3.6, it contains $k(x,\cdot)$ for every $x \in \mathcal{U}$, and thus:

$$\mathcal{E} \subseteq \mathcal{H} \tag{B.109}$$

Let $x, x' \in \mathcal{U}$. The reproducing property of $\mathcal{H}$ gives:

$$\langle k(x,\cdot) \mid k(x',\cdot)\rangle_{\mathcal{H}} = k(x,x') = \langle k(x,\cdot) \mid k(x',\cdot)\rangle_{\mathcal{H}_k} \tag{B.110}$$

and the two inner products thus coincide on $\mathcal{E}$.

Let $\psi \in \mathcal{H}$ be orthogonal to $\mathcal{E}$. Then:

$$\psi(x) = \langle \psi \mid k(x,\cdot)\rangle_{\mathcal{H}} = 0 \tag{B.111}$$

so $\psi = 0$, and $\mathcal{E}$ is dense in $\mathcal{H}$.

Let $\psi \in \mathcal{H}$, and let $(\psi_r)_{r\in\mathbb{N}^*}$ be a sequence of $\mathcal{E}$ that converges to ψ in $\mathcal{H}$. Since the two norms coincide on $\mathcal{E}$, $(\psi_r)_{r\in\mathbb{N}^*}$ is a Cauchy sequence of $\mathcal{E}$, which converges to some $\psi' \in \mathcal{H}_k$. By continuity of the two inner products, it comes that:

$$\begin{aligned}
\psi(x) &= \langle \psi \mid k(x,\cdot)\rangle_{\mathcal{H}} \\
&= \lim_{r\to+\infty} \langle \psi_r \mid k(x,\cdot)\rangle_{\mathcal{H}} \\
&= \lim_{r\to+\infty} \psi_r(x) \\
&= \lim_{r\to+\infty} \langle \psi_r \mid k(x,\cdot)\rangle_{\mathcal{H}_k} \\
&= \langle \psi' \mid k(x,\cdot)\rangle_{\mathcal{H}_k} \\
&= \psi'(x)
\end{aligned} \tag{B.112}$$

and:

$$\|\psi\|_{\mathcal{H}} = \lim_{r\to+\infty} \|\psi_r\|_{\mathcal{H}} = \lim_{r\to+\infty} \|\psi_r\|_{\mathcal{H}_k} = \|\psi'\|_{\mathcal{H}_k} \tag{B.113}$$

Therefore, $\psi = \psi' \in \mathcal{H}_k$, and its norm is the same in both spaces. Since $\mathcal{H}$ is complete, exchanging the roles of $\mathcal{H}$ and $\mathcal{H}_k$ gives the converse inclusion. Consequently, $\mathcal{H} = \mathcal{H}_k$, and the two inner products are equal.

B.14 Proof of Proposition 4.4

As established in section B.6, the weight matrix W of (**??**) and the Lagrange multiplier ξ of the unbiasedness constraints verify:

$$\mathcal{G}(\tilde{\mathbf{x}}) \begin{bmatrix} W \\ \xi \end{bmatrix} = \bar{k}(\tilde{\mathbf{x}}, x) \tag{B.114}$$

whose first block row is $G(\tilde{\mathbf{x}})\, W + g(\tilde{\mathbf{x}})\, \xi = k(\tilde{\mathbf{x}}, x)$. Let $e_{\tilde{\mathbf{x}}}$ and $e_{\tilde{\mathbf{x}}^r}$ denote the errors of the full and subsampled estimators:

$$\begin{cases} e_{\tilde{\mathbf{x}}} \triangleq z - \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) \\ e_{\tilde{\mathbf{x}}^r} \triangleq z - \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) \end{cases} \tag{B.115}$$

Since $e_{\tilde{\mathbf{x}}}$ is centred whatever the value of β, and ε is independent of $\tilde{\mathbf{z}}$, it comes that:

$$\mathbb{E}\left[\tilde{\mathbf{z}}\, e_{\tilde{\mathbf{x}}}^\top\right] = k(\tilde{\mathbf{x}}, x) - G(\tilde{\mathbf{x}})\, W = g(\tilde{\mathbf{x}})\, \xi \tag{B.116}$$

Therefore, $e_{\tilde{\mathbf{x}}}$ is uncorrelated with every linear combination $V^\top \tilde{\mathbf{z}}$ such that $V^\top g(\tilde{\mathbf{x}}) = 0$. The difference between the two interpolators can be decomposed as follows:

$$\hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) = e_{\tilde{\mathbf{x}}^r} - e_{\tilde{\mathbf{x}}} \tag{B.117}$$

Thus:

$$\begin{aligned} E_r(x) &= \mathbb{E}\left[(e_{\tilde{\mathbf{x}}^r} - e_{\tilde{\mathbf{x}}})\,(e_{\tilde{\mathbf{x}}^r} - e_{\tilde{\mathbf{x}}})^\top\right] \\ &= \mathbb{E}\left[e_{\tilde{\mathbf{x}}^r} e_{\tilde{\mathbf{x}}^r}^\top\right] + \mathbb{E}\left[e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}}^\top\right] - \mathbb{E}\left[e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}^r}^\top\right] - \mathbb{E}\left[e_{\tilde{\mathbf{x}}^r} e_{\tilde{\mathbf{x}}}^\top\right] \end{aligned} \tag{B.118}$$

Since both interpolators are unbiased, their difference is such a linear combination, and so:

$$\mathbb{E}\left[e_{\tilde{\mathbf{x}}}\,(e_{\tilde{\mathbf{x}}^r} - e_{\tilde{\mathbf{x}}})^\top\right] = 0 \tag{B.119}$$

The last two terms are thus both equal to $\mathbb{E}\left[e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}}^\top\right]$. Moreover, the second moments of the errors of unbiased interpolators are their covariances. Therefore:

$$E_r(x) = \mathbb{E}\left[e_{\tilde{\mathbf{x}}^r} e_{\tilde{\mathbf{x}}^r}^\top\right] - \mathbb{E}\left[e_{\tilde{\mathbf{x}}} e_{\tilde{\mathbf{x}}}^\top\right] = R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, g) - R_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, g) \tag{B.120}$$

Using Proposition 4.3 yields:

$$\begin{aligned} E_r(x) &= \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{k}(\tilde{\mathbf{x}}, x) - \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger\, \bar{k}(\tilde{\mathbf{x}}, x) \\ &= \bar{k}(x, \tilde{\mathbf{x}})\, \mathfrak{D}_G \bar{k}(\tilde{\mathbf{x}}, x) \end{aligned} \tag{B.121}$$

For $e_f(x)$, direct calculations give:

$$\begin{aligned} e_f(x) &= \left\| \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}, \tilde{\mathbf{z}}, g) - \hat{z}_{\mathrm{UK}}(x, \tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) \right\|_2 \\ &= \left\| \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}(\tilde{\mathbf{x}})^{-1} \bar{\mathbf{z}} - \bar{k}(x, \tilde{\mathbf{x}})\, \mathcal{G}_h(\tilde{\mathbf{x}}^r)^\dagger\, \bar{\mathbf{z}} \right\|_2 \\ &= \left\| \bar{k}(x, \tilde{\mathbf{x}})\, \mathfrak{D}_G \bar{\mathbf{z}} \right\|_2 \end{aligned} \tag{B.122}$$

B.15 Proof of Proposition 4.7

For the sake of brevity, we write G_r, g_r and M_r for $G(\tilde{\mathbf{x}}^r)$, $g(\tilde{\mathbf{x}}^r)$ and $M(\tilde{\mathbf{x}}^r)$. By Proposition A.2, the inverse of $A = \mathcal{G}(\tilde{\mathbf{x}}^r)$ is:

$$\mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} = \begin{bmatrix} G_r^{-1} - G_r^{-1} g_r \, M_r \, g_r^\top G_r^{-1} & G_r^{-1} g_r \, M_r \\ M_r \, g_r^\top G_r^{-1} & -M_r \end{bmatrix} \tag{B.123}$$

Let $x' \in \mathcal{U}$, and let y and t be two matrices with respectively $\tilde{N}^r d_z$ and $r_{\mathrm{UK}} d_z$ rows. It comes that:

$$\bar{k}(x', \tilde{\mathbf{x}}^r) \, \mathcal{G}(\tilde{\mathbf{x}}^r)^{-1} \begin{bmatrix} y \\ t \end{bmatrix} = k(x', \tilde{\mathbf{x}}^r) \, G_r^{-1} y - U(x') \, M_r \left(t - g_r^\top G_r^{-1} y \right) \tag{B.124}$$

The three expressions of (4.109) are obtained by applying (B.124) to every element x' of $\tilde{\mathbf{x}}^d$:

- Taking $y = k(\tilde{\mathbf{x}}^r, x)$ and $t = g(x)^\top$ gives the expression of $k_{d|r}(x)$, since $t - g_r^\top G_r^{-1} y$ is then equal to $U(x)^\top$.
- Taking the columns of $k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)$ and of $g(\tilde{\mathbf{x}}^d)^\top$ for y and t gives the expression of $S_{d|r} = D - B^\top A^{-1} B$ in the same way.
- Taking $y = \tilde{\mathbf{z}}^r$ and $t = 0$ gives the expression of $\tilde{\mathbf{z}}_{d|r}$, since $g_r^\top G_r^{-1} \tilde{\mathbf{z}}^r = M_r^{-1} \hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$.

Similarly, $S^0_{d|r}$ is the Schur complement of $G(\tilde{\mathbf{x}})$ with respect to G_r, and Proposition A.2 gives:

$$G(\tilde{\mathbf{x}})^{-1} = \begin{bmatrix} G_r^{-1} & 0 \\ 0 & 0 \end{bmatrix} + \begin{bmatrix} -G_r^{-1} k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ I_{\tilde{N}^d d_z} \end{bmatrix} \left(S^0_{d|r} \right)^{-1} \begin{bmatrix} -G_r^{-1} k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d) \\ I_{\tilde{N}^d d_z} \end{bmatrix}^\top \tag{B.125}$$

Multiplying this expression by $g(\tilde{\mathbf{x}})^\top$ on the left and by $g(\tilde{\mathbf{x}})$ on the right yields:

$$g(\tilde{\mathbf{x}})^\top G(\tilde{\mathbf{x}})^{-1} g(\tilde{\mathbf{x}}) = g_r^\top G_r^{-1} g_r + U(\tilde{\mathbf{x}}^d)^\top \left(S^0_{d|r} \right)^{-1} U(\tilde{\mathbf{x}}^d) \tag{B.126}$$

which is exactly (4.110).

Finally, $S^0_{d|r}$ is SPD as the Schur complement of the SPD matrix $G(\tilde{\mathbf{x}})$ with respect to G_r [7]. Since $U(\tilde{\mathbf{x}}^d) \, M_r \, U(\tilde{\mathbf{x}}^d)^\top \succeq 0$, $S_{d|r}$ is thus SPD as well. Moreover, adding $g(\tilde{\mathbf{x}}) \, \beta_0$ to the observations adds $U(\tilde{\mathbf{x}}^d) \, \beta_0$ to $\tilde{\mathbf{z}}^0_{d|r}$ and β_0 to $\hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g)$, which leaves $\tilde{\mathbf{z}}_{d|r}$ unchanged.

B.16 Proof of Propositions 4.11 and 4.12

For the sake of brevity, we write M, M_r and U_d for $M(\tilde{\mathbf{x}})$, $M(\tilde{\mathbf{x}}^r)$ and $U(\tilde{\mathbf{x}}^d)$. By Proposition 4.7, we have:

$$\begin{cases} k_{d|r}(x) = k^0_{d|r}(x) + U_d \, M_r \, U(x)^\top \\ S_{d|r} = S^0_{d|r} + U_d \, M_r \, U_d^\top \succeq S^0_{d|r} \end{cases} \tag{B.127}$$

and thus $\|\cdot\|_{S_{d|r}} \leq \|\cdot\|_{S^0_{d|r}}$. Applying the triangle inequality to Proposition 4.8 gives:

$$e_r(x) \leq \|k^0_{d|r}(x)\|_{S^0_{d|r}} + \left\|U_d\, M_r\, U(x)^\top\right\|_{S_{d|r}} \tag{B.128}$$

with equality when $k^0_{d|r}(x) = 0$.

By (4.110), the Woodbury formula (see Proposition A.5) applied to $S_{d|r}$ yields:

$$S^{-1}_{d|r} = \left(S^0_{d|r}\right)^{-1} - \left(S^0_{d|r}\right)^{-1} U_d\, M\, U_d^\top \left(S^0_{d|r}\right)^{-1} \tag{B.129}$$

and thus, using (4.110) once more:

$$\begin{aligned} U_d^\top S^{-1}_{d|r} U_d &= \left(M^{-1} - M_r^{-1}\right) - \left(M^{-1} - M_r^{-1}\right) M \left(M^{-1} - M_r^{-1}\right) \\ &= \left(M^{-1} - M_r^{-1}\right) M\, M_r^{-1} \\ &= M_r^{-1} - M_r^{-1} M\, M_r^{-1} \end{aligned} \tag{B.130}$$

The second term of (B.128) is therefore:

$$\begin{aligned} \left\|U_d\, M_r\, U(x)^\top\right\|^2_{S_{d|r}} &= \left\|U(x)\, M_r\, U_d^\top S^{-1}_{d|r} U_d\, M_r\, U(x)^\top\right\|_{\text{op}} \\ &= \left\|U(x)\,(M_r - M)\, U(x)^\top\right\|_{\text{op}} \end{aligned} \tag{B.131}$$

which proves Proposition 4.11.

For Proposition 4.12, it must be noted that $S^0_{d|r} - I_{\tilde{N}^d} \otimes \tilde{R}$ is the Schur complement of:

$$k(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}) + \begin{bmatrix} I_{\tilde{N}^r} & 0 \\ 0 & 0 \end{bmatrix} \otimes \tilde{R} \succeq 0 \tag{B.132}$$

with respect to $G(\tilde{\mathbf{x}}^r)$. Since this matrix is SPSD, so is its Schur complement [7], and thus:

$$S^0_{d|r} \succeq I_{\tilde{N}^d} \otimes \tilde{R} \succeq \tilde{\lambda}\, I_{\tilde{N}^d d_z} \tag{B.133}$$

Consequently, we have:

$$\begin{cases} \|k^0_{d|r}(x)\|_{S^0_{d|r}} \leq \dfrac{\|k^0_{d|r}(x)\|_{\text{op}}}{\sqrt{\tilde{\lambda}}} \\ M^{-1} \preceq M_r^{-1} + \dfrac{U_d^\top U_d}{\tilde{\lambda}} \end{cases} \tag{B.134}$$

where the second inequality follows from (4.110). Since the inversion reverses the order of SPD matrices, applying these inequalities to Proposition 4.11 concludes the proof.

B.17 Proof of Proposition 4.14

Let $N(K)$ denote the matrix of the operator norms of the blocks of a matrix K, and let v be a vector whose blocks have Euclidean norms gathered in u. Since $\|Kv\|_2 \leq \|N(K)\, u\|_2$, it comes that:

$$\|K\|_{\text{op}} \leq \|N(K)\|_{\text{op}} \leq \sqrt{\|N(K)\|_1\, \|N(K)\|_\infty} \tag{B.135}$$

where $\|N(K)\|_1$ and $\|N(K)\|_\infty$ are the largest column and row sums of $N(K)$ [7]. Applied to the symmetric matrix $G(\tilde{\mathbf{x}}^j)$, this inequality gives the blockwise Gershgorin bound:

$$\lambda_{\max}\left(G(\tilde{\mathbf{x}}^j)\right) \leq \lambda_j^+ \tag{B.136}$$

Similarly, applying it to (4.126) gives:

$$\begin{cases} \|k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\|_{\text{op}} \leq \sqrt{\tilde{N}^r \tilde{N}^d}\, \alpha \\ \|k(\tilde{\mathbf{x}}^r, x)\|_{\text{op}} \leq \sqrt{\tilde{N}^r}\, \kappa \\ \|k(\tilde{\mathbf{x}}^d, x)\|_{\text{op}} \leq \sqrt{\tilde{N}^d}\, \eta \\ \|g(\tilde{\mathbf{x}}^j)\|_{\text{op}} \leq \sqrt{\tilde{N}^j}\, g_{\max} \\ \|\tilde{\mathbf{z}}^j - g(\tilde{\mathbf{x}}^j)\, \beta_0\|_2 \leq \sqrt{\tilde{N}^j}\, \tilde{z}_{\max} \end{cases} \tag{B.137}$$

Since $G(\tilde{\mathbf{x}}^r)$ is SPD, it follows that:

$$\begin{aligned} \|k^0_{d|r}(x)\|_{\text{op}} &\leq \|k(\tilde{\mathbf{x}}^d, x)\|_{\text{op}} + \frac{\|k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\|_{\text{op}}\, \|k(\tilde{\mathbf{x}}^r, x)\|_{\text{op}}}{\lambda_r^-} \\ &\leq \sqrt{\tilde{N}^d}\, \frac{\lambda_r^- \eta + \alpha \tilde{N}^r \kappa}{\lambda_r^-} \end{aligned} \tag{B.138}$$

The same steps, applied to $U(\tilde{\mathbf{x}}^d)$ and $U(x)$, give:

$$\begin{cases} \|U(\tilde{\mathbf{x}}^d)\|_{\text{op}} \leq \sqrt{\tilde{N}^d}\, g_{\max}\, \rho_r \\ \|U(x)\|_{\text{op}} \leq g_{\max}\, \dfrac{\lambda_r^- + \tilde{N}^r \kappa}{\lambda_r^-} \end{cases} \tag{B.139}$$

Furthermore, for every unit vector $v \in \mathbb{R}^{r_{\text{UK}} d_z}$, the Rayleigh quotient [7] and the definitions of λ_r^+ and λ_r^g give:

$$v^\top M(\tilde{\mathbf{x}}^r)^{-1} v = \left(g(\tilde{\mathbf{x}}^r)\, v\right)^\top G(\tilde{\mathbf{x}}^r)^{-1} \left(g(\tilde{\mathbf{x}}^r)\, v\right) \geq \frac{\|g(\tilde{\mathbf{x}}^r)\, v\|_2^2}{\lambda_r^+} \geq \frac{\lambda_r^g}{\lambda_r^+} \tag{B.140}$$

and thus $\|M(\tilde{\mathbf{x}}^r)\|_{\text{op}} \leq \lambda_r^+ / \lambda_r^g$. Applying the triangle inequality to the expression of $k_{d|r}(x)$ given by Proposition 4.7, and combining these inequalities, yields the bound on $\|k_{d|r}(x)\|_{\text{op}}$.

Since $S_{d|r} \succeq S^0_{d|r}$ by Proposition 4.7, we have $\lambda_{\min}(S_{d|r}) \geq \lambda_{\min}(S^0_{d|r})$. Moreover, for every unit vector $v \in \mathbb{R}^{\tilde{N}^d d_z}$:

$$\begin{aligned} v^\top S^0_{d|r}\, v &= v^\top G(\tilde{\mathbf{x}}^d)\, v - \left(k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)\, v\right)^\top G(\tilde{\mathbf{x}}^r)^{-1} \left(k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)\, v\right) \\ &\geq \lambda_d^- - \frac{\|k(\tilde{\mathbf{x}}^r, \tilde{\mathbf{x}}^d)\|_{\text{op}}^2}{\lambda_r^-} \\ &\geq \frac{\lambda_r^- \lambda_d^- - \alpha^2 \tilde{N}^r \tilde{N}^d}{\lambda_r^-} \end{aligned} \tag{B.141}$$

Since this lower bound does not depend on v, it also bounds $\lambda_{\min}(S^0_{d|r})$. It is positive by assumption, and so:

$$\|S^{-1}_{d|r}\|_{\text{op}} = \frac{1}{\lambda_{\min}(S_{d|r})} \leq s_{\max} \tag{B.142}$$

Finally, since $\tilde{\mathbf{z}}_{d|r}$ is unchanged when a trend is added to the observations, Proposition 4.7 can be applied to the observations $\tilde{\mathbf{z}} - g(\tilde{\mathbf{x}})\,\beta_0$:

$$\begin{aligned}\tilde{\mathbf{z}}_{d|r} = \big(\tilde{\mathbf{z}}^d - g(\tilde{\mathbf{x}}^d)\,\beta_0\big) - k(\tilde{\mathbf{x}}^d, \tilde{\mathbf{x}}^r)\,G(\tilde{\mathbf{x}}^r)^{-1}\,(\tilde{\mathbf{z}}^r - g(\tilde{\mathbf{x}}^r)\,\beta_0) \\ - U(\tilde{\mathbf{x}}^d)\,(\hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) - \beta_0)\end{aligned} \tag{B.143}$$

The steps used for $k^0_{d|r}(x)$ bound the norm of the first two terms by $\sqrt{\tilde{N}^d}\,\rho_r\,\tilde{z}_{\max}$. Moreover, (B.137) gives:

$$\begin{aligned}\|\hat{m}_{\mathrm{UK}}(\tilde{\mathbf{x}}^r, \tilde{\mathbf{z}}^r, g) - \beta_0\|_2 &= \big\|M(\tilde{\mathbf{x}}^r)\,g(\tilde{\mathbf{x}}^r)^\top G(\tilde{\mathbf{x}}^r)^{-1}\,(\tilde{\mathbf{z}}^r - g(\tilde{\mathbf{x}}^r)\,\beta_0)\big\|_2 \\ &\leq \frac{\lambda_r^+}{\lambda_r^g}\,\frac{\sqrt{\tilde{N}^r}\,g_{\max}\,\sqrt{\tilde{N}^r}\,\tilde{z}_{\max}}{\lambda_r^-} \\ &= \frac{\tilde{N}^r g_{\max}\,\lambda_r^+}{\lambda_r^g\,\lambda_r^-}\,\tilde{z}_{\max}\end{aligned} \tag{B.144}$$

which concludes the proof by the triangle inequality.

BIBLIOGRAPHY

[1] O. Cappé, E. Moulines, and T. Rydén. *Inference in Hidden Markov Models*. Springer Series in Statistics. New York, NY: Springer, 2005.

[2] G. A. Einicke. "Nonlinear Prediction, Filtering and Smoothing". In: *Smoothing, Filtering and Prediction: Estimating the Past, Present and Future*. 2nd. Prime Publishing, 2019, pp. 293–322.

[3] H. L. Van Trees. "Classical Detection and Estimation Theory". In: *Detection, Estimation, and Modulation Theory, Part I*. New York, NY: John Wiley & Sons, 1968.

[4] S. Särkkä. "Bayesian Filtering Equations and Exact Solutions". In: *Bayesian Filtering and Smoothing*. Cambridge, UK: Cambridge University Press, 2013.

[5] M. S. Arulampalam, S. Maskell, N. Gordon, and T. Clapp. "A Tutorial on Particle Filters for Online Nonlinear/Non-Gaussian Bayesian Tracking". In: *IEEE Transactions on Signal Processing* 50.2 (2002), pp. 174–188.

[6] R. E. Kalman. "A New Approach to Linear Filtering and Prediction Problems". In: *Journal of Basic Engineering* 82.1 (1960), pp. 35–45.

[7] R. A. Horn and C. R. Johnson. "Positive Definite and Semidefinite Matrices". In: *Matrix Analysis*. 2nd. Cambridge, UK: Cambridge University Press, 2013.

[8] B. D. O. Anderson and J. B. Moore. "The Discrete-Time Kalman Filter". In: *Optimal Filtering*. Englewood Cliffs, NJ: Prentice-Hall, 1979.

[9] R. S. Bucy and P. D. Joseph. *Filtering for Stochastic Processes with Applications to Guidance*. New York, NY: Interscience Publishers, 1968.

[10] B. D. O. Anderson and J. B. Moore. "Computational Aspects". In: *Optimal Filtering*. Englewood Cliffs, NJ: Prentice-Hall, 1979.

[11] S. Särkkä. "Extended and Unscented Kalman Filtering". In: *Bayesian Filtering and Smoothing*. Cambridge, UK: Cambridge University Press, 2013.

[12] J. Havlík and O. Straka. "Performance Evaluation of Iterated Extended Kalman Filter with Variable Step-Length". In: *Journal of Physics: Conference Series* 659 (2015), p. 012022.

[13] B. M. Bell and F. W. Cathey. "The Iterated Kalman Filter Update as a Gauss-Newton Method". In: *IEEE Transactions on Automatic Control* 38.2 (1993), pp. 294–297.

[14] A. H. Jazwinski. "Nonlinear Filtering". In: *Stochastic Processes and Filtering Theory*. New York, NY: Academic Press, 1970.

[15] S. J. Julier and J. K. Uhlmann. "A New Extension of the Kalman Filter to Nonlinear Systems". In: *Proceedings of AeroSense: The 11th International Symposium on Aerospace/Defense Sensing, Simulation and Controls*. 1997, pp. 182–193.

[16] G. H. Golub and C. F. Van Loan. *Matrix Computations*. 4th. Baltimore, MD: Johns Hopkins University Press, 2013.

[17] S. J. Julier and J. K. Uhlmann. "Unscented Filtering and Nonlinear Estimation". In: *Proceedings of the IEEE* 92.3 (2004), pp. 401–422.

[18] R. van der Merwe and E. A. Wan. "The Square-Root Unscented Kalman Filter for State and Parameter-Estimation". In: *2001 IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. Vol. 6. 2001, pp. 3461–3464.

[19] G. Evensen. "The Ensemble Kalman Filter: Theoretical Formulation and Practical Implementation". In: *Ocean Dynamics* 53.4 (2003), pp. 343–367.

[20] P. L. Houtekamer and F. Zhang. "Review of the Ensemble Kalman Filter for Atmospheric Data Assimilation". In: *Monthly Weather Review* 144.12 (2016), pp. 4498–4532.

[21] N. Metropolis and S. Ulam. "The Monte Carlo Method". In: *Journal of the American Statistical Association* 44.247 (1949), pp. 335–341.

[22] F. Le Gland. "Méthodes de Monte Carlo". In: *Filtrage bayésien et approximation particulaire*. Cours SOD333. ENSTA Paris, 2021.

[23] C. P. Robert and G. Casella. "Monte Carlo Integration". In: *Monte Carlo Statistical Methods*. 2nd. New York, NY: Springer, 2004.

[24] H. Kahn and A. W. Marshall. "Methods of Reducing Sample Size in Monte Carlo Computations". In: *Journal of the Operations Research Society of America* 1.5 (1953), pp. 263–278.

[25] J. Geweke. "Bayesian Inference in Econometric Models Using Monte Carlo Integration". In: *Econometrica* 57.6 (1989), pp. 1317–1339.

[26] A. Doucet, S. Godsill, and C. Andrieu. "On Sequential Monte Carlo Sampling Methods for Bayesian Filtering". In: *Statistics and Computing* 10.3 (2000), pp. 197–208.

[27] A. Doucet, N. de Freitas, and N. Gordon. "An Introduction to Sequential Monte Carlo Methods". In: *Sequential Monte Carlo Methods in Practice*. New York, NY: Springer New York, 2001, pp. 3–14.

[28] P. Del Moral. *Feynman-Kac Formulae: Genealogical and Interacting Particle Systems with Applications*. Probability and its Applications. New York, NY: Springer, 2004.

[29] F. Le Gland. "Généralisation : distributions de Feynman-Kac". In: *Filtrage bayésien et approximation particulaire*. Cours SOD333. ENSTA Paris, 2021.

[30] N. J. Higham. "Computing a Nearest Symmetric Positive Semidefinite Matrix". In: *Linear Algebra and its Applications* 103 (1988), pp. 103–118.

[31] A. Kong, J. S. Liu, and W. H. Wong. "Sequential Imputations and Bayesian Missing Data Problems". In: *Journal of the American Statistical Association* 89.425 (1994), pp. 278–288.

[32] D. B. Rubin. "Using the SIR Algorithm to Simulate Posterior Distributions". In: *Bayesian Statistics 3*. Oxford, UK: Oxford University Press, 1988, pp. 395–402.

[33] N. J. Gordon, D. J. Salmond, and A. F. M. Smith. "Novel Approach to Nonlinear/Non-Gaussian Bayesian State Estimation". In: *IEE Proceedings F – Radar and Signal Processing* 140.2 (1993), pp. 107–113.

[34] T. Li, M. Bolić, and P. M. Djurić. "Resampling Methods for Particle Filtering: Classification, Implementation, and Strategies". In: *IEEE Signal Processing Magazine* 32.3 (2015), pp. 70–86.

[35] P. Kozierski, M. Lis, and J. Ziętkiewicz. "Resampling in Particle Filtering – Comparison". In: *Studia z Automatyki i Informatyki* 38 (2013), pp. 35–64.

[36] G. Kitagawa. "Monte Carlo Filter and Smoother for Non-Gaussian Nonlinear State Space Models". In: *Journal of Computational and Graphical Statistics* 5.1 (1996), pp. 1–25.

[37] D. Crisan and A. Doucet. "A Survey of Convergence Results on Particle Filtering Methods for Practitioners". In: *IEEE Transactions on Signal Processing* 50.3 (2002), pp. 736–746.

[38] N. Oudjane. "Stabilité et approximations particulaires en filtrage non linéaire. Application au pistage". PhD thesis. Université de Rennes 1, 2000.

[39] C. Snyder, T. Bengtsson, P. Bickel, and J. Anderson. "Obstacles to High-Dimensional Particle Filtering". In: *Monthly Weather Review* 136.12 (2008), pp. 4629–4640.

[40] B. Ristic, S. Arulampalam, and N. Gordon. "A Tutorial on Particle Filters". In: *Beyond the Kalman Filter: Particle Filters for Tracking Applications*. Boston, MA: Artech House, 2004.

[41] T. Li, S. Sun, T. P. Sattar, and J. M. Corchado. "Fight Sample Degeneracy and Impoverishment in Particle Filters: A Review of Intelligent Approaches". In: *Expert Systems with Applications* 41.8 (2014), pp. 3944–3954.

[42] S. Särkkä. "Particle Filtering". In: *Bayesian Filtering and Smoothing*. Cambridge, UK: Cambridge University Press, 2013.

[43] J. S. Liu and R. Chen. "Sequential Monte Carlo Methods for Dynamic Systems". In: *Journal of the American Statistical Association* 93.443 (1998), pp. 1032–1044.

[44] C. Musso, N. Oudjane, and F. Le Gland. "Improving Regularised Particle Filters". In: *Sequential Monte Carlo Methods in Practice*. New York, NY: Springer New York, 2001, pp. 247–271.

[45] M. Rosenblatt. "Remarks on Some Nonparametric Estimates of a Density Function". In: *The Annals of Mathematical Statistics* 27.3 (1956), pp. 832–837.

[46] E. Parzen. "On Estimation of a Probability Density Function and Mode". In: *The Annals of Mathematical Statistics* 33.3 (1962), pp. 1065–1076.

[47] B. W. Silverman. "The Kernel Method for Multivariate Data". In: *Density Estimation for Statistics and Data Analysis*. London: Chapman and Hall, 1986.

[48] V. A. Epanechnikov. "Non-Parametric Estimation of a Multivariate Probability Density". In: *Theory of Probability and Its Applications* 14.1 (1969), pp. 153–158.

[49] L. Devroye. "Nonuniform Random Variate Generation". In: *Handbooks in Operations Research and Management Science, Vol. 13: Simulation*. Amsterdam: Elsevier, 2006, pp. 83–121.

[50] M. K. Pitt and N. Shephard. "Filtering via Simulation: Auxiliary Particle Filters". In: *Journal of the American Statistical Association* 94.446 (1999), pp. 590–599.

[51] R. Douc, E. Moulines, and J. Olsson. "Optimality of the Auxiliary Particle Filter". In: *Probability and Mathematical Statistics* 29.1 (2009), pp. 1–28.

[52] T. Schön, F. Gustafsson, and P.-J. Nordlund. "Marginalized Particle Filters for Mixed Linear/Nonlinear State-Space Models". In: *IEEE Transactions on Signal Processing* 53.7 (2005), pp. 2279–2289.

[53] G. Casella and C. P. Robert. "Rao-Blackwellisation of Sampling Schemes". In: *Biometrika* 83.1 (1996), pp. 81–94.

[54] G. Hendeby, R. Karlsson, and F. Gustafsson. "A New Formulation of the Rao-Blackwellized Particle Filter". In: *2007 IEEE/SP 14th Workshop on Statistical Signal Processing (SSP)*. 2007, pp. 84–88.

[55] R. Chen and J. S. Liu. "Mixture Kalman Filters". In: *Journal of the Royal Statistical Society: Series B* 62.3 (2000), pp. 493–508.

[56] J. Ko and D. Fox. "GP-BayesFilters: Bayesian Filtering Using Gaussian Process Prediction and Observation Models". In: *Autonomous Robots* 27.1 (2009), pp. 75–90.

[57] C. Palmier, A. Giremus, P. Minvielle, C. P. Vacar, and G. Bourmaud. "Terrain-Aided SLAM with Limited-Size Reference Maps Using Gaussian Processes". In: *2023 31st European Signal Processing Conference (EUSIPCO)*. 2023, pp. 1–5.

[58] R. J. Adler and J. E. Taylor. "Gaussian Fields". In: *Random Fields and Geometry*. Springer Monographs in Mathematics. New York, NY: Springer, 2007.

[59] H. Cramér and H. Wold. "Some Theorems on Distribution Functions". In: *Journal of the London Mathematical Society* s1-11.4 (1936), pp. 290–294.

[60] I. J. Schoenberg. "Metric Spaces and Positive Definite Functions". In: *Transactions of the American Mathematical Society* 44.3 (1938), pp. 522–536.

[61] M. L. Stein. "Linear Prediction". In: *Interpolation of Spatial Data: Some Theory for Kriging*. Springer Series in Statistics. New York, NY: Springer, 1999.

[62] B. Matérn. *Spatial Variation*. Tech. rep. 5. Stockholm: Statens Skogsforskningsinstitut, 1960.

[63] M. Abramowitz and I. A. Stegun. *Handbook of Mathematical Functions with Formulas, Graphs, and Mathematical Tables*. Vol. 55. Applied Mathematics Series. Washington, D.C.: National Bureau of Standards, 1964.

[64] C. E. Rasmussen and C. K. I. Williams. "Covariance Functions". In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.

[65] G. Matheron. "Principles of Geostatistics". In: *Economic Geology* 58.8 (1963), pp. 1246–1266.

[66] C. E. Rasmussen and C. K. I. Williams. "Regression". In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.

[67] M. Ebden. *Gaussian Processes for Regression: A Quick Introduction*. arXiv: 1505.02965. 2015.

[68] G. S. Kimeldorf and G. Wahba. "A Correspondence Between Bayesian Estimation on Stochastic Processes and Smoothing by Splines". In: *The Annals of Mathematical Statistics* 41.2 (1970), pp. 495–502.

[69] D. G. Luenberger. "Local Theory of Constrained Optimization". In: *Optimization by Vector Space Methods*. New York, NY: John Wiley & Sons, 1969.

[70] G. Matheron. *Le krigeage universel*. Tech. rep. 1. Fontainebleau: École des Mines de Paris, 1969.

[71] P. S. Palar and K. Shimoyama. "On Efficient Global Optimization via Universal Kriging Surrogate Models". In: *Structural and Multidisciplinary Optimization* (2018).

[72] M. Syirojudin, E. Haryono, and S. Ahadi. "The Accuracy of Geostatistics for Regional Geomagnetic Modeling in an Archipelago Setting". In: *Scientific Reports* 12 (2022), p. 7441.

[73] H. Omre and K. B. Halvorsen. "The Bayesian Bridge Between Simple and Universal Kriging". In: *Mathematical Geology* 21.7 (1989), pp. 767–786.

[74] A. O'Hagan. "Curve Fitting and Optimal Design for Prediction". In: *Journal of the Royal Statistical Society, Series B* 40 (1978). With discussion, pp. 1–42.

[75] C. K. I. Williams and M. Seeger. "Using the Nyström Method to Speed Up Kernel Machines". In: *Advances in Neural Information Processing Systems 13*. 2001, pp. 682–688.

[76] M. K. Titsias. "Variational Learning of Inducing Variables in Sparse Gaussian Processes". In: *Proceedings of the 12th International Conference on Artificial Intelligence and Statistics (AISTATS)*. 2009, pp. 567–574.

[77] J. Mercer. "Functions of Positive and Negative Type, and their Connection with the Theory of Integral Equations". In: *Philosophical Transactions of the Royal Society A* 209 (1909), pp. 415–446.

[78] C. Eckart and G. Young. "The Approximation of One Matrix by Another of Lower Rank". In: *Psychometrika* 1.3 (1936), pp. 211–218.

[79] L. Mirsky. "Symmetric Gauge Functions and Unitarily Invariant Norms". In: *The Quarterly Journal of Mathematics* 11.1 (1960), pp. 50–59.

[80] R. J. Adler and J. E. Taylor. "Orthogonal Expansions". In: *Random Fields and Geometry*. Springer Monographs in Mathematics. New York, NY: Springer, 2007.

[81] N. Aronszajn. "Theory of Reproducing Kernels". In: *Transactions of the American Mathematical Society* 68.3 (1950), pp. 337–404.

[82] L. Dümbgen. *Reproducing Kernel Hilbert Spaces*. Reading notes, Universität Bern. 2021.

[83] H. Daumé III. *From Zero to Reproducing Kernel Hilbert Spaces in Twelve Pages or Less*. 2004.

[84] G. Kimeldorf and G. Wahba. "Some Results on Tchebycheffian Spline Functions". In: *Journal of Mathematical Analysis and Applications* 33.1 (1971), pp. 82–95.

[85] B. Schölkopf, R. Herbrich, and A. J. Smola. "A Generalized Representer Theorem". In: *Proceedings of the 14th Annual Conference on Computational Learning Theory (COLT)*. 2001, pp. 416–426.

[86] M. Kanagawa, P. Hennig, D. Sejdinovic, and B. K. Sriperumbudur. *Gaussian Processes and Kernel Methods: A Review on Connections and Equivalences*. arXiv: 1807.02582. 2018.

[87] C. E. Rasmussen and C. K. I. Williams. "Relationships between GPs and Other Models". In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.

[88] H. Wendland. *Scattered Data Approximation*. Vol. 17. Cambridge Monographs on Applied and Computational Mathematics. Cambridge: Cambridge University Press, 2004.

[89] A. J. Smola, A. Gretton, L. Song, and B. Schölkopf. "A Hilbert Space Embedding for Distributions". In: *Proceedings of the 18th International Conference on Algorithmic Learning Theory (ALT)*. 2007, pp. 13–31.

[90] H. Brezis. "Hilbert Spaces". In: *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Universitext. New York, NY: Springer, 2011.

[91] B. K. Sriperumbudur, A. Gretton, K. Fukumizu, B. Schölkopf, and G. R. G. Lanckriet. "Hilbert Space Embeddings and Metrics on Probability Measures". In: *Journal of Machine Learning Research* 11 (2010), pp. 1517–1561.

[92] L. Song, K. Fukumizu, and A. Gretton. "Kernel Embeddings of Conditional Distributions". In: *IEEE Signal Processing Magazine* 30.4 (2013), pp. 98–111.

[93] L. Song, J. Huang, A. J. Smola, and K. Fukumizu. "Hilbert Space Embeddings of Conditional Distributions with Applications to Dynamical Systems". In: *Proceedings of the 26th International Conference on Machine Learning (ICML)*. 2009, pp. 961–968.

[94] S. Grünewälder, G. Lever, L. Baldassarre, S. Patterson, A. Gretton, and M. Pontil. "Conditional Mean Embeddings as Regressors". In: *Proceedings of the 29th International Conference on Machine Learning (ICML)*. 2012.

[95] G. H. W. Gebhardt, A. Kupcsik, and G. Neumann. "The Kernel Kalman Rule". In: *Machine Learning* 108 (2019), pp. 2113–2157.

[96] M. Sun, M. E. Davies, I. K. Proudler, and J. R. Hopgood. "Adaptive Kernel Kalman Filter". In: *IEEE Transactions on Signal Processing* 71 (2023), pp. 713–726.

[97] C. Louis and S. Reynaud. "Method of Definition of a Navigation System". US 8452536 B2. Thales. 2013.

[98] C. E. Rasmussen and C. K. I. Williams. "Model Selection and Adaptation of Hyperparameters". In: *Gaussian Processes for Machine Learning*. Massachusetts Institute of Technology: the MIT Press, 2006.

[99] D. J. C. MacKay. "Comparison of Approximate Methods for Handling Hyperparameters". In: *Neural Computation* 11.5 (1999), pp. 1035–1068.

[100] K. B. Petersen and M. S. Pedersen. *The Matrix Cookbook*. Technical University of Denmark. 2012.

[101] J. Nocedal and S. J. Wright. "Quasi-Newton Methods". In: *Numerical Optimization*. 2nd. New York, NY: Springer, 2006. Chap. 6.

[102] D. L. Zimmerman and N. Cressie. "Mean Squared Prediction Error in the Spatial Linear Model with Estimated Covariance Parameters". In: *Annals of the Institute of Statistical Mathematics* 44.1 (1992), pp. 27–43.

[103] M. L. Stein. "Predicting With Estimated Parameters". In: *Interpolation of Spatial Data: Some Theory for Kriging*. Springer Series in Statistics. New York, NY: Springer, 1999.

[104] V. Bankey et al. *Digital Data Grids for the Magnetic Anomaly Map of North America*. Tech. rep. Open-File Report 02-414. https://pubs.usgs.gov/publication/ofr02414. U.S. Geological Survey, 2002.

[105] B. Hubert, K. Dahia, N. Merlinge, and A. Giremus. "Adaptive Kriging Particle Filter and its Application to Terrain-Aided Navigation". In: *2024 27th International Conference on Information Fusion (FUSION)*. 2024, pp. 1–7.

[106] J. H. Ward. "Hierarchical Grouping to Optimize an Objective Function". In: *Journal of the American Statistical Association* 58.301 (1963), pp. 236–244.

[107] G. N. Lance and W. T. Williams. "A General Theory of Classificatory Sorting Strategies: 1. Hierarchical Systems". In: *The Computer Journal* 9.4 (1967), pp. 373–380.

[108] J. MacQueen. "Some Methods for Classification and Analysis of Multivariate Observations". In: *Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability*. Vol. 1. 1967, pp. 281–297.

[109] S. P. Lloyd. "Least Squares Quantization in PCM". In: *IEEE Transactions on Information Theory* 28.2 (1982), pp. 129–137.

[110] K. Fukunaga and L. Hostetler. "The Estimation of the Gradient of a Density Function, with Applications in Pattern Recognition". In: *IEEE Transactions on Information Theory* 21.1 (1975), pp. 32–40.

[111] D. Comaniciu and P. Meer. "Mean Shift: A Robust Approach Toward Feature Space Analysis". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 24.5 (2002), pp. 603–619.

[112] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu. "A Density-Based Algorithm for Discovering Clusters in Large Spatial Databases with Noise". In: *Proceedings of the Second International Conference on Knowledge Discovery and Data Mining*. 1996, pp. 226–231.

[113] A. Murangira. "Nouvelles approches en filtrage particulaire. Application au recalage de la navigation inertielle". PhD thesis. Université de Technologie de Troyes, 2014.

[114] B. Hubert, N. Merlinge, K. Dahia, and A. Giremus. "A Clustered Approach to Adaptive Kriging Particle Filter for Magnetic Field-Aided Navigation". In: *2025 33rd European Signal Processing Conference (EUSIPCO)*. 2025, pp. 1–5.

[115] A. Gut. *Probability: A Graduate Course*. 2nd. Springer, 2013.

[116] M. L. Stein. "The Screening Effect in Kriging". In: *The Annals of Statistics* 30.1 (2002), pp. 298–323.

[117] F. Bachoc and R. Furrer. "On the Smallest Eigenvalues of Covariance Matrices of Multivariate Spatial Processes". In: *Stat* (2016).

[118] M. Kok and A. Solin. "Scalable Magnetic Field SLAM in 3D Using Gaussian Process Maps". In: *2018 21st International Conference on Information Fusion (FUSION)*. 2018, pp. 1353–1360.

[119] N. Wahlström, M. Kok, T. B. Schön, and F. Gustafsson. "Modeling Magnetic Fields Using Gaussian Processes". In: *2013 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 2013, pp. 3522–3526.

[120] A. Solin, M. Kok, N. Wahlström, T. B. Schön, and S. Särkkä. "Modeling and Interpolation of the Ambient Magnetic Field by Gaussian Processes". In: *IEEE Transactions on Robotics* 34.4 (2018), pp. 1112–1127.

[121] A. Solin and S. Särkkä. "Hilbert Space Methods for Reduced-Rank Gaussian Process Regression". In: *Statistics and Computing* 30 (2020), pp. 419–446.

[122] A. Rahimi and B. Recht. "Random Features for Large-Scale Kernel Machines". In: *Advances in Neural Information Processing Systems 20*. 2007, pp. 1177–1184.

[123] F. Liu, X. Huang, Y. Chen, and J. A. K. Suykens. "Random Features for Kernel Approximation: A Survey on Algorithms, Theory, and Beyond". In: *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44.10 (2022), pp. 7128–7148.

[124] Z. Qin, B. Chen, and N. Zheng. "Random Fourier Feature Kernel Recursive Least Squares". In: *2017 International Joint Conference on Neural Networks (IJCNN)*. 2017, pp. 2881–2886.

[125] B. Hubert, N. Merlinge, K. Dahia, and A. Giremus. "Learning Observation Model for SLAM via Random Fourier Features in a Kernel Kalman Filter". In: *2026 29th International Conference on Information Fusion (FUSION)*. 2026, pp. 1–8.

[126] T. W. Anderson. "The Multivariate Normal Distribution". In: *An Introduction to Multivariate Statistical Analysis*. 3rd. Hoboken, NJ: John Wiley & Sons, 2003.

[127] F. Zhang, ed. *The Schur Complement and Its Applications*. Numerical Methods and Algorithms. New York, NY: Springer, 2005.

[128] M. A. Woodbury. *Inverting Modified Matrices*. Tech. rep. 42. Princeton, NJ: Statistical Research Group, Princeton University, 1950.

[129] N. I. Akhiezer and I. M. Glazman. *Theory of Linear Operators in Hilbert Space*. New York, NY: Dover Publications, 1993.

[130] P. Guttorp and T. Gneiting. "On the Matérn Correlation Family". In: *Biometrika* 93.4 (2006), pp. 989–995.

[131] W. Rudin. "Integration on Product Spaces". In: *Real and Complex Analysis*. 3rd. International Series in Pure and Applied Mathematics. New York, NY: McGraw-Hill, 1987, pp. 160–174.

[132] H. Brezis. "Compact Operators. Spectral Decomposition of Self-Adjoint Compact Operators". In: *Functional Analysis, Sobolev Spaces and Partial Differential Equations*. Universitext. New York, NY: Springer, 2011.

[133] W. Rudin. "Sequences and Series of Functions". In: *Principles of Mathematical Analysis*. 3rd. International Series in Pure and Applied Mathematics. New York, NY: McGraw-Hill, 1976, pp. 143–171.